

Core Microeconomics Notes

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January 7, 2025

Abstract

Disclaimer: These notes most definitely contain errors and are indebted to the slides created by Oxford's Economics Department, as well as useful references such as Jehle and Reny's *Advanced Microeconomics*, Kreps' *A Course in Microeconomic Theory*, and of course (but sparingly) Mas-Colell, Whinston and Green's *Microeconomic Theory*.

How to use: These notes are designed to be comprehensive rather than concise, and contained worked examples and proofs of all relevant major theorems (contained in the technical appendix). Sometimes, full proofs and proof sketches are given for intuition. All theorems are boxed up and my notation is fairly standard (boldface for vectors, superscript T for transpose). Familiarity with linear algebra and multivariate calculus beyond the précis is only needed for technical proofs, which are optional for the exams (but I believe would be useful). Familiarity with results in consumer, producer theory and partial equilibrium from introductory microeconomics are assumed.

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1 General Equilibrium

1.1 Introduction to General Equilibrium

References: Simon Cowan's slides; Jehle and Reny, *Advanced Microeconomic Theory* Ch 5; Gravelle and Rees, *Microeconomics* Ch 12-13; Kreps, *A Course in Microeconomic Theory* Ch 6

Key takeaways on General Equilibrium

1. Understand how equilibrium obtains with multiple markets, how the good markets interact with factor (input) markets, and in the production economy
2. Assumptions of GE: When does GE occur, what happens when the structural assumptions (market power, complete markets) fail, what happens when preference assumptions are not satisfied
3. Theoretical Properties of GE: Existence (and how to find), Gross Substitutes Theorem for Uniqueness; and interpretation of GE as a tatonnement process and economic intuition as to *why* people trade or do not trade
4. Theoretical properties of excess demand (i.e. Walras Law) and economic intuition

Some basic notation:

1. $\mathcal{I} = \{1, 2, \dots\}$ refers to the set of agents in the economy
2. $\mathbf{e}^i$ refers to the endowment vector possessed by the i -th agent
3. $\mathbf{x}^i$ refers to a bundle of goods purchased by the i -th agent, represented by a vector i.e. $\mathbf{x} = (x_1, x_2, \dots, x_n)$
4. $\mathbf{p}$ refers to a vector of prices for a corresponding vector of goods i.e. $\mathbf{p} = (p_1, p_2, \dots, p_n)$
5. $F(\mathbf{e})$ refers to the set of feasible allocations in the economy, which is given by

$$F(\mathbf{e}) = \left\{ \mathbf{x} : \sum_{i \in \mathcal{I}} \mathbf{x}^i \leq \sum_{i \in \mathcal{I}} \mathbf{e}^i \right\}$$

Namely, that it contains all allocations of goods across individuals that, in total, do not exceed the total amount of available goods.

1.1.1 Existence of General Equilibrium

Assumptions for General Equilibrium

Assumption 1.1 (Conditions of the market). No frictions in the market: no externalities and no distortionary taxes

Assumption 1.2 (Market Power). Agents are price takers with no market power

Assumption 1.3 (Consumer Utility). Utility u_i is continuous (and differentiable), strongly monotonic, and (strictly) quasiconcave on $\mathbb{R}^n$. By **Proposition B4**, this implies convex preferences/indifference curves.

A1.3 has important implications. The assumption of utility being strongly monotonic implies that more goods are always better while quasiconcavity of utility implies that combinations of goods are always better (rather than buying only 1 type of good and nothing else).

Proposition 1.1 (Strict Quasi-concavity and demand). If utility u^i is strictly quasi-concave and satisfies **A1.3**, then there would be unique demand i.e. a unique solution $\mathbf{x}^i(\mathbf{p}, \mathbf{p} \cdot \mathbf{e}^i)$. to the problem:

$$\max u_i(\mathbf{x}) \text{ s.t. } \mathbf{p} \cdot \mathbf{x}^i \leq \mathbf{p} \cdot \mathbf{e}^i \quad (\text{CP})$$

Further, $\mathbf{x}^i(\mathbf{p}, \mathbf{p} \cdot \mathbf{e}^i)$ is continuous in $\mathbf{p}$ on $\mathbb{R}_{++}^n$

Proof. By **Proposition B4**, strict quasi-concavity implies strict convexity of preferences. Assume for contradiction that there are two solutions to CP. Then, by strict convexity and by strong monotonicity in **A1.3**, a linear combination of these two points would lie on an indifference curve representing higher utility. Then they cannot be solutions to the consumer problem, a contradiction.

To prove continuity, we take for granted the theorem of the maximum whereby since $\mathbf{p} \gg 0$, the solution $\mathbf{x}^i(\mathbf{p}, \mathbf{p} \cdot \mathbf{e}^i)$ is continuous ■

To understand the motivation behind general equilibrium, we look at two important definitions:

Definition 1.1 (Blocking Coalitions). Let $S \subset \mathcal{I}$ denote a coalition of consumers. We say that S blocks an allocation $\mathbf{x} \in F(\mathbf{e})$ if there exists an allocation $\mathbf{y}$ such that:

1. $\sum_{i \in \mathcal{I}} \mathbf{y}^i = \sum_{i \in \mathcal{I}} \mathbf{e}^i$
2. $\mathbf{y}^i \succsim^i \mathbf{x}^i$ for all $i \in S$, with at least one preference strict

Basically, this says that the consumers in S must be able to take what they themselves have and divide it up differently among themselves so that none is worse off and at least one is better off with their assignment under $\mathbf{x}$.

Definition 1.2 (The Core of an Exchange Economy). The core of an exchange economy with endowment $\mathbf{e}$, denoted $C(\mathbf{e})$, is the set of all unblocked feasible allocations

Since an equilibrium in an exchange economy would be an allocation that is both feasible and unblocked (since otherwise consumers will move away from the allocation if they can be made better off under another allocation), we want to be assured of the existence of such an allocation and ask under what conditions such an allocation would obtain.

Definition 1.3 (Excess Demand). The aggregate excess demand function for good k is the real-valued function,

$$z_k(\mathbf{p}) \equiv \sum_{i \in \mathcal{I}} x_k^i(\mathbf{p}, \mathbf{p} \cdot \mathbf{e}^i) - \sum_{i \in \mathcal{I}} e_k^i$$

The aggregate excess demand function is the vector-valued function,

$$\mathbf{z}(\mathbf{p}) \equiv (z_1(\mathbf{p}), \dots, z_n(\mathbf{p}))$$

When $z_k(\mathbf{p}) > 0$, the aggregate demand for good k exceeds the aggregate endowment of good k and so there is excess demand for good k . When $z_k(\mathbf{p}) < 0$, there is excess supply of good k .

Proposition 1.2 (Properties of Aggregate Excess Demand Functions). If for each consumer i , u^i satisfies **A1.3**, then for all $\mathbf{p} \gg 0$

1. Continuity: $\mathbf{z}(\cdot)$ is continuous at $\mathbf{p}$
2. Homogenous to degree zero: $\mathbf{z}(\lambda \mathbf{p}) = \mathbf{z}(\mathbf{p})$ for all $\lambda > 0$
3. Walras' law: $\mathbf{p} \cdot \mathbf{z}(\mathbf{p}) = 0$

The second property says that there is no money illusion; namely that if I double all the prices of all goods, the excess demand would remain the same. The third property (Walras' law) says that the value (i.e. price times quantity) of aggregate excess demand will always be zero at any set of positive prices. Let us prove the proposition.

Proof. **Continuity** follows immediately from **Proposition 1.1**.

Homogeneity follows from inspecting the budget constraint and to realise that individual demands and excess demands are homogenous of degree zero in prices i.e. if I increase prices by some scalar λ , my budget constraint does not increase and so my demand does not change. So, aggregate excess demand is also

homogenous of degree zero in prices.

Walras' law follows from the property that u^i is strongly monotonic, and so any solution to CP means that the budget constraint holds with equality. So,

$$\mathbf{p} \cdot \mathbf{x}^i \leq \mathbf{p} \cdot \mathbf{e}^i \implies \mathbf{p} \cdot \mathbf{x}^i = \mathbf{p} \cdot \mathbf{e}^i \implies \mathbf{p} \cdot (\mathbf{x}^i - \mathbf{e}^i) = 0$$

This implies that if I sum over the number of goods and agents, I have

$$\sum_{i \in \mathcal{I}} \sum_{k=1}^n p_k (x_k^i(\mathbf{p}, \mathbf{p} \cdot \mathbf{e}^i) - e_k^i) = 0$$

We can reverse the order of summation and because summation is distributive, we yield

$$\sum_{k=1}^n p_k \left(\sum_{i \in \mathcal{I}} x_k^i(\mathbf{p}, \mathbf{p} \cdot \mathbf{e}^i) - \sum_{i \in \mathcal{I}} e_k^i \right) = 0$$

From **Definition 1.1** the term in the parenthesis is the aggregate excess demand for good k , so we have

$$\sum_{k=1}^n p_k z_k(\mathbf{p}) = 0$$

which gives

$$\mathbf{p} \cdot \mathbf{z}(\mathbf{p}) = 0$$

and the claim is proved. ■

Remark: A linear algebraic interpretation of Walras Law is that excess demands are linearly dependent. In other words, for every k excess demands, you only need $k - 1$ demands to find the solution.

Definition 1.4 (Walrasian Equilibrium). A pair $(\mathbf{x}^*, \mathbf{p}^*)$ s.t. $\mathbf{x}^*, \mathbf{p}^* \in \mathbb{R}_{++}^n$ is called a Walrasian equilibrium if $\mathbf{z}(\mathbf{p}^*) \leq \mathbf{0}$ or equivalently when (in Kreps)

$$\sum_{i \in \mathcal{I}} \mathbf{x}^i \leq \sum_{i \in \mathcal{I}} \mathbf{e}^i$$

and $(\mathbf{p}, \mathbf{x}^i)$ solves CP. In other words, a Walrasian Equilibrium obtains when excess demand is 0 or the market clears.

Remark: Technically, $\mathbf{x}^*$ is an allocation matrix, not a vector. But it suffices to just write down the allocation vectors for each consumer.

Let us be clear how this relates to the blocking conditions and the feasibility conditions. The feasibility condition maps onto the excess demand condition by definition. Further, if $(\mathbf{p}, \mathbf{x}^i)$ solves (CP), then we know that the consumer's utility is maximised, so there is no other allocation $\mathbf{y}$ under which they can be made better off (so such an allocation is unblocked).

Now, we can turn to the main result: existence of the general equilibrium.

Theorem 1.1 (Existence of Walrasian Equilibrium). Consider an exchange economy. Suppose that the consumers' preferences are continuous, strongly monotonic and convex and so satisfies Proposition 1.1 and Proposition 1.2, and that $\mathbf{w} \gg \mathbf{0}$. Then, there exists a price vector $\mathbf{p}^* \gg \mathbf{0}$ such that $\mathbf{z}(\mathbf{p}^*) \leq \mathbf{0}$

Proof. Complete Proof is in the technical appendix. ■

Motivating Remarks: Some intuition about GE and existence can be imagined through the mechanism at which an economy reaches the equilibrium. One popular example of this is known as **tâtonnement** or the **Walrasian Auctioneer**. How the Walrasian Auctioneer works is through the following steps:

1. At time t , the Auctioneer stands up in front of the crowd and calls out a price vector $\mathbf{p}$
2. Everyone checks their preferences and reports their demand together as the vector-valued function $\mathbf{x}(\mathbf{p})$
3. The Auctioneer calculates the excess demand for each good k ,

$$z_k(\mathbf{p}) = \sum_{i \in \mathcal{I}} x_k^i(\mathbf{p}) - w_k$$

4. If $\sum_{i \in \mathcal{I}} z^i(\mathbf{p}) \leq 0$, everyone sighs a breadth of relief. Otherwise, at time $t + 1$, the Auctioneer adjusts prices according to the rule

$$\frac{dp_k(t)}{dt} = \alpha_k z_k(\mathbf{p}(t))$$

where $\alpha_k > 0$ is a constant that affects speed of adjustment. In words, this adjustment rule just says, increase prices for goods that are overdemanded i.e. $z_k(\mathbf{p}) > 0$ and reduces prices for goods that are underdemanded.

5. The stationary points/stable prices of this process are the Walrasian equilibrium prices. There is local stability if there is convergence from nearby prices, and global stability if there is convergence from any prices.

Key tool: Edgeworth box for Graphical Illustration

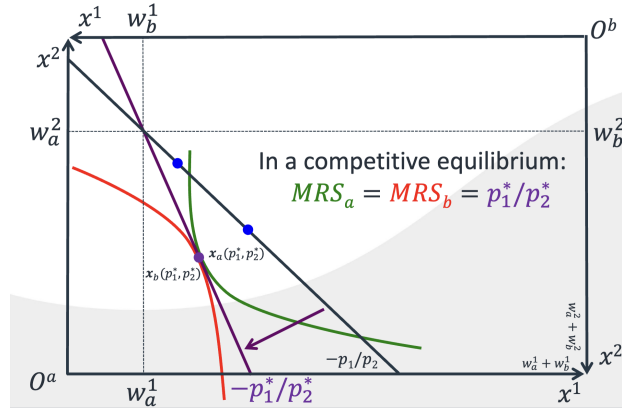


Fig 1.1: Market Clearing in Exchange Economy

This shows how the relative price $p = p_1/p_2$ adjusts so that there is a combination of goods which maximises each individual's utility subject to their budget constraint. Note that the relative price line must intersect the initial endowment in the **Edgeworth box**.

Remark: Some properties of the Edgeworth box for drawing properly in the exam

1. The relative price line must always intersect the initial endowment
2. When drawing multiple indifference curves, indifference curves cannot cross for well-behaved preferences (proof: See Varian)
3. The axes for the same good for different agents in a 2x2 economy form a 90 degree angle

1.1.2 Uniqueness of General Equilibrium

Now, we turn to the other property of Walrasian Equilibria: how many equilibria are there? In general, there are multiple WEA in an economy for a given specification of preferences and endowments so there is no global uniqueness. However, the number of WEA in an economy is also finite, so there is local uniqueness

(only one stationary point within a small neighbourhood of prices).

Example (Multiple Equilibria):

Consider an exchange economy with 2 agents and 2 goods, characterised by the utility functions:

$$u^a(x_1^a, x_2^b) = x_1^a - \frac{1}{8}x_1^{a-8} \quad \text{and} \quad u^b(x_1^b, x_2^b) = -\frac{1}{8}x_1^{b-8} + x_2^b$$

with endowments $(2, r)$ and $(r, 2)$ where $r = 2^{8/9} - 2^{1/9}$.

After solving for demands, we can construct the excess demand function for one good

$$z_1(p, 1) = (2^{8/9} - 2^{1/9})\left(\frac{1}{p} - 1\right) - p^{-8/9} + p^{-1/9}$$

The roots of the solution i.e. when $z_1(p, 1) = 0$ has three solutions $p^* \in \{\frac{1}{2}, 1, 2\}$.

But we have a sufficient condition for (global) uniqueness.

Definition 1.5 (Gross Substitutes). Goods i and j are gross substitutes at a price vector $\mathbf{p}$ if

$$\frac{\partial z_j(\mathbf{p})}{\partial p_i} > 0 \quad \forall j \neq i$$

Interpretation: An increase in the price of one good increases the excess demand of all other goods.

Theorem 1.2 (Gross substitutes implies unique equilibrium). If all goods are gross substitutes at all prices, then if $\mathbf{p}^*$ is an equilibrium price vector, it is the unique equilibrium price vector.

Proof. Suppose $\mathbf{p}'$ is some other equilibrium price vector. Since $\mathbf{p}^* \gg 0$, we can define $m = \max p'_i / p_i^* = 0$. By homogeneity and the fact that $\mathbf{p}^*$ is an equilibrium, we know that $\mathbf{z}(\mathbf{p}^*) = \mathbf{z}(m\mathbf{p}^*) = \mathbf{0}$. We know that for some price p_k , we have $mp_k^* = p'_k$ by the definition of m . We now lower each price $mp^{*'}_i$ other than p_k successively to p'_i . Since the price of each good other than k goes down in movement from $m\mathbf{p}^*$ to $\mathbf{p}'$, we must have the demand for good k going down. Thus $z_k(\mathbf{p}') < 0$ which implies $\mathbf{p}'$ cannot be an equilibrium. ■

1.1.3 Production Economies and General Equilibrium

Abstract Set Up

1. Firms are denoted $j \in \{1, \dots, m\}$
2. Language of net inputs: a production plan $\mathbf{y}^j \in \mathbb{R}^n$ has a positive entry if a good is a net output and a negative entry if a good is a net input
3. Set of production possibilities for firm j denoted Y^j dictated by the firm's production technology
4. A production plan $\mathbf{y}$ is efficient if there is no $\mathbf{y}' \in Y^j$ such that $\mathbf{y}' \geq \mathbf{y}$ and $\mathbf{y}' \neq \mathbf{y}$

Example: Suppose there are three goods y, x_1, x_2 and y is produced by using two goods x_1, x_2 in a Cobb-Douglas technology

$$Y^j = \{(y, -x_1, -x_2) \in \mathbb{R}^3 \mid x_1^\alpha x_2^{1-\alpha} \geq y\}$$

Firm j 's profit at competitive prices $\mathbf{p}$ is given by $\pi(\mathbf{p}, \mathbf{y}^j) = \mathbf{p} \cdot \mathbf{y}^j$.

Firm chooses $\mathbf{y}^{j*}$ by solving maximisation problem:

$$\max_{\mathbf{y}^j \in Y^j} \mathbf{p} \cdot \mathbf{y}^j$$

Then $\mathbf{y}^j(\mathbf{p})$ denotes the supply function, giving optimal $\mathbf{y}^{j*}$ at each $\mathbf{p}$.

The aggregate profit of all firms is given by

$$\sum_{j=1}^m \mathbf{p} \cdot \mathbf{y}^j(\mathbf{p})$$

Assumptions on Technology

Assumption 1.4 (Convex Production Possibility Set). For any two production plans $\mathbf{y}, \mathbf{y}'$

$$\mathbf{y}, \mathbf{y}' \in Y^j \implies \alpha \mathbf{y} + (1 - \alpha) \mathbf{y}' \in Y^j \quad \text{for all } \alpha \in (0, 1)$$

i.e. balanced input combinations are no less productive than unbalanced ones

Assumption 1.5 (Shut Down). $\mathbf{0} \in Y^j$ so inaction is possible

Assumption 1.6 (Free Disposal). If $\mathbf{y} \in Y^j$ and $\mathbf{y}' \leq \mathbf{y}$, then $\mathbf{y}' \in Y^j$.

Example: If $(y, x_1, x_2) = (2, -1, -1) \in Y^j$, then $(y', x'_1, x'_2) = (1, -2, -2) \in Y^j$ i.e. firms can destroy output or eliminate unnecessary inputs at no cost

Assumption 1.7 (Production is irreversible). If $Y = \sum_{j=1}^m Y^j$, then $\mathbf{y} \in Y \setminus \{\mathbf{0}\} \implies -\mathbf{y} \in Y$. Firms cannot cooperate to produce an infinite amount of goods i.e. if firm 1 produced $y = 1$ with $x = 1$ and firm 2 produced $x = 2$ with $y = 1$

Remark: If each Y^j satisfies assumptions 1.4-1.7, then the aggregate production possibility set Y , also satisfies Assumptions 1.4-1.7.

Consumers

1. Consumers are indexed $i \in \{1, \dots, I\}$ and u^i denotes the i -th consumer's utility function over the consumption set $\mathbb{R}_+^n$
2. Consumers are endowed with 1 unit of time
3. Utility is $u(x, R)$ where $R = 1 - L$ i.e. L is the proportion of hours on work while R is the proportion of hours on leisure
4. We consider a private ownership economy, where consumer i 's shares in firm j entitles him to some proportion $0 \leq \theta^{ij} \leq 1$ of the profits of firm j . These shares, summed over all consumers in the economy, must sum to 1

The economy is represented by the collection $(u^i, \mathbf{e}^i, \theta^{ij}, Y^j)_{i \in \mathcal{I}, j \in \mathcal{J}}$.

Definition 1.6 (Excess Demand in Production Economy). The aggregate excess demand function for good k is the real valued function

$$z_k(\mathbf{p}) = \sum_{i \in \mathcal{I}} x_k^i(\mathbf{p}, \mathbf{p} \cdot \mathbf{e}^i) - \sum_{j \in \mathcal{J}} y_k^j(\mathbf{p}) - \sum_{i \in \mathcal{I}} \mathbf{e}^i$$

The aggregate excess demand function is the vector-valued function

$$\mathbf{z}(\mathbf{p}) \equiv (z_1(\mathbf{p}), z_2(\mathbf{p}), \dots, z_n(\mathbf{p}))$$

Theorem 1.3 (Existence of Walrasian Equilibrium with Production). Consider the economy $(u^i, \mathbf{e}^i, \theta^{ij}, Y^j)_{i \in \mathcal{I}, j \in \mathcal{J}}$. If the consumers' preferences are continuous, strongly monotonic and convex, and each Y^j satisfies the assumptions on technology and $\mathbf{y} + \sum_{i \in \mathcal{I}} \mathbf{e}^i \gg \mathbf{0}$ for some aggregate production vector $\mathbf{y} \in \sum_{j \in \mathcal{J}} Y^j$, then there exists at least one price vector $\mathbf{p}^* \gg \mathbf{0}$, such that $\mathbf{z}(\mathbf{p}^*) = \mathbf{0}$

Graphical Illustration:

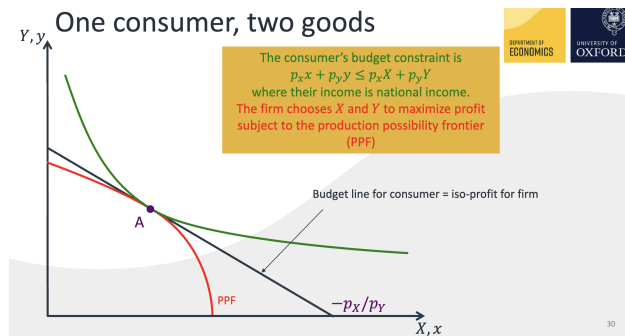


Fig. 2: Production Economy

Motivation: The basic motivation is that the consumer is trying to maximise utility and thus determines their labour (and leisure) production accordingly which in turn determines their income. The producer minimises cost to give their demands for inputs (labour and capital). They try to maximise profit. Then you construct the excess demands for each good in the economy (including labour and capital) to find the equilibrium price vector and allocation matrix.

1.2 International Trade and Multiple Industries

References: Simon Cowan's slides; Krugman and Obstfeld, *International Economics* Ch 3,4.

Key takeaways on Trade

1. The ability of the country to trade at any world prices other than autarkic prices makes the consumer better off
2. Direction of trade is irrelevant for improving consumer's utility
3. Specialisation is not necessary for utility improvement

When does everyone benefit from trade?

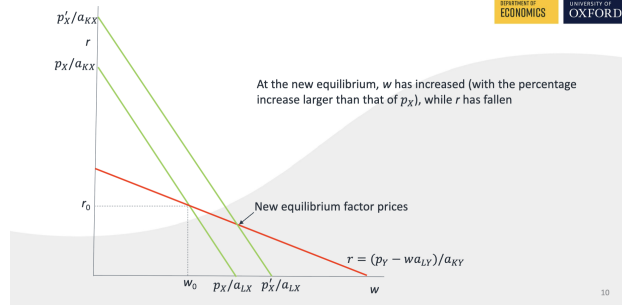
1. If (but not only if) there is equal ownership of all factors of production
2. Key is that national income rises with free trade, provided free trade prices differ from autarkic ones
3. Everyone can consume what they did with autarky, and they have additional opportunities for consumption

Abstract set up for Two Industries:

1. Two factors, capital K and labour L
2. Price of K is r and wage of labour is w
3. Two goods in the economy, X which is labour-intensive, and Y which is capital-intensive
4. For any wage-rental ratio w/r , optimal K/L is higher for good Y than for good X

Assumptions on Firms and Factors

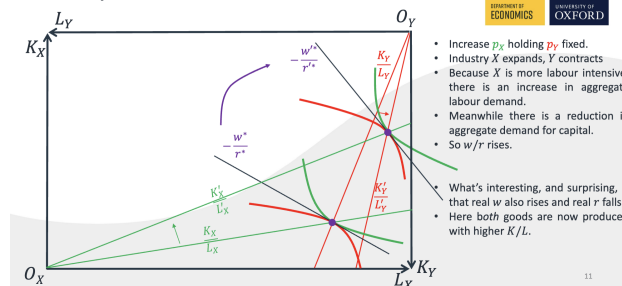
Effect of an increase in p_X



Therefore, if p_X increases, the factor price of the factor that X is more intensive (labour) in increases.

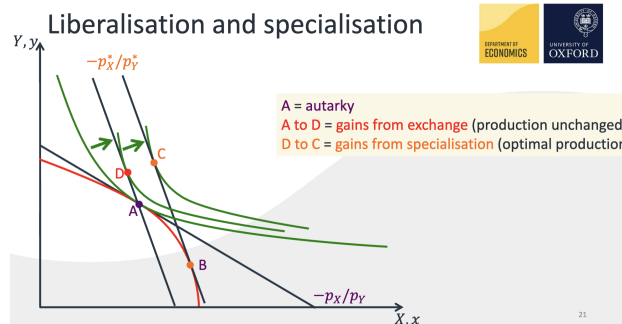
In a general equilibrium model we see the same effect:

Stolper-Samuelson Theorem



Takeaways from Trade:

- The ability of the country to trade at any world prices other than autarkic prices makes the consumer better off
- Direction of trade is irrelevant for improving consumer's utility
- Specialization is not necessary for improvement



Example: Trade and Intensity of Factors. Country A has two industries, each employing capital and labour to produce output, with constant-returns-to-scale production functions

$$y_1 = K_1^{\frac{1}{3}} L_1^{\frac{2}{3}}, \quad y = K_2^{\frac{2}{3}} L_2^{\frac{1}{3}}$$

Factor and product markets are competitive; the prices of the two products are p_1 and p_2 , and the prices of labour and capital are w and r .

1. By solving the cost minimisation problem for a firm in industry 1 producing an amount of output y_1 , show that:

- the optimal capital-labour ratio in industry 1 is: $\frac{K_1}{L_1} = \frac{w}{2r}$
- the marginal cost of y_1 is: $MC_1 = \left(\frac{w}{2/3}\right)^{\frac{2}{3}} \left(\frac{r}{1/3}\right)^{\frac{1}{3}}$
- and hence similarly: $\frac{K_2}{L_2} = \frac{2w}{r}, MC_2 = \left(\frac{w}{1/3}\right)^{\frac{1}{3}} \left(\frac{r}{2/3}\right)^{\frac{2}{3}}$

Which good is more labour intensive?

The firm's cost function is $C = rK_1 + wL_1$ and faces

$$\min C = rK_1 + wL_1 \quad \text{s.t.} \quad y_1 \geq K_1^{\frac{1}{3}} L_1^{\frac{2}{3}}$$

As usual, set up the Lagrangian as

$$\mathcal{L}(K_1, L_1, \lambda) = rK_1 + wL_1 - \lambda(K_1^{\frac{1}{3}} L_1^{\frac{2}{3}} - y_1)$$

The FOCs, solved, yields us

$$\frac{K_1}{L_1} = \frac{w}{2r}$$

Since we know that $c'(y_1) = \lambda$, we solve for λ which yields

$$\lambda = \left(\frac{w}{2/3}\right)^{\frac{2}{3}} \left(\frac{r}{1/3}\right)^{\frac{1}{3}}$$

Repeat for firm 2 and solve for λ . We get

$$\frac{K_2}{L_2} = \frac{2w}{r}$$

and

$$c'(y_2) = \lambda = \left(\frac{w}{1/3}\right)^{\frac{1}{3}} \left(\frac{r}{2/3}\right)^{\frac{2}{3}}$$

Good 1 is more labour intensive since $\frac{K_1}{L_1} < \frac{K_2}{L_2}$

2. The country has an endowment of 160 units of capital and 200 units of labour. Use the optimal capital-labour ratios above to determine how much capital and labour will be used in each industry, as functions of the factor price ratio. Illustrate the factor market equilibrium in an Edgeworth box for the case when $w/r = 1$. How would the output of the two industries change if the endowment of labour decreased?

This is just an exercise in algebra. You have $L_1 + L_2 = 200$, $K_1 + K_2 = 160$. Using the ratios, you yield

$$\frac{2w}{r} L_2 + \frac{w}{2r} L_1 = 160$$

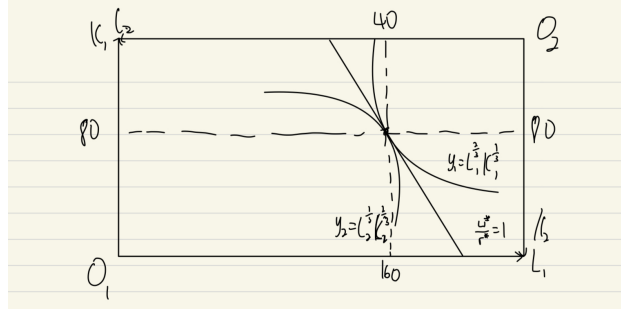
which is a system of two linearly independent equations with 2 variables. So, solving and repeating for K_1, K_2 gives us the following

$$\begin{aligned} L_1 &= \frac{800}{3} - \frac{320}{3} \left(\frac{r}{w}\right) \\ L_2 &= \frac{320}{3} \left(\frac{r}{w}\right) - \frac{200}{3} \\ K_1 &= \frac{400}{3} \left(\frac{w}{r}\right) - \frac{160}{3} \\ K_2 &= \frac{640}{3} - \frac{400}{3} \left(\frac{w}{r}\right) \end{aligned}$$

when $w/r = 1$, we get

$$K_1 = 80, K_2 = 80, L_1 = 160, L_2 = 40$$

and the edgeworth box is



If the endowment of labour decreased, since relative prices are fixed at $w/r = 1$ and both goods continue to be produced, by the Rybczynski theorem, a fall in supply of labour will lead to a fall in output of y_1 and an increase in output of y_2 since industry 1 uses labour more intensively.

3. Use the conditions that price equals marginal cost in both industries to show that there is a one-to-one relationship between relative factor prices and relative product prices:

$$\frac{w}{r} = \left(\frac{p_1}{p_2} \right)^3$$

How does the wage change if demand for product 1 increases? Explain this result intuitively. Again, this is an exercise in algebra. Set

$$p_1 = MC_1 = \left(\frac{w}{2/3} \right)^{2/3} \left(\frac{r}{1/3} \right)^{1/3}$$

$$p_2 = MC_2 = \left(\frac{w}{1/3} \right)^{1/3} \left(\frac{r}{2/3} \right)^{2/3}$$

Then,

$$\frac{p_1}{p_2} = \frac{\left(\frac{w}{2/3} \right)^{2/3} \left(\frac{r}{1/3} \right)^{1/3}}{\left(\frac{w}{1/3} \right)^{1/3} \left(\frac{r}{2/3} \right)^{2/3}} = \left(\frac{w}{r} \right)^{1/3}$$

So,

$$\frac{w}{r} = \left(\frac{p_1}{p_2} \right)^3$$

If demand for good 1 increases, the firm would need to produce more and thus demand for more factor inputs in which it is more intense in, resulting in the rise in demand for labour relative to capital, hence the relative price of wage rises

4. How do your results relate to the Stolper-Samuelson and Rybczynski Theorem?

The Rybczynski theorem shows that if labour supply falls, output in y_1 falls while output of y_2 rises given that y_1 uses labour more intensively. The Samuelson-Stolper theorem confirms the partial intuition above that the price increase in p_1 leads to a rise in the real return to labour in y_1 but also tells us that the real return to capital in y_1 reduces.

There is a second country B, identical to A except that it has a smaller endowment of labour. Consumers in the two countries have identical preferences: the relative demand for the two goods, $\frac{y_1}{y_2}$, is a function only of the relative prices, $\frac{p_1}{p_2}$.

5. Initially, both countries are both in autarkic equilibrium. Suppose that product prices in country A (determined by consumer demand) are equal: $p_1/p_2 = 1$. What can you say about relative product prices and relative factor prices in country B?

If B has less labour supply, then the wage rate will be higher since the labour market faces the same demand for labour. So,

$$\frac{w^B}{r^B} > \frac{w^A}{r^A}$$

Since $\frac{w}{r} = \left(\frac{p_1}{p_2}\right)^3$, and $\frac{w^B}{r^B} = \left(\frac{p_1^B}{p_2^B}\right)^3 > \frac{w^A}{r^A} = \left(\frac{p_1^A}{p_2^A}\right)^3 = 1$,

$$\left(\frac{p_1^B}{p_2^B}\right)^3 > 1 \implies p_1^B > p_2^B \implies w^B > r^B$$

6. Now suppose that the countries can trade freely with each other. Describe what happens to relative product and factor prices in the two countries. Which goods will be imported and exported by country A? What happens in country A to employment and output in the two industries?

The relative product and factor prices will converge i.e. $\frac{p_1^A}{p_2^A}$ will increase from 1 and $\frac{p_1^B}{p_2^B}$ will decrease s.t.

$$\frac{p_1^A}{p_2^A} = \frac{p_1^B}{p_2^B}$$

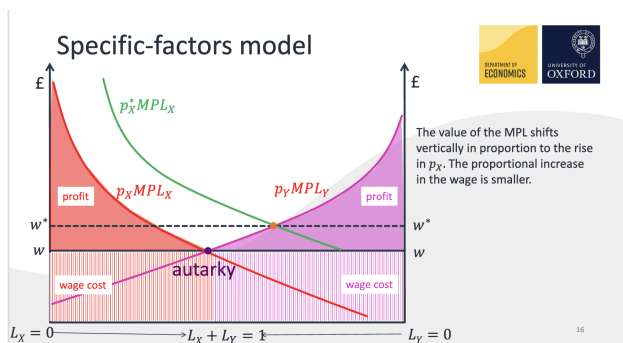
this is because A will specialise in A so p_1^A rises due to exporting good 1 while p_2^B rises since B will specialise in good 2 and export good 2. By the Samuelson-Stolper Theorem, since $\frac{p_1^A}{p_2^A}$ increases, w^A

increases while r^A decreases. Since $\frac{p_1^B}{p_2^B}$ falls, w^B falls while r^B rises. Country A will export good 1 and import good 2 while Country A will face higher employment since labour is used more intensively in its exported good 1. Output for y_1 will rise while output for y_2 will contract.

Specific Factors Model: A model for the short run. Our assumption has been that in the long run, K can be used for both goods 1 and 2. But consider that in the short run, K is not so substitutable i.e. K_X is specific to X and K_Y is specific to Y .

Equilibrium has the value

$$p_X MPL_X = w = p_Y MPL_Y$$



If the economy exports X during trade because p_X/p_Y is higher will lead to

- Increase production of X and decrease production of Y
- Raise the return on the specific factor in X (and hence real profits) and the reverse for Y
- Have effects on workers that depend on what they buy:
 - If a worker only buys y she is better off as the real wage in terms of X has risen
 - If a worker only buys x she is worse off because her real wage in terms of X has fallen (this is what encourages X industry to higher more workers)

Example. Production functions for food and cloth are

$$C(L_C) = 2L_C^{0.5}, \quad F(L_F) = 2L_F^{0.5}$$

There is one unit of capital in the cloth industry and one unit of land in the food industry. There are specific factors that earn profits. Total labour supply is 1 so $L_C + L_F = 1$. Labour moves freely between the industries. Price of C is p and price of F is normalised to 1, wage is w . Equilibrium in the labour market is

$$p_X MPL_X = w = p_Y MPL_Y \implies pL_C^{-0.5} = L_F^{-0.5} = w$$

So, $L_C = p^2 L_F$ and with $L_C + L_F = 1$, we yield

$$L_C = \frac{p^2}{1+p^2}, \quad L_F = \frac{1}{1+p^2}$$

Now we can find w as a function of p , obtaining $w = (1+p^2)^{0.5}$. The supply of the goods are now

$$C_s(p) = \frac{2p}{(1+p^2)^{0.5}}, \quad F_s(p) = \frac{2}{(1+p^2)^{0.5}}$$

Assume that utility is Cobb-Douglas with equal weights and half of national income is spent on each good so national income is the value of output

$$Y = pC_s + F_s = 2(1+p^2)^{0.5}$$

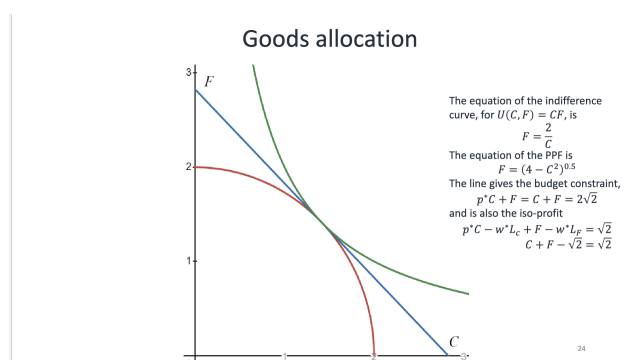
and $C_d(p) = \frac{Y}{2p} = \frac{(1+p^2)^{0.5}}{p}$, $F_d(p) = \frac{Y}{2} = (1+p^2)^{0.5}$. This derives from the standard Cobb-Douglas demand solutions and the fact that national income is the income of the aggregate worker. So, the equilibrium is

$$F_d(p) = F_s(p) \implies \frac{2}{(1+p^2)^{0.5}} = (1+p^2)^{0.5}$$

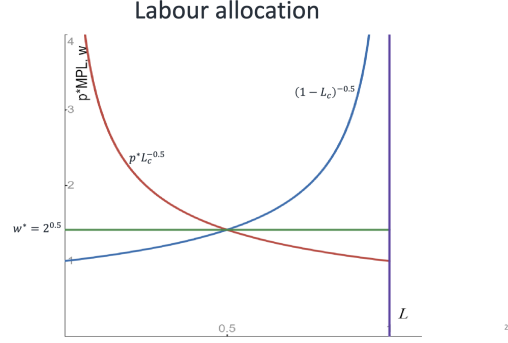
Hence

- Equilibrium price: $p^* = 1$
- Goods allocation: $C_s^* = 2^{0.5} = C_d^*, F_s^* = 2^{0.5} = F_d^*$
- Labour allocation: $L_C^* = L_F^* = \frac{1}{2}$
- Wage and profits: $w^* = 2^{0.5}, \pi_C^* = C_s^* - \frac{w^*}{2} = \frac{1}{2^{0.5}}, \pi_F^* = F_s^* - \frac{w^*}{2} = \frac{1}{2^{0.5}}$
- National income: $p^* C_d^* + F^* d = p^* C_s^* + F^* s = w^* + \pi_C^* + \pi_F^* = 2\sqrt{2}$

The goods allocation graph is



The labour allocation graph is



Suppose that the country can trade freely at the world price $p > 1$ and the country will export cloth and import food since $p > p^* = 1$. The demand for cloth is $L_C(w, p) = \left(\frac{p}{w}\right)^2$ increases and thus the real wage w/p in terms of cloth is lower. Since $w(p) = (1 + p^2)^{0.5}$, we have the profit functions

$$\pi_C(p) = pC_d(p) - w(p)C_s(p) = \frac{p^2}{(1 + p^2)^{0.5}}$$

$$\pi_F(p) = F_d(p) - w(p)F_s(p) = \frac{1}{(1 + p^2)^{0.5}}$$

Profits in food go down when p increases. The price of food is unchanged so the landowner can buy less food. The landowner can also buy less cloth (lower income, higher prices). Cloth profits are higher, despite the wage increasing. The capitalist, who owns the cloth firm, can buy more of both goods. Real cloth profit is higher in terms of both goods: we write labour demand in cloth as a function of p only i.e. $L(p)$

$$\frac{\pi_C(p)}{p} = C(L(p)) - \frac{w(p)}{p}L(p) > C(L(1)) - \frac{w(p)}{p}L(1) > C(L(1)) - w^*L(1) = \pi_C(1)$$

The first inequality follows by profit maximisation since lower profit with the old labour input at new real wage. The second inequality follows because $\frac{w(p)}{p} < w^*$. Even lower profit is gained if one pays the old real wage because that is just the old real profit. Because $p > 1$ profit in terms of food is also higher, with $\pi_C(p) > \frac{\pi_C(p)}{p} > \pi_C(1)$.

Effects of higher p on real wages

- A worker who buys only cloth is worse off: w/p is lower
- A worker who buys only food is better off: w rises

1.3 Finding General Equilibrium in different contexts

Cookbook Approach to finding General Equilibrium

1. Normalise the price of one good to 1
2. Find the demand and supply functions of each agent and for every good
3. Express demand and supply as functions of prices only: calculate value of the endowment or of profits and substitute into demand functions for income m
4. Calculate aggregate demand, aggregate supply, and aggregate endowment for each good
5. Obtain the excess demand for each good and set it to zero (market-clearing condition)
6. Since you have a market clearing condition for every good k : k equations with k unknowns. You must explain that by the existence theorem (that the consumer preferences and endowment satisfy assumptions), an equilibrium exists. You must also explain that by Walras Law, you only need to solve $k - 1$ of the conditions
7. Solve for the relative price
8. Once you've found equilibrium prices, put them into the demand and supply functions to find the allocations. Always specify both the equilibrium allocation AND the equilibrium price vector.

Remark: Since $\mathbf{p}^*$ is a price vector, any scalar multiple, $\lambda \mathbf{p}^*$ is also a solution to the problem, for all $\lambda \in \mathbb{R}_+$. This follows the intuition of the no money illusion condition, that excess demands are homogenous to degree 0.

1.3.1 Exchange Economy: Two Consumer-Two Good Case

Standard 2x2 Economy with Cobb-Douglas Preferences.

Consider an economy with two goods, 1 and 2, and two consumers a and b . The preferences of the consumers are represented by the utility functions:

$$u^a(x_1^a, x_2^a) = 2\ln x_1^a + 3\ln x_2^a \quad \text{and} \quad u^b(x_1^b, x_2^b) = 2\ln x_1^b + \ln x_2^b$$

Consumer a has an endowment $(w_1^a, w_2^a) = (20, 0)$ and Consumer b has an endowment $(w_1^b, w_2^b) = (0, 20)$.

Solution:

Step 1: Normalise the price of good 2 to 1. So $\rho = p_1/p_2$. Therefore, we write the budget constraints as

$$\rho x_1^a + x_2^a \leq m^a \quad \text{and} \quad \rho x_1^b + x_2^b \leq m^b$$

Step 2: Solve for demands through one of 2 methods.

1. Derive demands **by inspection:** Given that the utility function is Cobb-Douglas in nature, we know that the demands of the consumer are proportionate to the exponents on each good. Therefore, in this case, the demands are

$$\begin{aligned} x_1^a &= \frac{2m^a}{5\rho} \quad \text{and} \quad x_2^a = \frac{3m^a}{5} \\ x_1^b &= \frac{2m^b}{3} \quad \text{and} \quad x_2^b = \frac{m^b}{3} \end{aligned}$$

2. Derive demands via **method of the Lagrange Multipliers:** We write the consumers' maximisation problems as

$$\max u^a(x_1^a, x_2^a) \quad \text{s.t.} \quad \rho x_1^a + x_2^a \leq m^a, x_1^a \geq 0, x_2^a \geq 0 \quad (\text{CP-A})$$

and

$$\max u^b(x_1^b, x_2^b) \text{ s.t. } \rho x_1^b + x_2^b \leq m^b, x_1^b \geq 0, x_2^b \geq 0 \quad (\text{CP-B})$$

Now, we need to argue that CP-A (and CP-B) are equivalent to the following problem:

$$\max u^a(x_1^a, x_2^a) \text{ s.t. } \rho x_1^a + x_2^a = m^a$$

In other words, we need to argue that solution to CP must be when the budget constraint binds with equality, and that the positivity constraints will never bind.

We proceed with the argument that the budget constraint must bind with equality. First, the consumer's utility function is **continuous**, so it is possible for the consumer to use up all his income by consuming ϵ of some good to finish up his income. Second, the consumer's utility function, by assumption, satisfies **local non-satiation** (or **strict monotonicity**). Hence, we are assured that the consumer will maximise their utility by exhausting their budget i.e. the budget binds with equality.

Now, we argue that the weak positivity constraints since the consumer would choose strictly positive quantities even if unconstrained. This is because that the consumer's utility is given by the logarithms of goods 1 and 2. So if the consumer approaches 0 of either good, they approach negative infinity utility and hence experience infinite marginal utility if they instead consume $\epsilon > 0$ of either good.

Similarly, this applies to CP-B.

So, we can set up the Lagrangian for CP-A.

$$\mathcal{L}(x_1^a, x_2^a, m^a) = 2\ln x_1^a + 3\ln x_2^a - \lambda(\rho x_1^a + x_2^a - m^a)$$

We now get our $n + 1$ FOCs:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x_1^a} &= \frac{2}{x_1^a} - \lambda \rho = 0 \\ \frac{\partial \mathcal{L}}{\partial x_2^a} &= \frac{3}{x_2^a} - \lambda = 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= m^a - \rho x_1^a - x_2^a = 0 \end{aligned}$$

Solving both CP-A (and CP-B similarly), we get

$$\begin{aligned} x_1^a &= \frac{2m^a}{5\rho} \text{ and } x_2^a = \frac{3m^a}{5} \\ x_1^a &= \frac{2m^b}{3} \text{ and } x_2^a = \frac{m^b}{3} \end{aligned}$$

SOCs: since u^a is quasi-concave and the constraint is linear and thus quasi-convex, the solution solves CP-A and CP-B.

Step 3: Substitute incomes in terms of endowments. Our endowments here are

$$\begin{aligned} m^a &= \rho w_1^a + w_2^a = 20\rho \\ m^b &= \rho w_1^b + w_2^b = 20 \end{aligned}$$

Hence, our final demands in terms of relative price are

$$x_1^a = 8, x_2^a = 12\rho; x_1^b = 8\rho, x_2^b = 4$$

Remark: The good that I trade with is unconditional on relative price; while the good that I trade for is conditional on relative price.

Step 4: Write the excess demands. The definition of an excess demand function for good k is

$$z_k(\mathbf{p}) = \sum_{i \in \mathcal{I}} x_k^i(\mathbf{p}, \mathbf{p} \cdot \mathbf{e}^i) - \sum_{i \in \mathcal{I}} \mathbf{e}_k^i$$

$$\begin{aligned} z_1((\rho, 1)) &= 8 + \frac{8}{\rho} - 20 = \frac{8}{\rho} - 12 \\ z_2((\rho, 1)) &= 12\rho + 4 - 12 = 12\rho - 8 \end{aligned}$$

Remark: It is easy to verify that these demands satisfy Gross Substitutes, so there is a unique equilibrium.

Remark: If the excess demands are linearly dependent, then they satisfy Walras Law and you're probably correct.

Step 5: Set the excess demands to 0 and solve. Due to Walras' Law, if $n - 1$ markets clear, n markets clear. So we can just consider good 2.

$$z_2((\rho, 1)) = 12\rho - 8 = 0 \implies \rho = \frac{2}{3}$$

Remark: Any scalar of ρ is a valid solution so if you normalise the price of good 1, it is still okay.

Step 6: Substitute the relative prices into the demands, and write down the competitive equilibrium which is both the equilibrium price vector and the equilibrium allocation.

Our price vector is

$$\mathbf{p} = \left(\frac{2}{3}, 1\right)$$

Our allocations are

$$(x_1^a, x_2^a) = (8, 8) \text{ and } (x_1^b, x_2^b) = (12, 4) \quad \blacksquare$$

2x2 Pure Exchange Economy with Quasilinear utility functions.

Consider a pure exchange economy with two goods, x and y , and two consumers, a and b , that trade the goods. The preferences of consumer a are represented by the utility function

$$u_a(x_a, y_a) = x_a + \ln y_a$$

and the preferences of consumer b are represented by the utility function

$$u_b(x_b, y_b) = x_b + \ln y_b$$

There are two ways to solve this problem.

1. Shorter Way through contract curve condition.

We know that $MRS_a = \frac{u_x}{u_y} = y_a$ and $MRS_b = \frac{u_x}{u_y} = y_b$. Since we know that equilibrium occurs when $MRS_a = MRS_b$, this implies that $y_a = y_b$. Since $y_a + y_b = 4$, this means that $y_a = y_b = 2$. Since we normalise the price of good x to 1, our budget constraints are

$$\begin{aligned} x_a + 2\rho &= m_a \\ x_b + 2\rho &= m_b \end{aligned}$$

Since we know that MRS_a equals the slope of the budget line which is $1/\rho$, we know that $\rho = \frac{1}{2}$. Suppose consumer a has endowments $(4, 0)$ and b has endowments $(0, 4)$.

From our budget constraints, we have $x_a + 1 = 4 \implies x_a = 3$ and $x_b + 1 = 4 \implies x_b = 1$

2. Longer way: Consider consumer a 's maximisation problem as

$$\begin{aligned} \max u_a(x_a, y_a) &= x_a + \ln y_a \\ \text{subject to} \\ x_a + \rho y_a &\leq m_a, x_a \geq 0, y_a \geq 0 \end{aligned}$$

Arguing away the positivity constraints and that the budget constraint binds with equality, we construct the Lagrangian

$$\mathcal{L}(x_a, y_a, \lambda) = x_a + \ln y_a - \lambda(x_a + \rho y_a - m_a)$$

Our FOCs are:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x_a} &= 1 - \lambda = 0 \\ \frac{\partial \mathcal{L}}{\partial y_a} &= \frac{1}{y_a} - \lambda \rho = 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= m_a - x_a - \rho y_a = 0 \end{aligned}$$

which gives us $y = \frac{1}{\rho}, x_a = m_a - 1$. Note due to quasilinearity, below a certain amount of y , a will just consume y . Given additive separability, we find the condition below which consumer a will just consume y . There are two ways to do this:

First, we set the condition as

$$\frac{\partial u_a / \partial x_a}{1} \leq \frac{\partial u_a / \partial y_a}{\rho} \implies \frac{\partial u_a / \partial x_a}{\partial u_a / \partial y_a} \leq \frac{1}{\rho}$$

Since

$$\frac{\partial u_a / \partial x_a}{\partial u_a / \partial y_a} = y_a \implies y_a \leq \frac{1}{\rho}$$

The intuition is that y_a will only be consumed if the marginal utility per dollar is greater for y than for x . Hence, this means that $y_a = \frac{m_a}{\rho}$ and $x_a = 0$, when $m_a \leq 1$.

Alternatively, we know that it will be a corner solution if the budget constraint is steeper than the utility curve at all points. So, the slope of the budget line is $-\frac{1}{\rho}$ while the slope of the utility curve is just its MRS which is $-y_a$. Hence, we set

$$-y_a \geq -\frac{1}{\rho} \implies y_a \leq \frac{1}{\rho}$$

Again, we get $y_a = \frac{m_a}{\rho}$ and $x_a = 0$, when $m_a \leq 1$.

Our demands are piecewise

$$y_a = \begin{cases} m_a / \rho & m_a \leq 1 \\ 1 / \rho & m_a > 1 \end{cases}$$

and

$$x_a = \begin{cases} 0 & m_a \leq 1 \\ m_a - 1 & m_a > 1 \end{cases}$$

Since a has an endowment $(4, 0)$ such that $m_a = 4$, then $x_a = 3, y_a = \frac{1}{\rho}$.

For consumer b the problem is exactly the same with the same demand functions but since $m_b = 4\rho$, it is not clear whether it is a corner solution or not. So, we need to test it out by rewriting our demand functions first as

$$y_b = \begin{cases} m_b / \rho & \rho \leq 1/4 \\ 1 / \rho & \rho > 1/4 \end{cases}$$

and

$$x_b = \begin{cases} 0 & \rho \leq 1/4 \\ m_b - 1 & \rho > 1/4 \end{cases}$$

We have two cases to consider.

Case 1: $\rho > \frac{1}{4}$. We set excess demand to 0.

$$z_x(1, \rho) = 3 + 4\rho - 1 - 4 = 4\rho - 2 = 0$$

$$z_y(1, \rho) = \frac{1}{\rho} + \frac{1}{\rho} - 4 = 2/\rho - 4 = 0$$

which yields $\rho = \frac{1}{2}$. So, our demands are $(x_a, y_a) = (3, 2), (x_b, y_b) = (1, 2)$.

Case 2: $\rho \leq \frac{1}{4}$. Again we test if there is an equilibrium:

$$z_y(1, \rho) = 4 + \frac{1}{\rho} - 4 = 1/\rho \neq 0$$

So, no equilibrium.

Now, consider the endowments $(0, 1), (4, 3)$ respectively. We want to show that no trade is possible in this case. Here, we want to examine a 's demands and not b 's since $m_b = 4 + 3\rho > 1$. So, $x_b = 3 + 3\rho, y_b = 1/\rho$. We write consumer a 's demands as

$$x_a = \begin{cases} 0, & \rho \leq 1 \\ \rho - 1, & \rho > 1 \end{cases}$$

and

$$y_a = \begin{cases} 1, & \rho \leq 1 \\ 1/\rho, & \rho > 1 \end{cases}$$

Case 1: let $\rho > 1$. Then, excess demands are

$$z_x(1, \rho) = \rho - 1 + 3 + 3\rho - 4 = 4\rho - 2 = 0$$

$$z_y(1, \rho) = 2/\rho - 4 = 0$$

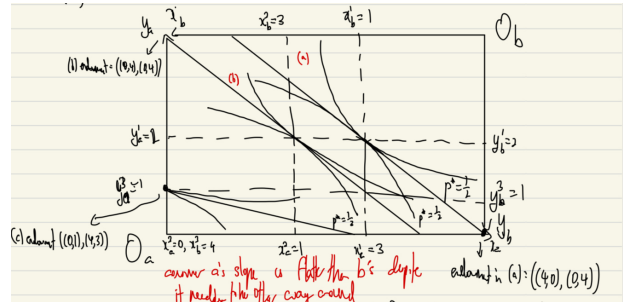
So, $\rho = 1/2 < 1$ which is a contradiction.

Case 2: $\rho \leq 1$ so $x_a = 0, y_a = 1$. So,

$$z_x(1, \rho) = 3\rho - 1 = 0$$

$$z_y(1, \rho) = 1/\rho - 3 = 0$$

Hence, $\rho = \frac{1}{3}$. But then, $(x_a, y_a) = (0, 1), (x_b, y_b) = (4, 3)$. Hence, there are no gains from trade since the equilibrium is at the corner of the Edgeworth box and there is nothing to trade with each other.



1.3.2 Production Economy: One Consumer-One Producer Case

Consider an economy with a single turnip farmer endowed with one unit of time (available for work and leisure) and a field. The farmer has preferences represented by the utility function

$$u(t, l) = \ln t + \ln(1 - l)$$

where t is the number of turnips consumed and l is the labour supplied.

There is only one industry in this economy – turnip production – which requires two inputs, labour and fields. Turnips are competitively produced with a production function

$$f(L, F) = L^{\frac{1}{2}} F^{\frac{1}{2}}$$

where L is the total labour used and F is the total number of fields used.

Assume that all three markets in this economy (turnips, labour and fields) are competitive. Normalise the turnip price to 1. Let w be the wage, and r the field's rental price.

We first argue that the farmer's constraint is $t \leq lw + r$. Since the price of turnips is 1, the farmer's expenditure on consumption is $1 \times t = t$. The farmer can spend up to the income he earns, so $t \leq m$ since the farmer's income is wage times time he works, and the rental price of the field which he is endowed with, his income is $m = lw + r$ and his constraint is

$$t \leq lw + r$$

The farmer's maximisation problem is hence

$$\max u(t, l) = \ln t + \ln(1 - t) \quad \text{s.t.} \quad t \leq lw + r$$

Since $u(t, l)$ is strongly monotonic, the solution binds with equality.

We set up the Lagrangian

$$\mathcal{L}(t, l, \lambda) = \ln t + \ln(1 - t) - \lambda(t - wl - r)$$

Our FOCs are

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial t} &= 1/t - \lambda = 0 \\ \frac{\partial \mathcal{L}}{\partial l} &= -1/(1 - l) + w\lambda = 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= wl + r - t = 0 \end{aligned}$$

We yield after some algebra, $l = \frac{w-r}{2w}$, $t = w(1 - \frac{w-r}{2w}) = \frac{w+r}{2}$. Labour is positive when the wage rate is greater than rental price.

Now, the firm faces

$$\min C = wL + rF \quad \text{s.t.} \quad f(L, F) \geq L^{\frac{1}{2}} F^{\frac{1}{2}}$$

Typical arguments allow us to construct

$$\mathcal{L}(L, F, \lambda) = wL + rF - \lambda(L^{\frac{1}{2}} F^{\frac{1}{2}} - q)$$

Solve for FOCs which yields $L = (\frac{r}{w})^{\frac{1}{2}} q$, $F = (\frac{w}{r})^{\frac{1}{2}} q$. The profit maximisation condition is

$$\max \pi(t) = t - 2(wr)^{\frac{1}{2}} t$$

since $\pi(t)$ is linear in t it will just produce as much as possible. Since $t = L^{\frac{1}{2}} F^{\frac{1}{2}}$ where $F = 1$, then we know that $t = (\frac{r}{w})^{\frac{1}{4}} t^{\frac{1}{2}} \implies t = (\frac{r}{w})^{\frac{1}{2}}$. Therefore, our market clearing conditions are $z_L(w, r) = 0 \implies L_s = L_d$, $z_F(w, r) = 0 \implies F_d = 1$, $z_t(w, r) = 0 \implies t_d = t_s$. Solving these gives $(p_t, w, r) = (1, \frac{\sqrt{3}}{2}, \frac{1}{2\sqrt{3}})$ and the consumer receives $(t, 1 - l, F) = (\frac{1}{\sqrt{3}}, \frac{2}{3}, 0)$ and the producer receives $(l, F) = (\frac{1}{3}, 1)$.

1.3.3 Production Economy: One Consumer-Two Producers and Trade

Equilibrium with Production and International Trade

Consider a small closed economy with two consumption goods, good 1 (meat) and good 2 (berries). There are two types of agent, h and g , and they have the same preferences over consumption, represented by the utility function

$$u(x_1, x_2) = \ln x_1 + \ln x_2$$

but there are twice as many type- h agents as type- g agents.

The only factors of production are their labour: when a type- h agent chooses to spend a fraction α of his day producing meat and the rest producing berries then his output is $(y_1^h, y_2^h) = (2\alpha, 2(1-\alpha))$; a type- g agent is more productive – when she chooses to spend a fraction β of her day producing meat and the rest producing berries, then her output is $(y_1^g, y_2^g) = (3\beta, 12(1-\beta))$.

Normalise the price of one unit of berries (good 2) to 1, and let p be the price of one unit of meat (good 1).

1. What would the agents choose for α and β if $p < 1$?

What would the agents choose for α and β if $p > 4$?

Argue that those values of p cannot be part of an equilibrium.

If $p < 1$, then h types choose $\alpha = 0$ since $\frac{d(2\alpha p)}{d\alpha} = 2p < \frac{d(2(1-\alpha))}{d(1-\alpha)} = 2$. Hence, the marginal gain of dedicating more time to good 2 will always be more than good 1. So, $\alpha = 0$. Also, $\beta = 0$, since $\frac{d(3\beta p)}{d\beta} = 3p < \frac{d(12(1-\beta))}{d(1-\beta)} = 12$. So the maximise income, $\beta = 0$. This cannot be an equilibrium since the utility of each individual would be $\lim_{x_1^i \rightarrow 0} u^i(x_1^i, x_2^i) \rightarrow -\infty$ as $x_1^1, x_1^g = 0$. So the price, $p \rightarrow \infty$ which

contradicts the price behaviour.

If $p > 4$, $\alpha = \beta = 1$ since $\frac{d(2\alpha p)}{d\alpha} = 2p > \frac{d(2(1-\alpha))}{d(1-\alpha)} = 2$ and $\frac{d(3\beta p)}{d\beta} = 3p > \frac{d(12(1-\beta))}{d(1-\beta)} = 12$. If $\alpha = 1, \beta = 1$ $(x_1^h, x_2^g) = (0, 0)$. So, utility will be $-\infty$ at the limit and the willingness to pay for an $\varepsilon > 0$ of x_2^i is infinite, so the relative price p tends to 0 i.e. $p \rightarrow 0$ contradicting price behaviour.

Now assume that $4 > p > 1$. (Check later that these inequalities are strict).

2. Taking p as given, calculate what each agent will produce daily.

Imagine that the agents sell their output to the market at prices p and 1, producing ‘income’, and then decide on how much of each good they actually want to consume, resulting in ‘expenditure’.

Calculate what each agent will demand.

So, from the above inequality, $\frac{d(2\alpha p)}{d\alpha} = 2p > \frac{d(2(1-\alpha))}{d(1-\alpha)} = 2$ so $\alpha = 1$ and $\beta = 0$, since $\frac{d(3\beta p)}{d\beta} = 3p < \frac{d(12(1-\beta))}{d(1-\beta)} = 12$. Hence, $(x_1^h, x_2^h) = (2, 0)$ and $(x_1^g, x_2^g) = (0, 12)$. So, their incomes are $m^h = 2p, m^g = 12$. Their utility functions are Cobb-Douglas, so the normal demand solutions are

$$\begin{aligned} x_1^h &= \frac{1}{2} \frac{m^h}{p} = \frac{1}{2} \cdot \frac{2p}{p} = 1 \\ x_2^h &= \frac{1}{2} \frac{m^h}{1} = \frac{1}{2} \cdot \frac{2p}{1} = p \end{aligned}$$

and

$$\begin{aligned} x_1^g &= \frac{1}{2} \frac{m^g}{p} = \frac{1}{2} \cdot \frac{12}{p} = 6/p \\ x_2^g &= \frac{1}{2} \frac{m^g}{1} = \frac{1}{2} \cdot \frac{12}{1} = 6 \end{aligned}$$

So, their incomes are $m^h = 2p, m^g = 12$. Their utility functions are Cobb-Douglas, so the normal demand solutions are

$$\begin{aligned} x_1^h &= \frac{1}{2} \frac{m^h}{p} = \frac{1}{2} \cdot \frac{2p}{p} = 1 \\ x_2^h &= \frac{1}{2} \frac{m^h}{1} = \frac{1}{2} \cdot \frac{2p}{1} = p \end{aligned}$$

and

$$x_1^g = \frac{1}{2} \frac{m^g}{p} = \frac{1}{2} \cdot \frac{12}{p} = 6/p$$

$$x_2^g = \frac{1}{2} \frac{m^h}{1} = \frac{1}{2} \cdot \frac{12}{1} = 6$$

3. Remembering that there are twice as many type-h agents as type-g agents, find the value of p that equates demand and supply in the meat market, and confirm that $4 > p > 1$.
 Check that with this value of p , demand and supply are equated in the market for berries.
 Let n be no. of g-type agents and $2n$ be no. of h-type agents. Consequently, we construct our excess demands

$$z_1(p, 1) = \frac{6n}{p} + 2n - 4n = 0$$

$$z_2(p, 1) = 2pn + 6n - 12n = 0$$

By Walras law, if $n - 1$ markets clear, the n -th market clears. Solving gives us $p = 3$.

4. Show that in this equilibrium, type-h agents each consume 1 unit of meat and 3 unit of berries, whereas type-g agents each consume 2 units of meat and 6 units of berries.
 Since $p = 3$, $(x_1^h, x_2^h) = (1, p) = (1, 3)$ and $(x_1^g, x_2^g) = (6/p, 6) = (2, 6)$.

The hunter gatherers now have the possibility of opening up their economy to free trade. In world markets, 1 unit of meat can be exchanged for 2 units of berries, and the country would be a price-taker.

5. Using world prices, calculate what each agent would produce daily. By considering whether each type of agent would become better or worse off, what do you think would be the likely outcome of a referendum on whether or not to open up the economy?
 The world price $p^w = 2 < p = 3$ at equilibrium. Since $1 < p^w = 2 < 4$, h-types will still produce $(2, 0)$ and g-types will still produce $(0, 12)$. Since $x_1^h = 1, x_2^h = p^w = 2$, then $(x_1^h, x_2^h) = (1, 2)$ and $x_1^g = 6/p^w = 3, x_2^g = 6$, then $(x_1^g, x_2^g) = (3, 6)$.
 Type h workers will be worse off since $(1, 2) < (2, 3)$, the amount of meat and berries they receive prior trade more. This is likely because their income, $2p$ is less as world prices for meat fell. Type g workers is better off since $(3, 6) > (2, 6)$ because their income is unchanged at 12 while price for meat fell so their purchasing power increased. Since there are twice as many h types as g types and we assume that there is majority rule and workers vote in their self-interest, the referendum would lead to the economy not opening.

2 Welfare Economics and Social Choice

2.1 Social Welfare Functions and Pareto Efficiency

References: Simon Cowan's slides; Jehle and Reny, *Advanced Microeconomic Theory* Ch 6

Key takeaways on Social Welfare Functions

1. Understand concepts of Pareto efficiency and the Pareto criterion; understand Utilitarian, Pareto and Rawlsian social welfare functions and whether they rank all kinds of bundles
2. The amount of information required by different SWFs: whether they require full comparability of utilities or only ordinal rankings of bundles
3. Kaldor-Hicks improvements vs Pareto improvements
4. How to prove pareto efficiency

Definition 2.1 (Utility Possibility Frontier). Consider utility profile of individuals at allocation $\mathbf{x}$ where $\mathbf{u}(\mathbf{x}) = (u_1(\mathbf{x}), \dots, u_I(\mathbf{x}))$. The utility possibility frontier U describes the utility of agents when we reallocate resources. Formally,

$$U = \{\mathbf{u}(\mathbf{y}) : \sum_{i \in \mathcal{I}} \mathbf{y}^i = \sum_{i \in \mathcal{I}} \mathbf{x}^i\}$$

Deriving the UPF: Consider a fixed endowment $X = x_1 + x_2$ and suppose $u^i(x_i) = x_i^\beta$. Then, it follows that $u^2(u^1) = (X - x_1(u^1))^\beta = (X - u^{1\frac{1}{\beta}})^\beta$. So the UPF is strictly concave if $\beta < 1$ and linear if $\beta = 1$.

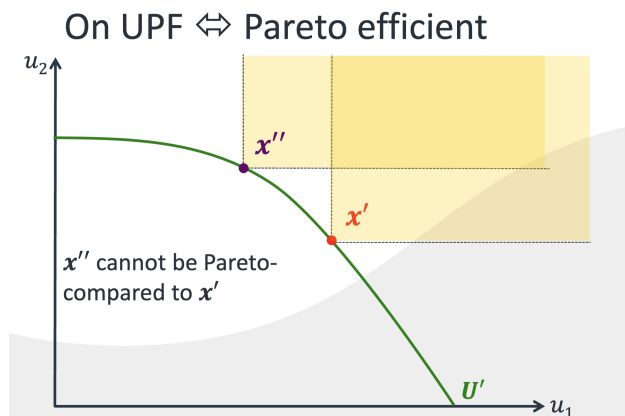


Fig 2.1: Utility Possibility Frontier

Proposition 2.1. If $\mathbf{x}$ is inside the UPF, it is not Pareto efficient and is dominated by another allocation. If $\mathbf{x}$ is on boundary of the UPF, then it is Pareto efficient.

The 2-individual case is easy to prove by geometrical argument; the n -individual case can be generalised by using the concept of balls.

Proof. Notice that any Pareto improvement is equivalent to an outward movement in the first quadrant (upper-right) of a circle. Therefore, to say that an allocation $\mathbf{x}$ is Pareto efficient is equivalent to saying that such a quadrant does not exist between the allocation $\mathbf{x}$ and the boundary of the UPF. But if $\mathbf{x}$ is inside the UPF, then there exists a circle with radius $\varepsilon > 0$ around $\mathbf{x}$ that does not include allocations on boundary of the UPF. So such a quadrant must exist and $\mathbf{x}$ is Pareto improvable. ■

Remark: Not all movements from an allocation inside the UPF to an allocation on the boundary of the UPF is a Pareto improvement. Instead, they are all a Kaldor-Hicks improvement. See Fig 2.2.

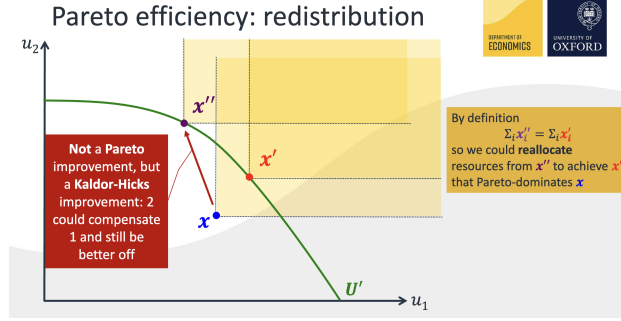


Fig 2.2: Kaldor-Hicks Improvement

Definition 2.2 (Kaldor-Hicks Improvement and Efficiency). An allocation $\mathbf{y}$ is said to be a Kaldor-Hicks improvement on an allocation $\mathbf{x}$ if those that are made better off can hypothetically compensate those who are made worse off and lead to a Pareto-improving outcome. Formally, an allocation $\mathbf{y}$ is a Kaldor-Hicks improvement on $\mathbf{x}$ if there exists an allocation $\mathbf{y}'$ s.t.

$$\sum_{i \in \mathcal{I}} \mathbf{y}^i = \sum_{i \in \mathcal{I}} \mathbf{y}'^i$$

and $\mathbf{y}'$ Pareto-dominates $\mathbf{x}$. An allocation $\mathbf{y}$ is said to be Kaldor-Hicks efficient if there are no such allocations i.e. an allocation is not Kaldor-Hicks improvable.

Characterisation 2.1 (Social Welfare Function): A social welfare function is a function that aggregates individual preferences in order to rank social states according to their desirability. There are three main social functions we examine (we will assume two individuals for simplicity):

1. **Rawls' Maximin Function:**

$$W(\mathbf{u}) = \min\{u_1, u_2\}$$

Informational requirements: only ordinal ranking of utility matters

2. **Bentham's Utilitarian Function:**

$$W(\mathbf{u}) = u_1 + u_2$$

Informational requirements: cardinal comparisons of utility matters

3. **Atkinson's Inequality Adverse Function:**

$$W(\mathbf{u}) = u_1^\alpha u_2^{1-\alpha}$$

Worked Examples

1. Consumption Efficiency and SWF I

In an exchange economy with two consumers and two goods, consumers A and B have identical well-behaved preferences represented by the utility function $u = \ln x + 2 \ln y$. In the initial allocation, consumer A has $(11, 3)$ (that is, 11 units of good x and 3 units of good y), and consumer B has $(5, 9)$

- (a) Explain why the initial allocation is not Pareto efficient. Illustrate this situation using an Edgeworth box diagram.

Answer (to be updated)

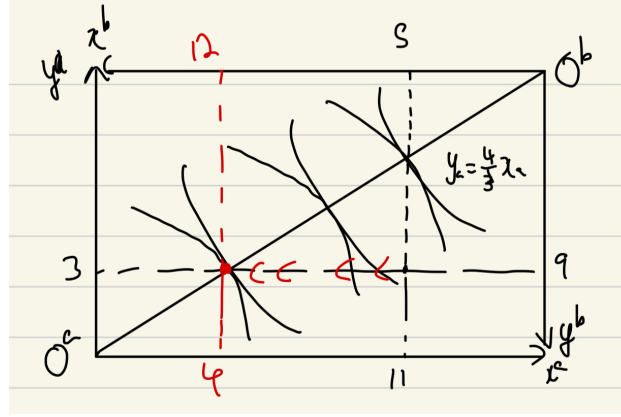
- (b) Suppose that good y cannot be transferred between the consumers. Find the transfer (of good x) that would result in a Pareto efficient allocation. Mark this allocation in your diagram.

A Pareto efficient allocation is when $MRS_a = MRS_b$, and when the resources in the economy are exhausted. Therefore, when $MRS_a = MRS_b$, then

$$\frac{u_x^a}{u_y^a} = \frac{u_x^b}{u_y^b} \implies \frac{1/x^a}{2/y^a} = \frac{1/x^b}{2/y^b} \implies \frac{y^a}{x^a} = \frac{y^b}{x^b}$$

Let $y^a = 3, y^b = 9$, then $3x^b = 9x^a$ which implies $x^b = 3x^a$. Since $x^a + x^b = 16, 4x^a = 16$. Thus, $x^a = 4, x^b = 12$.

So A should transfer 7 units of x to B s.t. $(x^a, y^a) = (4, 3), (x^b, y^b) = (12, 9)$. We can illustrate this via the Edgeworth box from (a):



- (c) Can the initial allocation and the allocation in part (b) be ranked using the Pareto criterion? Explain why or why not.

No the two allocations cannot be ranked. The Pareto criterion says that an allocation pareto dominates another if it is a pareto improvement over the other i.e. the new allocations makes someone better off without making others worse off. Since allocation 2 is a transfer of 7 units of x from A to B , A is made worse off while B is made better off. So allocation 2 is not a pareto improvement over allocation 1 and cannot be ranked by the Pareto criterion.

- (d) Now suppose that good y can also be transferred between the consumers. Find the allocation that would be chosen by a social planner that aimed to maximise (i) the sum of utilities of the two consumers, (ii) the minimum of the utilities of the two consumers. Comment briefly on your answers.

Let $(x^b, y^b) = (16 - x^a, 12 - y^a)$. Then $u^b = \ln(16 - x^a) + \ln(12 - y^a)$.

- i. The utilitarian criterion is to

$$\max u = u^a + u^b = \ln x^a + 2\ln y^a + \ln(16 - x^a) + 2\ln(12 - y^a)$$

The FOCs are:

$$\begin{aligned} \frac{\partial u}{\partial x^a} &= \frac{1}{x^a} - \frac{1}{16 - x^a} = 0 \\ \frac{\partial u}{\partial y^a} &= \frac{2}{y^a} - \frac{2}{12 - y^a} = 0 \end{aligned}$$

We conclude from that FOCs that

$$\begin{aligned} 16 - x^a = x^a &\implies x^a = x^b = 8 \\ 12 - y^a = y^a &\implies y^a = y^b = 6 \end{aligned}$$

The social planner thus chooses the allocations

$$(x^a, y^a) = (8, 6) \text{ and } (x^b, y^b) = (8, 6)$$

- ii. The Rawlsian criterion is to

$$\max \{\min\{u^a, u^b\}\}$$

The condition to maximise is when $u^a = u^b$ i.e.

$$\ln x^a + 2\ln y^a = \ln(16 - x^a) + 2\ln(12 - y^a)$$

which is when $x^a = x^b = 8, y^a = y^b = 6$.

The two answers are the same because there are identical preferences which exhibit decreasing marginal utility in both goods x and y . Hence, utilitarians would distribute goods equally to maximise utility which precisely maximises the minimum as well, and thus will be chosen by a Rawlsian social planner

2.2 First and Second Fundamental Welfare Theorems

References: Simon Cowan's slides; Jehle and Reny, *Advanced Microeconomic Theory* Ch 6; Kreps, *A Course in Microeconomic Theory* Ch 6; Mas-Colell, Whinston and Green, *Microeconomic Theory* Ch 18

Key takeaways on Welfare Theorems

1. Understand the assumptions that underpin first and second welfare theorems (similar to existence of GE + on preferences)
2. How to use Second Welfare Theorem to find a redistribution of endowments to support a new competitive equilibrium
3. Finding the contract curve, and the MRS = MRS condition

Definition 2.3 (Pareto-Efficient Allocations). A feasible allocation, $\mathbf{x} \in F(\mathbf{e})$, is Pareto-efficient if there is no other feasible allocation $\mathbf{y} \in F(\mathbf{e})$ such that $\mathbf{y}^i \succsim^i \mathbf{x}^i$ for all consumers, i , with at least one preference strict i.e. it is not Pareto-dominated by another allocation.

Note the parallel with blocking coalitions.

Definition 2.4 (Walrasian Equilibrium Allocations). Let $\mathbf{p}^*$ be a Walrasian equilibrium for some economy with initial endowments $\mathbf{e}$, and let

$$\mathbf{x}(\mathbf{p}^*) \equiv (\mathbf{x}^1(\mathbf{p}^*, \mathbf{p}^* \cdot \mathbf{e}^1), \dots, \mathbf{x}^I(\mathbf{p}^*, \mathbf{p}^* \cdot \mathbf{e}^I))$$

where component i gives the n-vector of goods demanded and received by consumer i at prices $\mathbf{p}^*$. Then $\mathbf{x}(\mathbf{p}^*)$ is called a Walrasian equilibrium allocation. We denote the set of WEAs as $W(\mathbf{e})$

Intuition for First Fundamental Welfare Theorem and finding set of Pareto-optima:

Suppose my welfare function is $W(\mathbf{u}) = \alpha u_1 + \beta u_2$ (or any convex function as in Fig 2.3) and you fix the resource constraint or endowment. Having derived the utility possibility frontier U , society's optimisation problem is simply:

$$\max W(\mathbf{u}) = \alpha u_1 + \beta u_2 \text{ s.t. } W(\mathbf{u}) \geq U \quad (\text{SP})$$

Or as illustrated,

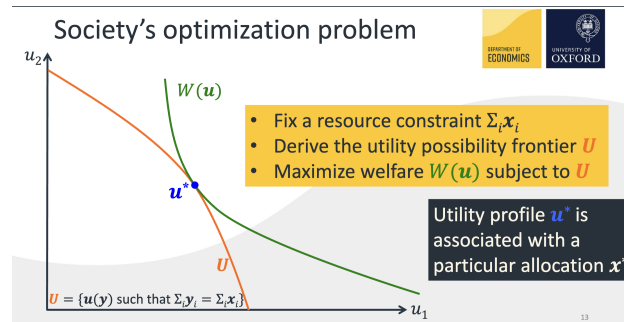


Fig 2.2: Society's Problem

Example (Case of two individuals): Suppose we have Alice and Bob, and we want to find the set of Pareto-optimal allocations. First, we fix Bob's utility at $u^b(\mathbf{x}) = \bar{u}$. Then, we have the maximisation problem:

$$\begin{aligned} \max & u^a(x_1^a, x_2^a) \\ \text{s.t.} & \\ & u^b(x_1^b, x_2^b) \geq \bar{u}, \\ & x_1^a + x_1^b = w_1^a + w_1^b, \\ & x_2^a + x_2^b = w_2^a + w_2^b \end{aligned}$$

This gives us the following Lagrangian (the last two constraints makes the first constraint bind with equality):

$$\mathcal{L}(x_1^a, x_2^a, x_1^b, x_2^b) = u^a(x_1^a, x_2^a) + \lambda(u^b(x_1^b, x_2^b) - \bar{u}) - \mu_1(x_1^a + x_1^b - w_1^a - w_1^b) - \mu_2(x_2^a + x_2^b - w_2^a - w_2^b)$$

where λ, μ_1, μ_2 are the Lagrangian multipliers on the utility and resource constraints. Solving for FOCs, we yield for all $i \in \{a, b\}$:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x_1^i} &= \frac{\partial u^i}{\partial x_1^i} - \mu_1 = 0 \\ \frac{\partial \mathcal{L}}{\partial x_2^i} &= \frac{\partial u^i}{\partial x_2^i} - \mu_2 = 0 \end{aligned}$$

Then, by the definition of the Marginal Rate of Substitution,

$$\begin{aligned} MRS^a &= \frac{\partial u^a / \partial x_1^a}{\partial u^a / \partial x_2^a} = \frac{\mu_1}{\mu_2} \\ MRS^b &= \frac{\partial u^b / \partial x_1^b}{\partial u^b / \partial x_2^b} = \frac{\mu_1}{\mu_2} \end{aligned}$$

By assumption, since Alice and Bob have Cobb-Douglas utility functions s.t.

$$\begin{aligned} u^a(x_1^a, x_2^a) &= (x_1^a)^\alpha (x_2^a)^{1-\alpha} \\ u^b(x_1^b, x_2^b) &= (x_1^b)^\beta (x_2^b)^{1-\beta} \end{aligned}$$

and we have endowment vectors $\mathbf{w}^a = (1, 0)$, $\mathbf{w}^b = (0, 1)$, we can find the set of Pareto efficient allocations by setting:

$$MRS^a = MRS^b = \frac{\alpha}{1-\alpha} \frac{x_2^a}{x_1^a} = \frac{\beta}{1-\beta} \frac{x_2^b}{x_1^b}$$

and we use the resource constraints $x_1^b = 1 - x_1^a$ and $x_2^b = 1 - x_2^a$. This gives us x_2^a as a function of x_1^a along the contract curve. ■

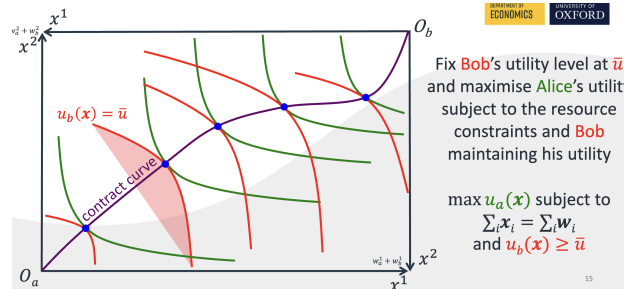


Fig. 2.3: Finding Pareto Efficient Allocations

To prove the FWT, we will first prove two lemmas.

Lemma 2.1. Let $\mathbf{p}^*$ be a Walrasian equilibrium for some economy with initial endowments $\mathbf{e}$. Let $\mathbf{x}(\mathbf{p}^*)$ be the associated WEA. Then $\mathbf{x}(\mathbf{p}^*) \in F(\mathbf{e})$

Proof. Quite trivially, this follows from the definition of $F(\mathbf{e})$ and **Definition 1.4** ■

Lemma 2.2. Suppose that u^i is strictly increasing on $\mathbb{R}_+^n$, that consumer i 's demand is well-defined at $\mathbf{p} \geq \mathbf{0}$ and equal to $\hat{\mathbf{x}}^i$, and that $\mathbf{x}^i \in \mathbb{R}_+^n$.

1. If $u^i(\mathbf{x}^i) > u^i(\hat{\mathbf{x}}^i)$, then $\mathbf{p} \cdot \mathbf{x}^i > \mathbf{p} \cdot \hat{\mathbf{x}}^i$
2. If $u^i(\mathbf{x}^i) \geq u^i(\hat{\mathbf{x}}^i)$, then $\mathbf{p} \cdot \mathbf{x}^i \geq \mathbf{p} \cdot \hat{\mathbf{x}}^i$

This is just the fact that utility is strictly increasing with respect to the amount of goods consumed.

Proof. If $u^i(\mathbf{x}^i) > u^i(\hat{\mathbf{x}}^i)$, we can conclude by the strict monotonicity of u^i , that $\mathbf{x} > \hat{\mathbf{x}}^i$. It follows immediately that $\mathbf{p} \cdot \mathbf{x}^i > \mathbf{p} \cdot \hat{\mathbf{x}}^i$. Suppose (i) holds. Assume for contradiction that (ii) does not hold. Then $u^i(\mathbf{x}^i) \geq u^i(\hat{\mathbf{x}}^i)$ but $\mathbf{p} \cdot \mathbf{x}^i < \mathbf{p} \cdot \hat{\mathbf{x}}^i$. Consequently, beginning with $\mathbf{x}^i$, we may increase the amount of every good consumed by a small enough amount so that the resulting bundle $\bar{\mathbf{x}}^i$ remains strictly less expensive than $\hat{\mathbf{x}}^i$. But because u^i is strictly increasing, we then have $u^i(\bar{\mathbf{x}}^i) > u^i(\mathbf{x}^i) \geq u^i(\hat{\mathbf{x}}^i)$ and $\mathbf{p} \cdot \bar{\mathbf{x}}^i < \mathbf{p} \cdot \hat{\mathbf{x}}^i$. But this contradicts (i) with $\mathbf{x}^i$ replaced by $\bar{\mathbf{x}}^i$ ■

Theorem 2.3 (First Fundamental Welfare Theorem). If $(\mathbf{x}^*, \mathbf{p}^*)$ is a competitive equilibrium in the exchange economy, then $\mathbf{x}^*$ is Pareto efficient.

Proof. Unlike the lectures, we will prove this in the general case with multiple agents. The proof proceeds in two steps. The first step is to observe that all allocations in the Core i.e. $C(\mathbf{e})$ are Pareto efficient. The second is to show that all Walrasian equilibrium allocations are in the core.

Step 1: By **Definition 1.2**, we note that all core allocations are unblocked. By **Definition 1.1**, an unblocked allocation implies that there does not exist a coalition of customers $S \subset \mathcal{I}$ with an alternative allocation $\mathbf{y}$ such that it makes them better off without making anyone worse off i.e. no other feasible allocation $\mathbf{y} \in F(\mathbf{e})$ such that $\mathbf{y}^i \succsim^i \mathbf{x}^i$ for all consumers, i , with at least one preference strict, with is just the definition of a Pareto Efficient Allocation.

Step 2: Suppose that $\mathbf{x}(\mathbf{p}^*)$ is a WEA and assume $\mathbf{x}(\mathbf{p}^*) \neq C(\mathbf{e})$. Because $\mathbf{x}(\mathbf{p}^*)$ is a WEA, we know from **Lemma 2.1** that $\mathbf{x}(\mathbf{p}^*) \in F(\mathbf{e})$, so $\mathbf{x}(\mathbf{p}^*)$ is feasible. However, because $\mathbf{x}(\mathbf{p}^*) \neq C(\mathbf{e})$, we can find a coalition $S \subset \mathcal{I}$ and another allocation $\mathbf{y}$ such that

$$\sum_{i \in S} \mathbf{y}^i = \sum_{i \in S} \mathbf{e}^i \quad (\text{P.1})$$

and

$$u^i(\mathbf{y}^i) \geq u^i(\mathbf{x}^i(\mathbf{p}^*, \mathbf{p}^* \cdot \mathbf{e}^i)) \text{ for all } i \in S \quad (\text{P.2})$$

with at least one inequality strict. (P.1) implies

$$\mathbf{p}^* \cdot \sum_{i \in S} \mathbf{y}^i = \mathbf{p}^* \cdot \sum_{i \in S} \mathbf{e}^i \quad (\text{P.3})$$

From (P.2) and **Lemma 2.2**, we know that for each $i \in S$, we must have

$$\mathbf{p}^* \cdot \mathbf{y}^i \geq \mathbf{p}^* \cdot \mathbf{x}^i(\mathbf{p}^*, \mathbf{p}^* \cdot \mathbf{e}^i) = \mathbf{p}^* \cdot \mathbf{e}^i \quad (\text{P.4})$$

with *at least one inequality strict*. This is why when we sum over all consumers in S , we obtain

$$\mathbf{p}^* \cdot \sum_{i \in S} \mathbf{y}^i > \mathbf{p}^* \cdot \sum_{i \in S} \mathbf{e}^i$$

contradicting (P.3). So, $\mathbf{x}(\mathbf{p}^*) \in C(\mathbf{e})$.

Since all Walrasian equilibrium allocations are in the core, and every allocation in the core is Pareto efficient, all Walrasian equilibrium allocations are Pareto efficient allocations. ■

We now turn to what Reny and Jehle call "an affirmative answer to the following question: is a system that depends on decentralised, self-interested decision making by a large number of consumers capable of sustaining the socially best allocation of resources, if we could just agree on what that was. Under the conditions stated before, the Second Welfare Theorem says yes, as long as socially best requires, at least, Pareto efficiency."

Theorem 2.4 (Second Fundamental Welfare Theorem). Consider an exchange economy $(u^i, e^i)_{i \in \mathcal{I}}$ with aggregate endowment $\sum_{i=1}^{\mathcal{I}} e^i \gg \mathbf{0}$, and with preferences which are continuous, convex and monotonic. Suppose that $\bar{\mathbf{x}}$ is a Pareto-efficient allocation for $(u^i, e^i)_{i \in \mathcal{I}}$, then $\bar{\mathbf{x}}$ is a Walrasian Equilibrium Allocation for some Walrasian equilibrium $\bar{\mathbf{p}}$ after redistribution of initial endowments to any allocation $\mathbf{e}^* \in F(\mathbf{e})$ such that $\bar{\mathbf{p}} \cdot \mathbf{e}^{*i} = \bar{\mathbf{p}} \cdot \bar{\mathbf{x}}^i$ for all $i \in \mathcal{I}$.

Remark: The Second Fundamental Welfare Theorem, roughly speaking, says that under the prevailing assumptions of preferences, the set of Pareto efficient allocations implies the set of Walrasian equilibrium allocations. So, we can have any outcome as a competitive equilibrium, so long as it is Pareto efficient. Taken together, both Welfare Theorem, say that there is a bijection between the set of Pareto efficient allocations and the set of Walrasian equilibrium allocations.

Consequences of the Second Fundamental Welfare Theorem:

- Any Pareto efficient allocation can be implemented as a market outcome
- All the caveats of the 1st FWT apply but even more stringent
- Large redistribution of resources would be required i.e. Lump sum redistribution required
- Huge informational demands on the social planner

Procedure for SWT Questions: Consider some desired allocation by the social planner $(x_a, y_a), (x_b, y_b)$ with starting endowments $(w_x^a, w_y^a), (w_x^b, w_y^b)$

1. **Satisfy the SWT:** show that the desired allocation by the social planner $(x_a, y_a), (x_b, y_b)$ is pareto efficient, and that the utility functions satisfy the properties required by the SWT
2. **Find the supporting price:** Use the condition that $MRS_a = -\frac{p_x}{p_y}$ to find the supporting price ratio, which is also the slope of the budget line from a 's perspective
3. **Write the Budget Line equation:** $\rho x_a + y_a = \mathbf{w} \cdot \mathbf{p}$. Alternatively, write $y - y_a = \rho(x - x_a)$ (my notation here is a bit confused, technically $y \equiv y_a$ and the other y_a is the fixed quantity in the desired allocation with the same for x_a)
4. **Solve for the last variable:** Substitute the values for x_a, y_a, ρ as well as the fixed endowment by the question and solve

Worked Examples

1. Welfare Theorems

Consider an economy with two goods, 1 and 2, and two consumers, a and b . The preferences of consumers are represented by the utility functions:

$$u^a(x_1^a, x_2^a) = 2\ln x_1^a + 3\ln x_2^a \quad \text{and} \quad u^b(x_1^b, x_2^b) = 2\ln x_1^b + \ln x_2^b$$

- (a) If the economy is composed of 20 units of goods 1 and 12 units of good 2, find the set of all Pareto efficient allocations.

To find the set of all pareto efficient allocations, we find the contract curve whereby $MRS_a = MRS_b$. Therefore,

$$\frac{u_{x_1}^a}{u_{x_2}^a} = \frac{u_{x_1}^b}{u_{x_2}^b} \implies \frac{2/x_1^a}{3/x_2^a} = \frac{2/x_1^b}{1/x_2^b} \implies \frac{2x_2^a}{3x_1^a} = \frac{2x_2^b}{x_1^b}$$

since $x_1^a + x_1^b = 20$ and $x_2^a + x_2^b = 12$, we have

$$\frac{x_2^a}{3x_1^a} = \frac{12 - x_2^b}{20 - x_1^a}$$

After some algebra, we yield the contract curve equation

$$\begin{aligned} 20x_2^a - x_2^a x_1^a &= 36x_1^a - 3x_2^a x_1^a \\ \implies x_2^a(20 + 2x_1^a) &= 36x_1^a \\ \implies x_2^a &= \frac{18x_1^a}{10 + x_1^a} \end{aligned}$$

Hence, the set of all pareto efficient allocations are

$$S = \left\{ x_1^a, x_2^a, x_1^b, x_2^b \mid x_2^a = \frac{18x_1^a}{10 + x_1^a}, x_1^a + x_1^b = 20, x_2^a + x_2^b = 12 \right\}$$

- (b) Suppose that consumers a has the ownership of the 20 units of good 1, while consumer b has the ownership of the 12 units of good 2 (and none of good 1). Compute the equilibrium allocation of this private economy and then show that the allocation is Pareto efficient. Were all these steps necessary to conclude that the resulting equilibrium is Pareto efficient?

Let us normalise the price of x_1 to 1, resulting in the relative price of x_2 as ρ .

Therefore, a 's budget constraint is

$$x_1^a + \rho x_2^a \leq 20$$

while b 's budget constraint is

$$x_1^b + \rho x_2^b \leq 12\rho$$

By inspection, since a and b 's utility functions are Cobb-Douglas, demands are

$$\begin{aligned} x_1^a &= \frac{2}{5}(20) = 8 \quad \text{and} \quad x_2^a = \frac{3}{5\rho}(20) = \frac{12}{\rho} \\ x_1^b &= \frac{2}{3}(12\rho) = 8\rho \quad \text{and} \quad x_2^b = \frac{1}{3\rho}(12\rho) = 4 \end{aligned}$$

By Walras' Law, if $n - 1$ markets clear, then the n -th market clears. So our market clearing condition is just excess demand for x_1 set to 0, i.e. $z_1(1, \rho) = 0$. Hence,

$$z_1(1, \rho) = 8\rho + 8 - 20 = 8\rho - 12 = 0$$

So, $\rho = \frac{3}{2}$. The equilibrium is therefore the price vector $(1, \frac{3}{2})$ and the allocations $(x_1^a, x_2^a) = (8, 8)$ and $(x_1^b, x_2^b) = (12, 4)$.

To show that the allocation is pareto efficient, we show that the allocations are a member of the pareto efficient set S . Substitute $x_1^a = 8$ into the contract curve, then

$$x_2^a = \frac{18(8)}{10 + 8} = 8$$

Further, $x_1^b = 20 - x_1^a = 20 - 8 = 12$, while $x_2^b = 12 - x_2^a = 12 - 4 = 8$. So the allocations satisfy the membership criterion for the set of pareto efficient allocations, verifying that the equilibrium is pareto efficient.

These steps were not necessary since we can appeal to the first fundamental welfare theorem that if $(\mathbf{X}^*, \mathbf{p}^*)$ is a competitive equilibrium, allocation $\mathbf{X}^*$ is pareto efficient.

- (c) Show that the allocation $(x_1^a, x_2^a) = (5, 6), (x_1^b, x_2^b) = (15, 6)$ is also Pareto efficient. Can it be obtained as a Walrasian equilibrium of some private economy? If so, what will be the equilibrium prices of that economy?

Like (b), we show that $(x_1^a, x_2^a) = (5, 6), (x_1^b, x_2^b) = (15, 6)$ is part of the Pareto efficient set. So, we substitute $x_1^a = 5$ into the contract curve equation:

$$x_2^a = \frac{18(5)}{10 + 5} = \frac{18}{3} = 6$$

Thus,

$$x_1^b = 20 - 5 = 15 \text{ and } x_2^b = 12 - 6 = 6$$

thus verifying the membership criterion. So the new allocation is part of the Pareto efficient set. By the second welfare theorem, the new Pareto efficient allocation can be part of an equilibrium since preferences are convex, continuous and monotonic, so any Pareto efficient allocation can be supported as a competitive equilibrium by some price vector after some redistribution of resources.

We know that at equilibrium, $MRS_a = -\frac{p_1}{p_2}$. Hence

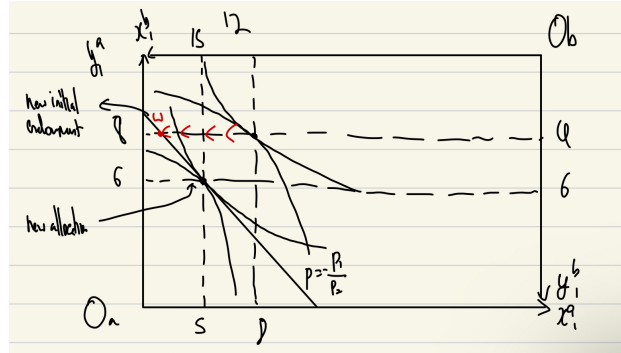
$$MRS_a = -\frac{2x_2^a}{3x_1^a} = -\frac{2 \times 6}{3 \times 5} = -\frac{12}{15} = -\frac{p_1}{p_2}$$

Since it is the price of x_1 that is normalised to 1, $\rho = \frac{p_2}{p_1} = \frac{5}{4}$. Hence,

$$\mathbf{p} = \left(1, \frac{5}{4}\right)$$

- (d) A social planner would like to implement the allocation in Part (c) using lump-sum transfers but is unable to transfer good 2. How should she transfer good 1 between a and b so that her desired allocation will be part of the resulting Walrasian equilibrium?

Assuming the initial endowment is the equilibrium in part (b), she should transfer good 1 from a and b such that the new initial endowment lies on the budget line. This can be represented on the Edgeworth box



we know that from (c) $\frac{p_1}{p_2} = \frac{4}{5}$. The budget line equation from a 's perspective is

$$(x_2^a - 6) = -\frac{4}{5}(x_1^a - 5) \implies x_2^a = -\frac{4}{5}x_1^a + 10$$

Since x_2^a is fixed at 8, $x_1^a = -\frac{5}{4}(8 - 10) = \frac{5}{2}, x_2^b = 20 - \frac{5}{2} = 17.5$.

The initial endowment to support the new Pareto efficient allocation is

$$(w_1^a, w_2^a) = (2.5, 8) \text{ and } (w_1^b, w_2^b) = (17.5, 4)$$

whereby the social planner transfers 5.5 units of good 1 from consumer a to b .

2.3 Preference Aggregation and Social Choice

References: Simon Cowan's slides; Jehle and Reny, *Advanced Microeconomic Theory* Ch 6; Kreps, *A Course in Microeconomic Theory* Ch 5

Key takeaways on Social Choice Theory

Social Welfare Functions/Rankings

1. Ideas of transitivity, completeness, reflexivity
2. Understand Arrow's conditions for a good social choice ranking as well as Condorcet Paradox
3. Arrow's Impossibility Theorem
4. How Single Peaked Preferences defeats unrestricted domain but avoids dictatorship (Black's Theorem)

Social Choice Function

1. Different from SWF by it selecting one social state
2. Median Voter Theorem and proof idea
3. Strategy-proofness vs dictatorship; the Gibbard-Sattherthwaite Theorem

Preliminaries/Abstract Set-up:

1. Let $X = \{x, y, z, \dots\}$ be the set of available social states
2. Let $\mathcal{I} = \{1, 2, \dots, I\}$ be the number of individuals in a society
3. Let the i -th individual have a complete and transitive preference relation $\succsim_i$ on X for all $i \in \mathcal{I}$.
4. A society's preference profile is $(\succsim^1, \succsim^2, \dots, \succsim^I)$
5. A society's preference ordering is denoted $\succsim^*$ s.t. if $x \succsim^* y$, then society prefers x to y

Definition 2.5 (Completeness). A binary relation $\succsim$ is complete if for all x and y in X , either $x \succsim y$ or $y \succsim x$

Definition 2.6 (Transitivity). A binary relation $\succsim$ is transitive if for any three elements x, y, z in X , if $x \succsim y$ and $y \succsim z$, then $x \succsim z$

Definition 2.7 (Preference Relation). The binary relation $\succsim$ on X is called a preference relation if it is both complete and transitive

Definition 2.8 (Social Welfare Function). A social welfare function (or preference aggregation rule) is a function $f : \succsim^1 \times \succsim^2 \times \dots \times \succsim^I \rightarrow \succsim^*$

In other words, a SWF is a map assigning to each preference profile $(\succsim^1, \succsim^2, \dots)$ a preference ordering $\succsim^*$ for society. In general, we want the SWF to be complete and transitive (Counterexample: Condorcet's Voting Paradox).

Remark: A Social Welfare Function is not a preference ordering on X . This is why there exist scenarios whereby transitivity (and completeness) is violated.

We now turn to the most important result in social choice theory: Arrow's impossibility Theorem. To begin, we start by considering certain "conditions" that we would want a preference aggregation rule to have.

Arrow's Axioms

Let us consider a social welfare function or a preference aggregation rule f . Then, f must fulfill the following axioms:

Axiom 2.1 (Unrestricted Domain (U)). The domain of f must include all possible combinations of individual preference relations on X

Axiom 2.2 (Weak Pareto Principle (WP)). For any pair of alternatives x, y in X , if $x \succ^i y$ for all i , then $x \succ^* y$

Axiom 2.3 (Independence of Irrelevant Alternatives (IIA)). Consider two preference profiles and two social states x, y in X . If each individual i ranks x and y in the same way in both preference profiles, even if the other orderings are different, then f should produce the same social ranking of x vs y (i.e. if $x \succsim^* y$) under both preference profiles.

Axiom 2.4 (Non-dictatorship (D)). There is no individual i such that for all x and y in X , $x \succ^i y$ implies $x \succ^* y$ regardless of the preferences $\succsim^j$ of all other preferences $j \neq i$

Interpretation of the Axioms:

Axiom U says that f is able to generate a social preference ordering regardless of what the individuals' preference relations happen to be. It formalises the principle that the ability of a mechanism to make social choices should not depend on society's members holding any particular sorts of views. This condition, together with transitivity, rules out majority voting as an appropriate mechanism because it sometimes fails to produce a transitive social ordering $\succsim^*$.

Axiom WP is the most straightforward. It says that society should prefer x to y if every member of society prefers x to y .

Axiom IIA is the most difficult. Roughly, it says that the social ranking of x and y should depend only on the individual rankings of x and y . Note that the preference profiles are allowed to differ in their other rankings over pairs other than x and y . To see its reasonableness, suppose IIA fails. Then, suppose that individuals rank z below x and y but some prefer x to y and others y to x . Now suppose that given these preferences, f generates a social preference of $x \succ^* y$. Suppose the vote is postponed to a few hours later, in which individuals now rank z over the other two social states. IIA says it is unreasonable for y to now be preferred over x and that f should still generate $x \succ^* y$.

Axiom D is mild because it says there should be no single individual who dictates the social ranking.

Theorem 2.5 (Arrow's Impossibility Theorem). If there are at least three social states in X , then there is no social welfare function f that simultaneously satisfies axioms U, WP, IIA, and D. Specifically, if f satisfies U, WP and IIA, then there will be an individual i such that for any two social states $x, y \in X$, $x \succ^i y$ if and only if $x \succ^* y$.

Proof. Proved in Appendix. ■

Remark: Note that Arrow's theorem can be cast as a 'possibility' result. Namely, that we can always yield a dictatorial social rule if we satisfies W, U and IIA.

Solutions to Arrow's Theorem:

1. Relaxing any of the axioms (reasonable?)
2. Have two social states and not three

Some Preference Aggregation Rules:

1. Dictator Rule
2. Majority Rule
3. Scoring Rule

We will consider more closely majority rule with the following theorems.

Theorem 2.6 (May's Theorem). Consider the problem of finding a ranked social choice function f that satisfies anonymity (treats all voters equally), neutrality (treats all candidates equally) and positive responsiveness (a voter changing his preference to x cannot cause x to lose). Then majority rule is the unique social choice function that satisfies these requirements.

Remark: IIA, U and W are implicitly satisfied because there are only two social states to choose from. Anonymity implies non-dictatorship.

Majority rule can also work under a specific class of preferences (i.e. satisfy all of Arrow's requirements as well as not violate transitivity).

Definition 2.9 (Single Peaked Preferences). A group $\mathcal{I}$ is said to have single peaked preferences over X if there exists an ordering $>$ of outcomes such that for every agent $i \in \mathcal{I}$:

$$\begin{aligned} x_k < x_j \leq x_i^* &\implies x_j \succ^i x_k \\ x_k > x_j \geq x_i^* &\implies x_j \succ^i x_k \end{aligned}$$

i.e. there is an ideal point x_i^* for an agent i , and he will always prefer the option closest to the bliss point than those further away.

Considering such a class of preferences gives us Black's Theorem, which says that majority rule no longer violates transitivity, while satisfying Arrow's requirements.

Theorem 2.7 (Black's Theorem). If social states can be ordered along a single dimension and voters have single-peaked preferences, then majority rule induces a transitive social preference ordering and selects the median voter's social state over others.

Remark: A trivial limitation of Black's Theorem is that there must be an odd number of citizens. Further, even though the median voter's preference is always picked, this does not mean that D is violated. This is because the median voter is characterised by having the median social state; in contrast, in Arrow's theorem, it is because of a specific voter's position in the preference profile (i.e. being the 5th voter or 10th voter) that allows him to dictate the social preference ordering.

Why the median social state? Consider the following game where by citizen's preferences are single peaked uniformly over a one-dimensional tax policy space $t \in [0, 1]$. So, their utility function can be characterised as $u_i(t) = -(t - b_i)^2$ where b_i is the i -th citizen's bliss point. Then, suppose candidates only want to win and can simultaneously commit to policies in advance. To win the majority would simply be to pick the median voter's preference.

Theorem 2.8 (Downs' Median Voter Theorem). If preferences are single-peaked, and if there are two candidates who can commit in advance to policies and only care about winning, then there is a unique equilibrium

$$p_1^* = p_2^* = b_m$$

where b_m denotes the median voter's bliss point and p_i denotes the i -th candidates' policy.

Proof. (Sketch) Assume that voters' ideal points $\{b_j\}$ are continuously distributed on the interval $[0, 1]$, with the cumulative distribution function $F(\cdot)$. Suppose candidate 1 sets $p_1 < b_m = p_2^*$. Then, halfway between

p_1 and b_m are indifferent voters i.e.

$$-(p_1 - b_i)^2 = -(b_m - b_i)^2 \implies b_i = \frac{p_1 + b_m}{2} < b_m$$

Then, all individuals with $b_j < b_i$ vote for candidate 1 while all with $b_j > b_i$ vote for candidate 2. As $b_i < b_m$, and we know that b_m is the median voter, then candidate 1 receives less than 50% of the vote. This means that candidate 1 will lose because candidate 2 will have more than 50% by definition. But candidate 1 wants to win, so they will not set $p_1 < b_m$. Likewise for candidate 2. Notice that if $p_1 > b_m$, we can use the exact argument for candidate 1 as candidate 2. ■

Social Choice Functions instead of Preference Aggregation: An Alternative

Now, instead of thinking about a preference aggregation rule that would hopefully generate a complete and transitive ordering of social states, we consider a **social choice function** $c(\cdot)$ which maps all profiles of individual preferences on X into a choice from X , and whose range is all of X .

One desirable characteristic of a social choice function would be the notion of strategy-proofness. That is, we want each individual to report their preferences honestly. To do so requires us to think of a mechanism that makes the *dominant strategy* when reporting preferences to be to report their actual preferences. In other words, an individual cannot be made better off if they report preferences that are not their own.

Definition 2.10 (Strategy-Proof Social Choice Function). A social choice function $c(\cdot)$ is strategy-proof when, for every individual, i , for every pair $\succsim^i$ and $\succsim_L^i$ of his preferences, and for every profile $\{\succsim^{-i}\}$ of others' preferences, if $c(\succsim^i, \{\succsim^{-i}\}) = x$ and $c(\succsim_L^i, \{\succsim^{-i}\}) = y$, then $x \succsim^i y$.

In words, this definition just says that I will always actually prefer the social state that the strategy-proof social choice function generates when I truthfully report my preferences even if others are lying about their preferences.

But there is an important result which shows that strategy-proofness is not all that desirable.

Theorem 2.9 (The Gibbard-Satterthwaite Theorem). If there are at least three social states, then every strategy-proof social choice function is dictatorial.

Proof. (Sketch) I will not prove this here. But we can outline an extremely brief idea of the proof. The first step is to show that strategy-proofness implies two properties: (i) that the social choice function is monotonic (the social choice does not change, when individual preferences change so that every individual strictly prefers the social choice to any distinct social state it was originally at least as good as i.e. weakly preferred to); (ii) that it is pareto-efficient (whenever x is at the top of every individual's ranking, the social choice is x). The second step is to show that these two properties, in the same vein as Arrow's impossibility theorem, imply the existence of a dictator. ■

Remark: The implication of the Gibbard-Satterthwaite Theorem is that, in a rich enough setting, it is impossible to design a non-dictatorial system in which social choices are made based on self-reported preferences without introducing the possibility that individuals can gain by lying. Therefore, any election or social preference aggregation can be manipulated by voters if they have full knowledge about the preferences of others.

The way to escape this is to impose a domain restriction known as *quasi-linearity* – this will be discussed briefly in the optional section on social choice, but requires one to be acquainted with game theory.

Remark: Note that the Gibbard-Satterthwaite Theorem does not apply to social welfare functions! Only to social choice functions.

2.3.1 Optional Information on Social Choice

Axiom 2.5 (Anonymity (A)). Let $\bar{\mathbf{u}}$ be a utility N -vector and let $\tilde{\mathbf{u}}$ be another vector obtained from $\bar{\mathbf{u}}$ after some permutation of its elements (like a preference profile). Then $W(\bar{\mathbf{u}}) = W(\tilde{\mathbf{u}})$.

Condition A simply states that people should be treated symmetrically. So, the ranking of social states should not depend on the identity of people involved and only the level of welfare or preference. Note that this is a stronger formulation of the axiom D and is used by May's theorem.

Axiom 2.6 (Hammond Equity (HE)). Let $\bar{\mathbf{u}}$ and $\tilde{\mathbf{u}}$ be two distinct utility N -vectors and suppose that $\bar{u}^k = \tilde{u}^k$ for all k except i and j . If $\bar{u}^j > \tilde{u}^j > \tilde{u}^i > \bar{u}^i$, then $W(\tilde{\mathbf{u}}) > W(\bar{\mathbf{u}})$.

Condition HE is obviously controversial because it expresses the idea that society has a preference towards the equality of welfare amongst individuals.

Why does Downs'/Black's Theorem fail when social states are ordered along two dimensions i.e. along a unit circle?

Theorem 2.10 (McKelvey–Schofield chaos theorem). Let the policy space be multi-dimensional, and assume individuals have single-peaked preferences. Then, majority rule is in general unstable i.e. there is no Condorcet winner. If there is no Condorcet winner, then any point in the space can be reached from any other point by a sequence of majority votes.

3 Applied Welfare: Externalities and Public Goods

3.1 Applied Welfare: Measuring Welfare, Welfare and Tax, Cost-Benefit Analysis

References: Simon Cowan's slides; Jehle and Reny, *Advanced Microeconomic Theory* Ch 1

Takeaways on Applied Welfare

1. Interpretations of Compensating and Equivalent Variation to measure welfare; money-metric measures of welfare
2. Quasilinear utility framework and its significance in analysis
3. Ramsey Optimal Taxation and derivation
4. Simple Cost-Benefit analysis through welfare analysis

3.1.1 Measuring Welfare

Consider the problem of measuring the effect of a price change from $\mathbf{p}_0$ to $\mathbf{p}_1$.

1. If utility is known: then we use the indirect utility function $v(\mathbf{p}, m)$ by putting demand functions into the utility function.

Example. Consider that we have a utility function $u(x_1, x_2) = \frac{1}{2}\ln x_1 + \frac{1}{2}\ln x_2$ with demands

$$x_1(\mathbf{p}, m) = \frac{m}{2p_1}, \quad x_2(\mathbf{p}, m) = \frac{m}{2p_2}$$

The indirect utility function becomes

$$v(\mathbf{p}, m) = \frac{1}{2}\ln\left(\frac{m}{2p_1}\right) + \frac{1}{2}\ln\left(\frac{m}{2p_2}\right) = \ln m - \frac{1}{2}\ln p_1 - \frac{1}{2}\ln p_2 - \ln 2$$

Problems: usually, we do not observe utility, and that utility is an ordinal concept but we want a cardinal measure. Further, there are problems interpersonal comparisons between persons (related to ordinal property of utility functions)

2. Money-metric measures of welfare through the expenditure function: The expenditure function is derived by first finding the Hicksian demand function

$$\min \mathbf{p} \cdot \mathbf{x} \quad \text{subject to} \quad u(\mathbf{x}) \geq \bar{u}$$

where the solution $\mathbf{x}^* = \mathbf{h}(\mathbf{p}, \bar{u})$. We construct the expenditure function as

$$e(\mathbf{p}, \bar{u}) = \mathbf{p} \cdot \mathbf{x}^* = m$$

Definition 3.1 (Compensating Variation). The Compensating Variation is the amount of money that must be taken/given from the consumer, given a price change $\mathbf{p}_0$ to $\mathbf{p}_1$ happens, to ensure they reach the old utility:

$$v(\mathbf{p}_1, m - CV) = v(\mathbf{p}_0, m)$$

If the new prices generate higher utility, CV is positive and is a measure of welfare gain.

Remark: Suppose all prices are multiplied by $\lambda > 0$. The old utility is obtained if income is multiplied by λ , so

$$\begin{aligned} m - CV &= \lambda m \\ CV &= (1 - \lambda)m \end{aligned}$$

Definition 3.2 (Equivalent Variation). The Equivalent Variation is the change in income at the old prices $\mathbf{p}_0$ that would give the same utility as with the new prices $\mathbf{p}_1$

$$v(\mathbf{p}_1, m) = v(\mathbf{p}_0, m + EV)$$

Remark: When all prices are multiplied by $\lambda > 0$:

$$EV = \frac{1 - \lambda}{\lambda} m$$

Multiplying all prices by λ has the same effect as dividing income by λ , so $m + EV = m/\lambda$.

Illustration:

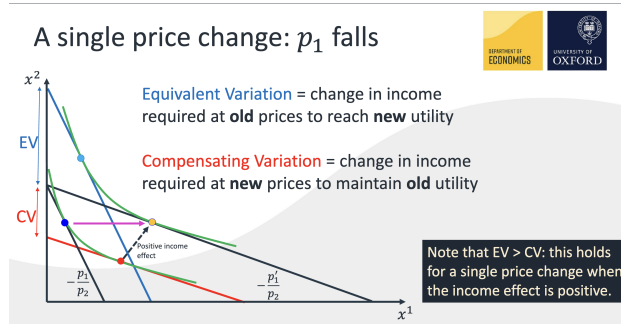


Fig 3.1: Graphical illustration of CV and EV

Using the expenditure function, which gives us a money-metric measure of welfare, we have the EV and CV as:

- CV: Income - Expenditure(New Prices, Old Utility)
- EV: Expenditure(Old Prices, New Utility) - Income

Remark: If a good is normal (the income effect is positive) then if its price changes,

$$CV < \Delta CS < EV$$

- ΔCS can be a good approximation if the income effect is small because either the budget share or the income elasticity is small.
- ΔCS is the correct measure when utility is quasi-linear so there is no income effect

Developing the Quasilinear framework and its significance

Definition 3.3 (Quasilinear Utility). An agent exhibits quasilinear utility if his utility function is

$$U(x, y) = \phi(x) + y$$

where by $\phi(x)$ is concave in x and U is linear in y where y usually denotes money.

Remark. The importance of Quasilinearity in money allows the social planner to add the utilities of agents together since they are money-metric measures of welfare. In general, this is not possible because the concept of utility is ordinal rather than cardinal.

Remark. There is a Lagrangian analysis of Quasilinearity which is to show that when you maximise an agent's utility function subject to another agent's utility, the shadow price i.e. λ of the other agent's utility is 1. So, you can make a 1-1 comparison between your utility and another agent's utility. Let us call this maximisation problem (UP)

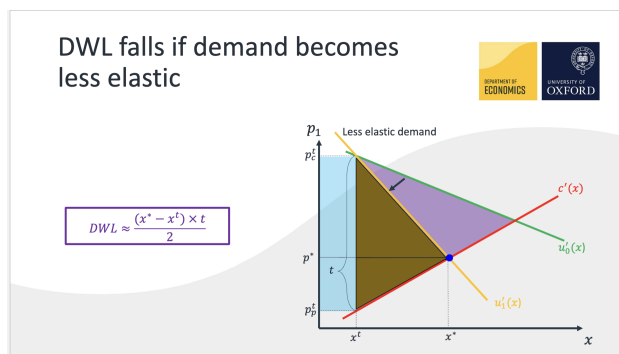
Proposition 3.1. Quasilinearity in money is equivalent to the shadow price of violating the constraint in (UP) being equal to 1

3.1.2 Taxation and Welfare

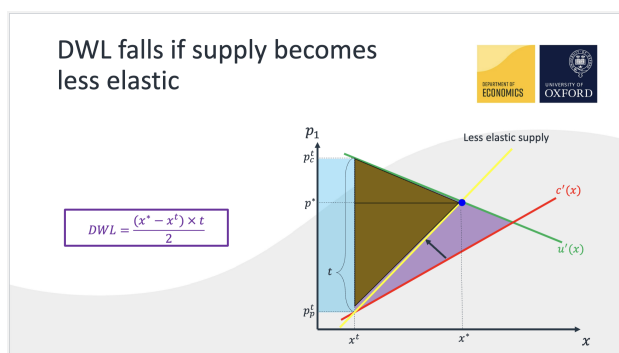
Ramsey's Optimal Taxation: Suppose the social planner wants to minimise deadweight loss of taxation on all goods, while obtaining G as revenue. Formally, this is

$$\sum_{k=1}^n DWL_k \quad \text{s.t.} \quad \sum_{k=1}^n t_k x_k \geq G$$

If we assume all supply elasticities are infinite, then Ramsey taxation says tax the goods with high demand elasticities and high supply elasticities less. Illustration:



and



3.1.3 Cost-Benefit Analysis

What should you value

Basically, come up with (1) tangible costs such as direct investment and operational cost, (2) intangible costs such as congestion and disruption, (3) tangible benefits such as incomes generated by related investment, consumption and (4) intangible benefits such as improved local environments or health benefits.

How should you value

- Revealed Preference Methods
 1. Estimate demand directly from market data: if one can observe demand at various prices and can make a ceteris paribus assumption this can be done
 2. Estimate demand indirectly using proxies:

- (a) Travel Cost Method: employs access costs i.e. transportation and time costs to access facilities as a price for the good; can build econometric models of relationship between number of visits and travel costs and other determinants of demand
- (b) Hedonic pricing: Use property prices to value regional intangible interventions (e.g. neighborhood amenities); can build econometric model such as property price = $f(\text{House characteristics}_k, \text{Neighbourhood}_m, \text{Intervention})$

However, need to account for market failure such as in a monopoly, since the appropriate valuation of a resource is the MC not p . Furthermore, we need to identify opportunity costs through shadow prices. For instance, in labour, the cost of employing a worker for an hour, if that worker is otherwise employed the shadow price is just the market wage which is the marginal product in an alternative job.

- Stated preference methods: nothing much to say its literally just surveys. Want to use a truth-telling mechanism

Social discount factor.

In examining cost and benefits, we want to calculate the net present value i.e.

$$NPV = \sum_{t=0}^{\infty} \frac{B_t - C_t}{(1+r)^t}$$

We want to choose r properly which is the social discount factor i.e. society's value of future consumption relative to the present.

We need to think about intertemporal utility since intertemporal welfare is the sum of all discounted utility flows from consumption c_t over time $t \in \{0, 1, \dots\}$. Hence

$$W(c_0, c_1, \dots) = \sum_{t=0}^{\infty} \frac{1}{(1+\rho)^t} u(c_t)$$

where ρ is the pure rate of time preference – the rate at which future utility is discounted.

Proposition 3.2 (Ramsey's social discount rate). In a two period model, $r = \rho + \eta g$

where ρ is pure rate of time preference, η is elasticity of marginal utility of consumption and g is the rate of consumption growth.

Proof. Remark: This is more a derivation rather than an actual proof.

With two periods,

$$W(c_1, c_2) = u(c_1) + \frac{1}{1+\rho} u(c_2)$$

Let $\eta = -\frac{cu''(c)}{u'(c)}$. Also,

$$u(c) = \begin{cases} \frac{c^{1-\eta}-1}{1-\eta} & \text{for } \eta \neq 1 \\ \ln(c) & \text{otherwise} \end{cases}$$

Differentiating $W(c_1, c_2(c_1))$ and with the FOC gives us

$$u'(c_1) + \frac{1}{1+\rho} u'(c_2) \frac{dc_2}{dc_1} = 0$$

and thus

$$\frac{dc_2}{dc_1} = -\frac{(1+\rho)u'(c_1)}{u'(c_2)} = -(1+\rho) \left(\frac{c_2}{c_1}\right)^\eta$$

The slope of the production possibility frontier that allows reductions in c_1 to be transformed into increases in c_2 is $-(1+r)$ where r is the real discount rate for consumption, since that is the rate of trade off between c_1 and c_2 by definition. Therefore,

$$1+r = (1+\rho) \left(\frac{c_2}{c_1} \right)^\eta$$

Taking logarithms and using the fact that $\frac{c_2}{c_1} = 1+g$, we get

$$\ln(1+r) = \ln(1+\rho) + \eta \ln(1+g)$$

For small x , $\ln(1+x) \approx x$. So, we yield

$$r = \rho + \eta g$$

■

3.2 Public Goods

References: Simon Cowan's slides; Mas-Collel, Whinston and Green, *Microeconomic Theory* Ch 11; Varian, *Microeconomic Analysis* Ch 23

Takeaways on Public Goods

1. Reasons for underprovision; Samuelson Condition for optimal provision and derivation; Equilibrium with Public Goods
2. Lindahl pricing as a solution to optimal provision
3. Clarke-Groves Mechanism as a strategyproof mechanism to incentivise agents to truthfully reveal values; The Clarke tax and interpretation of Clarke-Groves mechanism
4. How to prove strategyproofness for mechanisms to induce truthtelling

Definition 3.4 (Classical Taxonomy of Goods). The following is the classical taxonomy of goods based on the properties of excludability and rivalrousness (might be a spectrum)

- A good is non-rivalrous iff it is non-depletable/low congestion by the use of an additional unit of it; A good is non-excludable iff one cannot prevent another from consuming it.
- A good is a **Public Good** iff it is both non-rivalrous and non-excludable
- A good is a **Common Resource** iff it is non-excludable and rivalrous
- A good is a **Club Good** iff it is excludable and non-rivalrous
- A good is a **Private Good** iff it is excludable and rivalrous

3.2.1 Underprovision for Public Goods

The Public Good problem is that of underprovision; that actors which only consider their private costs contribute less than the optimal provision.

Theorem 3.1 (Samuelson Condition I). Consider a quasilinear utility function of a representative agent $i \in \mathcal{I}$, $U_i(Q) = y_i + \phi(Q)$ and a cost function $c(Q)$. The Samuelson condition provides the optimal level of public good provision:

$$\sum_{i=1}^{\mathcal{I}} U'_i(Q) = c'(Q)$$

Proof. The social planner's maximisation problem is

$$\max_{Q \geq 0} \sum_{i=1}^{\mathcal{I}} U_i - c(Q)$$

The summation is justified by U_i 's quasilinearity in money y_i . The constraint does not bind because it will always be better to choose $Q = \epsilon > 0$ than any other negative value of Q . The first order condition, immediately, gives us the Samuelson condition:

$$\sum_{i=1}^{\mathcal{I}} U'_i(Q) = c'(Q)$$

■

Proposition 3.3 (Underprovision under individual maximisation). Consider a quasilinear utility function of a representative agent $i \in \mathcal{I}$, $U_i(Q) = y_i + \phi(Q)$ and a cost function $c(Q)$. If individuals individually solve for their maximisation problem

$$\max U_i(Q) - c(Q)$$

the overall provision $\sum_{i=1}^{\mathcal{I}} q_i^* < Q^*$.

Intuition: The reason why people underprovide collectively when individually maximise utility is because they do not consider the marginal benefit they provide to others. The Samuelson condition maximises the aggregate surplus because the social planner internalises the 'positive externalities' from the provision of an additional public good.

Theorem 3.2 (Samuelson Condition II). Consider a utility function of a representative agent that faces a utility function over two goods where one is public and one is private i.e. $U_i(x, y) = v_i(x) + u_i(y)$ and faces a budget constraint of the form $p_x x + p_y y = m$. Then, the Samuelson condition is

$$\sum_{i \in \mathcal{I}} \frac{\partial U_i / \partial x}{\partial U_i / \partial y} = \frac{p_x}{p_y}$$

In other words, the sum of the marginal rates of substitutions across all agents should equal the rate of marginal transformation

$$\sum_{i \in \mathcal{I}} MRS_i = MRT$$

Proof. We prove this only in the two agent case. To internalise the externality, here we merge the people and choose the quantities of private goods and the overall level of the provision of the public good $\{x_1, x_2, g\}$ according to social preferences

$$W(u_1(x_1, g), u_2(x_2, g))$$

where W is some increasing function of the individuals' preferences. We then choose x_1, x_2, g to optimise social welfare, subject to the way that the public good is provided

$$g = g_1 + g_2$$

and the social budget constraint

$$p_x(x_1 + x_2) + p_g g = m_1 + m_2$$

This must give us a Pareto optimum because if we are maximising joint welfare it will be impossible to find a Pareto improvement. Here is the Lagrangian

$$\mathcal{L}(x_1, x_2, g, \lambda) = W(u_1(x_1, g), u_2(x_2, g)) - \lambda[p_x(x_1 + x_2) + p_g g - m_1 - m_2]$$

Here, the FOCs are (by chain rule)

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial x_1} &= \frac{\partial W}{\partial u_1} \frac{\partial u_1}{\partial x_1} - \lambda p_x = 0 \\ \frac{\partial \mathcal{L}}{\partial x_2} &= \frac{\partial W}{\partial u_2} \frac{\partial u_2}{\partial x_2} - \lambda p_x = 0 \\ \frac{\partial \mathcal{L}}{\partial g} &= \frac{\partial W}{\partial u_1} \frac{\partial u_1}{\partial g} + \frac{\partial W}{\partial u_2} \frac{\partial u_2}{\partial g} - \lambda p_g = 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= p_x(x_1 + x_2) + p_g g - m_1 - m_2 = 0\end{aligned}$$

Take $\frac{\partial \mathcal{L}}{\partial x_1}$

$$\frac{\partial W}{\partial u_1} \frac{\partial u_1}{\partial x_1} - \lambda p_x = 0$$

and rearrange

$$\frac{\partial W}{\partial u_1} = \lambda \frac{p_x}{\frac{\partial u_1}{\partial x_1}}$$

Similarly

$$\frac{\partial W}{\partial u_2} = \lambda \frac{p_x}{\frac{\partial u_2}{\partial x_2}}$$

Here is the $\mathcal{L}_g$ condition:

$$\left[\frac{\partial W}{\partial u_1} \right] \frac{\partial u_1}{\partial g} + \left[\frac{\partial W}{\partial u_2} \right] \frac{\partial u_2}{\partial g} = \lambda p_g$$

Sub in

$$\left[\lambda \frac{p_x}{\frac{\partial u_1}{\partial x_1}} \right] \frac{\partial u_1}{\partial g} + \left[\lambda \frac{p_x}{\frac{\partial u_2}{\partial x_2}} \right] \frac{\partial u_2}{\partial g} = \lambda p_g$$

Simplify

$$\left[\frac{\frac{\partial u_1}{\partial g}}{\frac{\partial u_1}{\partial x_1}} \right] + \left[\frac{\frac{\partial u_2}{\partial g}}{\frac{\partial u_2}{\partial x_2}} \right] = \frac{p_g}{p_x}$$

Which is

$$|MRS_{x_1, g}^1| + |MRS_{x_2, g}^2| = \frac{p_g}{p_x}$$

This characterises the socially-optimal level of provision. This is the Samuelson condition. ■

Contrast this with the individually optimal choice: Consider person 1. She has to choose $\{x_1, g_1\}$. Bt her utility depends, in part, on g_2 . If 2 chooses to contribute more, she gets the benefit. If 2 is expected to provide enough to satisfy 1 then 1 will free ride. Consider 1's problem: to choose the amount of the private good x_1 and her contribution to the public good g_1 according to her preferences given what she can afford and the way the public good is produced

$$\max_{x_1, g_1} u_1(x_1, g) \text{ s.t. } p_x x_1 + p_g g_1 = m_1 \text{ and } g = g_1 + g_2$$

We can take the second constraint and substitute it into the objective function

$$\max_{x_1, g_1} u_1(x_1, g_1 + g_2) \text{ s.t. } p_x x_1 + p_g g_1 = m_1$$

Solving using the method of the Lagrangian multiplier

$$\mathcal{L}(x_1, g_1, \lambda) = u_1(x_1, g_1 + g_2) - \lambda[p_x x_1 + p_g g_1 - m_1]$$

FOCs:

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial x_1} &= \frac{\partial u_1}{\partial x_1} - \lambda p_x = 0 \\ \frac{\partial \mathcal{L}}{\partial g_1} &= \frac{\partial u_1}{\partial g_1} - \lambda p_g = 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= p_x x_1 + p_g g_1 - m_1 = 0\end{aligned}$$

Taking the ratio of $\frac{\partial \mathcal{L}}{\partial x_1}$ and $\frac{\partial \mathcal{L}}{\partial g_1}$ gives the usual MRS relative price conditions

$$\frac{\frac{\partial u_1}{\partial x_1}}{\frac{\partial u_1}{\partial g_1}} = |MRS_{x_1, g_1}^1| = \frac{p_x}{p_g}$$

This describes the trade-off between her private consumption x_1 and her contribution to the public good g_1 . We are interested in the trade off with respect to the consumption of the public good g . Recall that $u(x_1, g_1 + g_2)$. The marginal utility of g_1 is (by the chain rule)

$$\frac{\partial u_1}{\partial g_1} = \frac{\partial u_1}{\partial (g_1 + g_2)} \frac{\partial (g_1 + g_2)}{\partial g_1}$$

And since $g_1 + g_2 = g$, we have

$$\frac{\partial u_1}{\partial (g_1 + g_2)} = \frac{\partial u_1}{\partial g}$$

and

$$\frac{\partial (g_1 + g_2)}{\partial g_1} = 1$$

So

$$\frac{\partial u_1}{\partial g_1} = \frac{\partial u_1}{\partial g}$$

This implies that she gets the same benefit from a change in the level of provision of the public good whether she pays for it or someone else does. It is this which gives rise to the free-rider problem (since she incurs a cost by contributing).

Example (Public Goods and Voluntary Contribution): A village has n residents, who each obtain utility from private goods, and a public good, flood defences. Resident i has utility function $U_i = y_i + aQ - \frac{Q^2}{100}$ where Q is the level of flood defences, and y_i is individual's i expenditure on private goods. Flood defence costs c per unit. We want to find the competitive outcome and the optimal outcome.

Solution.

Step 1: Finding the Pareto Optimal level through the Samuelson Condition

The utility function is

$$U_i = y_i + aQ - \frac{Q^2}{100}$$

Hence, the private marginal benefit is

$$\frac{dU_i}{dQ} = a - \frac{Q}{50}$$

The Samuelson condition is derived from the pareto optimal condition

$$\max_{Q \geq 0} \sum_{i=1}^n U_i(Q) - cQ$$

The positivity constraint $Q \geq 0$ does not bind because picking $Q = \epsilon > 0$ will yield a higher value than any $Q \leq 0$. So, our maximisation problem is

$$\max_Q \sum_{i=1}^n U_i(Q) - cQ$$

This gives us

$$\sum_{i=1}^n U'_i(Q) - c = 0 \implies \sum_{i=1}^n U'_i(Q) = c$$

which is the Samuelson condition. Specifically, it gives us the optimal value of Q ,

$$\sum_{i=1}^n \left(a - \frac{Q}{50} \right) = c \implies na - \frac{Qn}{50} = c$$

which gives us

$$Q = 50a - \frac{50c}{n}$$

The Samuelson condition maximises the aggregate surplus, i.e. utility net of cost.

Step 2: Finding the reaction function of an individual

We define $Q = \sum_{i=1}^n q_i$. The individual person 1 faces

$$\max U_1(Q) - cq_1$$

. So, given the definition of Q ,

$$\max_{q_1} U_1 \left(q_1 + \sum_{i=2}^n q_i \right) - cq_1 = y_1 + a \left(q_1 + \sum_{i=2}^n q_i \right) - \frac{(q_1 + \sum_{i=2}^n q_i)^2}{100}$$

Our FOC is

$$\frac{\partial U_1}{\partial q_1} = c$$

This gives us

$$a - \frac{q_1 + \sum_{i=2}^n q_i}{50} = c$$

This gives us the reaction function

$$q_1 = 50(a - c) - \sum_{i=2}^n q_i$$

Step 3: Using the reaction function to find symmetric equilibrium

Now, each i 's reaction function is $q_i = 50(a - c) - \sum_{i \neq j}^n q_j$. Since the equilibrium is symmetric, $q_1 = q_2 = \dots = q_n$, we yield

$$\begin{aligned} nq_i &= 50(a - c) \\ q_i &= \frac{50}{n}(a - c) \end{aligned}$$

So, $Q = nq_i = 50(a - c)$. ■

Remark: The solution Q does not depend on the number of individuals since as n increases, each person contributes less from their reaction function being decreasing in n since they rely on others more. Hence, the total remains the same.

Equilibrium with Public Goods

Note that in General Equilibrium, we assume that all goods have markets (i.e. markets are complete). However, public goods might be underprovided due to free-ridership so the positive externality has no market.

To illustrate this, we look at the following example:

Example (Equilibrium and Public Goods): Two students, a and b , share a house. Heat for this house is a pure public good. Student a feels the cold. Her (derived) benefit from x units of heat is:

$$\phi^a(x) = 20x - \frac{x^2}{2}$$

Student b is a hardier soul. Her (derived) benefit from x units of heat is:

$$\phi^b(x) = 15x - x^2$$

The heating system runs on prepayment cards. Each student can provide units of heat by purchasing cards from a single profit-maximising heat supply firm. This firm's cost of supplying q units of heat is:

$$c(q) = 2q^2$$

Let x^a denote the units of heat purchased by student a , and x^b the units of heat purchased by student b . Total heat provision is then $x = x^a + x^b$. Assume that both students, as well as the firm, take the market price of heat as given. Assume, also, that each student takes her housemate's purchase of heat as given. We want to find the free market level of provision of heat and the pareto efficient level of heat.

Solution.

Step 1: Consumer case

We assume that there is a competitive (Nash-Cournot) equilibrium involving price p^* , and student a 's purchase of the public good, x^{a*} , must maximise her utility given x^{b*} . Let $x = x^a + x^b$. Consumer a 's maximisation problem is

$$\max_{x^a} (\phi^a(x) - px^a) = 20(x^a + x^b) - \frac{(x^a + x^b)^2}{2} - 2(x^a)^2$$

The FOC is

$$\begin{aligned} \frac{\partial(\phi^a - px^a)}{\partial x^a} &= 20 - (x^a + x^b) - p = 0 \\ x^a &= 20 - x^b - p \end{aligned}$$

b 's problem is

$$\max_{x^b} (\phi^b - px^b) = 15(x^a + x^b) - (x^a + x^b)^2 - 2(x^b)^2$$

FOC:

$$\begin{aligned} \frac{\partial(\phi^b - px^b)}{\partial x^b} &= 15 - 2(x^a + x^b) - p = 0 \\ x^b &= \frac{15}{2} - x^a - \frac{p}{2} \end{aligned}$$

No, only student a will purchase heat. This is because student b will free ride on student a 's purchase of heat and will be satisfied by it, since it has a lower marginal utility of heating. So, the first unit of heating student b attempts to purchase will have a higher price than he is willing to have.

Step 2: Firm Case

Now we consider the firm's problem involving the price p^* and it must choose q^* to maximise profits. The firm's profit function is

$$\pi(q) = p^*q - 2q^2$$

The firm's maximisation problem:

$$\max_q \pi(q) = p^*q - 2q^2$$

FOC:

$$\begin{aligned}\frac{d\pi}{dq} &= p^* - 4q = 0 \\ q &= \frac{p^*}{4}\end{aligned}$$

Step 3: Solving for GE

The market clearing condition is

$$x^a + x^b = q$$

Substituting the demands from (a) gives us

$$\begin{aligned}20 - x^b - p + x^b &= \frac{p}{4} \\ 20 &= \frac{5p}{4} \\ p^* &= 16\end{aligned}$$

Thus, $q^* = \frac{p^*}{4} = 4$.

Step 4: Assessing and finding pareto efficient level of provision

$q^* = 4$ is not pareto efficient because it is a result of individual maximisation. The social planner's problem is

$$\begin{aligned}\max(\phi^a(x) + \phi^b(x) - px) &= 20x - \frac{x^2}{2} + 15x - x^2 - px \\ &= 35x - \frac{3x^2}{2} - px\end{aligned}$$

This is justified because we assume that the utilities functions are quasilinear and one-one in money. Hence, we can add them together. Our FOC is

$$\begin{aligned}\frac{\partial(\phi^a + \phi^b - px)}{\partial x} &= 35 - 3x - p = 0 \\ x &= \frac{35}{3} - \frac{p}{3}\end{aligned}$$

Since the market clearing condition is $x = q$, this means that

$$\begin{aligned}\frac{35}{3} - \frac{p}{3} &= \frac{p}{4} \\ 35 &= \frac{7p}{4} \\ p^* &= 20\end{aligned}$$

Hence, $q^o = x^o = 5$ ■

3.2.2 Solutions to underprovision

The central problem with public good provision by the state is that the government does not know how much an agent values the public good. Therefore, the solutions focus on revealing the agents valuations of the public goods.

1. **Clarke-Groves Mechanism:** The Clarke-Groves Mechanism has three components

- Each agents reports a "bid" for the public good, b_i . This may or may not be their true net value.
- The public good is provided if $\sum_i b_i \geq 0$, and it is not provided if $\sum_i b_i < 0$

- Each agent i receives a sidepayment equal to the sum of other bids $\sum_{j \neq i} b_j$, if the public good is provided. (If the sum is positive, agent i receives it; if it is negative, agent i must pay this amount.)

Proposition 3.4 (Strategy-Proofness of the Clarke-Groves Mechanism). The Clarke-Groves Mechanism is strategy-proof.

Proof. Proof is in the Appendix ■

Remark. This means that agents report their true net value regarding the provision of the public good. So the government achieves revelation.

Corollary 3.3 (Dictatorship). The Clarke-Groves Mechanism gives rise to dictatorial households

Proof. A trivial application of the Gibbard-Satterwhite Theorem. ■

Intuition: Consider a simple case where the externality is either produced ($h = 1$) or not ($h = 0$). The consumer receives y if $h = 0$ and $y - c^*$ if $h = 1$; the firm receives x if $h = 0$ and $x + b^*$ if $h = 1$. The government's maximisation problem is $\max_h \{b^* - c^*, 0\}$. The consumer's maximisation problem is $\max_c \{y - c^*, y\}$ while the firm's maximisation problem is $\max_b \{x + b^*, x\}$. Clearly, the firm will overreport b to infinity and consumer will overreport c to infinity. If the government subsidises the consumer by b and taxes the firm by c if the externality is allowed, the consumer seeks to now maximise $b - c^*$ and the firm $b^* - c$, which converges to the government's problem. So, the C-G mechanism attempts to align the consumer's and firm's problem with the government's.

Example (How to do calculate provision): The *gross* valuations of a public good for the four households of a small economy are 25, 20, 16 and 2 respectively. The cost of providing the public good is 60 and, if provided, each household will contribute 15 towards the cost. Should the public good will be provided? Use a Clarke-Groves mechanism to determine who is pivotal and how much tax they should pay. What can you say about the net benefit (after the contribution to the cost and the tax) of the pivotal households? How do your answers change when the gross valuations are 5, 10, 14 and 28? *Solution:* Since we know that the Clarke Groves mechanism is strategyproof and thus truthtelling is dominant, the total value reported is

$$v = (25 - 15) + (20 - 15) + (16 - 15) + (2 - 15) = 3 > 0$$

So the public good will be provided and should be provided since the households collectively value is positively and will be better off with it produced. ■

Example (How to prove strategy-proofness): Consider the same problem described in the intuition segment. We want to prove that truth-telling is weakly dominant.

Proof. Firm's Case:

Step 1: The firm will not over-report b because reporting b^* weakly dominates reporting $b > b^*$. We prove this by trichotomy. Consider some report c . If $c > b > b^*$, then the actions b and b^* yield x as the same payoff. If $b > c > b^*$, then the firm will be better off reporting b^* since the externality will be allowed but their payoff is $x + b^* - c < x$ where x is the payoff if firm reported b^* since it is less than c . Lastly, if $b > b^* > c$, then the payoffs are equivalent because the payoff is just $x + b^* - c$. So, by trichotomy, we establish that the report of b^* weakly dominates $b > b^*$.

Step 2: The firm will not under-report b because reporting b^* weakly dominates reporting $b^* > b$. We also prove this by trichotomy. Consider some arbitrary report c . If $c > b^* > b$ or $b^* > b > c$, then the actions b and b^* yield the same payoffs x or $x + b^* - c$ respectively. If $b^* > c > b$, then the payoff is $x < x + b^* - c$, so the firm is better off to report b^* . Claim established for the firm.

Consumer's Case: We prove both cases by trichotomy.

Step 3: The consumer will not over-report c . Consider some report b . If $c > c^* > b$ or $b > c > c^*$, then either action yields equal payoffs y or $y - c^* + b$. If $c > b > c^*$, then choosing c yields y which is lower than $y - c^* + b$. So the consumer is better off reporting c^* .

Step 4: The consumer will not under-report c either. Consider some arbitrary b . If $c^* > c > b$ or $b > c^* > c$, then either action yields equal payoffs y or $y - c^* + b$ respectively. If $c^* > b > c$, then the consumer is better off choosing c^* since their payoff will be y which is higher than $y - c^* + b$.

So, the consumer's choice of c^* weakly dominates any other choice of c . ■

Clarke Tax: Suppose the government cannot make net transfers. Clarke proposed an additional tax of $t_i = \max\{\sum_{j \neq i} r_j, 0\}$. An agent pays money to the government iff they are pivotal. The tax net of payment is $t_i - p_i \geq 0$.

- If $r_i + \sum_{j \neq i} r_j > 0$ and $\sum_{j \neq i} r_j < 0$, $t_i - p_i = -p_i = -\sum_{j \neq i} r_j > 0$
- If $r_i + \sum_{j \neq i} r_j < 0$ and $\sum_{j \neq i} r_j > 0$, $t_i - p_i = t_i = \sum_{j \neq i} r_j > 0$

If i is pivotal, then $t_i - p_i$ equals their effects on all others welfare. The goal of the Clarke tax therefore is for the agent to internalise the externality he imposes on others.

2. **Lindahl Pricing:** The Lindahl pricing has the following steps

- Each consumer reports their marginal utility function, $u'_i(x)$. We assume truth-telling
- The government adds the marginal utilities together, choosing x^* which satisfies the Samuelson condition and sets a unit tax for i equal to their marginal utility at the optimum:

$$t_i = u'_i(x^*)$$

- By construction, the total tax revenue equals the cost of the public good since

$$\sum_i^n t_i = \sum_i^n u'_i(x^*) = c$$

The second equality follows from the Samuelson condition.

Note that this is why the Lindahl pricing mechanism is called personalised Lindahl price. The word 'tax' is a misnomer; it is the entire price that the consumer pays. Here is a graphical illustration of the two individual case:

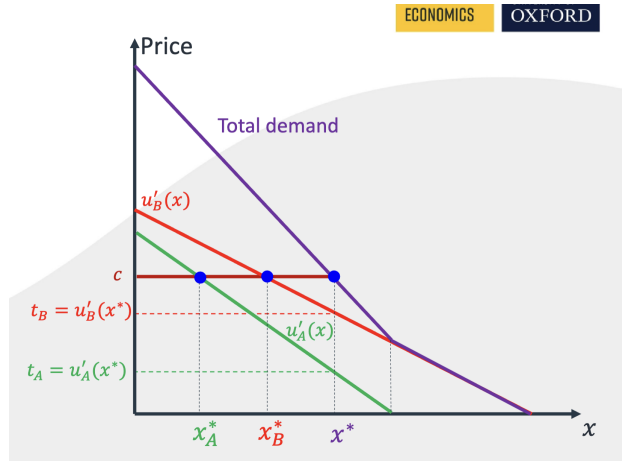


Fig 3.2: Lindahl Pricing

Benefits:

- Efficient
- Balanced Budget

Costs:

- High informational demands on the government
- Not strategyproof: people will report their value such that the Lindahl pricing would give them their individually optimal consumption

Example. College 1 and College 2 want to establish a joint library, with books available to members of both colleges. The preferences of both colleges are the same, and are represented by the utility function $u_i(B, x_i) = \frac{1}{3}\ln B + \frac{2}{3}\ln x_i$ where $b_1 + b_2 = B$ is the total number of books in the joint library and x_i is the money that college i has for other uses. Each college has an initial endowment of \$600 and no books. The price of books is $p_b = 1$ and the price of money is $p_x = 1$.

1. Suppose that each college decides, independently of the other, how many books to purchase and put in the joint library.

- (a) Show that, if college 2 were to buy b_2 books, where $b_2 < 300$, college 1 would buy $b_1 = 200 - \frac{2}{3}b_2$ books.

College i faces

$$\max u_i(B, x_i) = \frac{1}{3}\ln(b_i + b_{-i}) + \frac{2}{3}\ln x_i \quad \text{s.t.} \quad b_i + x_i \leq 600$$

The Lagrangian is

$$\mathcal{L}(b_i, x_i, \lambda) = \frac{1}{3}\ln(b_i + b_{-i}) + \frac{2}{3}\ln x_i - \lambda(b_i + x_i - 600)$$

Our FOCs are

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial b_i} &= \frac{1}{3} \cdot \frac{1}{b_i + b_{-i}} - \lambda = 0 \\ \frac{\partial \mathcal{L}}{\partial x_i} &= \frac{2}{3} \frac{1}{x_i} - \lambda = 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= 600 - b_i - x_i = 0 \end{aligned}$$

We yield the condition $2b_i + 2b_{-i} = x_i$ which gives us $b_i + 2b_i + 2b_{-i} = 600$. Hence,

$$b_1 = 200 - \frac{2}{3}b_2$$

- (b) How many books in total would there be in the library when each college chooses independently? By symmetry, $b_2 = 200 - \frac{2}{3}b_1$. Solving both gives us $b_1 = b_2 = 360 - 240 = 120$. So, $b_1 + b_2 = 240$

2. Consider now the optimal joint provision of books.

- (a) Write down the condition that an optimal provision of books must satisfy. The Samuelson condition is

$$MRS_1 + MRS_2 = MRT \implies \frac{x_1}{2(b_1 + b_2)} + \frac{x_2}{2(b_1 + b_2)} = 1$$

- (b) What is the optimal number of books and why is it different from the number of books purchased when each college decides independently of the other?

Since $x_1 = 600 - b_1, x_2 = 600 - b_2$, we yield

$$\frac{600 - b_1}{2(b_1 + b_2)} + \frac{600 - b_2}{2(b_1 + b_2)} = 1 \implies b_1 + b_2 = 400$$

The Samuelson condition exhausts all marginal benefit and internalises the externality.

3. Suppose now that the university wants to help colleges to implement the socially optimal number of books.

- (a) Describe formally how the Lindahl pricing mechanism would work in this situation. (You may assume that in the social optimum each college pays for half the optimal number of books).
The Lindahl pricing has two steps: (1) The College reports MRS_i assuming truth telling, (2) The government adds the MRS together, choosing b_1^*, b_2^* satisfying

$$MRS_1 + MRS_2 = MRT$$

and sets a unit price p_i equal to their MRS_i at the optimum B^* . Since $b_1 = b_2 = 300$ by assumption, then $MRS_i = p_i = 1/2$. So, the new constraint is $x_i + \frac{1}{2}B = 600$. So, each college faces

$$\max \frac{1}{3} \ln(B) + \frac{2}{3} \ln(600 - 0.5B)$$

Our FOC is

$$\frac{1}{3} \cdot \frac{1}{B} - \frac{2}{3} \cdot \frac{1/2}{600 - 0.5B} = 0 \implies B = 600 - 0.5B$$

So, $1.5B = 600 \implies B = 400$. So, every college will choose $B = 400$ as their optimal bundle.

Remark: In the Lindahl framework, the college does not choose b_i but rather the total provision B .

- (b) Describe formally why the notion of strategy-proofness is relevant in this type of problem.
If the utility functions are private information then Lindahl pricing won't work: each is tempted to say that they value books less at the margin than is true, because this then reduces the individual price they pay.
- (c) The university decides to let colleges act independently, and pays for a subsidy for books so the new price is below 1. What price will ensure that colleges buy the optimal number of books?
If we assume a general p for the price of books, then the Nash Equilibrium is $b_1 = \frac{200}{p} - \frac{2}{3}b_2$. By symmetry,

$$b_1 = b_2 = \frac{120}{p}$$

Since $b_1^* = b_2^* = 200, p = \frac{3}{5}$. So the subsidy is $s = 1 - p = \frac{2}{5}$

3.3 Negative Externalities

References: Simon Cowan's slides; Mas-Collé, Whinston and Green, *Microeconomic Theory* Ch 11; Varian, *Microeconomic Analysis* Ch 23

Takeaways on Negative Externalities

1. How to derive Pareto optimal amount of transactions; reasons for overprovision; drawing marginal social costs or benefit curves
2. General Equilibrium with externalities
3. Solutions: Pigouvian tax, Coasean bargaining and property rights; informational demands for different solutions
4. Applications to Climate Change

General Equilibrium with an externality

In our standard analysis of General Equilibrium models, we assume that markets are complete. However, negative externalities are unpriced by the market and thus do not have a market for them. This is why General Equilibrium models fail to yield a Pareto efficient outcome i.e. the MRS condition or the contract

curve fails.

The following example illustrates that when there are n agents, the externality does not get internalised or price into the market and this is why the market becomes inefficient. However, when there are only two agents, bilateral bargaining allows them to internalise the externality. **Example.** Consider a pure exchange economy with two goods, x and y , and *two equally large groups of consumers* a and b . The preferences of each type a consumers are represented by the utility function $u_a(x_a, y_a, x_b) = \ln x_a + 2\ln y_a - \ln x_b$, and they each have an endowment of $(6, 0)$, whereas the preferences of each type b consumer are represented by the utility function $u_b(x_b, y_b) = \ln x_b + 2\ln y_b$, and they each have an endowment $(0, 6)$.

The Walrasian equilibrium can be found as follows: Normalise the price of good y to 1, so the price of good x is $\rho = \frac{p_1}{p_2}$

The budget constraint for type a is:

$$\rho x_a + y_a \leq 6\rho$$

For type b is:

$$\rho x_b + y_b \leq 6$$

By inspection, since type b consumers exhibit Cobb-Douglas utility functions, their solution to:

$$\max u_b(x_b, y_b) = \ln x_b + 2\ln y_b \quad \text{s.t.} \quad \rho x_b + y_b \leq 6, x_b \geq 0, y_b \geq 0$$

are $x_b = \frac{6}{3\rho} = \frac{2}{\rho}, y_b = \frac{2 \cdot 6}{3} = 4$.

Type a agents faces the problem:

$$\max u_a(x_a, y_a, x_b) = \ln x_a + 2\ln y_a - \ln x_b \quad \text{s.t.} \quad \rho x_a + y_a \leq 6\rho, x_a \geq 0, y_a \geq 0$$

Due to continuity and strict monotonicity in x_a, y_a , the solution to CP-A must lie on the budget line i.e. when constraint binds with equality. We know that the weak positivity constraints do not bind because choosing $x_a \leq 0$ or $y_a \leq 0$ will induce negative infinite utility, so choosing $\epsilon > 0$ will create an infinite amount of marginal utility.

Because there are $2n$ agents in the economy, we assume that the agents do not internalise the externality since each agent's consumption is so small relative to the entire economy. Hence, the problem for consumer a is

$$\max u_a(x_a, y_a) = \ln x_a + 2\ln y_a - \ln x_b \quad \text{s.t.} \quad \rho x_a + y_a = 6\rho$$

The Lagrangian is

$$\mathcal{L}(x_a, y_a) = \ln x_a + 2\ln y_a - \ln x_b - \lambda(\rho x_a + y_a - 6\rho)$$

Our FOCs are

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x_a} &= \frac{1}{x_a} - \lambda\rho = 0 \\ \frac{\partial \mathcal{L}}{\partial y_a} &= \frac{2}{y_a} - \lambda = 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= 6\rho - \rho x_a - y_a = 0 \end{aligned}$$

From the FOCs we yield:

$$\frac{1}{\rho x_a} = \frac{2}{y_a} \implies y_a = 2\rho x_a$$

Substituting into the constraint:

$$6\rho = 3\rho x_a \implies x_a = 2, y_a = 4\rho$$

Because we have n a-type consumers and n b-type consumers, our excess demands are

$$z_x(\rho, 1) = \sum_1^n (2 + \frac{2}{\rho} - 6)$$

$$z_y(\rho, 1) = \sum_1^n (4\rho + 4 - 6)$$

By Walras law, if $n-1$ markets clear, the n th market clears; so

$$z_y(\rho, 1) - 4n\rho - 2n = 0 \implies \rho = \frac{1}{2}$$

The Walrasian Equilibrium is the price vector $\mathbf{p} = (1/2, 1)$, with the allocations $(x_a, y_a) = (2, 2), (x_b, y_b) = (4, 4)$.

Note that the Walrasian equilibrium here is NOT pareto efficient. We can see this because the underlying MRS condition is violated. Let $x_b := 6 - x_a$ i.e. pricing in the externality. So $u_a(x_a, y_a) = \ln x_a + 2\ln y_a - \ln(6 - x_a)$ The MRS for a becomes

$$\text{MRS}_a = \frac{\partial u_a / \partial x_a}{\partial u_a / \partial y_a} = \frac{y_a}{x_a} + \frac{y_a}{6 - x_a}$$

The MRS for b is

$$\text{MRS}_b = \frac{y_b}{x_b}$$

It is easy to verify that under $(x_a, y_a) = (2, 2), (x_b, y_b) = (4, 4)$,

$$\text{MRS}_a \neq \text{MRS}_b$$

So the FWT fails under the presence of externality.

Assume now that there is *only one consumer of each type*. Since there are only 2 consumers in the economy, bilateral bargaining can occur and consumer a can perceive the negative externality from b 's consumption of x ; the negative externality would be internalised into the market and priced in so b can compensate a for its consumption of x_b . The outcome would be different in that consumer a would have more x_a and consumer b would have less x_b . The relative price of x compared to y would fall since consumer b consumes less x , since his initial endowment of $x_b = 0$, the bargaining for less x results in a lower relative price. The increase in consumption of x by a does not change the relative price of x since he is endowed with all of x as $x_a = 6$ initially.

If the externality from x_b is internalised, we can take $x_b = 6 - x_a$. Then, a 's maximisation problem becomes

$$\max u^a(x_a, y_a, x_b) = \ln x_a + 2\ln y_a - \ln(6 - x_a) \quad \text{s.t.} \quad \rho x_a + y_a \leq 6\rho, x_a \geq 0, y_a \geq 0$$

For the analogous reasons above, the positivity constraints do not bind and the solution must lie when the budget constraint binds with equality due to continuity and local non-satiation.

Our Lagrangian is

$$\mathcal{L}(x_a, y_a) = \ln x_a + 2\ln y_a - \ln(6 - x_a) - \lambda(\rho x_a + y_a - 6\rho)$$

Our FOCs are

$$\frac{\partial \mathcal{L}}{\partial x_a} = \frac{1}{x_a} + \frac{1}{6 - x_a} - \lambda\rho = 0$$

$$\frac{\partial \mathcal{L}}{\partial y_a} = \frac{2}{y_a} - \lambda = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 6\rho - \rho x_a - y_a = 0$$

From the FOCs, we yield

$$\frac{1}{\rho x_a} + \frac{1}{\rho(6-x_a)} = \frac{2}{y_a} \implies \frac{6}{\rho(x_a(6-x_a))} = \frac{2}{y_a}$$

So,

$$y_a = \rho \frac{x_a(6-x_a)}{3}$$

Substituting into the constraint, we yield

$$18 = 3x_a + 6x_a - x_a^2 \implies (x_a - 3)(x_a - 6) = 0$$

So, either $x_a = 3$ or $x_a = 6$. But $x_b = \frac{2}{\rho} \neq 0$, so $x_a \neq 6$. We construct our excess demands and the market clearing condition using Walras Law,

$$z_x(\rho, 1) = 3 + \frac{2}{\rho} - 6 = 0$$

$$z_y(\rho, 1) = 3\rho + 4 - 6 = 0$$

so, $\rho = 2/3$. The allocations become $(x_a, y_a) = (3, 2)$ and $(x_b, y_b) = (3, 4)$, with the price vector $\mathbf{p} = (2/3, 1)$. ■

Example: Externalities and Gains from Trade.

Two students, a and b live in college. Student a has only one hobby: listening to “*****” music. Student a ’s music system can produce noise levels of up to 100 decibels. Her utility depends on the loudness of the sound, D , and the amount of money she has, m^a . Her preferences are represented by the utility function:

$$u^a(m^a, D) = 10D^{\frac{1}{2}} + m^a$$

Student b lives next door to student a and is irritated by the loud music he has to put up with. His preferences are represented by the utility function:

$$u^b(m^b, D) = 10(100 - D)^{\frac{1}{2}} + m^b$$

where m^b denotes the amount of money he has.

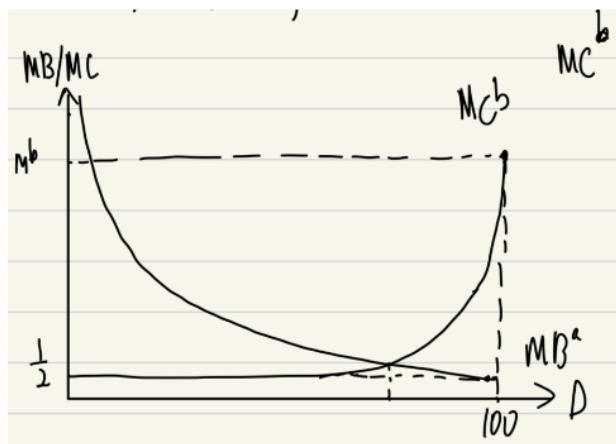
The utility functions are

$$u^a(m^a, D) = 10D^{\frac{1}{2}} + m^a \quad \text{and} \quad u^b(m^b, D) = 10(100 - D)^{\frac{1}{2}} + m^b$$

Their marginal benefit and cost curves are

$$MB^a = \frac{\partial u}{\partial D} = 5D^{-\frac{1}{2}}$$

$$MC^b = -\frac{\partial u}{\partial D} = 5(100 - D)^{-\frac{1}{2}}$$



The pareto efficient level of D would be when $MC^b = MB^a$ since no one can be made better off without making someone else worse off. This can be achieved by maximising

$$W(m^a, m^b, D) = u^a + u^b$$

The first order condition is

$$\frac{\partial W}{\partial D} = \frac{\partial u^a}{\partial D} + \frac{\partial u^b}{\partial D} = 0$$

which implies

$$\frac{\partial u^a}{\partial D} = -\frac{\partial u^b}{\partial D}$$

This yields,

$$5D^{-\frac{1}{2}} = -5(100 - D)^{-\frac{1}{2}}$$

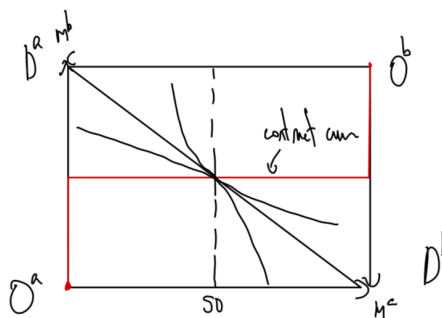
So, $D = 50$ is the pareto optimal level of noise.

Student b has £100. What is the maximum “bribe” he is willing to pay to student a to turn down her music to a Pareto-efficient level? Would £50 be sufficient?

Since we operate under a quasilinear utility framework where u^a is linear in m^a , we can assume that a unit rise in money is a unit rise in utility. Student b would be willing to pay a value b in between his total cost and a ’s total benefit over $100 \geq D \geq 50$. Formally, this is the interval S where

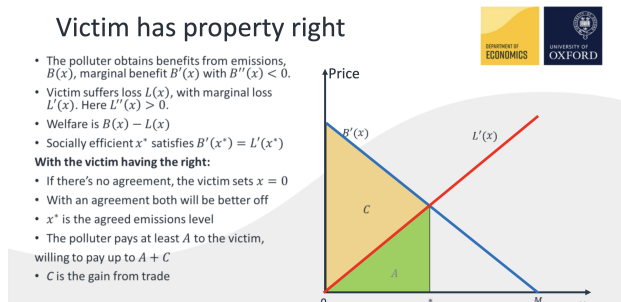
$$\int_{50}^{100} 5(100 - D)^{-\frac{1}{2}} dD \geq S \geq \int_{50}^{100} 5D^{-\frac{1}{2}} dD \implies 10\sqrt{50} \geq S \geq 50 - 10\sqrt{2}$$

Since $50 > 50 - 10\sqrt{2}$, 50 is sufficient. The edgeworth box to illustrate the problem is

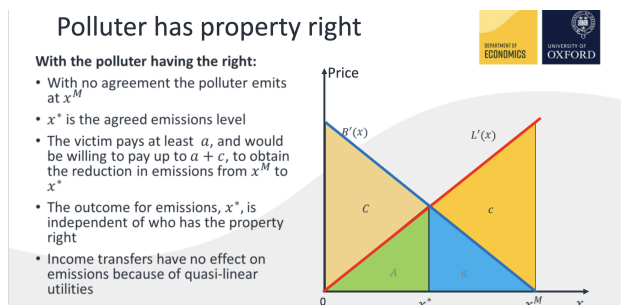


Solutions

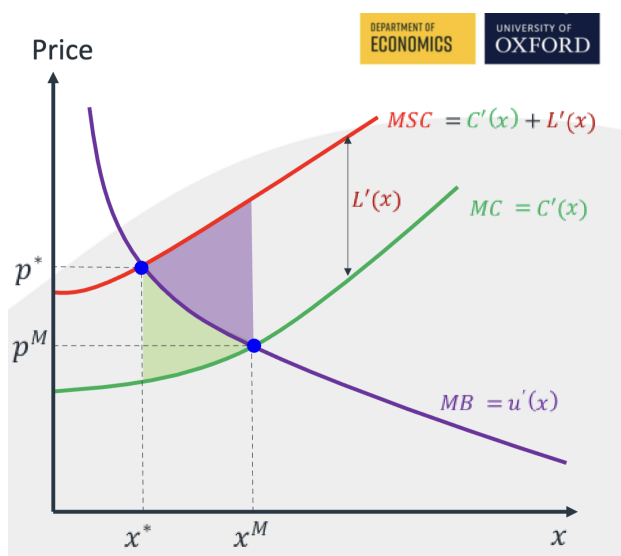
1. Coasean Bargaining. The Coase Theorem states that irrespective of the allocation of property rights, as long as rights are clearly specified and are enforced) frictionless bargaining produces an efficient outcome in presence of externalities. Consider the victim case



and the polluter case



2. Taxes: Under a perfectly competitive market, we have a firm that chooses to maximise $px - C(x)$ and a consumer with utility function $u(x)$. The consumers suffer a loss $L(x)$. So, the marginal social cost is $MSC = C'(x) + L'(x)$.



Aggregate welfare is $W = u(x) - C(x) - L(x)$. Society's FOC is

$$\frac{dW}{dx} = u'(x) - C'(x) - L'(x) = 0 \implies u'(x) = C'(x) + L'(x)$$

So, $x^M > x^*$ is too high. Green area is the reduction in $u(x) - C(x)$ to move to x^* . So the green plus purple area is the total gain from less of the externality, while the purple area is the net gain for society to optimise to x^* .

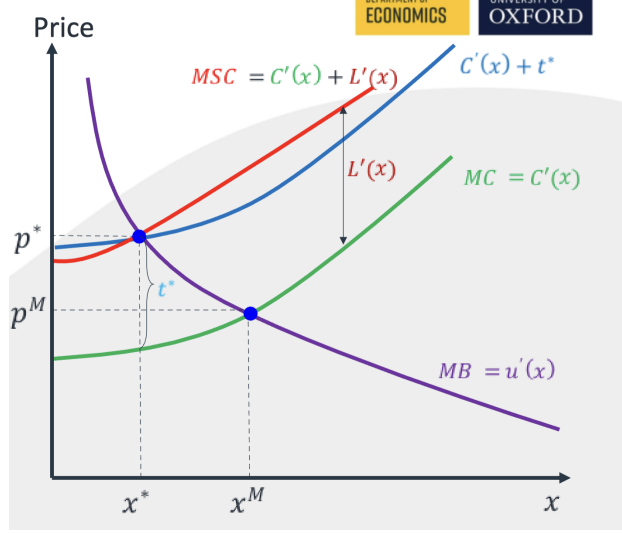
A Pigouvian tax is a unit tax such that you tax the firm's output. So the firm faces

$$\pi = px - C(x) - tx$$

FOC is

$$p = C'(x) + t$$

Consumer receives tax revenue tx . At the social optimum we need $t^* = L'(x^*)$ i.e. the tax to equal marginal external cost evaluated at the optimum. Illustration:

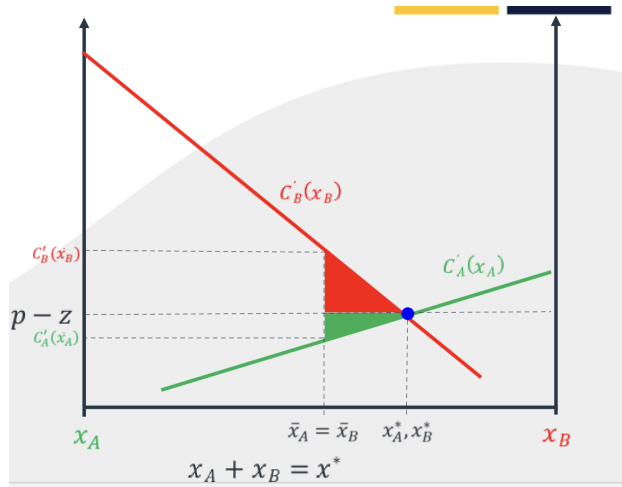


3. Quotas: we consider quotas through the cap and trade system. Let us assume one consumer, one third party and n firms who emit pollution. Let us assume that emissions = output for simplicity. The government sets maximum level of output at x^* . Firm i is given permission of $\bar{x}_i$ with $\sum_i \bar{x}_i = x^*$. Permits can be traded between firms. If firm i wants to emit $x_i > \bar{x}_i$, it must purchase $\bar{x} - x_i$ additional permits at price z . The permit price equals the marginal external cost evaluated at optimum x^* i.e. $z = L'(x^*)$.

The Cap and Trade system induces efficient allocation across emitters. Each firm aims to maximise $\pi_i = px_i - C_i(x_i) - z(x_i - \bar{x}_i)$. The FOC is

$$\frac{d\pi}{dx_i} = p - C'_i(x_i) - z = 0.$$

An illustration of how the trading works:



Firm A wants to produce more and thus gains the green triangle wrt the beginning permit allocations. Firm B is willing to sell permits and thus gains the red triangle wrt the beginning permit allocations.

Proposition 3.5. Permit price z equals the marginal external cost evaluated at the optimum x^* . In other words,

$$z = L'(x^*)$$

Let us assume that $C_i(x_i) = \frac{a_i x_i^2}{2}$ so that $C'_i(x_i) = a_i x_i$. Because $\sum_i x_i = \sum_i \bar{x}_i = x^*$ the retail price for emissions is optimal i.e. $p = p^*$. Society's optimal output is defined by

$$p^* = a_i x_i + L'(x^*) \implies x_i^* = \frac{p^* - L'(x^*)}{a_i}$$

So, the aggregate demand is

$$\sum_i x_i^* = (p^* - L'(x^*)) \sum_i \left(\frac{1}{a_i} \right) = x^*$$

The firm's FOC is

$$p^* - a_i x_i - z = 0 \implies x_i = \frac{p^* - z}{a_i}$$

So, the aggregate supply is

$$\sum_i x_i = (p^* - z) \sum_i \left(\frac{1}{a_i} \right) = x^*$$

Since aggregate demand = aggregate supply,

$$(p^* - z) \sum_i \left(\frac{1}{a_i} \right) = (p^* - L'(x^*)) \sum_i \left(\frac{1}{a_i} \right)$$

By inspection we have $z = L'(x^*)$.

Deciding between taxes and quotas Note the two policies seem to be dual of one another

- Taxes: Set the Pigouvian tax t^* equal to the external marginal cost at x^* to induce the optimal quantity x^*
- Quota: Set the optimal quantity x^* which requires the price of the permits z^* to equal to the external marginal cost at x^*

Intuitively, it seems that quotas should be equivalent to taxes. So what makes a government prefer one over the other? The answer is uncertainty over the information the government has – the government uses its expectation of the social benefit from the tax or the quota. But often, it does not know how flat or steep the MSC is etc. So, the government will choose a policy that minimises the expected deadweight loss for error given their informational constraint.

Prices vs Quantities (Weitzman, 1974). The main thesis of this paper is that uncertainty over marginal costs and benefits creates preference for taxes vs quotas. This gives rise to the following proposition:

Proposition 3.6. Let Δ be the relative advantage of taxes over quantities and $B(\cdot)$ be the social benefit function and $C(\cdot)$ be the social cost function. $\Delta > 0$ if $B'' > |C''|$ and $\Delta < 0$ if $B'' < |C''|$.

Now, we dive into the paper in detail (only the problem statement is essential; the derivation of the result is not).

1. Problem Statement.

The social planner's problem is to maximise social welfare by choosing an optimal q i.e. it faces

$$\max_q W(q) = B(q) - C(q)$$

The solution q^* must therefore satisfy

$$B'(q^*) = C'(q^*)$$

We also know that $p^* \equiv B'(q^*) = C'(q^*)$. In the ideal world, the planner announcing price p^* and the firm thereby maximising $p^* q - C(q)$ and the planner establishing a quota q^* makes no difference

However, in reality, the government does not know the social cost and benefit functions. In effect, they face a stochastic function for both costs and benefits:

$$B(q, \eta) \quad \text{and} \quad C(q, \theta)$$

where θ, η are your stochastic parameters. For all intents and purposes, you can treat $\theta, \eta \sim N(\mu, \sigma)$ but it does not really matter.

An ideal instrument of central control would be a contingency message whose instructions depend on which state of the world is revealed by η and θ . The ideal ex ante quantity signal and price signal are in the form of the following schedule as functions of θ and η

$$B(q^*(\theta, \eta), \eta) = C(q^*(\theta, \eta), \theta) = p^*(\theta, \eta)$$

By employing either ideal signal, the ex ante uncertainty has in effect been eliminated ex post and we are right back to the case where there is no theoretical difference between price and quantity control modes. It is infeasible for the center to transmit an entire schedule of ideal prices or quantities. A contingency message is a complicated, specialised contract which is expensive to draw up and hard to understand.

Given the informational failure, the social planner can only hope to maximise their expected welfare. In other words, they want to choose $\hat{q}$ s.t.

$$E[B(\hat{q}, \eta) - C(\hat{q}, \theta)] = \max_q E[B(q, \eta) - C(q, \theta)]$$

By linearity of expectations and the first order condition, we yield that the choice $\hat{q}$ satisfies

$$E[B_1(\hat{q}, \eta)] = E[C_1(\hat{q}, \theta)]$$

Now, when the price instrument is announced as p , production is adjusted such that $q = h(p, \theta)$ which maximises profits given η and θ . In other words,

$$ph(p, \theta) - C(h(p, \theta), \theta) = \max_q pq - C(q, \theta) \quad (1)$$

implying that $p = C'(h(p, \theta), \theta)$. If planners are rational, they want to choose a price instrument $\tilde{p}$ to maximise the expected difference between social benefits and costs i.e.

$$E[B(h(\tilde{p}, \theta), \eta) - C(h(\tilde{p}, \theta), \theta)] = \max_p E[B(h(p, \theta), \eta) - C(h(p, \theta), \theta)]$$

So, $\tilde{p}$ must satisfy

$$E[B_1(h(\tilde{p}, \theta), \eta) \cdot h'(\tilde{p}, \theta)] = E[C_1(h(\tilde{p}, \theta), \theta) \cdot h'(\tilde{p}, \theta)]$$

Since we know that $p = C_1(h(p, \theta), \theta)$, we can write $\tilde{p}$ as

$$\tilde{p} = \frac{E[B_1(h(\tilde{p}, \theta), \eta) \cdot h'(\tilde{p}, \theta)]}{E[h'(\tilde{p}, \theta)]}$$

Corresponding to the ex ante price $\tilde{p}$ is the ex post profit maximising quantity $\tilde{q}(\theta) \equiv h(\tilde{p}, \theta)$. In the presence of uncertainty, price and quantity instruments transmit central control in quite different ways. It is important to note that by choosing a specific mode for implementing an intended policy, the planners are at least temporarily locking themselves into certain consequences.

In all likelihood, for some choice of $\hat{q}$ and $\tilde{p}$ we have

$$\begin{aligned} B_1(\hat{q}, \eta) &\neq C_1(\hat{q}, \theta) \\ B_1(\tilde{q}(\theta), \eta) &\neq C_1(\tilde{q}(\theta), \theta) \end{aligned}$$

Neither instrument yields the optimum ex post. The relevant question becomes which one comes closest or minimises the potential deadweight loss.

2. Solution. We want to introduce what we call the relative advantage of prices over quantity, defined by

$$\Delta = E[B'(\tilde{q}(\theta), \eta) - C'(\tilde{q}(\theta), \theta) - (B'(\hat{q}, \eta) - C'(\hat{q}, \theta))]$$

This basically just says Δ is the difference between the net welfare generated by using taxes vs net welfare generated by using quotas.

Now for the more involved part – but it is really quite simple. We want to characterise the social costs and benefit functions through second order Taylor approximations to give the functions more tractability. So we have

$$\begin{aligned} C(q, \theta) &\approx a(\theta) + (C' + \alpha(\theta))(q - \hat{q}) + \frac{C''}{2}(q - \hat{q})^2 \\ B(q, \eta) &\approx b(\eta) + (B' + \beta(\eta))(q - \hat{q}) + \frac{B''}{2}(q - \hat{q})^2 \end{aligned}$$

$a(\theta), b(\eta), \alpha(\theta), \beta(\eta)$ are stochastic functions while C', C'', B', B'' are fixed coefficients. Note that WLOG, $E[\alpha(\theta)] = E[\beta(\eta)] = 0$. Since θ and η are independently distributed,

$$E[\alpha(\theta) \cdot \beta(\eta)] = 0.$$

Note that the stochastic functions

$$\begin{aligned} a(\theta) &\approx C(\hat{q}, \theta) \\ b(\eta) &\approx B(\hat{q}, \eta) \end{aligned}$$

translate different values of θ and η into pure vertical shifts of the cost and benefit curves. Differentiating with respect to q ,

$$\begin{aligned} C_1(q, \theta) &\approx (C' + \alpha(\theta)) + C'' \cdot (q - \hat{q}), \\ B_1(q, \eta) &\approx (B' + \beta(\eta)) + B'' \cdot (q - \hat{q}). \end{aligned}$$

Employing the above equations, the following interpretations are available for the fixed coefficients of the Taylor approximations:

$$\begin{aligned} C' &\approx E[C_1(\hat{q}, \theta)], \\ B' &\approx E[B_1(\hat{q}, \eta)], \\ C'' &\approx C_{11}(q, \theta), \\ B'' &\approx B_{11}(q, \eta). \end{aligned}$$

We yield

$$B' = C'.$$

It is apparent that stochastic changes in $\alpha(\theta)$ represent pure unbiased shifts of the marginal cost function. The variance of $\alpha(\theta)$ is precisely the mean square error in marginal cost:

$$\sigma^2 = E[(C_1(q, \theta) - E[C_1(q, \theta)])^2] = E[\alpha(\theta)^2].$$

Analogous comments hold for the marginal benefit function where we have

$$E[(B_1(q, \eta) - E[B_1(q, \eta)])^2] = E[\beta(\eta)^2].$$

Furthermore

$$h(p, \theta) \approx \hat{q} + \frac{p - C' - \alpha(\theta)}{C''}.$$

implying

$$h_1(p, \theta) \approx \frac{1}{C''}.$$

Using the previous equation with the expression of $\tilde{p}$ and cancelling out C'' yields

$$\tilde{p} \approx E[B_1(h(\tilde{p}, \theta), \eta)].$$

Replacing q in the expression for B_1 by the expression for $h(\tilde{p}, \theta)$ and plugging into the previous equation, the following equation is obtained after using mean zero of $\alpha(\theta), \beta(\theta)$

$$\tilde{p} \approx B' + \frac{B''}{C''}(\tilde{p} - C').$$

From $B' = C'$ and the condition $B'' < 0 < C''$, the previous equation implies

$$\tilde{p} \approx C'.$$

Combining we yield,

$$\tilde{q}(\theta) \approx \hat{q} - \frac{\alpha(\theta)}{C''}.$$

Now alternately substitute $q = \hat{q}$ and $q = \tilde{q}(\theta)$ from previous into the taylor approximations. Then plugging the resulting values into Δ , using mean zero and independence, and collecting terms,

$$\Delta \approx \frac{\sigma^2 B''}{2C''^2} + \frac{\sigma^2}{2C''}.$$

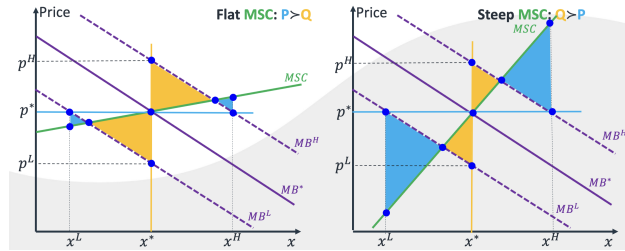
What this says is that if prices are better than quantity i.e. $\Delta > 0$, then $\frac{\sigma^2 B''}{2C''^2} > -\frac{\sigma^2}{2C''}$ which implies

$$B'' > |C''|$$

and also if quantity is better than prices i.e. $\Delta < 0$, then $\frac{\sigma^2 B''}{2C''^2} < -\frac{\sigma^2}{2C''}$ implying

$$B'' < |C''|$$

Here is some graphical intuition for Weitzman's result:



Example. A firm is deciding how much pollution to emit. Its profit depends on pollution h , a managerial efficiency parameter η , and the amount of money w_f that it has. Its profit function is $\pi(h; \eta) = w_f + \eta h - 2h^2$. The pollution affects a consumer living nearby. The consumer's preferences are represented by the utility function $\phi(h) = w_c - h^2$, where w_c denotes the amount of money she has. Assume that $\eta = 24$

1. The firm chooses to

$$\max \pi(h) = w_f + 24h - 2h^2$$

Our FOC is

$$\frac{\partial \pi}{\partial h} = 24 - 4h = 0$$

Firm chooses $h = 6$.

2. Given that $\phi(h)$ is linear in w_c , and that a 1 unit rise in w_c implies 1 unit increase in $\phi(h)$, $\phi(h)$ is measured in money. So we can aggregate welfare as

$$W(h) = \phi(h) + \pi(h)$$

To find the optimal h is to

$$\max W(h)$$

Our FOC is

$$\frac{\partial W}{\partial h} = \frac{\partial \pi}{\partial h} + \frac{\partial \phi}{\partial h} = 24 - 4h - 2h = 0$$

The optimal quota is $h^* = 4$

3. With tax per unit of h , the firm's problem becomes

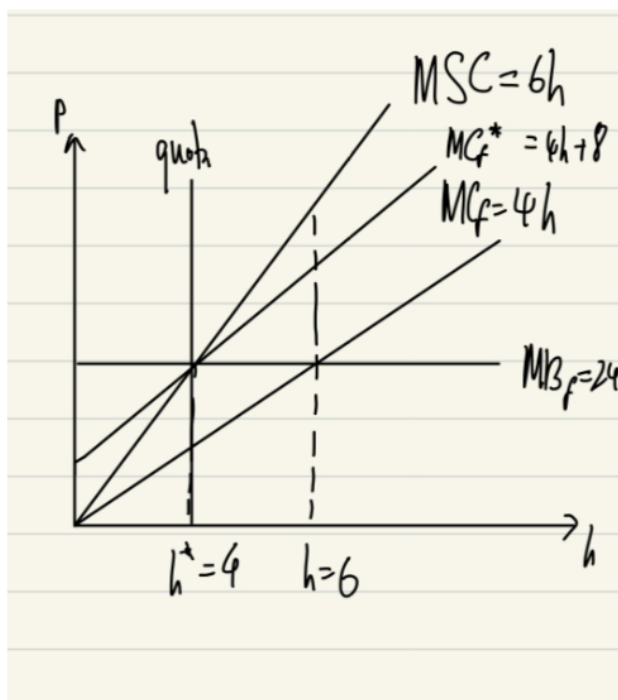
$$\max \pi(h) = w_f = 24h - 2h^2 - ht$$

where our FOC is

$$\frac{\partial \pi}{\partial h} = 24 - t - 4h = 0$$

Since we want $h^* = 4$, we get $24 - 16 = t$, which implies $t^* = 8$.

The diagram with the MC and MB for the firm and the MSC can be illustrated as:



Assume that $\eta = 36$ now.

1. if $\eta = 36$, then the optimal h is when

$$\max W(h) = \pi(h) + \phi(h)$$

Our First Order Condition is

$$\frac{\partial W}{\partial h} = \frac{\partial \pi}{\partial h} + \frac{\partial \phi}{\partial h} = 36 - 4h - 2h = 0$$

This gives $6h = 36$, which yields $h = 6$. If the government sets $h^* = 4$, the deadweight loss is

$$DWL = \frac{1}{2}(36 - 24)(2) = 12$$

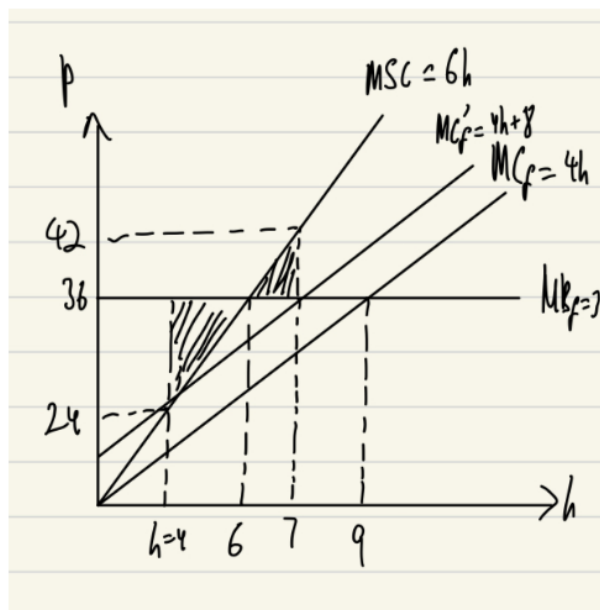
2. Let the firm's maximisation problem with tax is

$$\max \pi(h) = w_f + 36h - 2h^2 - 8h$$

which gives

$$\frac{d\pi}{dh} = 36 - 4h - 8 = 0 \implies h = 7$$

Accordingly, we illustrate the deadweight loss on a diagram:



So the deadweight loss if there is a tax at the level in (a) where $t = 8$ is

$$DWL = \frac{1}{2}(42 - 36)(1) = 3$$

So the tax at the level in (a) is better since the DWL under tax is lower.

Applications to Climate Change

Only thing interesting in Climate Change is the Nordhaus v. Stern v. Broome plans. ChatGPT gives the following:

The Nordhaus, Stern, and Broome frameworks for addressing climate change represent differing philosophical, economic, and policy-oriented approaches. Each plan is grounded in distinct assumptions about how to value future climate damages, economic trade-offs, and ethical considerations. Below is a comparative analysis.

1. William Nordhaus: Cost-Benefit Optimization

Approach: Economic optimization using *Integrated Assessment Models (IAMs)*.

Key Framework:

- Nordhaus emphasizes finding an economically *optimal path* for greenhouse gas reductions using the DICE (Dynamic Integrated Climate-Economy) model.
- Focuses on balancing the costs of mitigating climate change with the **discounted benefits** of avoiding future damages.
- Advocates for a **carbon tax** as an efficient mechanism to internalize the cost of emissions.

Discount Rate:

- Uses a *relatively high discount rate* (3–4%), reflecting the preference for current generations over distant future generations.
- Leads to a **gradual approach** to emissions reduction, with limited immediate action but increasing mitigation over time.

Criticism:

- Critics argue that Nordhaus's discount rate undervalues catastrophic risks to future generations.
- Gradualism in his approach has been called inadequate given the urgency of climate action.

2. Nicholas Stern: Climate Action as a Moral Imperative

Approach: Aggressive mitigation to prevent catastrophic risks, emphasizing **ethical obligations**.

Key Framework:

- Stern, in the Stern Review (2006), positions climate change as the *largest market failure in history*.
- Advocates for **strong and immediate action** to limit warming to manageable levels.
- Incorporates **tipping points** and *non-linear damages* from extreme climate events, justifying rapid action.

Discount Rate:

- Uses a *very low discount rate* (1.4%), based on the principle that future generations' welfare should be weighted nearly as highly as current generations'.
- This increases the **present value of future damages**, justifying immediate, large-scale investments.

Criticism:

- Economists argue that Stern's low discount rate is inconsistent with observed economic behavior.
- The proposed scale of action faces practical and political feasibility challenges.

3. John Broome: Ethics and Responsibility

Approach: Climate action as a **moral and ethical duty**, grounded in *utilitarianism* and broader philosophical reasoning.

Key Framework:

- In *Climate Matters: Ethics in a Warming World*, Broome highlights the **moral wrongs** of greenhouse gas emissions, which disproportionately harm future generations and vulnerable populations.
- Advocates for **personal responsibility** (e.g., carbon offsets) alongside government and collective action.
- Emphasizes **global equity** and redistribution, highlighting richer countries' greater moral obligations to act.

Discount Rate:

- Broome questions the validity of using traditional discount rates in intergenerational contexts, as it undermines the principle of treating future people equally.

Criticism:

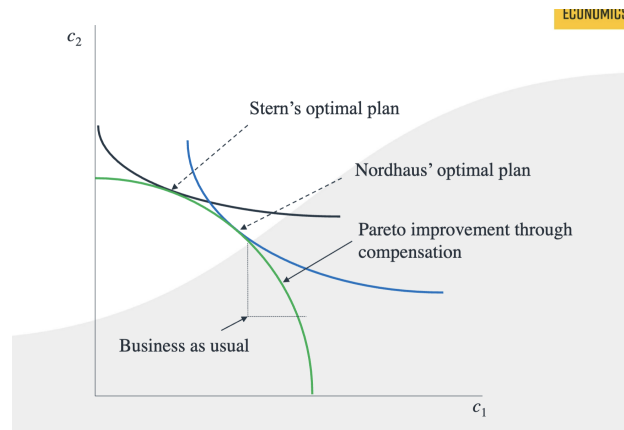
- Broome's framework is criticized as idealistic and difficult to translate into practical policy.
- Focusing on individual morality can detract from the systemic changes required for large-scale mitigation.

Comparison Table

Aspect	Nordhaus	Stern	Broome
Primary Focus	Economic optimization	Urgency of large-scale action	Ethical responsibility
Policy Recommendation	Gradual carbon tax	Rapid investment in green energy	Individual and collective responsibility
Discount Rate	3–4% (High)	1.4% (Low)	Rejects traditional discounting
Ethical Considerations	Secondary	Important	Central
Risk Management	Probabilistic (expected values)	Incorporates tipping points	Emphasizes unequal harm
Criticism	Too slow, undervalues future risks	Unrealistic scale of action	Idealistic, impractical

Table 1: Comparison of Nordhaus, Stern, and Broome Approaches

So, the graph illustrates the trade offs between future and current wealth based on discount rates.



The only other interesting thing is application of Weitzman's paper. Since there is a steep MSC curve it is better to control quantities directly – speaks in favour of cap-and-trade. Weitzman model is static, but climate policy is dynamic: we learn about science all the time and you can adjust prices or quantities accordingly. May be politically easier to gradually increase taxes anyway.

4 Game Theory

4.1 Basic Game Theory: Nash Equilibrium, Iterated Deletion of Dominated Strategies, Mixed Strategies

References: Simon Cowan's slides; Kreps, *A Course in Microeconomic Theory*, Ch11-12; Mas-Collel, Whinston and Green, *Microeconomic Theory* Ch 11; Osborne and Rubinstein, *A Course in Game Theory* Ch 2-4

Takeaways on Basic Game Theory

1. Definitions of dominance, Nash Equilibrium, weak dominance etc; awareness of basic theorems in Game Theory i.e. existence of mixed nash equilibrium
2. Presentation of normal form games as a matrix
3. Solution concepts to pure and mixed Nash Equilibrium; interpretation of solution concepts

4.1.1 Pure Strategies

Definition 4.1 (Strategic Game). A strategic game consists of

- A set of players N
- For each player $i \in N$, a nonempty set of available actions A_i to player i
- For each player $i \in N$, a preference relation $\succsim^i$ on $A_i = \times_{j \in N} A_j$

Assumption 4.1 (Perfect Information). An agent i has perfect information regarding other players' utility functions/payoffs u_{-i} and actions a_{-i} , and their own payoff functions

Normal-form Representation of a single-shot, simultaneous game:

Example (Prisoner's Dilemma):

		Player Y	
		C	D
Player X	C	(5, 5)	(1, 6)
	D	(6, 1)	(4, 4)

Interpretation: One common interpretation of a strategic game is that it models an event that occurs only once and each player only knows the details of the game and the fact that all players are "rational", while the players choose their actions simultaneously and independently. Under this interpretation, each player is unaware of what the other person chooses at the time of choosing his own action. There is no information (except the primitives of the model) on which a player can base his expectation of the other players' behaviour.

Motivating the Idea of Iterated Elimination of Strictly Dominated Actions

If players are rational, we want to find strategy profiles which are consistent with this assumption of rationality, to restrict our search in predicting the eventual outcome of the game. This gives rise to the idea of rationalisability of strategies.

Definition 4.2 (Rationalisability). An action $a_i \in A_i$ is rationalisable in the strategic game $\langle N, (A_i), (u_i) \rangle$ if for each $j \in N$, there is a set $Z_j \subseteq A_j$ such that

- $a_i \in Z_i$

- every action $a_j \in Z_j$ is a best response to a belief $u_j(a_j)$ of player j whose support is a subset of Z_{-j}

This basically says that for every player j in the game, there is a subset of all available actions, called Z_j , that denotes the collection of all rationalisable actions which must satisfy: (1) the condition that player i 's rational strategy a_i to be included in this set Z_i ; (2) the condition that every action $a_j \in Z_j$ for player j must be a best response to some belief that player j has about what other actions other players $-j$ might take based on Z_{-j} ; and by implication (3) For an action to be rationalisable, each player's belief must align with the rationalisable actions of the others. Essentially, if player i is choosing a_i , it is because they believe that others are choosing actions within their respective sets Z_{-i} which themselves are rational.

Definition 4.3 (Never Best Response). An action of player i in a strategy game is a never-best response if it is not a best response to any belief of player i (i.e. any strategy undertaken by players $-i$)

Definition 4.4 (Strictly Dominated Strategies). The action $a_i \in A_i$ of player i in the strategic game $\langle N, (A_i), (u_i) \rangle$ is strictly dominated if there exists another strategy $\alpha_i \in A_i$ such that $u(a_{-i}, \alpha_i) > u(a_{-i}, a_i)$ for all $a_{-i} \in A_{-i}$.

Proposition 4.1. An action of a player in a finite strategic game is a never-best response if and only if it is strictly dominated

IEDS Procedure:

1. Consider a player i 's strategy $a_i \in A_i$. If there is another strategy α_i such that for all $a_{-i} \in A_{-i}$, $u_i(\alpha_i, a_{-i}) > u_i(a_i, a_{-i})$, then eliminate that strategy a_i
2. Repeat the procedure in the reduced game for either another strategy by the same player, or by a different player

Definition 4.5 (Nash Equilibrium). A Nash Equilibrium of a strategic game $\langle N, (A_i), (\succsim_i) \rangle$ is a profile $a^* \in A$ of actions with the property that for every player $i \in N$ we have

$$(a_{-i}^*, a_i^*) \succsim_i (a_{-i}^*, a_i) \text{ for all } a_i \in A_i$$

Thus, for a^* to be a Nash equilibrium, it must be that no player has an action yielding an outcome that he prefers to that generated when he chooses a_i^* , given that every other player j chooses his equilibrium action a_j^* . Briefly, no player can profitably deviate, given the actions of the other players.

Relationship between Nash Equilibrium and IEDS

We consider finite two player games but the following propositions will generalise to any finite games.

Proposition 4.2. Any game will have at most one weakly dominant solution

Proof. It is impossible for a to weakly dominate a_1 and a_1 to weakly dominate a ■

Proposition 4.3. If (a^*, b^*) is a weakly dominant solution, then (a^*, b^*) is a Nash Equilibrium

Proof. The strategy a^* weakly dominates every other strategy in A such that

$$u_1(a^*, b^*) \geq u(a, b^*)$$

for all $a \in A$ and a^* is the unique best response to b^* . Similarly, b^* is the unique best response to a^* . Hence, (a^*, b^*) is a Nash Equilibrium ■

The converse fails: not all Nash Equilibria are weakly dominant solutions such as in *Battle of the sexes*.

Standard Games and solutions:

1. Prisoner's Dilemma
2. Battle of the Sexes/Bach or Stravinsky
3. Stag Hunt Game
4. Hawk Dove Game

4.1.2 Mixed Strategies

Our focus are on mixed strategies. Some optional (technical) definitions.

Definition 4.6 (Mixed extension of the strategic game). The mixed extension of the strategic game $\langle N, (A_i), (u_i) \rangle$ is the strategic game $\langle N, (\Delta(A_i)), (U_i) \rangle$ in which $\Delta(A_i)$ is the set of probability distributions over A_i and $U_i : \times_{j \in N} \Delta(A_j) \rightarrow \mathbb{R}$ assigns to each $\alpha \in \times_{j \in N} \Delta(A_j)$ the expected value under u_i of the lottery over A that is induced by α (so that $U_i(\alpha) = \sum_{a \in A} (\prod_{j \in N} \alpha_j(a_j)) u_i(a)$ if A is finite)

Definition 4.7. A mixed strategy Nash equilibrium of a strategic game is a Nash equilibrium of its mixed extension.

Remark: In the exam, always assume that the game is a mixed strategy extension of the strategic game unless stated otherwise (i.e. mixed strategies are possible answers)!!

Theorem 4.1 (Existence of Nash Equilibria). Every finite strategic game has a finite strategy Nash Equilibrium

Proof. Proof is in the appendix ■

Another theorem to help you in your answers for most instances.

Theorem 4.2 (Wilson's Oddness Theorem). Generically, finite (strategic-form) games have an odd number of mixed Nash equilibria.

The most important result to help us find mixed strategy Nash equilibrium is the following:

Lemma 4.3. Let $G = \langle N, (A_i), (u_i) \rangle$ be a finite strategic game. Then $\alpha^* \in \times_{i \in N} \Delta(A_i)$ is a mixed strategy Nash equilibrium of G if and only if for every player $i \in N$ every pure strategy in the support of α_i^* is a best response to α_{-i}^* .

Proof. First, suppose that there is an action a_i in the support of α_i^* that is not a best response to α_{-i}^* . Then by linearity of U_i in α_i since $U_i(\alpha) = \sum_{a_i \in A_i} \alpha_i(a_i) U_i(\alpha_{-i}, e(a_i))$ where $e(a_i)$ is the degenerate mixed strategy of player i that attaches probability 1 to $a_i \in A_i$, player i can increase his payoff by transferring probability from a_i to an action that is a best response; hence α_i^* is not a best response to α_{-i}^* . Second, suppose that there is a mixed strategy α'_i that gives a higher expected payoff than does α_i^* in response to α_{-i}^* . Then again by linearity of U_i at least one action in the support of α'_i must give a higher payoff than some action in the support of α_i^* , so that not all actions in the support of α^* are best responses to α_{-i}^* . ■

What the Lemma says is that **every action in the support of any player's equilibrium mixed strategy yields the player the same payoff.**

Now, we have three examples each telling an important story.

Example: The Simple Case

Consider the following game:

	<i>Left</i>	<i>Right</i>
<i>Up</i>	3 5	1 1
<i>Down</i>	0 0	5 3

The Pure Nash Equilibria are {Up, Left} and {Down, Right} since neither player can profitably and unilaterally deviate from their pre-existing strategy. Now, we want to find the mixed strategy Nash equilibrium. Let the probability that Column plays right be $\alpha(R)$ and the probability that Row plays Up be $\alpha(U)$. We consider Row's expected payoffs:

1. If Row chooses Up, their expected payoff is

$$E[\text{Up}] = 5(1 - \alpha(R)) + 1 \cdot \alpha(R) = 5 - 5\alpha(R) + \alpha(R) = 5 - 4\alpha(R)$$

2. If Row chooses Down, their expected payoff is

$$E[\text{Down}] = 0 \cdot (1 - \alpha(R)) + 3 \cdot \alpha(R) = 3\alpha(R)$$

We use the foregoing lemma to set $E[\text{Up}] = E[\text{Down}]$ which gives us

$$3\alpha(R) = 5 - 4\alpha(R) \implies \alpha(R) = \frac{5}{7}$$

and so $\alpha(L) = 1 - \alpha(R) = \frac{2}{7}$. We repeat the procedure for Column:

1. If Column chooses Left:

$$E[\text{Left}] = 3\alpha(U) + 0 \cdot (1 - \alpha(U)) = 3\alpha(U)$$

2. If Column chooses Right:

$$E[\text{Right}] = 1 \cdot \alpha(U) + 5(1 - \alpha(U)) = 5 - 4\alpha(U)$$

So, $3\alpha(U) = 5 - 4\alpha(U) \implies \alpha(U) = \frac{5}{7}$ and $\alpha(D) = 1 - \alpha(U) = \frac{2}{7}$. ■

Remark: The expected payoff for a mixed strategy is usually worse than a player's average payoff across two pure strategy Nash equilibrium because it is randomising between strategies that are not best responses to another player's foregoing strategy, inducing a possibly lower payoff unlike in the two NEs which are mutually best responses.

Example: (Uncountably) Infinite Mixed Strategies

Consider the following game:

	<i>Left</i>	<i>Right</i>
<i>Up</i>	v 0	2 2
<i>Down</i>	0 3	3 0

and set $v = 2$. Clearly, the pure Nash Equilibrium is {Up, Right} since these are mutual best responses. Consider the expected payoffs for Row:

1. If Row chooses Up:

$$E[\text{Up}] = 0 + 2(1 - \alpha(L)) = 2(1 - \alpha(L))$$

2. If Row chooses Down:

$$E[\text{Down}] = 3\alpha(L)$$

So, we get $2(1 - \alpha(L)) = 3\alpha(L) \implies \alpha(L) = \frac{2}{5}$ Let us consider the expected payoffs for Column:

1. If Column chooses Left:

$$E[\text{Left}] = 2\alpha(U)$$

2. If Column chooses Right:

$$E[\text{Right}] = 2\alpha(U) + 3(1 - \alpha(U))$$

We yield $2\alpha(U) = 2\alpha(U) + 3(1 - \alpha(U)) \implies \alpha(U) = 1$. But if $\alpha(U) = 1$, then $\alpha(D) := 1 - \alpha(U) = 0$. This means that $E[\text{Up}] \geq E[\text{Down}]$. So our solution for $\alpha(L)$ is an inequality rather than just a unique probability. Hence, we get

$$2(1 - \alpha(L)) \geq 3\alpha(L) \implies \alpha(L) \leq \frac{2}{5}$$

which is satisfied by our initial answer but also shows that there are uncountably infinite number of mixed Nash equilibria whereby $\alpha(L) \leq \frac{2}{5}$. ■

Remark: Honestly, I think it's quite hard to see this – you just have to check when one player will always play the same strategy.

Example: Pure Strategies Dominated by Mixed Strategies

Consider the following game:

	<i>Left</i>	<i>Centre</i>	<i>Right</i>
<i>Top</i>	7 5	2 3	0 1
<i>Bottom</i>	1 2	3 3	6 4

It seems that the iterated deletion of strictly dominated strategies cannot help us at first glance. However, consider the convex combination of Left and Right with some probability $\alpha \in (0, 1)$. Now, we need to choose an α such that $7\alpha > 2$ and $\alpha + 6(1 - \alpha) > 3$. This simplifies to $\alpha > \frac{2}{7}$ and $\alpha < \frac{3}{5}$. So, for any $\alpha \in (\frac{2}{7}, \frac{3}{5})$, a mixed strategy over Left and Right will strictly dominate Center for Column. ■

Remark: This will only happen for games that are of larger dimension than 2 by 2 and is relatively easy to eyeball with some intuition over convex combinations.

4.2 Dynamic Games

References: Simon Cowan's slides; Kreps, *A Course in Microeconomic Theory*, Ch11-12; Mas-Colell, Whinston and Green, *Microeconomic Theory* Ch 11; Osborne and Rubinstein, *A Course in Game Theory* Ch 6

Takeaways on Dynamic Games

1. Definition of Dynamic Games, Subgame Perfect Equilibria
2. Presentation of extensive form games as a game tree
3. Solution concepts via Backward induction; interpretation of solution concepts

A dynamic or an extensive game is a detailed description of the sequential structure of decision problems encountered by the players in a strategic situation. There is perfect information in such a game if each

player, when making any decision, is perfectly informed of all the events that have previously occurred. We will make this assumption for simplicity. The below are optional technical definitions

Definition 4.8 (Extensive Game). An extensive game with perfect information has the following components:

- A set N of players
- A set H of sequences (finite or infinite) that satisfies the following three properties
 1. The empty sequence $\emptyset$ is a member of H
 2. If $(a^k)_{k=1,\dots,K} \in H$ (where K may be infinite) and $L < K$, then $(a^k)_{k=1,\dots,L} \in H$
 3. If an infinite sequence $(a^k)_{k=1}^\infty$ satisfies $(a^k)_{k=1,\dots,L} \in H$ for every positive integer L then $(a^k)_{k=1}^\infty \in H$

A history is terminal if it is infinite and denote this set of terminal histories Z

- A function P that assigns to each nonterminal history a member of N . (P is the player function, $P(h)$ being the player who takes an action after the history h)
- For each player $i \in N$ a preference relation $\succsim_i$ on Z .
- After any nonterminal history h player $P(h)$ chooses an action from the set

$$A(h) = \{a : (h, a) \in H\}$$

Perfect recall and information allows us to use the strategy of **backward induction** i.e. seeing what will be done in period t to determine what will be done in period $t - 1$ and so on and so forth.

Definition 4.9 (Strategy in Extensive games). A strategy of player $i \in N$ in an extensive game with perfect information $\langle N, H, P, (\succsim_i) \rangle$ is a function that assigns an action in $A(h)$ to each nonterminal history $h \in H \setminus Z$ for which $P(h) = i$.

The lecture notes do not make the following distinction but I think it is important to show how the notion of the subgame perfect equilibrium is a refinement of the Nash equilibrium in ways beyond the fact that SPNE only applies to sequential games.

Definition 4.10 (Nash Equilibrium). A Nash equilibrium of an extensive game with perfect information $\langle N, H, P, (\succsim_i) \rangle$ is a strategy profile s^* such that for every player $i \in N$ we have

$$O(s_{-i}^*, s_i^*) \succsim_i O(s_{-i}^*, s_i) \quad \text{for every strategy } s_i \text{ of player } i$$

whereby $O(s)$ is the outcome of s i.e. the terminal history that results when each player $i \in N$ follows the precepts of s_i

Now, we introduce the refinement of the concept of a Nash Equilibrium.

Definition 4.11 (Subgame). The subgame of the extensive game with perfect information $\Gamma = \langle N, H, P, (\succsim_i) \rangle$ that follows the history h is the extensive game $\Gamma(h) = \langle N, H|_h, P|_h, (\succsim_i|_h) \rangle$ where $H|_h$ is the set of sequences h' of actions which $(h, h') \in H$, $P|_h$ is defined by $P|_h(h') = P(h, h')$ for each $h' \in H|_h$ and $\succsim_i|_h$ is defined by $h' \succsim_i|_h h''$ iff $(h, h') \succsim_i (h, h'')$

The upshot of this is that the equilibrium we now define requires that the action prescribed by each player's strategy be optimal given other player's strategies after every history.

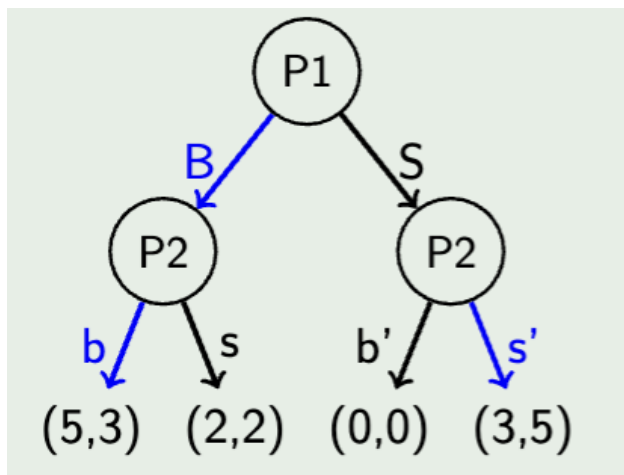
Definition 4.12 (Subgame Perfect Equilibrium). A subgame perfect equilibrium of an extensive game with perfect information $\Gamma = \langle N, H, P, (\succsim_i) \rangle$ is a strategy profile s^* such that for every player $i \in N$ and every nonterminal history $h \in H \setminus Z$ for which $P(h) = i$ we have

$$O_h(s_{-i}^*|_h, s_i^*|_h) \succsim_i|_h O_h(s_{-i}^*|_h, s_i^-|_h)$$

Equivalently, we can define a SPNE to be a strategy profile s^* in Γ for which any history h the strategy profile $s^*|_h$ is a Nash equilibrium of the subgame $\Gamma(h)$. In short, an SPNE plans out the full contingent strategy of i for any other possible historical actions of players $-i$.

The notion of SPNE eliminates Nash Equilibria in which the player's threats are not credible.

Example. Consider the two representations of the extensive form game.



Extensive Form representation

and

	bb'	bs'	sb'	ss'
B	<u>5</u> , <u>3</u>	<u>5</u> , <u>3</u>	<u>2</u> , 2	2, 2
S	0, 0	3, <u>5</u>	0, 0	<u>3</u> , <u>5</u>

Strategic Form representation

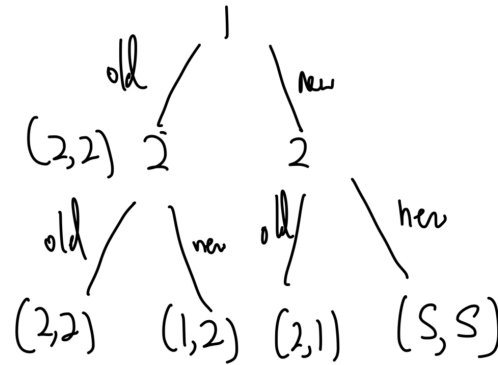
A short remark on how to interpret the strategic form representation: A strategy (bb') says that if player 1 chooses B then b will be played and thus the payoff is $(5, 3)$ (so it enters the cell (B/bb') and if player 2 chooses S , then b' will be played and thus the payoff is $(0, 0)$.

We can see that bb' is not sequentially rational because if player 1 chooses S , player 2 will choose s' since $O(S, s') \succsim_i |_h O(S, b')$. So the SPNE eliminates sequentially irrational actions i.e. noncredible threats which requires specifying the full contingent plan. So the unique SPNE in this case is (B, bs') .

Example. This example shows what happens when there are indifferent outcomes during SPNE. Consider the extensive game whereby player 1 moves first and player 2 moves second and they can either use new or old technology. The benefit to the player for playing new or old depends on what the other does:

- For old: the benefit is simply the number that stay as old
- For new: the benefit is 3 times the number that move to new

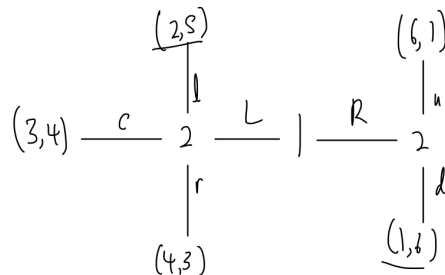
Moving to new also induces a cost of 1. So, the extensive form representation is



Clearly, player 1 will always choose new since $(5, 5) \succ_i (2, 2)$. For player 2, if player 1 chooses new, he will also choose new. The interesting part is what happens if player 1 chooses old since the payoff between old and new is the same. In this case, player 2's possible strategies include choosing any $\lambda \in (0, 1)$ such that λ is the probability old is played and $1 - \lambda$ the probability that new is played. In other words, the SPNE (through abuse of notation) can be described as $\{\text{new}, ((\text{old}, \text{new}), \text{new})\}$ whereby (old, new) is mixed over all uncountably infinite possible choices of $(\lambda, 1 - \lambda)$. Therefore, the number of SPNE is NOT 2.

Remark: this tells you what happens when there are actions that yield indifferent outcomes. Another thing to note is that when outcomes are strictly preferred to one another the SPNE tends to be unique.

Example. This example shows how mixed strategies are accounted for in SPNE (and often eliminated as candidate NEs). Consider the following extensive form game: At stage 1, player 1 can play either Left (L) or Right (R). If player 1 plays L , then, at stage 2, player 2 can play left (ℓ) resulting in a payoff of $(2, 5)$ or center (c) with payoffs $(3, 4)$ or right (r) with payoffs $(4, 3)$. If player 1 plays R , then player 2 can play up (u) with payoffs $(6, 1)$ or play down with payoffs $(1, 6)$. Let us illustrate the game in extensive form:



The SPNE here is clearly $\{L, (\ell, d)\}$. However, assume player 2 chooses to mix its strategies between u and d . If $p \geq 1/5$ to play u , then player 1 will now choose R since $x + 4/5 \geq 2$ when $x \geq 6/5$. But this threat of mixing is not credible because once R is played, player 2 will immediately choose to play d instead of mixing the strategies. So, SPNE eliminates possible NEs that include Mixed Nash Equilibrium.

4.3 Repeated Games

References: Simon Cowan's slides; Kreps, *A Course in Microeconomic Theory*, Ch11-12; Mas-Collel, Whinston and Green, *Microeconomic Theory* Ch 11; Osbourne and Rubinstein, *A Course in Game Theory* Ch 8

Takeaways on Repeated Games

1. Infinitely vs finitely repeated games
2. Folk Theorems and discounting rates on payoffs
3. Solution concepts to repeated games

A finitely repeated game for our purposes is basically a single-shot game that is repeatedly played up to T times. The following theorem is basically all you need to know for a finitely repeated game.

Theorem 4.4. If the strategic game G has a unique Nash equilibrium payoff profile then for any value T the action profile chosen after any history in any subgame perfect equilibrium of the T -period repeated game of G is a Nash equilibrium of G .

Proof. The outcome in any subgame that starts in period T of any subgame perfect equilibrium of the repeated game is a Nash equilibrium of G . Thus each player's payoff in the last period of the game is independent of history/ Consequently in any subgame that starts in period T the action profile is a Nash equilibrium of G . An inductive argument completes the proof. ■

Basically, in a repeated prisoner's dilemma the SPNE is to always play defect in every round. The intuition is that at the last stage, the optimal play is to play defect for both. So, we exclude the last stage for the second last stage, and see that the same obtains. This happens for all stages until the first stage.

An infinitely repeated game for our purposes is basically a single-shot game that is repeatedly played infinitely. For infinitely repeated games, we usually use a discount rate δ to discount future payoffs.

We have the following lemma that motivates our class of grim trigger strategies.

Lemma 4.5. A strategy profile is a subgame perfect equilibrium of the δ -discounted infinitely repeated game of G if and only if no player can gain by deviating in a single period after any history.

What this motivates is that for games with unique Nash Equilibrium, to sustain 'cooperation' one should threaten to punish deviation by playing the Nash Equilibrium strategy after deviation. The intuition is two-fold: (i) this is a credible threat for the punisher because if the game has a unique Nash equilibrium, then under deviation, playing the unique Nash equilibrium strategy will force the punished player to play that same strategy which yields a greater payoff for the punisher than continuing to play the initial strategy while the punished player deviates; (ii) the punished player will be stuck in this situation since the best response will involve playing the NE strategy.

So, we have a grim trigger strategy of the following form:

Definition 4.13 (Grim Trigger Strategy). Under a suitable δ sufficiently close to 1, there a grim trigger strategy is a strategy that constitutes an SPNE that involves perpetual cooperation by threatening to punish deviation by playing the single-shot Nash Equilibrium, in an infinitely-repeated single shot game with a unique Nash Equilibrium

Example. Consider the following game.

	<i>Left</i>	<i>Centre</i>	<i>Right</i>
<i>Top</i>	8 8	9 5	7 3
<i>Middle</i>	5 9	6 6	4 1
<i>Bottom</i>	3 7	1 4	2 2

If the game is played infinitely often, discuss for which values of the common discount factor $\delta \in (0, 1)$, the following strategies constitute a Nash equilibria of the repeated game: the Row player plays *Top* and the Column player plays *Left* unless either player has deviated, in which case they both switch to the equilibrium actions of the one-shot game forever.

If Row and Column play $\{\text{Top, Left}\}$ infinitely, their payoffs are

$$\lim_{n \rightarrow \infty} (8 + \delta 8 + \delta^2 8 + \cdots + \delta^n 8) = \frac{8}{1 - \delta}$$

If a player switches, they will switch to Center or middle to yield a payoff of $9 > 8$ for the next round. Then, the trigger punishment strategy occurs s.t. their payoff from switching is

$$\lim_{n \rightarrow \infty} (9 + 6\delta + \cdots + 6\delta^n) = 9 + \frac{6\delta}{1 - \delta}$$

Without Loss of Generality, we can assume that the first game is when they deviate, as we can treat discard the first k games before deviation as both players yield the same payoff. The reason why we can still do an infinite sum for the payoffs is because the number of games are defined on $\mathbb{N}$, so by Cantor's theorem, we can form a bijection from a set with countably infinite elements to countably infinite $+k$ elements.

So to find δ ,

$$\begin{aligned} \frac{8}{1 - \delta} &\geq 9 + \frac{6\delta}{1 - \delta} \\ 8 &\geq 9(1 - \delta) + 6\delta \\ 3\delta &\geq 1 \end{aligned}$$

Hence, $1 > \delta \geq \frac{1}{3}$.

The grim trigger strategy is subgame perfect. An SPE requires that a player's strategies form a NE in every subgame of the repeated game, including subgames that may not be reached in the equilibrium path.

(1) Cooperation path: As long as both players are cooperating, each player's optimal strategy is to stick to either *Top* or *Left* respectively, given (b) since the Present Discounted Value of payoffs are higher.

(2) Punishment Phase: If a deviation occurs, we trigger the punishment phase where both players play $\{\text{Middle, Middle}\}$ forever. Given both players play $\{\text{Middle, Middle}\}$ forever, there is no incentive to deviate either since $\{\text{Middle, Middle}\}$ is itself a Nash Equilibrium in a one-shot game. So, no player can benefit unilaterally from deviation.

So, the strategy is SPE since it prescribes an optimal play in every subgame.

5 Industrial Organisation and Competition Policy

5.1 Game Theory and Industrial Organisation

5.1.1 Price Setting Game

The Bertrand Game

Assumption 5.1 (Homogeneous Goods). We assume that firms sell homogenous goods i.e. they are perfect substitutes.

Assumption 5.2 (Price competition). We assume that firms compete on price, not quantity.

Assumption 5.3 (Quantity Assumptions). We assume that there are no capacity constraints and no fixed costs and all firms have identical marginal costs c

Assumption 5.4 (Market Behaviour). If all firms sell at the same price, then each firm receives total demand divided by number of firms; if a firm sells at a lower price, then all demand goes to the firm

Given the assumptions, the firm k 's best response function at stage i is to choose recursively $p_{i+1}^k = p_i - \epsilon > c$ where $\epsilon > 0$. Intuitively, this is because a small reduction by ϵ in the price sold but getting all the demand will make you better off and firm k knows that firms $-k$ dominant strategy would be to undercut the price subject to the marginal cost. Formally, the firm k faces the following options:

- $p_k > p_{-k}$ which results in $\pi_k(q) = 0$
- $p_k = p_{-k}$ which results in $\pi_k(q) = p_k \cdot q/n$
- $p_k - \epsilon < p_{-k}$ which results in $\pi_k(q) = p_k \cdot q$. This implication is permissible because we choose an infinitesimally small ϵ sufficiently close to 0.

Hence, their best response would be to undercut the price inductively until $p^k = c$. This is symmetric about all n firms. So the Bertrand-Nash equilibrium is $\{c, c, \dots, c\}$.

Some important things to note:

1. The Bertrand-Nash Equilibrium is independent of the number of firms i.e. the same competitive outcome occurs with two firms as with n firms
2. The Bertrand-Nash Equilibrium is unique.

Proof. Observe that the price setting strategy $p_k = c$ weakly dominates all other strategies. We know that the Bertrand-Nash Equilibrium is a solution to the problem. We also know that any game has at most one weakly dominant solution since it is impossible for a strategy a_i weakly dominates a_{-i} and for a_{-i} to weakly dominate a_i . ■

3. Fitting with stylised facts: ‘premise’ is correct because firms tend to compete on prices rather than quantity, but ‘conclusion’ is false because (1) firms do not set price equal to marginal cost and make zero profits, (2) industries exist with firms having different marginal cost i.e. it is not the case that a firm with a lower marginal cost does not get any demand
4. The zero profit result only holds for *fixed* marginal costs. If costs are instead **convex**, positive profit will be made since there is only zero profit on the last unit of production.

The Bertrand Game with Differentiated Goods

We change the homogeneity assumption to include differentiated products. There are two kinds of differentiation:

- **Vertical Differentiation:** Goods vary only in quality; if goods are sold at the same price, everyone would choose the higher quality product

- **Horizontal Differentiation:** Goods vary in some dimension other than just quality and usually imperfect substitutes; this will be the focus

This gives us the following results that differ from the standard Bertrand Game: (1) price is higher than marginal cost, (2) firms with heterogeneous costs exist in the industry

Example (Bertrand Game with Differentiated Goods): Suppose that two firms produce horizontally differentiated products. The firms compete for a single period and choose prices simultaneously. Each firm has the same constant marginal cost of production $c \geq 0$, and fixed costs are zero. The linear demand functions are given by:

$$\begin{aligned} q_1(p_1, p_2) &= 100 - \alpha p_1 + \beta p_2 \\ q_2(p_1, p_2) &= 100 - \alpha p_2 + \beta p_1 \end{aligned}$$

Assume that $\alpha > \beta > 0$ and that $100 > \alpha c$. Find the Symmetric Bertrand-Nash Equilibrium.

Solution. The BR function for firm 1 is derived from

$$\begin{aligned} \max_{p_1} \pi_1(p_1) &= q_1 \cdot p_1 - c q_1 \\ &= 100p_1 - \alpha p_1^2 + \beta p_2 p_1 - c(100 - \alpha p_1 + \beta p_2) \end{aligned}$$

Our FOC:

$$\frac{d\pi_1}{dp_1} = 100 - 2\alpha p_1 + \beta p_2 + \alpha c = 0$$

This yields

$$p_1^* = \frac{50}{\alpha} + \frac{\beta}{2\alpha} p_2 + \frac{c}{2}$$

By symmetry, $p_2 = \frac{50}{\alpha} + \frac{\beta}{2\alpha} p_1 + \frac{c}{2}$.
So, substituting p_2 into p_1 yields

$$\begin{aligned} p_1 &= \frac{50}{\alpha} + \frac{\beta}{2\alpha} \left(\frac{50}{\alpha} + \frac{\beta}{2\alpha} p_1 + \frac{c}{2} \right) + \frac{c}{2} \\ &= \frac{50}{\alpha} + \frac{50\beta}{2\alpha^2} + \frac{\beta^2}{4\alpha^2} p_1 + \frac{\beta c}{4\alpha} + \frac{c}{2} \\ \left(1 - \frac{\beta^2}{4\alpha^2}\right) p_1 &= \frac{200\alpha + 100\beta + \beta\alpha c + 2\alpha^2 c}{4\alpha^2} \\ (4\alpha^2 - \beta^2) p_1 &= 200\alpha + 100\beta + \beta\alpha c + 2\alpha^2 c \\ (2\alpha - \beta)(2\alpha + \beta) p_1 &= 100(2\alpha + \beta) + \alpha c(2\alpha + \beta) \\ p_1^* &= \frac{100 + \alpha c}{2\alpha - \beta} \end{aligned}$$

By symmetry,

$$p_2^* = p_1^* = \frac{100 + \alpha c}{2\alpha - \beta}$$

since $\alpha > \beta > 0$, $2\alpha - \beta > 0$. Since $c \geq 0$, $100 - \alpha c > 0$, we conclude that $p_1^* = p_2^* > 0$. To verify that the quantity is positive, we derive the quantity to be

$$q_1^* = q_2^* = 100 + (\beta - \alpha) \left(\frac{100 + \alpha c}{2\alpha - \beta} \right)$$

We know that $\beta - \alpha < 0$ and that $\left(\frac{100 + \alpha c}{2\alpha - \beta} \right) > 0$, so we must show that

$$100 > (\alpha - \beta) \left(\frac{100 + \alpha c}{2\alpha - \beta} \right)$$

Since we know that $100 > \alpha c$, we conclude that $100 + \alpha c < 200$. We also know that

$$\frac{200(\alpha - \beta)}{2\alpha - \beta} = 100 - \frac{100\beta}{2\alpha - \beta} < 100$$

This gives us

$$(\alpha - \beta) \left(\frac{100 + \alpha c}{2\alpha - \beta} \right) < \frac{200(\alpha - \beta)}{2\alpha - \beta} < 100$$

So, $q_1 = q_2 > 0$ at equilibrium. ■

Remark: In the example above, β represents cross-price sensitivity or redirected demand from an increase in price from the other good and α represents own-price sensitivity or loss in demand from an increase in price of own good. Hence $\alpha > \beta$ because intuitively, a rise in price would lead to some consumers going to the other good but also some consumers just exiting the market. Also, note two comparative statics:

- As α increases, there is a downward pressure on demand for both goods which decreases price since α is negatively related to q_i . So equilibrium price falls
- When β increases, a rise in the other good's price increases your demand by a greater amount. This leads to a rise in price for your good. Hence, the equilibrium price would be higher as a rise in price for the one good would lead to a rise in demand and thus price for the other good.

Bertrand Game with Capacity Constraints (Optional):

5.1.2 Quantity Setting Game

Consider a duopoly, with two firms producing goods that are perfect substitutes; they play a quantity-setting game for one period only. The firms differ in their cost functions: the total cost to firm k to producing a quantity q_k is $c_k q_k$, for $k = 1, 2$. The inverse demand function in the market is

$$p(q) = a - q \text{ where } q = q_1 + q_2$$

The Cournot Game

Let $q = q_1 + q_2$. Firm i 's maximisation problem is

$$\max_{q_i} \pi(q) = p(q)q_i - c_i q_i$$

Our FOC is

$$\begin{aligned} \frac{d\pi}{dq_i} &= a - (q_i + q_{-i}) + q_i(-1) - c_i = 0 \\ a - 2q_i - q_{-i} - c_i &= 0 \end{aligned}$$

So, firm i 's reaction function is

$$q_i = \frac{1}{2}(a - q_{-i} - c_i)$$

Hence, firm 1's and 2's best response functions are

$$\begin{aligned} q_1(q_2) &= \frac{1}{2}(a - q_2 - c_1) \\ q_2(q_1) &= \frac{1}{2}(a - q_1 - c_2) \end{aligned}$$

respectively. To find the Nash Equilibrium, both must be playing best responses. So substituting firm 2's BR function into firm 1's gives us

$$\begin{aligned}
q_1 &= \frac{1}{2}(a - \frac{1}{2}(a - q_1 - c_2) - c_1) \\
&= \frac{1}{2}(a - \frac{1}{2}a + \frac{1}{2}q_1 + \frac{1}{2}c_2 - c_1) \\
&= \frac{1}{4}a + \frac{1}{4}q_1 + \frac{1}{4}c_2 - \frac{1}{2}c_1 \\
\frac{3}{4}q_1 &= \frac{1}{4}a + \frac{1}{4}c_2 - \frac{1}{2}c_1 \\
q_1 &= \frac{1}{3}(a + c_2 - 2c_1)
\end{aligned}$$

By symmetry,

$$q_2 = \frac{1}{3}(a + c_1 - 2c_2)$$

Derivation of Profits with identical costs: Let the market demand be $p(Q) = a - bQ$ with n firms facing constant marginal costs c , where $Q := \sum_{i=1}^n q_i$. Then, each firm faces the problem

$$\max \pi_i(q_i) = (p - c)q_i = (a - bQ - c)q_i$$

So, our FOC is

$$\frac{d\pi_i}{dq_i} = a - c - 2bq_i - b \sum_{k \neq i}^n q_k = 0$$

This yields

$$q_i = \frac{a - c}{2b} - \frac{1}{2} \sum_{k \neq i}^n q_k = 0 \quad \text{by symmetry} \implies q_i = \frac{a - c}{b(n + 1)}$$

The profit equation is

$$\pi_i(q_i) = (p - c)q_i = \left(\frac{(a - c)(n + 1) - n(a - c)}{n + 1} \right) \cdot \frac{a - c}{b(n + 1)} = \frac{(a - c)^2}{b(n + 1)^2}$$

Direct and Indirect Effects of increases in costs. We must consider both the direct and indirect effects of an increase in cost. Consider the following question:

Show that firm 1's equilibrium profit, π_1^C , is given by

$$\pi_1^C(q_1^C, q_2^C) = (a - q_1^C - q_2^C - c_1)q_1^C$$

Hence show that a marginal change in its own cost parameter, c_1 , affects its profit through three channels: (i) the direct effect of the change in its own costs, (ii) an indirect effect via the change in its own quantity, and (iii) a second indirect effect via the change in the other firm's quantity.

Explain why the indirect effect via the marginal change in its own quantity is 0.

Hence show that the total effect of the change is $d\pi_1^C/dc_1 = -\frac{4}{3}q_1^C$.

Similarly, find the effect on firm 1's profit of a marginal change in the other firm's cost parameter, c_2 . Which of the three effects are zero in this case?

Is the size of the change in firm 1's profit greater when its own cost changes, or when the other firm's cost changes? Why?

Solution. Let $\pi_1^C(q_1^C, q_2^C, c_1) = \max \pi_1(q_1, q_2, c_1)$.

From the envelope theorem, we yield

$$\frac{d\pi_1^C(q_1^C, q_2^C, c_1)}{dc_1} = \frac{d\pi_1(q_1^C(c_1))}{dc_1}$$

From the total differential, this gives us

$$\frac{d\pi_1(q_1^C(c_1))}{dc_1} = \frac{\partial\pi_1}{\partial c_1}(q_1^C(c_1), c_1) + \frac{\partial\pi_1}{\partial q_1}(q_1^C(c_1), c_1) \frac{dq_1}{dc_1} + \frac{\partial\pi_1}{\partial q_2}(q_1^C(c_1), c_1) \frac{dq_2}{dc_1}$$

For (i), we know that

$$\frac{\partial\pi_1}{\partial c_1}(q_1^C(c_1), c_1) = -q_1^C$$

For (ii), since π_1 is at the optimum value at q_1^C , the derivative $\frac{\partial\pi_1}{\partial q_1} = 0$, hence the indirect effect by their quantity is

$$\frac{\partial\pi_1}{\partial q_1}(q_1^C(c_1), c_1) \frac{dq_1^C}{dc_1} = 0$$

For (iii), we yield

$$\frac{\partial\pi_1}{\partial q_2}(q_1^C(c_1), c_1) \frac{d\pi_1}{dq_2^C} = -q_1^C \cdot \frac{1}{3}$$

So, the total derivative is

$$\frac{d\pi_1(q_1^C(c_1))}{dc_1} = -q_1^C + 0 - \frac{1}{3}q_1^C = -\frac{4}{3}q_1^C$$

Applying the same, we know that both

$$\frac{\partial\pi_1}{\partial c_2} = 0$$

and because it is at the optimum,

$$\frac{\partial\pi_1}{\partial q_1}(q_1^C(c_2), c_2) \frac{dq_1^C}{dc_2} = 0$$

So, both of the above changes (i.e. direct effect of cost change and indirect effect through own quantity produced) are 0. But then

$$\frac{\partial\pi_1}{\partial q_2}(q_2^C(c_2), c_2) \frac{d\pi_1}{dc_2} = \frac{2}{3}q_1^C$$

So, from the envelope theorem and the total derivative, we yield

$$\frac{d\pi_1^C(q_1^C, q_2^C, c_2)}{dc_2} = \frac{d\pi_1(q_1^C(c_2))}{dc_2} = \frac{d\pi_1}{dc_2}(q_1^C(c_2), c_2) = \frac{2}{3}q_1^C$$

The size of the firm 1's profit change is greater when its own cost changes because their own cost affects their production directly, while the other firm's cost increasing reduces competition but does not directly result in the amount they are able to provide (since the partial derivative with respect to c_2 is zero).

The Stackelberg Game

If firm 1 moves first, it will be larger than q_1^C since firm 2 has to follow whatever firm 1 does and hence firm 1 can produce even more as an SPE.

Suppose firm 2 observes q_1 and thus faces

$$\max_{q_2} \pi(q_1, q_2) = p(q)q_2 - c_2q_2$$

From (a) we know that firm 2's best response function is

$$q_2 = \frac{1}{2}(a - q_1 - c_2)$$

Firm 1 predicts this because this is firm 2's best response and thus faces the revised maximisation problem

$$\begin{aligned} \max_{q_1} \pi(q_1) &= a - (q_1 + \frac{1}{2}(a - q_1 - c_2))q_1 - c_1q_1 \\ &= \frac{1}{2}aq_1 - \frac{1}{2}q_1^2 + \frac{1}{2}c_2q_1 - c_1q_1 \end{aligned}$$

Our FOC is

$$\frac{d\pi}{dq_1} = \frac{1}{2}a - q_1 + \frac{1}{2}c_2 - c_1 = 0$$

So,

$$q_1^S = \frac{1}{2}(a + c_2 - 2c_1) > q_1^C$$

while

$$\begin{aligned} q_2^S &= \frac{1}{2}\left(a - \frac{1}{2}(a + c_2 - 2c_1) - c_2\right) \\ &= \frac{1}{4}(a + 2c_1 - 3c_2) \end{aligned}$$

The General Case: Consider n firms in the market, where firm 1 is the leader and there are $n - 1$ followers. Output of the leader is q_1 and the output of the followers are $q_2, \dots, q_n$. Each firm, being a profit maximiser, seeks to maximise

$$(a - q_i - Q_{-i})q_i$$

The FOC is immediately $a - 2q_i - Q_{-i} = 0$, which gives us

$$q_i = \frac{a - Q_{-i}}{2}$$

Since the Nash equilibrium is symmetric about the followers, $q_2 = \dots = q_n = q$ and $Q_{-i} = q_1 + (n - 2)q$. Hence, we yield

$$q = \frac{a - q_1 - (n - 2)q}{2} \implies q = \frac{a - q_1}{n}$$

The leaders problem is to choose q_1 to maximise

$$(a - q_1 - (n - 1)\frac{a - q_1}{n})q_1 = \frac{1}{n}(a - q_1)q_1$$

The FOC is $\frac{1}{n}(a - 2q_1) = 0$ and the profit function is strictly concave, so

$$q_1^s = \frac{a}{2}$$

Thus, the output of the Stackelberg leader does not depend on the number of firms, and equals the monopoly output.

Remark: Consumer prefers a lower price so they would prefer the firm with the lower cost to move first as that firm would produce more. part (c) already tells us what happens if one of the firm moves first. So, our price would be

$$\begin{aligned} p &= a - \frac{1}{2}(a + c_2 - 2c_1) - \frac{1}{4}(a + 2c_1 - 3c_2) \\ &= \frac{1}{4}a + \frac{1}{4}c_2 + \frac{1}{2}c_1 \end{aligned}$$

By symmetry, if firm 2 moves first, then

$$p = \frac{1}{4}a + \frac{1}{4}c_1 + \frac{1}{2}c_2$$

Without loss of generality, assume $c_1 > c_2$, then they would prefer firm 2 to move first as the price would be lower and the quantity would be higher. This is because the firm with the lower cost would be more willing to produce more due to the profit maximisation condition $MR = MC$ implying that a greater output would be needed to let the marginal revenue equalise with a lower marginal cost. Hence, this would yield a lower price overall for consumers.

Entry Deterrence Game

Consider a market in which firm 1 is an incumbent and firm 2 is a potential entrant. The total cost to firm 1 of producing a quantity q_1 is $2q_1$, whereas the total cost to firm 2 of entering then producing a quantity q_2 is $F + 4q_2$, where $F \geq 0$ is its fixed cost of entry. The inverse demand function in the market is

$$p(q) = 40 - q \text{ where } q = q_1 + q_2$$

Assume that the firms move sequentially, firm 1 first: firm 1 chooses its quantity, q_1 , observed by firm 2; firm 2 then decides whether or not to enter the market, and if it does so, it chooses its quantity, q_2 . (Assume that firm 2 enters the market only if it makes a *strictly* positive profit.)

We consider the following conditions:

1. $F = 0$.

The firm 2's best response is derived from

$$\max \pi_2(q_2) = (40 - q_1 - q_2)q_2 - F - 4q_2$$

The FOC is:

$$\frac{d\pi_2}{dq_2} = 40 - q_1 - 2q_2 - 4 = 0$$

This yields the BR function

$$q_2(q_1) = \frac{1}{2}(36 - q_1)$$

Observing this, firm 1 faces

$$\begin{aligned} \max \pi_1(q_1) &= (40 - q_1 - \frac{1}{2}(36 - q_1))q_1 - 2q_1 \\ &= (22 - \frac{1}{2}q_1)q_1 - 2q_1 \end{aligned}$$

FOC:

$$\frac{d\pi_1}{dq_1} = 22 - q_1 - 2 = 0$$

So, the quantity produced is $q_1 = 20$.

The profit functions are

$$\begin{aligned} \pi_1(q_1) &= (22 - 10)(20) - 20(2) = 240 - 40 = 200 \\ \pi_2(q_2) &= 12(8) - F - 4(8) = 66 - 32 - F = 34 - F \end{aligned}$$

When $F = 0$, $\pi_2(q_2) = 34$

2. $F = 4$ i.e. Fixed Cost is low

Firm 2's profit function is

$$\pi_2(q_2) = (40 - q_1 - q_2)q_2 - F - 4q_2$$

substituting firm 2's BR

$$\begin{aligned} \pi_2(q_1) &= (40 - q_1 - \frac{1}{2}(36 - q_1)) \cdot \frac{1}{2}(36 - q_1) - F - 2(36 - q_1) \\ &= \frac{1}{2}(22 - \frac{1}{2}q_1)(36 - q_1) - F - 2(36 - q_1) \\ &= 324 - 18q_1 + \frac{1}{4}q_1^2 - F \end{aligned}$$

Firm 1 chooses q_1 s.t. $\pi_2(q_1) \leq 0$ so,

$$\frac{1}{4}q_1^2 - 18q_1 + 324 - F \leq 0$$

This yields

$$\frac{18 - \sqrt{18^2 - 4(324 - F)(\frac{1}{4})}}{1/2} \leq q_1 \leq \frac{18 + \sqrt{18^2 - 4(324 - F)(\frac{1}{4})}}{1/2}$$

This gives us the general case

$$36 - 2\sqrt{F} \leq q_1 \leq 36 + 2\sqrt{F}$$

So if $F = 4$,

$$32 \leq q_1 \leq 40$$

would deter entry. If entry is deterred $q_2 = 0$, so

$$\max \pi_1(q_1) = (40 - q_1)q_1 - 2q_1$$

The FOC is

$$\frac{d\pi_1}{dq_1} = 40 - 2q_1 - 2 = 0$$

This yields

$$q_1 = 19$$

Since deterring entry is only from $32 \leq q_1 \leq 40$. Maximum profit is $q_1 = 32$. This gives us

$$\pi_1(32) = 8(32) - 2(32) = 192$$

So firm 1 will not deter entry & play the Stackelberg game i.e. $q_1 = 20$, $q_2 = 8$.

3. $F = 36$ i.e. Fixed Cost is medium

If $F = 36$, entry deterrence is from

$$36 - 12 \leq q_1 \leq 36 + 12$$

This yields

$$24 \leq q_1 \leq 48$$

Again, max profit while $q_2 = 0$, $q_1 = 24$,

$$\pi_1(24) = 16 \cdot 24 - 2 \cdot 24 = 336 > 200$$

So firm 1 produces $q_1 = 24$, $q_2 = 0$ i.e. firm 2 will not enter

4. $F = 100$ i.e. Fixed cost is high

Entry deterring condition while $F = 100$ is

$$16 \leq q_1 \leq 56$$

So, firm 1 chooses monopoly maximising quantity $q_1 = 19$, $q_2 = 0$.

Remark: If the cost is high enough, firm 1 can act as an unconstrained monopolist but if not, they must overproduce to keep out potential entrants. The intuition is just because you are a monopoly does not mean that the monopolist does not face competitive constraints

5.2 Collusion

5.2.1 Applications of Folk Theorem to induce Collusion

For the next few sections, we will assume the following set up: There are n firms in an industry and they have identical cost functions $c(q_k) = 2q_k$, $k = 1, 2, \dots, n$; the inverse demand function in the industry is $p(q) = 10 - q$, where $q = \sum_{k=1}^n q_k$.

In order to figure out the payoffs in the immediate round following deviation, we compute the monopoly profit: If $n = 1$, firm faces the problem $\max \pi(q) = p(q)q - 2q$. Our FOC is

$$\frac{d\pi(q)}{dq} = p'(q)q + p(q) - 2 = 0$$

which gives us $8 = 2q$ and thus $q^M = 4$. The monopoly price is thus, $p^M = p(4) = 10 - 4 = 6$. The firm earns

$$\pi^M = \pi(4) = 6 \cdot 4 - 2 \cdot 4 = 16$$

in each period

Collusion in Quantity Setting Game

As an alternative, each of them considers playing the following *trigger* strategy: supply q^M/n in each period (the resulting market price being determined by demand) as long as all the other firms have done so in the past; if *any* firm deviates, then supply the ‘Cournot quantity’ for evermore.

If each of them adopts the above strategy, show that the maximum profit that any one firm could make in the period it deviated is $4(1 + \frac{1}{n})^2$.

What is the amount of profit that that firm would make in each period of the punishment phase?

When $n = 2$, how large does δ have to be for trigger strategies to constitute a subgame perfect equilibrium in which collusion is sustained?

What if $n = 3$?

Solution. The maximum profit any one firm can yield is by acting as a monopoly for the residual demand. Let the quantity of the deviating firm be q_d . So, we get the new demand function

$$p(q) = 10 - q_d - (n - 1) \cdot \frac{q^M}{n}$$

We can assume only one firm deviates because we are finding the maximum profit any one firm can yield from deviation. Since $q^M = 4$,

$$p(q) = 10 - q_d - \frac{4(n - 1)}{n} = 6 - q_d + \frac{4}{n}$$

Profit for the deviating firm is thus

$$\begin{aligned} \pi_d(q) &= p(q)q_d - 2q_d \\ &= (6 - q_d + \frac{4}{n})q_d - 2q_d \\ &= 4q_d - q_d^2 + \frac{4q_d}{n} \end{aligned}$$

The firm thus faces

$$\max \pi_d(q) = 4q_d - q_d^2 + \frac{4q_d}{n}$$

Our FOC:

$$\frac{d\pi_d(q_d)}{dq_d} = 4 - 2q_d + \frac{4}{n} = 0$$

This yields

$$q_d = 2 + \frac{2}{n}$$

Our profit is thus

$$\begin{aligned} \pi_d &= \left(6 - q_d + \frac{4}{n}\right) \left(2 + \frac{2}{n}\right) \\ &= \left(2 + \frac{2}{n}\right) \left(2 + \frac{2}{n}\right) = 4 \left(1 + \frac{1}{n}\right)^2 \end{aligned}$$

which verifies the claim.

The profit in each stage of punishment is just the Cournot-Nash Equilibrium: the firm k maximises

$$\pi_k(q_k) = \left(10 - q_k - \sum_{j \neq k} q_j\right) q_k - 2q_k$$

FOC:

$$\frac{d\pi_k}{dq_k} = 10 - 2q_k - \sum_{j \neq k} q_j - 2 = 0$$

By symmetry, assume $q_1 = \dots = q_n$. So,

$$\begin{aligned} 10 - 2q_k - (n-1)q_k &= 2 \\ (n+1)q_k &= 8 \\ q_k &= \frac{8}{n+1} \end{aligned}$$

Our profit becomes

$$\begin{aligned} \pi_k(q_k) &= \left(10 - \frac{8n}{n+1}\right) \left(\frac{8}{n+1}\right) - 2 \cdot \frac{8}{n+1} \\ &= \left(8 - \frac{8n}{n+1}\right) \left(\frac{8}{n+1}\right) \\ &= \frac{64}{(n+1)^2} \end{aligned}$$

To find the δ to maintain a collusive outcome, we must find δ s.t.

$$\sum_{k=0}^{\infty} \delta^k \cdot \frac{16}{n} \geq 4 \left(1 + \frac{1}{n}\right)^2 + \lim_{k \rightarrow \infty} \left(\delta \cdot \frac{64}{(n+1)^2} + \dots + \delta^k \cdot \frac{64}{(n+1)^2} \right)$$

which reduces to

$$\frac{16/n}{1-\delta} \geq 4 \left(1 + \frac{1}{n}\right)^2 + \delta \cdot \frac{64/(n+1)^2}{1-\delta}$$

When $n = 2$,

$$\begin{aligned} \frac{8}{1-\delta} &\geq 9 + \delta \cdot \frac{64/9}{1-\delta} \\ 8 &\geq 9(1-\delta) + \delta \cdot \frac{64}{9} \\ \delta(9 - 64/9) &\geq 1 \implies \delta \geq \frac{9}{17} \end{aligned}$$

When $n = 3$,

$$\frac{16/3}{1-\delta} \geq \frac{4 \cdot 16}{9} + \frac{4\delta}{1-\delta}$$

This yields

$$\begin{aligned} 12 &\geq 16(1-\delta) + 9\delta \\ 7\delta &\geq 4 \\ \delta &\geq \frac{4}{7} \end{aligned}$$

Collusion in Price Setting Game

Each of them considers playing the following *trigger* strategy: charge p^M in each period (the resulting market quantity being determined by demand) as long as all the other firms have done so in the past; if *any* firm deviates, then charge marginal cost evermore.

If each of them adopts the above strategy, what is the upper limit on the amount of profit that any one firm could make in the profit it deviated?

What is the amount of profit that that firm would make in each period of the punishment phase?

How large does δ have to be for these trigger strategies to constitute a subgame perfect equilibrium in which

collusion is sustained?

Solution. If a firm deviated, it would choose an infinitesimally small $\epsilon > 0$ such that $p_k = p^M - \epsilon$. By assumption of the price setting game, the firm now gets all the demand at price p^M and hence receives the monopoly profit from (a) i.e. $\pi_k = 16$.

Since the strategy they revert to is the Bertrand Nash Equilibrium strategy whereby $p_k = c'(q)$ for all firms, their individual profit during each phase of punishment is $\pi_k = q \cdot (2 - 2) = 0$.

For the strategy to be a SPE, we need to prove two things. First, that the equilibrium path whereby collusion is sustained is the best response for each firm. Second, if a firm deviates, the each stage of the punishment phase is also a Nash Equilibrium. The second follows immediately from the fact that the punishment phase is the Bertrand Nash Equilibrium. To find the discount rate δ such that collusion is the best response for each firm is to find δ s.t.

$$\sum_{k=0}^{\infty} \delta^k \left(\frac{4}{n} \cdot 6 - \frac{4}{n} \cdot 2 \right) \geq 16 + \lim_{k \rightarrow \infty} [\delta(2-2) \frac{8}{n} + \delta^2(2-2) \frac{8}{n} + \dots + \delta^k(2-2) \frac{8}{n}]$$

which reduces to

$$\begin{aligned} \sum_{k=0}^{\infty} \delta^k \left(\frac{16}{n} \right) &\geq 16 \\ \frac{16/n}{1-\delta} &\geq 16 \\ \frac{1}{n} &\geq 1-\delta \\ \delta &\geq \frac{n-1}{n} \end{aligned}$$

So, if $\delta \geq \frac{n-1}{n}$ then cooperation is the equilibrium path and the best response for each firm in each period. Thus, the strategy is an SPE.

Remark: Hence show that when $n = 2$ there are values of δ for which collusion is sustainable in the price-setting game of part (b) but not in the quantity-setting game of part (c), and that when $n = 3$ this conclusion is reversed.

Briefly comment.

Solution. In the Bertrand Game, when $n = 2$, $\delta \geq \frac{1}{2}$. In the Cournot Game, when $n = 2$, $\delta \geq \frac{9}{17}$. So, when

$$\frac{9}{17} > \delta \geq \frac{1}{2}$$

collusion is sustainable in the price setting game but not the quantity setting game, as the payoffs for deviation would be greater than the payoff for collusion.

When $n = 3$, to sustain collusion in the Bertrand game, $\delta > \frac{2}{3}$ while in the Cournot game, $\delta \geq \frac{4}{7}$. Hence, when

$$\frac{2}{3} > \delta \geq \frac{4}{7}$$

collusion is sustainable in the quantity setting game but not the price setting game.

The observation is that as the number of firms increases, the minimum δ to sustain collusion in the Bertrand game increases while the minimum δ to sustain collusion in the Cournot game increases. This is because in the Cournot Game, the punishment profits decreases as more firms enter the market since the punishment phase payoffs is $\frac{64}{(n+1)^2}$. On the contrary, the punishment profits in the Bertrand game is 0 and is invariant with the number of firms. So, a lower discount rate is required for the collusion to be maintained in the Cournot game as the payoffs in the punishment phase decrease automatically (and at a faster rate than the split monopoly profits decrease) when more firms decrease; a higher discount rate is required for collusion in the Bertrand game since the shared monopoly payoffs decrease without the punishment payoffs decreasing.

The General Case: Assume that there are n firms, identical marginal cost c , no fixed costs, identical products, no capacity constraints. $\Pi(p^m) > 0$ is the industry profit when all firms set the collusive price p^m . The payoffs are thus $\Pi(p^m)/n$ during collusion and $\Pi(p^m)$ during deviation. Punishment payoff is 0 given the grim trigger strategy. Discount factor is $\delta = \frac{1}{1+r}$ where $r > 0$. Collusion is sustainable with grim trigger strategies if

$$\frac{\Pi(p^m)}{n}(1 + \delta + \delta^2 + \dots) > \Pi(p^m)$$

$$\frac{\Pi(p^m)}{n(1 - \delta)} > \Pi(p^m)$$

This gives us

$$\delta > \frac{n-1}{n} = 1 - \frac{1}{n}$$

Remark: RHS rises as n increases so more difficult to sustain while LHS rises as interest rate r falls.

Frequency of Period: Assume that the goods are now sold every other period. So, we have sustainable collusion if

$$\frac{\Pi(p^m)}{n}(1 + \delta^2 + \delta^4 + \dots) > \Pi(p^m)$$

$$\frac{\Pi(p^m)}{n(1 - \delta^2)} > \Pi(p^m)$$

This gives us

$$\delta^2 > \frac{n-1}{n} = 1 - \frac{1}{n}$$

Since $\delta < 1$, $\delta^2 < \delta$, so it is harder to sustain collusion when goods are sold every other period.

Ease of Detection: Suppose a firm cheating on a tacit agreement can earn profits for two periods before being detected and punished. The present value of cheating is $\Pi(p^m)(1 + \delta)$ while the PV of colluding is $\frac{\Pi(p^m)}{n(1-\delta)}$. Since

$$(1 - \delta)(1 + \delta) = 1 - \delta^2$$

Collusion is sustainable if

$$\delta^2 > 1 - \frac{1}{n}$$

Same as two period frequency of sales.

5.2.2 Policy and collusion

Real world examples

5.3 Mergers

Types of Mergers

- Vertical Mergers: Merger between companies that operate in different parts of the supply chain
- Horizontal Mergers: Merger between companies that operate in the same part of the supply chain

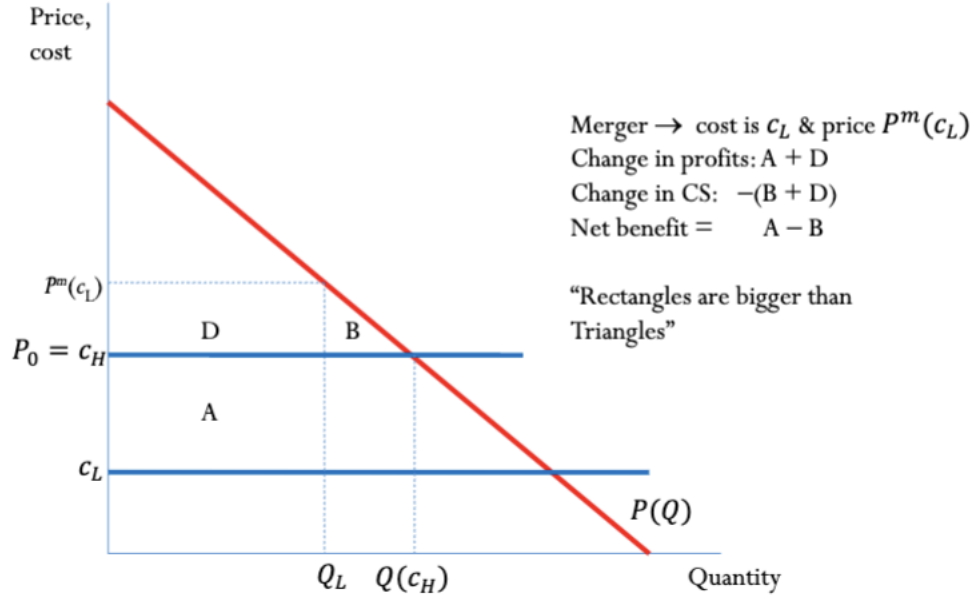
Issues with Mergers

- Anticompetitive Effects
 1. Unilateral effects: Equilibrium faces less competition i.e. higher price and lower quantity
 2. Coordinated effects: Collusion

- Potential economic benefits
 1. Efficiency gains
 2. Avoidance of Double Margins

Merger under Bertrand Case

We assume that there are n firms that compete through the Bertrand game. The change in equilibrium is illustrated accordingly:



Before the merger, the price is $p_0 = c_H$. After the merger, we can say that the cost falls to c_L because of economies of scale. Price increases to $p_m(c_L) > c_L$ and the change in profits is $A + B$ while the change in consumer surplus is $-(B + D)$ so the net change in welfare is $A - B$ where $A > B$.

So, under the Bertrand case, there are efficiency gains to be made, even if the price is higher and the quantity is restricted. Note: the analysis may change if c_H does not fall to c_L and instead we replace the constant marginal cost with a convex marginal cost function $c(q)$.

Merger under the Cournot Case

We assume that n symmetric firms with marginal costs $c > 0$ compete on quantity, facing a market demand of $p(Q) = a - bQ$. We know that profit for each firm is

$$\pi_i(n) = \frac{(a - c)^2}{b(n + 1)^2}$$

Proposition 5.1. If $n > 2$, then merger is unprofitable under the Cournot scheme.

Proof. If two firms merge, there are $n - 1$ firms each with profit

$$\pi_j(n - 1) = \frac{(a - c)^2}{bn^2}$$

Merger is only profitable iff the following equation satisfies:

$$\pi(n - 1) = \frac{(a - c)^2}{bn^2} > 2\pi(n) = 2 \frac{(a - c)^2}{b(n + 1)^2} \implies \frac{1}{n^2} > \frac{2}{(n + 1)^2}$$

which holds iff $n \leq 2$. ■

Intuition: The merged firm cuts the initial joint output of its two members since they have to return back to the Cournot equilibrium with $n - 1$ firms i.e. $q(n - 1) = \frac{a-c}{bn} < 2q(n) = 2 \cdot \frac{a-c}{b(n+1)}$. This induces its rivals to increase their outputs, which harms the merged firm. In equilibrium the merged firm loses more from the induced reduction in output than it gains from the higher margin. The beneficiaries are the non-merging firms, whose outputs and profits increase. When $n = 2$ there are no non-merging firms, and the merging firms capture all the market power gains. Merger to monopoly from duopoly is always profitable.

Stackelberg Competition: Avoiding the Cournot Paradox

Mergers under quantity competition can still be profitable above 2 firms if merging gives the additional advantage of moving first (i.e. being a Stackelberg leader).

Mergers under Differentiated Bertrand

Some general intuition: Assume that firms $1, \dots, n$ want to maximise profits and trades off higher profit margins against loss of sales (higher price). Suppose firms 1 and 2 want to merge and raise prices. If firm 1 raises their price, they lose some demand but some of this lost demand is diverted to the other firms. A subset of this lost demand that is diverted to 2 will contribute to its joint profits so both firm 1 and 2 will want to raise prices. The greater the diversion of demand to B the more the merged firm will want to raise p_1 . Best responses are upward-sloping (as a function of price), so others will also raise their prices. A merger is very likely to be profitable.

Farrell and Shapiro (2010): Practical Rule of Thumb for Diversion effect

Firm A will want to increase its price in a merger with B if B's pre-merger margin $\times$ Diversion Ratio of A/B $>$ Fall in A's marginal cost. The diversion ratio, D_{AB} , is the fraction of the sales lost by A that go to B. Hence

$$D_{AB} = \frac{\partial q_B / \partial p_A}{-\partial q_A / \partial p_A}$$

For a linear demand function $Q(p_A, p_B) = \frac{1}{1-b^2}(1-b-p_A+bp_B)$, $D_{AB} = b$. Sometimes, we can also use market shares: Assume that sales for other firms rise in proportion to their existing market shares and that total sales remain constant. So $D_{AB} = \frac{s_B}{1-s_A}$.

Remark: the larger the existing market share of the partner, the greater the price-increasing effect of the merger.

Formal Derivation of Result.

Assume firm A's profit function is $\pi_A(\mathbf{p}) = (p_A - c_A)q_A(\mathbf{p})$ where $\mathbf{p} = (p_A, p_B)$. The Nash Equilibrium prices are $\mathbf{p}^*$. The FOC evaluated at $\mathbf{p}^*$ is

$$q_A(\mathbf{p}^*) + (p_A - c_A) \frac{\partial q_A(\mathbf{p}^*)}{\partial p_A} = 0$$

The marginal cost reduction for firm A with merger is k . The profit function for the merged firm is

$$\pi_A + \pi_B = (p_A - c_A + k)q_A(\mathbf{p}) + (p_B - c_B)q_B(\mathbf{p}) = (p_A - c_A)q_A(\mathbf{p}) + kq_A(\mathbf{p}) + (p_B - c_B)q_B(\mathbf{p})$$

Note that the first term is the pre-merger profit function, the second term is the cost-savings function and the third term is B's profit.

Differentiating this with respect to p_A and evaluating it at $\mathbf{p}^*$ gives us

$$\begin{aligned} \frac{\partial \pi_A + \pi_B}{\partial p_A} &= q_A(\mathbf{p}^*) + (p_A^* - c_A) \frac{\partial q_A(\mathbf{p}^*)}{\partial p_A} + k \frac{\partial q_A(\mathbf{p}^*)}{\partial p_A} + (p_B^* - c_B) \frac{\partial q_B(\mathbf{p}^*)}{\partial p_A} \\ &= 0 - \frac{\partial q_A(\mathbf{p}^*)}{\partial p_A} [-k + (p_B^* - c_B)D_{AB}] \end{aligned}$$

Note that the first 2 terms go to zero because it is the derivative of the pre-merger profit function evaluated at the optimum. Note that $-\frac{\partial q_A(\mathbf{p}^*)}{\partial p_A} > 0$, so the upward pressure on price (i.e. wanting to raise p_A to raise

joint profit) is if and only iff $(p_B^* - c_B)D_{AB} > k$. ■

Vertical Mergers: Avoiding Double Marginalisation

Consider 2 firms, a retailer R who sets the retail price p and a supplier S who sets the wholesale price w . The retailer faces the constant marginal cost w and a market demand $Q(p) = 1 - p$. So the maximising retail price is $p = (1 + w)/2$ which solves $\max(p - w)(1 - p)$. Now, the supplier faces the marginal manufacturing cost c and wants to maximise profit. So it sets w to maximise $\pi_S = (w - c)(1 - w)$ (assume that the demand for supply is also $Q_S(w) = 1 - p(w) = (1 - w)/2$). So, $w = (c + 1)/2$. So $p = (3 + c)/4$. The total profit is

$$\pi = \pi_R + \pi_S = ((3+c)/4 - (c+1)/2)(1 - (3+c)/4) + ((c+1)/2 - c)(1 - (c+1)/2) = \frac{(1-c)^2}{16} + \frac{(1-c)^2}{8} = \frac{3(1-c)^2}{16}$$

If the firms merge, the integrated monopolist faces a marginal cost $c < w$ since $c < 1$. The monopolist faces the problem $\max(p^M - c)(1 - p^M)$ which gives the solution $p^M = \frac{1+c}{2}$. The joint profit is

$$\pi^M = (1 - (c+1)/2)((c+1)/2 - c) = \frac{(1-c)^2}{4} > \frac{3(1-c)^2}{16}$$

Measuring Market Power and Estimations

Definition 5.1 (Lerner's Index). The Lerner Index measures market power through the percentage markup of price over marginal cost and. It is given by

$$L = \frac{P - MC}{P}$$

Remark: The Lerner Index assumes that market power is proportionate to the strength of markup.

Lerner Index for different market structures

- Monopoly: $\frac{1}{\varepsilon}$
- Perfect Competition: 0
- Bertrand Competition: 0
- Symmetric Cournot: $\frac{1}{n\varepsilon}$

where $\varepsilon := \frac{p}{q} \frac{dq}{dp}$ i.e. the price elasticity of demand

Definition 5.2 (Herfindahl Index). The Herfindahl Index measures market power through the sum of squared market shares. It is given by

$$H = \sum_{i=1}^n \left(\frac{q_i}{Q} \right)^2 = \sum_{i=1}^n s_i^2$$

HHI for Different market structures

- Monopoly: 1
- Perfect Competition: tends to 0
- Symmetric Bertrand: $\frac{1}{n}$
- Symmetric Cournot: $\frac{1}{n}$

Remark: A merger leads to an increase in H, but this need not reduce welfare.

Mergers and Collusions Can affect in two ways:

1. Reducing the number of firms: Collusion is sustainable if

$$\delta > 1 - \frac{1}{n}$$

in the price setting game. If n decreases, then $1 - \frac{1}{n}$ decreases as well, implying that a lower discount rate is needed and hence is easier to sustain

2. Might make the industry less symmetric and hence harder to sustain collusion in asymmetric industries.

6 Decisions under Risk

6.1 Expected Utility Theory

References: Simon Cowan's slides; Jehle and Reny, *Advanced Microeconomic Theory*, Ch 2; Kreps, *A Course in Microeconomic Theory*, Ch 3; Mas-Colell, Whinston and Green, *Microeconomic Theory* Ch 6; Gravelle and Rees, *Microeconomics* Ch 17 (Skip G), 19

Takeaways from Expected Utility Theory

1. Definition of a Lottery, Expected Value, Certainty Equivalent and Risk Premium
2. Axioms of choices made under uncertainty: transitivity, completeness, continuity, reduction, independence
3. Expected Utility Function as a representable utility function; criticisms of Expected Utility Theory

6.1.1 Lotteries, Risk Premiums and Expected Value

Definition 6.1 (Simple Lotteries). A Simple (2-goods) Lottery is denoted by either

$$p \circ x \oplus (1 - p) \circ y$$

or equivalently

$$\mathbb{L} = [p, 1 - p; x, y]$$

which means x obtains with probability p and y obtains with probability $1 - p$. Generally, it is denoted by

$$\mathbb{L} = [p_1, \dots, p_n; x_1, \dots, x_n]$$

where $x_i \in X, p_i \geq 0, \sum_{i=1}^n p_i = 1$

Definition 6.2 (Expected Value of a Lottery). The Expected Value of a Lottery is a convex combination of the prizes whereby the coefficients of the prizes is the probability of the prize occurring i.e.

$$EV(\mathbb{L}) = \sum_{i=1}^n p_i x_i$$

We make the following assumptions about Lotteries:

Assumption 6.1. $1 \circ x \oplus (1 - 1) \circ y \sim x$. Getting a prize with probability one is the same as getting the prize for certain

Assumption 6.2. $p \circ x \oplus (1 - p) \circ y \sim (1 - p) \circ y \oplus p \circ x$. The consumer does not care about the order in which the lottery is described

Definition 6.3 (Certainty Equivalent). The certainty equivalent of any simple Lottery $\mathbb{L}$ over wealth levels is an amount of wealth, CE, offered with certainty, such that $u(\mathbb{L}) \equiv u(\text{CE})$.

Example. Suppose $u(w) \equiv \ln(w)$. Let $\mathbb{L} = [\frac{1}{2}, \frac{1}{2}; (w_0 + h), (w_0 - h)]$. The certainty equivalent must satisfy

$$u(\text{CE}) = EU(\mathbb{L}) = \sum_{i=1}^n p_i u(x_i)$$

which is

$$\ln(\text{CE}) = (1/2)\ln(w_0 + h) + (1/2)\ln(w_0 - h) = \ln(w_0^2 - h^2)^{1/2}$$

Thus, $CE = (w^2 - h^2)^{1/2} < EV(\mathbb{L})$ ■

Generally, we have

$$CE = u^{-1} \left(\sum_{i=1}^n p_i u(x_i) \right)$$

Definition 6.4 (Risk Premium). The Risk Premium is an amount of wealth, P , such that $u(\mathbb{L}) \equiv u(EV(\mathbb{L}) - P)$. Clearly, then

$$P \equiv EV(\mathbb{L}) - CE$$

Example. From the CE example, we have $P = EV(\mathbb{L}) - CE = w_0 - (w^2 - h^2)^{1/2}$ ■

Definition 6.5 (Compound Lotteries). A compound lottery is denoted by

$$\mathbb{C} = [p_1, \dots, p_n; \mathbb{L}_1, \dots, \mathbb{L}_n]$$

where $\mathbb{L}_i$ are simple lotteries.

Assumption 6.3 (Reduction to Simple Lotteries). There exists a simple lottery $\mathbb{L}_C$ s.t. a compound lottery $\mathbb{C}$ is equivalent to $\mathbb{L}_C$. In other words, $q \circ (p \circ x \oplus (1-p) \circ y) \oplus (1-q) \circ y \sim (qp) \circ x \oplus (1-qp) \circ y$. A consumer's perception of a lottery depends only on the effective probabilities of receiving the various prizes.

Intuitively, what this assumption means is that every compound lottery $\mathbb{C}$ can be understood as a simple lottery $\mathbb{L}_C$ by simply enumerating all possible prizes and their corresponding probabilities.

Example. Let $\mathbb{L} = [\frac{1}{10}, \frac{9}{10}; w_0, w_1]$ and $\mathbb{C} = [\frac{1}{2}, \frac{1}{2}; [1; w_0], \mathbb{L}]$. Then,

$$\mathbb{L}_C = \left[\frac{11}{20}, \frac{9}{20}; w_0, w_1 \right]$$

since $1 \times \frac{1}{2} + \frac{1}{10} = \frac{11}{20}$ and $\frac{1}{2} \times \frac{9}{10} = \frac{9}{20}$. ■

6.1.2 The Expected Utility Theorem

We want our preferences to respect basic properties of rationality even under uncertainty. The following are our axioms of choice under uncertainty:

Axiom 6.1 (Completeness). For any two lotteries $\mathbb{L}_1$ and $\mathbb{L}_2$ in $\mathcal{L}$, either $\mathbb{L}_1 \succsim \mathbb{L}_2$ or $\mathbb{L}_2 \succsim \mathbb{L}_1$

Axiom 6.2 (Transitivity). For any three lotteries, $\mathbb{L}_1, \mathbb{L}_2, \mathbb{L}_3 \in \mathcal{L}$, if $\mathbb{L}_1 \succsim \mathbb{L}_2$ and $\mathbb{L}_2 \succsim \mathbb{L}_3$, then $\mathbb{L}_1 \succsim \mathbb{L}_3$.

Our next axiom is new. Without getting into technical details, let us assume without loss that the elements of X have been indexed so that $x_1 \succsim x_2 \succsim \dots \succsim x_n$ i.e. X is the set of all possible prizes (that a lottery may or may not include). It seems plausible that no lottery is better than receiving x_1 with certainty and no lottery is worse than receiving x_n with probability 1. So, for any lottery $\mathbb{L}$, it seems plausible that $[1, 0; x_1, x_n] \succsim \mathbb{L}$ and $\mathbb{L} \succsim [0, 1; x_1, x_n]$. So, the next axiom of continuity says that if indifference does not hold at either extreme, then it must hold for some intermediate value of p^*

Axiom 6.3 (Continuity). For any three lotteries $\mathbb{L}_1, \mathbb{L}_2, \mathbb{L}_3 \in \mathcal{L}$, if $\mathbb{L}_1 \succsim \mathbb{L}_2 \succsim \mathbb{L}_3$, then there exists a $p^* \in (0, 1)$ s.t.

$$\mathbb{L}_2 \sim [p^*, 1 - p^*; \mathbb{L}_1, \mathbb{L}_3]$$

or sometimes,

For any lottery $\mathbb{L} \in \mathcal{L}$, there exists a probability $p^* \in [0, 1]$, such that $\mathbb{L} \sim [p^*, (1 - p^*); x_1, x_n]$

Axiom 6.4 (Monotonicity). For all probabilities $\alpha, \beta \in [0, 1]$,

$$[\alpha, (1 - \alpha); x_1, x_n] \succsim [\beta, (1 - \beta); x_1, x_n] \iff \alpha \geq \beta$$

Axiom 6.5 (Independence). For any three lotteries $\mathbb{L}_1, \mathbb{L}_2, \mathbb{L}_3 \in \mathcal{L}$,

$$\mathbb{L}_1 \succsim \mathbb{L}_2 \iff [p, 1-p; \mathbb{L}_1, \mathbb{L}_3] \succsim [p, 1-p; \mathbb{L}_2, \mathbb{L}_3]$$

What this says is that adding mixing two lotteries with another lottery should not affect preferences.

Definition 6.6 (Expected Utility Property). The utility function $u : \mathcal{L} \rightarrow \mathbb{R}$ has the expected utility property if, for every $\mathbb{L} \in \mathcal{L}$,

$$u(\mathbb{L}) = \sum_{i=0}^n p_i u(x_i)$$

where $[p_1, \dots, p_n; x_1, \dots, x_n]$ is the simple lottery induced by $\mathbb{L}$

Theorem 6.1 (Existence of an Expected Utility Function/Expected Utility Theorem). If $(\mathcal{L}, \succsim)$ satisfy Axioms 6.1-6.5 and the assumption of reduction, there is a utility function u defined on $\mathcal{L}$ that satisfies the expected utility property:

$$u(\mathbb{L}) = \sum_{i=0}^n p_i u(x_i)$$

Proof. Proof is in the appendix. ■

Remark. This is also known as a Von Neumann-Morgenstern utility function.

Proposition 6.1. The Expected Utility function satisfies completeness, transitivity, continuity and independence.

Proof. Let $u(\cdot)$ be our VNM function s.t. $u : \mathcal{L} \rightarrow \mathbb{R}$.

Completeness: Consider two arbitrary lotteries $\mathbb{L}_1, \mathbb{L}_2 \in \mathcal{L}$. Then, we know that $\mathbb{L}_1 \succsim \mathbb{L}_2 \iff u(\mathbb{L}_1) \geq u(\mathbb{L}_2)$. From our definition of $u(\cdot)$, either $u(\mathbb{L}_1) \geq u(\mathbb{L}_2)$ or $u(\mathbb{L}_1) \leq u(\mathbb{L}_2)$ since the range of $u(\cdot)$ is defined on the reals. Hence, either $\mathbb{L}_1 \succsim \mathbb{L}_2$ or $\mathbb{L}_2 \succsim \mathbb{L}_1$, for any $\mathbb{L}_1, \mathbb{L}_2 \in \mathcal{L}$.

Transitivity: consider any three lotteries $\mathbb{L}_1, \mathbb{L}_2, \mathbb{L}_3 \in \mathcal{L}$. Without loss, assume $\mathbb{L}_1 \succsim \mathbb{L}_2$ and $\mathbb{L}_2 \succsim \mathbb{L}_3$. This implies that $u(\mathbb{L}_1) \geq u(\mathbb{L}_2)$ and $u(\mathbb{L}_2) \geq u(\mathbb{L}_3)$. We know that the ordering $\geq$ on $\mathbb{R}$ is transitive, so $u(\mathbb{L}_1) \geq u(\mathbb{L}_3)$ which implies $\mathbb{L}_1 \succsim \mathbb{L}_3$.

Independence: Let us consider 3 lotteries $\mathbb{L}_1, \mathbb{L}_2, \mathbb{L}_3 \in \mathcal{L}$. Without loss, assume $\mathbb{L}_1 \succsim \mathbb{L}_2$. Now, we consider two composite lotteries $\mathbb{L} = \alpha\mathbb{L}_1 + (1-\alpha)\mathbb{L}_3$ and $\mathbb{L}' = \alpha\mathbb{L}_2 + (1-\alpha)\mathbb{L}_3$. By the expected utility property, we have

$$\begin{aligned} u(\mathbb{L}) &= \alpha u(\mathbb{L}_1) + (1-\alpha)u(\mathbb{L}_3) \\ u(\mathbb{L}') &= \alpha u(\mathbb{L}_2) + (1-\alpha)u(\mathbb{L}_3) \end{aligned}$$

Now, $(1-\alpha)u(\mathbb{L}_3) = (1-\alpha)u(\mathbb{L}_3)$. Since $\mathbb{L}_1 \succsim \mathbb{L}_2 \iff u(\mathbb{L}_1) \geq u(\mathbb{L}_2) \iff \alpha u(\mathbb{L}_1) \geq \alpha u(\mathbb{L}_2)$ for $\alpha \geq 0$, we know that

$$\alpha u(\mathbb{L}_1) + (1-\alpha)u(\mathbb{L}_3) \geq \alpha u(\mathbb{L}_2) + (1-\alpha)u(\mathbb{L}_3)$$

and since

$$\alpha u(\mathbb{L}_1) + (1-\alpha)u(\mathbb{L}_3) \geq \alpha u(\mathbb{L}_2) + (1-\alpha)u(\mathbb{L}_3) \iff \alpha\mathbb{L}_1 + (1-\alpha)\mathbb{L}_3 \succsim \alpha\mathbb{L}_2 + (1-\alpha)\mathbb{L}_3$$

we have $\alpha\mathbb{L}_1 + (1-\alpha)\mathbb{L}_3 \succsim \alpha\mathbb{L}_2 + (1-\alpha)\mathbb{L}_3$, preserving preferences over $\mathcal{L}$ as desired.

Continuity: Satisfied due to linearity in p . So, if we consider the graphs of the lotteries involved with the prizes involved, they will either (1) intersect at some $p^* \in [0, 1]$ or be the same line. ■

Remark. Note that the expected utility function is not the only rational preference under uncertainty. There are others (Osbourne and Rubinstein give a discussion on this). One example is $U_{max}(\mathbb{L})$ which gives the utility of the largest prize with a strictly positive probability (Q1 of PS6).

Corollary 6.2. The following statements are equivalent:

1. The preferences $\succsim$ are rational and satisfy reduction, continuity and independence
2. The preferences $\succsim$ correspond to some expected utility function $EU(\mathbb{L})$

In general, we know that utility functions are not unique. In particular, we know that a preference ordering is preserved over monotonic transformations of utility functions. Does this hold for expected utility functions? The next theorem says that utility functions are unique up to affine transformations.

Theorem 6.3 (Uniqueness of Expected Utility function). Suppose that the VNM utility function $u(\cdot)$ represents $\succsim$. Then, the VNM utility function, $v(\cdot)$ represents those same preferences if and only if for some scalar α and some scalar $\beta > 0$,

$$v(\mathbb{L}) = \alpha + \beta u(\mathbb{L})$$

for all lotteries $\mathbb{L}$. That is, VNM utility functions are unique up to positive affine transformations.

Proof. Proof is in the technical appendix. ■

How good is expected utility as a theory? We must examine three areas: (a) Are there theoretical reasons to distrust these models? (b) How good are they empirically? (c) If there are problems with the models, what are the alternatives? Here are some criticisms by Kreps.

Criticisms of EU theory

1. Theoretical Reasons: Problem arises from the in which the models are used in applications: only a limited portion of the consumer's overall decision problem is modeled, and it is assumed that the consumer's choice behaviour on the limited problem conforms to this sort of model. Even though the consumer's overall choice behaviour might conform to this sort of model, choice behaviour can fail to conform on pieces examined in isolation. Two of the most important reasons for this have to do with portfolio effects and the temporal resolution of uncertainty
 - Portfolio effects: analogous to complement/substitute effects in the usual theory of consumer choice under certainty (aka price of corn is a key variable is important for determining the demand for wheat). With lotteries, suppose we asked the consumer how he felt about lotteries posed by taking positions in the stock of General Motors. According to these models, we would describe the lotteries of probability distribution over the dollar returns from the positions (possibly having the consumer give his subjective assessment of those distribution), and we would expect the consumer's preferences concerning gambles are given by the expected utility they generate. But if this consumer already has a position in the stock of General Motors, or a position in a competing stock like Ford, we would expect his preferences to be affected. Put another way, if we had two gambles with the exact same probability distribution, but one involved the stock of General Motors and the other involved the stocks of General Mills, because of the differences in the correlation in the returns between Ford and General Motors and between Ford and General Mills, we expect the consumer to feel differently about them. We would only see this if we modeled the entire portfolio choice problem of the individual – including in one model all the separate gamble that make up the one grand lottery that this individual faces. But we cannot look at preferences between gambles involving the stock of General Motors and the stock of General Mills in isolation and expect the model to work. We cannot incant *ceteris paribus* here with much conviction since holding fixed the consumer's net position in Ford is not the issue but the differences in correlation between the gambles being investigated and left out of the model. The (marginal) probability distributions of the gambles under consideration are not sufficient descriptions of the objects under choice; two gamble with the same marginal distribution (but different joint distributions with the consumer's unmodeled portfolio) would rank differently.
 - Temporal resolution of uncertainty: Consider an example. Suppose a coin will be flipped. If it comes up heads, you'll be given \$10,000; if it comes up tails, you get \$0. Or you can just receive \$3000. One complication arises is that you'll only get the money one year from now. But consider a further complication. We have in mind three gambles that you might be given: the first is just

as above, a coin flip gamble, where the coin is flipped today and you are told the result today. The second is as above, a coin flip gamble, where the coin is flipped next year and you are told the result then. And the third, the coin is flipped today, but you will be told the result next year. How would you rank this?

If you conform to standard models, then you would rank them identically since they all have the same probability distribution on prizes. But most people would rank the first better than the next two and indifferent between the second two. This is because (a) getting the information sooner is valuable to you and (b) the important time is when the uncertainty resolves for you, the decision maker, and not when uncertainty is "physically" resolved.

This has an important consideration when we think about the choices we make over time: it depends in part on resources you have now and the resources you think you'll have later. If the uncertainty in this gamble is resolved for you now, you can use that information better to fit your consumption choices now and next year to your (now better-known) resources. If the uncertainty resolves for you next year, you will have to temporize in the meantime. Timing of resolution of uncertainty can matter.

2. Solutions to Theoretical Problem: Include in our models all the relevant portions of the consumer's overall choice problem which is fairly unpalatable since putting everything in makes the model unwieldy and often intractable. Or you could restrict attention only to those pieces for which the standard models do apply, which is also unpalatable because the conditions required for the standard models to apply are quite stringent. A better alternative is to look for models of choice that are correct for pieces of a bigger model. For example, in dealing with portfolio effects, if you choose a "states of the world" model, then you can have additive separability across states and (even) some comparability of the various state-dependent utility functions. In dealing with problems in temporal resolution of uncertainty, you can have preferences that are "locally linear in probabilities"; recent work by Machina (1982,1984) shows that this is enough to get back some of the standard results in the economic of uncertainty.
3. Empirical evidence in simple settings:

6.2 Risk Aversion

Definition 6.7 (Risk Attitudes). Let $u(\cdot)$ be an individual's VNM utility function for lottery over non-negative levels of wealth. Then for the simple lottery $\mathbb{L} = [p_1, \dots, p_n; w_1, \dots, w_n]$, the individual is said to be

1. risk averse at $\mathbb{L}$ if $u(\text{EV}(\mathbb{L})) > u(\mathbb{L})$
2. risk neutral at $\mathbb{L}$ if $u(\text{EV}(\mathbb{L})) = u(\mathbb{L})$
3. risk loving at $\mathbb{L}$ if $u(\text{EV}(\mathbb{L})) < u(\mathbb{L})$

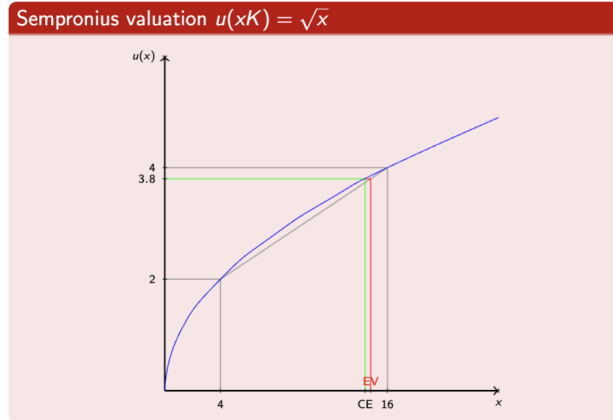
If for every non-degenerate (a simple lottery is non-degenerate if it assigns strictly positive probability to at least two distinct wealth levels) simple lottery $\mathbb{L}$, the individual is, for example, risk averse at $\mathbb{L}$, then the individual is said simply to be risk averse (or risk averse on $\mathcal{L}$ for emphasis). Similarly, an individual can be defined to be risk neutral and risk loving (on $\mathcal{L}$).

Proposition 6.2. The following are equivalent:

- For every lottery, $\text{EU}(\mathbb{L}) \leq u(\text{EV}(\mathbb{L}))$
- For every lottery, $\text{CE}(\mathbb{L}) \leq \text{EV}(\mathbb{L})$
- For every lottery, $\text{RP}(\mathbb{L}) \geq 0$
- u is concave

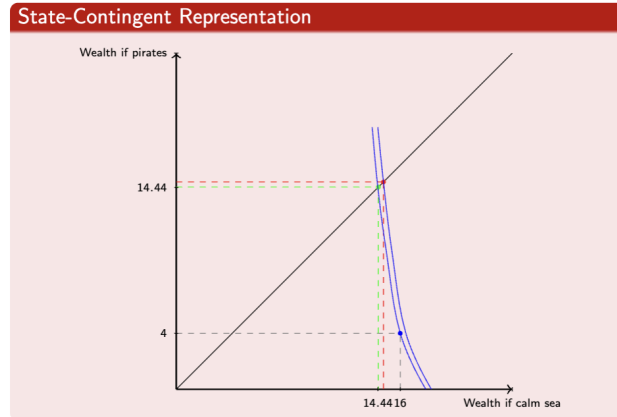
Graphical Representations of Risk Aversion

1. Standard Representation



Not much to talk about

2. State Contingent Representation



Basically, just treat the 2 states as if they are two different goods. Note that the state-contingent representation is important to show why smoothing out one's income is always better through mentioning convex combinations for risk averse agents.

Definition 6.8 (Arrow-Pratt Measure of Absolute Risk Aversion). Given a twice-differentiable VNM utility function $u(\cdot)$, the Arrow-Pratt measure (coefficient) of absolute risk aversion is given by

$$R_a(w) \equiv -\frac{u''(w)}{u'(w)}$$

Remark: the sign of this measure immediately tells us the basic attitude towards risk: $R_a(w)$ is positive, negative or zero as the agent is risk averse, risk loving.

Intuition: The Arrow-Pratt measure can be (informally) motivated as follows. We know that risk-neutrality is equivalent to the linearity of $u(x)$, that is, to $u''(x) = 0$ for all x . Therefore, it seems logical that the degree of risk aversion be related to the curvature of $u(\cdot)$. One possible measure of curvature of the VNM utility function $u(\cdot)$ at x is $u''(x)$. However, this is not adequate because it is not invariant to a linear transformation of the utility function. To make it invariant, the simplest modification is to use $-\frac{u''(x)}{u'(x)}$.

Proposition 6.3 (An Affine transformation of VNM utility functions retain the same Arrow-Pratt Measure). Let $u(\cdot)$ be a VNM utility function and $v = \alpha + \beta u$ be an affine transformation of $u(\cdot)$. Then, v and u have the same Absolute Arrow-Pratt Measure

Proof. Trivial. Since $v(\cdot) = \alpha + \beta u(\cdot)$, $v'(\cdot) = \beta u'(\cdot)$ and $v''(\cdot) = \beta u''(\cdot)$. Then

$$-\frac{v''(\cdot)}{v'(\cdot)} = -\frac{\beta u''(\cdot)}{\beta u'(\cdot)} = -\frac{u''(\cdot)}{u'(\cdot)}$$

■

Remark: The measure is a function of wealth and is typically evaluated at the wealth level of the individual. The effectiveness of the Arrow-Pratt measures are indeed more risk averse in a behaviourally significant respect: they have lower certainty equivalents and are willing to accept fewer lotteries.

Theorem 6.4 (Arrow-Pratt Theorem). The following statements are equivalent:

1. For every lottery, the risk premium created by u_1 is larger than that of u_2
2. u_1 is more concave than u_2 (that is, $u_1 = f(u_2)$, for a function f concave)
3. The Absolute Arrow-Pratt Measure of u_1 is, as a function, greater than that of u_2

Example. Consider 2 VNM utility functions such that the utility over outcomes are $u_1(x) = \sqrt{x}$ and $u_2 = x^{\frac{2}{3}}$. Then,

$$A_1(x) = -\frac{u_1''(x)}{u_1'(x)} = -\frac{-(1/4)x^{-3/2}}{(1/2)x^{-1/2}} = \frac{1}{2x}$$

and

$$A_2(x) = -\frac{u_2''(x)}{u_2'(x)} = -\frac{-(2/9)x^{-4/3}}{(2/3)x^{-1/3}} = \frac{1}{3x}$$

So, $A_1(x) > A_2(x)$. Notice that $CE_2(x) > CE_1(x)$ as well and that u_1 is a concave transformation of u_2 as we can write

$$x^{\frac{1}{2}} = (x^{\frac{2}{3}})^{\frac{3}{4}}$$

so u_1 would exhibit more risk aversion. ■

Definition 6.9 (Measure of Relative Risk Aversion). Given a twice-differentiable VNM utility function $u(c)$, the Arrow-Pratt measure (coefficient) of relative risk aversion is given by

$$R_r(w) \equiv -\frac{u''(w)}{u'(w)} \cdot w$$

Remark: it is just the Arrow Pratt measure of relative risk aversion by wealth.

Special Properties of Utility Functions: CARA, CRRA and DARA

Definition 6.10 (CARA utilities). A utility function has constant absolute risk aversion of $A(x) = a$ if they are characterised by the following

$$\begin{aligned} u_a(x) &= -\frac{1}{a}e^{-ax} \quad \text{whenever } a \neq 0 \\ u_a(x) &= x \quad \text{whenever } a = 0 \end{aligned}$$

Definition 6.11 (CRRA utilities). A utility function has constant relative risk aversion of $R(x) = r$ if they are characterised by the following

$$\begin{aligned} u_r(w) &= \frac{1}{1-r}x^{1-r} \quad \text{whenever } r \neq 1 \\ u_r(w) &= \ln x \quad \text{otherwise} \end{aligned}$$

Remark: CRRA means that risk aversion is scale invariant – the degree of risk aversion does not change when wealth and payoffs are scaled proportionately.

Definition 6.12 (DARA utilities). A utility function exhibits decreasing absolute risk aversion if for its Arrow-Pratt measure of absolute risk aversion, $\frac{dA(x)}{dx} < 0$

Remark: utility functions like $x^{\frac{1}{2}}$ and $x^{\frac{2}{3}}$ are DARA utilities.

Taylor Approximations of the Risk Premium. One method of approximating the Risk Premium is through the Taylor Approximation.

Example. Nero has wealth $x + \epsilon$, where $\epsilon \sim N(0, \sigma^2)$. We have $u(\text{CE}(\mathbb{L})) = u(x - \text{RP}(\mathbb{L})) = \text{EU}(\mathbb{L})$. From the Taylor Expansion, we have

$$\begin{aligned} u(x - \text{RP}(\mathbb{L})) &\approx u(x) - \text{RP}(\mathbb{L})u'(x) \\ \text{EU}(\mathbb{L}) &\approx u(x) + u'(x) \cdot 0 + \frac{1}{2}u''(x) \cdot \sigma^2 = u(x) + \frac{\sigma^2}{2}u''(x) \end{aligned}$$

So, we have

$$\text{RP}(\mathbb{L}) \approx \frac{1}{2}\sigma^2 A(x)$$

Definition 6.13 (First Order Stochastic Dominance). A lottery $\mathbb{L}_a$ first order stochastically dominates $\mathbb{L}_b$ if, for every y :

$$F_{\mathbb{L}_a}(y) \leq F_{\mathbb{L}_b}(y)$$

Proposition 6.4. If $\mathbb{L}_1$ FOSD $\mathbb{L}_2$, then for every expected utility maximiser, $\mathbb{L}_1 \succ \mathbb{L}_2$

Definition 6.14 (Second Order Stochastic Dominance). Given lotteries with the same expected values, lottery $\mathbb{L}_a$ second order stochastically dominates $\mathbb{L}_b$ if there exists a monetary value y^* such that

$$\begin{aligned} F_{\mathbb{L}_a}(y) &\leq F_{\mathbb{L}_b}(y) \text{ whenever } y \leq y^* \\ F_{\mathbb{L}_a}(y) &\geq F_{\mathbb{L}_b}(y) \text{ whenever } y \geq y^* \end{aligned}$$

More generally, we say $\mathbb{L}_a$ SOSD $\mathbb{L}_b$ iff for all $y \in [0, 1]$

$$\int_{-\infty}^y F_{\mathbb{L}_a}(y)dy \leq \int_{-\infty}^y F_{\mathbb{L}_b}(y)dy$$

with

$$\int_{-\infty}^y F_{\mathbb{L}_a}(y)dy < \int_{-\infty}^y F_{\mathbb{L}_b}(y)dy$$

for some y

Remark: The former definition is a special case of the general definition for CDFs that cross only once from below.

Proposition 6.5. If $\mathbb{L}_1$ SOSD $\mathbb{L}_2$, then for every risk averse expected utility maximiser, $\mathbb{L}_1 \succ \mathbb{L}_2$

Definition 6.15 (Mean-preserving Spread). A lottery $\mathbb{L}_2$ is a mean preserving spread of $\mathbb{L}_1$ iff

$$\int_{-\infty}^y F_{\mathbb{L}_2}(y)dy - \int_{-\infty}^y F_{\mathbb{L}_1}(y)dy \geq 0 \text{ with the inequality being strict for some } y$$

and

$$\lim_{y \rightarrow \infty} \left(\int_{-\infty}^y F_{\mathbb{L}_2}(y)dy - \int_{-\infty}^y F_{\mathbb{L}_1}(y)dy \right) = 0$$

The second condition is equivalent to saying that they have the same mean. This is because we can express the mean of some random variable X as

$$E[X] = \int_{-\infty}^{\infty} 1 - F(X) dX = 1 - \int_{-\infty}^{\infty} F(X) dX$$

Remark: The definition of a mean preserving spread is just to say that a lottery $\mathbb{L}_2$ is a mean preserving spread of $\mathbb{L}_1$ iff $E[\mathbb{L}_1] = E[\mathbb{L}_2]$ and $\sigma_2^2 > \sigma_1^2$.

Proposition 6.6. If $\mathbb{L}_2$ is a mean preserving spread of $\mathbb{L}_1$ iff every risk averse utility maximiser prefers $\mathbb{L}_1$ to $\mathbb{L}_2$

Example. The outcomes of lottery L_a are uniformly distributed on the unit interval, that is, the associated pdf $f(x)$ takes the value 1 on $[0, 1]$, and 0 elsewhere. What is the mean outcome, μ_a ? Prove that the variance of the outcomes, σ_a^2 , is $\frac{1}{12}$.

Let X be a r.v. denoting the outcome of the lottery. So, $X \sim U(0, 1)$. Then the mean

$$\mu_a = \frac{1 + 0}{2} = \frac{1}{2}$$

From the definition of $\text{Var}(X) = E[X^2] - (E[X])^2$. By Law of the Unconscious Statistician, we have

$$\begin{aligned} \sigma_a^2 &= \int_0^1 x^2 \cdot 1 dx - \frac{1}{4} \\ &= \left[\frac{1}{3} x^3 \right]_0^1 - \frac{1}{4} \\ &= \frac{1}{3} - \frac{1}{4} = \frac{1}{12} \end{aligned}$$

Consider another function $g(x)$ that takes the value $c(x - \frac{1}{2})^2$ on $[0, 1]$, and 0 elsewhere. What value of c ensures that g is a pdf, that is $\int_0^1 g(x) dx = 1$?

What is the mean outcome of the lottery L_b whose outcomes are distributed according to the pdf $g(x)$? What is the variance of the outcome of this lottery?

By Law of the Unconscious Statistician,

$$\begin{aligned} \int_0^1 c(x - \frac{1}{2})^2 \cdot 1 dx &= c \int_0^1 (x^2 - x + \frac{1}{4}) dx \\ &= c \left[\frac{1}{3} x^3 - \frac{1}{2} x^2 + \frac{x}{4} \right]_0^1 \\ &= c \left(\frac{1}{12} \right) = 1 \end{aligned}$$

This yields $c = 12$.

The mean is

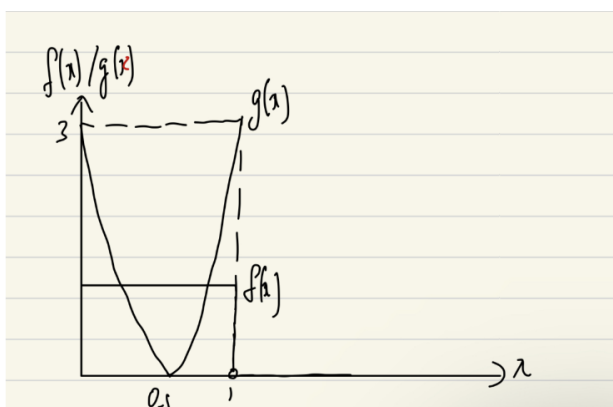
$$\begin{aligned} \mu &= \int_0^1 x \cdot g(x) dx \\ &= 12 \int_0^1 x \cdot (x - \frac{1}{2})^2 dx \\ &= 12 \int_0^1 (x^3 - x^2 + \frac{x}{4}) dx \\ &= 12 \left[\frac{x^4}{4} - \frac{x^3}{3} + \frac{x^2}{8} \right]_0^1 \\ &= 12 \left[\frac{1}{24} \right] = \frac{1}{2} \end{aligned}$$

Variance is

$$\begin{aligned}
 \sigma^2 &= \int_0^1 x^2 \cdot g(x) - \mu \\
 &= 12 \int_0^1 (x^4 - x^3 + \frac{x^2}{4}) dx - \frac{1}{4} \\
 &= 12 \left[\frac{x^5}{5} - \frac{x^4}{4} + \frac{x^3}{12} \right]_0^1 - \frac{1}{4} \\
 &= \frac{12}{30} - \frac{1}{4} = \frac{3}{20}
 \end{aligned}$$

Show that the lottery L_b is a mean-preserving spread of lottery L_a . (Compare the two pdfs, $f(x)$ and $g(x)$, graphically.)

Their pdfs are



So,

$$E[L_a] = E[L_b] = \frac{1}{2}$$

and

$$\sigma_a^2 = \frac{1}{12} < \sigma_b^2 = \frac{3}{20}$$

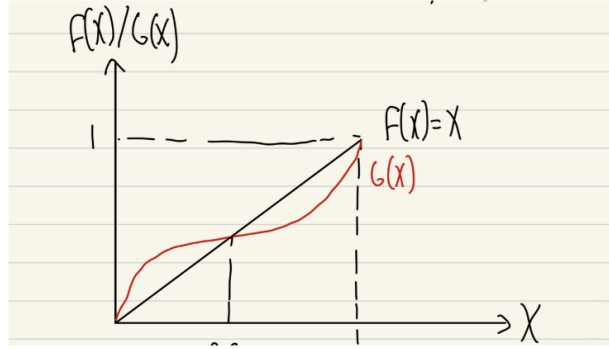
So L_b is a mean preserving spread of L_a . $f(x)$ is flat while $g(x)$ has a U-shape, indicating higher variance.

The CDF of $f(x)$ is $F(X) = X$, $0 \leq X \leq 1$. Calculate the CDF, $G(X) = \Pr[x \leq X]$, associated with $g(x)$. Sketch these two CDFs and confirm that the lottery L_a second-order stochastically dominates lottery L_b .

$F(X) = X$ and

$$\begin{aligned}
 G(X) &= \int 12(x - \frac{1}{2})^2 dx \\
 &= 12 \left[\frac{1}{3}x^3 - \frac{1}{2}x^2 + \frac{x}{4} \right] \\
 &= 4X^3 - 6X^2 + 3X, \quad 0 \leq X \leq 1
 \end{aligned}$$

As established, the means of both lotteries are the same. So,



whenever $x \leq 0.5$, $F_{\mathbb{L}_a}(x) \leq F_{\mathbb{L}_b}(x)$. And whenever $x \geq 0.5$, $F_{\mathbb{L}_a}(x) \geq F_{\mathbb{L}_b}(x)$. So $\mathbb{L}_a$ SOSD $\mathbb{L}_b$. SOSD is equivalent to a mean-spreading spread.

Does one of the lotteries L_a and L_b first-order stochastically dominate the other?

No, since when $x \leq 0.5$, $F_{\mathbb{L}_a}(x) \leq F_{\mathbb{L}_b}(x)$ and when $x \geq 0.5$, $F_{\mathbb{L}_a}(x) \geq F_{\mathbb{L}_b}(x)$, violating the condition for FOSD

6.3 Applications

We consider three main applications: (i) insurance, (ii) risk pooling and (iii) risk sharing. The difference between risk pooling and risk sharing is that if individuals enter separate lotteries, then that is risk pooling while if people enter the same lottery, that is risk sharing.

1. Insurance

Example. Perdita is a risk averse expected utility maximiser. Her initial wealth is w and her preferences over monetary outcomes, y , are summarised by the utility function $u(y)$. She is living in uncertain times – she might incur a loss of $L > 0$, or she might lose nothing at all. The loss occurs with probability π , but insurance is available at a cost of pq for an amount q of cover, i.e. she can pay a premium of pq and, in the event of a loss, she will receive a reimbursement of q .

What can you say about the signs of u' and u'' ? What does this imply for the comparison of $u'(w - L)$ with $u'(w)$? If Perdita is risk averse, this implies that $u(\cdot)$ is concave. So $u'(y) > 0$, $u''(y) < 0$. Since $w - L < w$, and $u''(y) < 0$, $u'(y)$ is decreasing in y . Hence

$$u'(w - L) > u'(w)$$

Briefly explain why the case in which $p = \pi$ is called *actuarially fair*.

Definition 6.16. An insurance is called actuarially fair when the cost of insurance equals the expected payout

The insurance is essentially a lottery

$$\mathbb{L} = [\pi, (1 - \pi); q, 0]$$

Hence

$$E(\mathbb{L}) = \pi q = pq$$

where pq is the amount that is insured.

When the amount of insurance cover she buys is q , what is her expected utility?

Show that the first-order condition for her optimal choice, q^* , is

$$\pi(1 - p)u'(w - L + (1 - p)q^*) - (1 - \pi)pu'(w - pq^*) = 0$$

Her expected utility is

$$u(q) = \pi u(w - L + q - pq) + (1 - \pi)u(w - pq)$$

She faces

$$\max u(q) = \pi u(w - L + q - pq) + (1 - \pi)u(w - pq)$$

Our FOC is

$$\frac{du}{dq} = \pi(1 - p)u'(w - L + (1 - p)q^*) - (1 - \pi)pu'(w - pq^*) = 0$$

this is because of the chain rule.

By using the FOC, or otherwise, show that when the insurance premium is actuarially fair she will fully insure, i.e. $p = \pi \implies q^* = L$.

When $p = \pi$, $p(1 - p)u'(w - L + (1 - p)q^*) = p(1 - p)u'(w - pq^*)$, which gives

$$u'(w - L + (1 - p)q^*) = u'(w - pq^*)$$

since u' is a one-one function,

$$w - L + (1 - p)q^* = w - pq^*$$

Then $q^* = L$.

What if $p > \pi$?

If $p > \pi$,

$$\pi(1 - p)u'(w - L + (1 - p)q^*) = (1 - \pi)pu'(w - pq^*)$$

Since we know that $\pi(1 - p)$ falls while $(1 - \pi)p$ rises, it must be that

$$u'(w - L + (1 - p)q^*) > u'(w - pq^*)$$

Since $u(\cdot)$ is bijective and decreasing, we yield

$$w - L + (1 - p)q^* < w - pq^* \implies q^* < L$$

In this case, buying insurance costs you something in terms of expected value. If $p = \pi$, then insurance is costless in expected value to trade income between the good state and the bad state.

2. Risk Pooling: Basically, risk pooling is when two people undertake lotteries and they agree to give the other some amount if they do well. We give two examples to see how to answer such questions.

Example: Simple Case. Sam and Janet are risk averse, each deriving utility $\ln w$ from final wealth w .

There is a lottery that would leave Sam with final wealth of either 5 or 20, each equally likely; there is a second lottery that would leave Janet with final wealth of either 4 or 25, again each equally likely. If Sam and Janet have the same initial wealth, explain why they would either both accept, or both decline, their separate gamble.

Let $\mathbb{L}_S = [\frac{1}{2}, \frac{1}{2}; 5, 20]$ and $\mathbb{L}_J = [\frac{1}{2}, \frac{1}{2}; 4, 25]$. For Sam, $\ln \text{CE} = \frac{1}{2} \ln 5 + \frac{1}{2} \ln 20 = \frac{1}{2} \ln 100 = \ln 10$. Hence, $\text{CE} = 10$. For Jane, we have $\ln \text{CE} = \frac{1}{2} (\ln 4 + \ln 25) = \ln 10$. So, $\text{CE} = 10$. Their decision condition is enter lottery is $\text{CE} > w$ and do not enter if $\text{CE} < w$. Since they both have the same starting wealth and CE, they will either both enter or not enter the lottery.

If the two lotteries are independent, would you advise them to each participate in their lottery, and share the costs and proceeds equally?

The simple lottery for each person would be

$$\mathbb{L} = \left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}; 4.5, 12, 15, 22.5 \right]$$

We want to find the certainty equivalent, which is from

$$\ln(\text{CE}) = \frac{1}{4} [\ln 4.5 + \ln 12 + \ln 15 + \ln 22.5] = \ln(18225)^{\frac{1}{4}}$$

So, we have

$$\text{CE} = 18225^{0.25} \approx 11.6 < 12$$

So, both of them will not enter the lottery as its CE is less than their initial wealth.

If the two lotteries are perfectly negatively correlated, why would you advise them to each participate in their lottery (again sharing the costs and proceeds equally), even if they were both much more risk averse?

There are two possibilities now, $w \in \{12, 15\}$. So,

$$\mathbb{L} = \left[\frac{1}{2}, \frac{1}{2}; 12, 15 \right]$$

Therefore,

$$\ln(\text{CE}) = \frac{1}{2} \ln(12 \cdot 15) = \frac{1}{2} \ln 180$$

So, $\text{CE} = 13.42 > 12$. Hence, they will both enter the lottery.

Suppose that $u(\cdot)$ is more risk averse function than $\ln$. I will claim that as long as $u(\cdot)$ is concave and risk averse, then they will accept the proposed risk pooling technique. We know that

$$u(\text{CE}) = \frac{1}{2}u(12) + \frac{1}{2}u(15)$$

We want to check if $u(\text{CE}) > u(12)$. Given concavity, we know that $u(\cdot)$ is increasing when it is non-negative. So

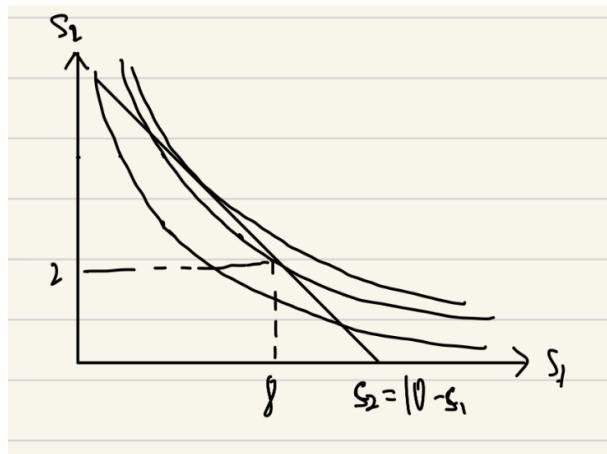
$$\begin{aligned} & u(15) > u(12) \\ \implies & \frac{1}{2}u(15) > \frac{1}{2}u(12) \\ \implies & \frac{1}{2}u(12) + \frac{1}{2}u(15) > u(12) \\ \implies & u(\text{CE}) > u(12) \end{aligned}$$

which proves our claim.

Example: Bilateral Insurance Contracts. Arthur is risk averse, and his income tomorrow depends on which of the two possible states occurs; each state is equally likely. His income will be 8 if state 1 occurs, but only 2 if state 2 occurs.

Draw Arthur's indifference curves in state-contingent income space, and explain how he could be better off if he were able to buy insurance.

Arthur's indifference curves are



The insurance is characterised by the following equations $s_1 = 8 - T$ and $s_2 = 2 + T$. So, we have $s_2 = 2 + (8 - s_1) = 10 - s_1$. So, by smoothing out his payoffs across states, Arthur achieves a convex combination of payoffs with respect to his original indifference curve at points $(8, 2)$ and $(2, 8)$, and thus yields a higher payoff $(8 - T, 2 + T)$ which lies on a higher indifference curve.

Norma is risk neutral. Her income will be 3 if state 1 occurs, and 7 if state 2 occurs. Suppose that they can write contracts of the form "Arthur will give Norma an amount x if and only if state 1 occurs, and Norma will give Arthur an amount y if and only if state 2 occurs." What shape are Norma's indifference curves? From Arthur's point of view, what is the best contract that Norma would find acceptable, and why?

Her indifference curves are linear in shape. Norma's acceptability condition is (by linearity)

$$\frac{1}{2}(3 + 7) = 5 = \frac{1}{2}(3 + x) + \frac{1}{2}(7 - y) = \frac{1}{2}(10 + x - y)$$

This gives us $x = y$. Arthur faces the problem

$$\max_{x,y} \frac{1}{2}u(8 - x) + \frac{1}{2}u(2 + y)$$

since $x = y$, this reduces to

$$\max_x \frac{1}{2}u(8 - x) + \frac{1}{2}u(2 + x)$$

Our first order condition is

$$\frac{d(\frac{1}{2}u(8 - x) + \frac{1}{2}u(2 + x))}{dx} = -\frac{1}{2}u'(8 - x) + \frac{1}{2}u'(2 + x) = 0$$

This gives us

$$\begin{aligned} u'(8 - x) &= u'(2 + x) \\ 8 - x &= 2 + x \\ 6 &= 2x \\ x &= 3 \end{aligned}$$

So, $y = x = 3$. This is because Norma is indifferent between not taking the contract and taking the contract in this case, and Arthur has maximised his expected utility.

Now assume that Arthur's utility as a function of his income m is $\ln m$. From Norma's point of view, what is the best contract that Arthur would find acceptable?

Arthur's acceptability condition is

$$\frac{1}{2}\ln 8 + \frac{1}{2}\ln 2 = \ln 4 = \frac{1}{2}\ln(8-x) + \frac{1}{2}\ln(2+y)$$

This yields

$$16 = (8-x)(2+y)$$

After some algebra we get,

$$x(2+y) = 8y \implies x = \frac{8y}{2+y}$$

Norma faces

$$\max_{x,y} \frac{1}{2}(3+x) + \frac{1}{2}(7-y) \quad \text{s.t.} \quad x = \frac{8y}{2+y}$$

This reduces to

$$\max_y \frac{1}{2} \left(3 + \frac{8y}{2+y} \right) + \frac{1}{2}(7-y)$$

Our FOC is

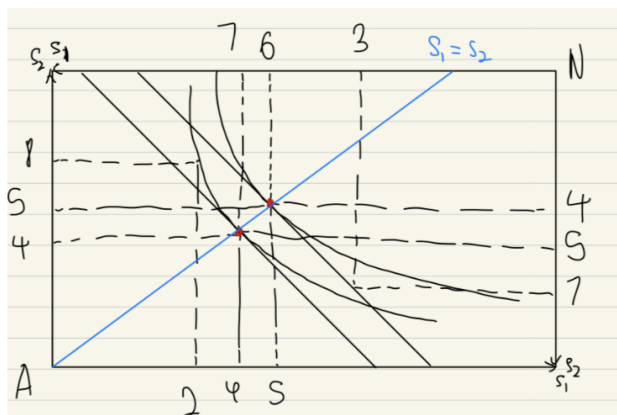
$$\frac{du}{dy} = \frac{8(2+y) - 8y}{(2+y)^2} - 1 = 0$$

So,

$$16 = (2+y)^2 \implies y = 2 \implies x = 4$$

Illustrate your results from part (b) and part (c) in an Edgeworth box, and highlight the set of efficient risk-sharing contracts.

The edgeworth box from (b) and (c) is (in state-contingent representation):



The set of efficient contracts is when $y = 6 - x$ or equivalently, when $s_1 = s_2$. Formally, we choose

$$\{y \in \mathbb{R} : s_1 = s_2, y = 6 - x\}$$

This follows from the $MRS = MRS$ condition.

3. Risk Sharing

Example: Comparing Risk Pooling and Risk Sharing

Bill's initial wealth is 22. He can invest it all in a risky project that has a 50:50 chance of succeeding. If he invests and the project succeeds, his final wealth will be 40, but if it fails, his final wealth will be 10. Bill's preferences over monetary outcomes, y , are summarised by the utility function $u(y) = \ln y$. For him, what is the certainty equivalent of this project? What is the risk premium? Why would he

reject the opportunity to invest in the project?

Our lottery is

$$\mathbb{L} = \left[\frac{1}{2}, \frac{1}{2}; 10, 40 \right]$$

So, we want to find the CE.

$$\begin{aligned} \text{CE} &= \frac{1}{2} \ln 10 + \frac{1}{2} \ln 40 \\ &= \frac{1}{2} \ln 400 \\ &= \ln 20 \end{aligned}$$

So the CE is 20. To find our risk premium we have $\text{RP} = \text{EV} - \text{CE}$, which is

$$\text{RP} = \frac{1}{2}(40 + 10) - \text{CE} = 5$$

Since his current wealth is greater than the certainty equivalent of the lottery, he will reject the lottery.

Ben has the same preferences and initial wealth as Bill, and suggests to him that they each invest 11 in the project and divide the proceeds equally, each ending with his remaining wealth plus either 40/2 after a success or 10/2 after a failure.

Should Bill accept this offer to share the risk?

Under the new scheme, their individual lotteries are

$$\mathbb{L} = \left[\frac{1}{2}, \frac{1}{2}; 16, 31 \right]$$

To find the CE,

$$\ln(\text{CE}) = \frac{1}{2} \ln 16 + \frac{1}{2} \ln 31 \approx 22.3 > 22$$

Since CE is greater than initial wealth, they should take the lottery.

Ben now finds an opportunity of his own to invest in a project with the same characteristics as Bill's. (The success or failure of one project is independent of that of the other.) Obviously, he wouldn't do it on his own, but they could agree that each of them take on their project and then divide the proceeds equally between them. Should they pool their risks and invest in the projects?

The lottery for each is

$$\mathbb{L} = \left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}; 10, 25, 40 \right]$$

The CE is derived from

$$\ln(\text{CE}) = \frac{1}{4} \ln 10 + \frac{1}{4} \ln 40 + \frac{1}{2} \ln 25$$

After some algebra, we reach

$$\text{CE} = \sqrt{500} = 10\sqrt{5} > 22$$

They should pool their risks.

If things were scaled up, so that the project payoffs became 400 or 100, and initial wealth was 220, would their choices change?

No, nothing will change since $u(x) = \ln x$ exhibits the CRRA property, since

$$A(x)x = -\frac{u''(x)}{u'(x)}x = 1$$

So, the degree of risk aversion does not change when wealth and payoffs are scaled proportionately, in this case by a scalar of 100.

Example. Janet has CARA preferences. She has initial wealth w and is offered the opportunity to buy a lottery ticket. If she buys it, her final wealth will be either $w + 4$ or $w - 2$, each equally likely. She is indifferent between buying the ticket and not buying it.

Explain whether she is risk averse, risk neutral, or risk loving. Draw this situation in state-contingent income space, with the ‘good’ outcome on the horizontal axis and the ‘bad’ outcome on the vertical axis.

If indifferent between buying and not buying, then Jane satisfies $CE = w$. So, $u(CE) = u(w) = \frac{1}{2}u(w - 2) + \frac{1}{2}u(w + 4)$. We know she is definitely not risk neutral, since $E[\mathbb{L}] = w + 1 > w$, and a risk neutral person would not be indifferent between a lottery with a higher expected payoff and a lower certain payoff. So, given CARA preferences, we know that her utility is of the form

$$u(w) = -\frac{1}{a}e^{-aw}$$

where a is the Absolute Arrow-Pratt coefficient of risk aversion. Given the indifference condition, we have

$$\begin{aligned} -\frac{1}{a}e^{-aw} &= \frac{1}{2} \cdot -\frac{1}{a}e^{-a(w-2)} + \frac{1}{2} \cdot -\frac{1}{a}e^{-a(w+4)} \\ e^{-aw} &= \frac{1}{2}e^{-aw} \cdot e^{2a} + \frac{1}{2}e^{-aw} \cdot e^{-4a} \end{aligned}$$

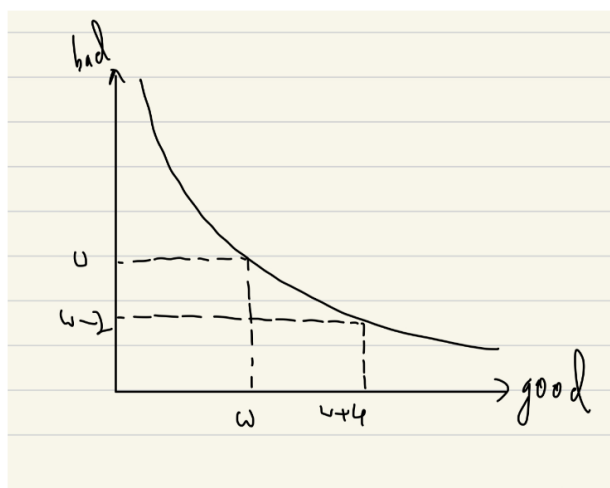
This gives us the equation

$$2 = e^{2a} + e^{-4a} \implies 2e^{4a} = e^{6a} + 1$$

Let $x = e^{2a}$, and we have the polynomial $2x^2 - x^3 = 1 \implies (x - 1)(x^2 - x - 1) = 0$. But $x \neq 1$ since if $e^{2a} = 1$, then $a = 0$ which contradicts our assumption that Jane is not risk neutral. So, $x = \frac{1+\sqrt{5}}{2}$ since $x > 0$. Hence

$$e^{2a} = \frac{1+\sqrt{5}}{2} \implies 2a = \ln\left(\frac{1+\sqrt{5}}{2}\right) \implies a = \frac{\ln(1+\sqrt{5}) - \ln 2}{2} > 0$$

So, since the Absolute Arrow-Pratt Coefficient is positive, Jane is risk averse. Her indifferent curves are thus convex to the origin in state-contingent representation:



She offers her friend Sam (who has the same initial wealth and preferences as she does) the following proposition: they buy the ticket together, and share the cost and proceeds equally. Should Sam accept? Illustrate your answer in a diagram.

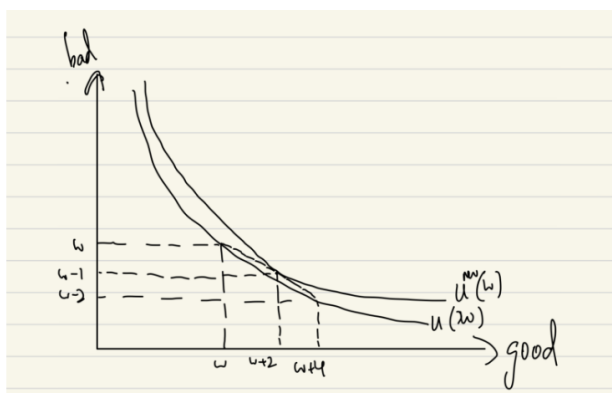
The initial arrangement is the lottery

$$\mathbb{L}_1 = \left[\frac{1}{2}, \frac{1}{2}; w-2, w+4 \right]$$

The new arrangement is the lottery

$$\mathbb{L}_2 = \left[\frac{1}{2}, \frac{1}{2}; w-1, w+2 \right]$$

Notice that the points $(w-1, w+2)$ on the state-contingent utility graph is a convex combination of the initial points due to the first arrangement (w, w) and $(w-2, w+4)$. A convex combination of points on a convex graph will always lie 'above' the convex graph itself, thus lying on a higher indifference curve as shown:

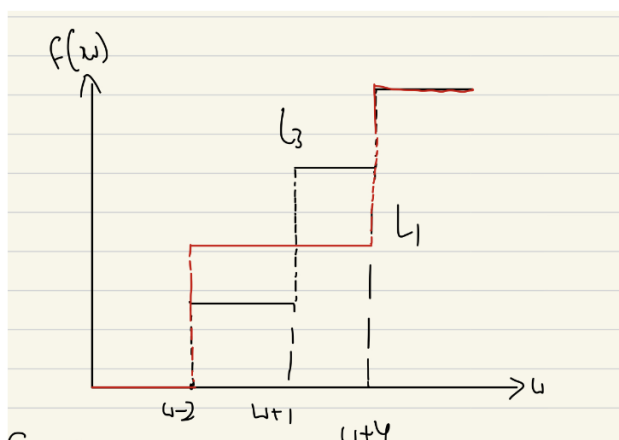


Sam has another idea: they buy two tickets (whose outcomes are independent), again sharing the costs and proceeds equally. Is this better than buying no tickets?

Let us call this new lottery $\mathbb{L}_3 = \left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}; w-2, w+1, w+4 \right]$. Note that the expected value of the first and third lotteries are the same since

$$E[\mathbb{L}_3] = \frac{1}{4}(w-2) + \frac{1}{2}(w+1) + \frac{1}{4}(w+4) = w+1 = E[\mathbb{L}_1]$$

Sketching the CDFs of $F_{\mathbb{L}_1}(w)$ and $F_{\mathbb{L}_3}(w)$, we have



Notice that

$$\begin{aligned} F_{\mathbb{L}_3}(w) &\leq F_{\mathbb{L}_1}(w) \quad \text{whenever } w \leq w + 1 \\ F_{\mathbb{L}_3}(w) &\geq F_{\mathbb{L}_1}(w) \quad \text{whenever } w \geq w + 1 \end{aligned}$$

So, $\mathbb{L}_3$ SOSD $\mathbb{L}_1$, and since Janet and Sam are risk averse, $\mathbb{L}_3 \succ \mathbb{L}_1$. Given $\mathbb{L}_1 \sim [1; w]$, by transitivity of preferences (as an EU maximiser is rational), $\mathbb{L}_3 \succ [1; w]$. So, this is better than buying no tickets as both Sam and Janet prefer this to their initial wealth

7 Information Economics

7.1 Adverse Selection

Definition 7.1 (Asymmetric Information). Asymmetric information is defined as any situation in which some subset of agents has more information than others

Asymmetric information usually leads to the issue of adverse selection

Definition 7.2. Market situation in which one of the parties is informed while the other is not and, for the uninformed party, the fact that the other party wants to contract is bad news.

Usually, we consider cases of asymmetric information where buyers and sellers have differentiated information. In turn, we consider three classical cases:

1. Market for Second-hand Goods: Sellers have more information about the actual quality of their goods; Buyers only have information regarding how much the seller would sell a good of x quality for, and the average value of the goods
2. Labour Market: Sellers have more information (i.e. workers) about their education level; Buyers (i.e. firms) only has information over the kinds of people that purchase education and beliefs about what education conveys
3. Insurance Market: Buyers have more information about their private information (i.e. state of health and conditions); Sellers have information over what buyer would buy the high or low risk premium

7.1.1 Akerlof's Market for Lemons Model

We first introduce Akerlof's original idea and then illustrate the principle using examples.

Model Set Up. Let us assume the following

1. A large number of sellers own used cars (implying perfect competition)
2. Cars can be of different quality, denoted $q \in [0, 100]$
3. There are the same number of cars for each quality
4. The sellers value each car i at their respective quality level q_i
5. A large number of buyers want to buy cars
6. The value of these cars is $1.5q_i$
7. Only sellers know what q_i is

Proposition 7.1 (Market for Lemons). Under the above set up, the equilibrium price is 0. In other words, total market failure occurs.

Proof. Let us assume, without loss of generality, that p is the price. Buyers know that only sellers that sell $p \geq q$ cars would sell, so they reduce the size of the market to exclude all sellers with $q > p$. They also know that the average quality of the cars in this new subset of the market must therefore be $p/2$. So, the prevailing price is what the buyers would want to pay which is $3p/4$. Most definitely now, the buyers would also be aware that only sellers that sell $3p/4 \geq q$ cars would be willing to sell. However, the average quality becomes $3p/8$ and this happens again and again. As $n \rightarrow \infty$, the price converges to 0 i.e. $\lim_{n \rightarrow \infty} p_n \rightarrow 0$. ■

Usually, what is tested in the exam is not complete market failure, but partial market failure. Specifically, partial unraveling means that some sellers are unable to sell to buyers who would otherwise have bought their products if there were perfect information.

We can see this in the next 2 examples: a simple case and a more complex case.

Example: The Simple Case. Three qualities of second hand-bicycles are available in equal numbers. Type X consumers and type Y consumers have different values for each quality of bicycle:

Quality	Type X value	Type Y value
<i>High</i>	90	110
<i>Medium</i>	80	85
<i>Low</i>	70	60

If a social planner were allocating the bicycle, which type of consumer would own which quality bicycle? The social planner will allocate Medium and High quality bikes to Type Y and low quality bikes to Type X .

If *both* sellers and buyers know the quality of each bicycle, what will be the market equilibrium outcome? The socially efficient outcome will obtain since they will offer money and accept based on their respective values. So Y will own high and medium quality bikes while X will own low quality bikes.

If *neither* sellers *nor* buyers know the quality of any bicycle, what will be the market equilibrium outcome? Y 's expected value is

$$EV_Y = \frac{1}{3}(110 + 85 + 60) = 85$$

X 's expected value is

$$EV_X = \frac{1}{3}(90 + 80 + 70) = 80$$

Since $EV_Y > EV_X$, X will sell all bicycles to Y

Finally, suppose that buyers do not know the quality of any particular bicycle for sale, but sellers do know that the quality of one that they own (However, buyers do know the average quality of bicycles that would be traded at any given price.) What will be the market equilibrium outcome?

Again, $EV_Y = 85$. But Y knows that X will not sell High so they will update their belief to consider only medium and low i.e. $EV_Y = \frac{1}{2}(85 + 60) = 72.5 < 80$. So, Y will only buy low quality bikes from X who would only sell low quality bikes to Y but this is not possible since Y 's value for low bikes is less than X 's. No trade occurs.

Comment on the efficiency, or otherwise, of the outcomes in settings (a)-(d).

The outcomes of (a) and (b) are the same since they are both socially efficient though (b) does not need a social planner. The overall 'utility' from trade will in both scenarios is 265. In (c), only Y has the bicycles which means the total 'utility' is 255, and so is inefficient. (d) is the least efficient with a total utility of 240. The reason for this is that the asymmetry of information induces adverse selection and therefore prevents trade, but also that Y 's valuation of the high and medium quality bikes combined is just much higher than the comparative on X 's side. In other words, if Y were the seller, (d) would be as efficient as (c).

Example: Case where there are different numbers of buyers and sellers. For qualities of a certain brand of car are produced in equal numbers. There are two groups of potential customers, none of whom wants to own more than a single car. Three times as many are type X as are type Y . The two types place different values on each grade of car, summarised in the table below. The price of a new car is £ 16,000, and in the eyes of each consumer no depreciation occurs in the first six months after the car has left the factory.

Quality	Type X value (£,000)	Type Y value (£,000)
<i>A</i>	21	18
<i>B</i>	18	17
<i>C</i>	15	16
<i>D</i>	11	12

Who buys new cars, assuming that they discover the grade only after the purchase?
X's expected value would be

$$EV_X = \frac{1}{4}(21000 + 18000 + 15000 + 11000) = 16250 > 16000$$

So, all of Type *X* would buy new cars.
Y's expected value would be

$$EV_Y = \frac{1}{4}(18000 + 17000 + 16000 + 12000) = 15750 < 16000$$

So, none of Type *Y* buyers would buy.

Analyse the market for six month resales, continuing to assume that consumers are able to judge the quality of a specific car only after purchase, but that they do know the average quality of used cars traded – i.e. if potential buyers offer a price of £ 20,000, they know that any type *Y* owners would sell, but that the type *X* owners would sell only if their car was grade *B*, *C*, or *D*. (Each potential trader decides whether or not to trade a single used car at the prevailing price and no later resales are possible).

So, only Type *X* buyers will buy new cars from (a), so they are suppliers in the market. Given Type *Y*'s EV is 15750, he is only willing to offer up to 15750 to a seller who is a Type *X*. But he knows that *X* values *A* and *B* cars higher than 15750, so *X* will only sell *C*, *D* cars. Knowing this, *Y* updates his beliefs and his new EV is

$$EV_Y = \frac{1}{2}(16000 + 12000) = 14000$$

But now, *Y* knows that *X* would not sell *C* cars either, so *Y*'s new expected value is 12000. Notice that only $\frac{1}{4}$ of *X* own *D* type cars. So even though the total number of *X* is three times more than *Y*, the number of *D* type sellers is less. So, since there are less sellers than buyers, *Y* must offer the maximum amount that it is willing to purchase at which is 12000. So, *Y* benefits.

If both buyers and sellers knew the quality of every used car traded, who would be better off than in part (b), and who would be worse off?

If both sellers and buyers knew the quality of cars ex ante, then *Y* buyers will buy *C* and *D* cars from *X* sellers since they value it more. WLOG let us assume that there are $12n$ sellers equally distributed over each type and $4n$ buyers. If *Y* buyers are buying from both *C* and *D*, there are only $2n$ buyers in each market. So, this implies that there are more sellers than buyers. Therefore, *Y* will offer the minimum amount possible i.e. his actual valuation and the price vector of $\mathbf{p} = (p_c, p_d) = (15000, 11000)$. So *D* sellers are doing worse comparatively than before while *C* sellers are doing the same.

Procedure for how to solve these questions: The recommended approach for showing the equilibrium price is the following procedure

1. Propose an arbitrary market price range p
2. Determine who is willing to sell at that price p
3. Buyers compute their expected value of a purchase at that price p
4. Verify consistency with the internally proposed market price

7.1.2 Spence's Signalling Model

Model Set Up

1. There are two types of workers, H -types or θ_H and L -types or θ_L , with different levels of productivity
2. Workers know their own type, but firms do not (they only know the expected productivity of a worker and the probability that the worker is of a certain type)
3. The population of workers are split according to (θ_H, θ_L) in $(\lambda, 1 - \lambda)$ proportions
4. Firms pay workers according to their marginal productivity and always break even on the wage schedule
5. There is an education that a worker can take, with differentiated costs for a worker based on his type: $c(\theta_H)$ and $c(\theta_L)$ with $c(\theta_L) > c(\theta_H)$. Education is a pure signal i.e. it does not increase productivity (the full model assumes that there is a level s of education that exists on some continuum rather than taking the value of 0 or 1)
6. Firms have beliefs over what education conveys, based on their expectations of who undertakes education (this is crucial)
7. Agents have a utility function that is his wage less his signalling cost (quasilinear in money):

$$u(y, s) = y - c_i s$$

There are two outcomes in Spence's model: **the separating equilibrium** and **the pooling equilibrium**.

Remark: Questions often say *fully characterise the market equilibrium*. This means you need to write three things

1. The total market output
2. The wage for θ_H workers
3. The wage for θ_L workers

Suppose that there are high ability and low ability workers, denoted θ_H and θ_L , and each worker chooses an education signal $s = 1$ or $s = 0$ in $(\lambda, 1 - \lambda)$ proportions. We can characterise the pooling and separating equilibrium accordingly.

Definition 7.3 (Separating Equilibrium). An equilibrium is a separating equilibrium if the firm pays workers via a wage schedule $\{w(1), w(0)\}$ where $w(1) > w(0)$

Definition 7.4 (Pooling Equilibrium). An equilibrium is a pooling equilibrium if all workers choose to not educate i.e. choose $s = 0$ the firm pays all workers a wage $\lambda w(1) + (1 - \lambda)w(0)$, that is according to the expected marginal output of a worker.

Definition 7.5 (Informationally Consistent Equilibrium). In an informationally consistent equilibrium (ICE) the employer offers a wage schedule which induces each type of employee A to choose a level of s which is consistent with his beliefs about the relationship between s and θ

Analysis.

Suppose that the wage schedule is a simple step function: for $s < s^*$, the wage is w_L and for $s \geq s^*$, the wage is w_H . It will be an ICE if it induces the θ_L type to choose $s < s^*$ and the θ_H type to set $s \geq s^*$. If A sets $s < s^*$ he will just choose $s = 0$ since otherwise will just increase his signalling cost without increasing his wage. Similarly, if A chooses $s \geq s^*$, he will choose s^* . It will be rational for the low productivity type to set $s = 0$ if

$$w_L \geq w_H - c_L s^* \implies s^* \geq (w_H - w_L)/c_L \equiv s_0$$

A high productivity type will choose to set $s = s^*$ and get the wage w_H rather than set $s = 0$ and receive w_L provided

$$w_H - c_H s^* \geq w_L \implies s_1 \equiv (w_H - w_L)/c_H \geq s^*$$

Since $c_L > c_H$, the equations imply $s_1 \geq s_0$. Hence, if $s^* \in [s_0, s_1]$, the step function wage schedule constitutes a fully revealing separating ICE in which each type reveals his productivity and the employer's beliefs about the relationship between the signal and productivity already confirmed.

Conditions for Pooling Equilibrium

The conditions for the pooling equilibrium are the same as the conditions for which separating equilibrium occurs.

Suppose the employer believes that

- if $s > s_1$, A is high productivity for sure
- If $s \leq s_1$, A is low productivity with probability π and high productivity with probability $1 - \pi$

If P then offers a wage of w_H if $s > s_1$, and a wage of $\bar{w}$ if $s \leq s_1$ where $\bar{w} := \pi w_H + (1 - \pi)w_L$, both types will set $s = 0$, and P will break even ex ante and never have her beliefs contradicted.

Most importantly, a pooling equilibrium occurs if the signalling cost is too high such that the higher wage does not offset their increased signalling cost relative to being paid the average wage. In other words,

$$\pi w_H + (1 - \pi)w_L \geq w_H - c_H s$$

θ_L types have no choice but to not educate so no check is needed.

Is Signalling Socially Wasteful?

Since P breaks even in all ICE, she is indifferent between whether the equilibrium is pooling or separating. θ_L types are better off in the pooling equilibrium than in any separating equilibrium. The high productivity type may be better or worse off than in the pooling equilibrium depending on the level of s^* required to qualify for the higher wage (more means less likely). His wage will be higher in such a separating equilibrium, but not sufficiently so to offset the signalling cost compared with the pooling equilibrium where he sets $s = 0$. The larger s^* (or equivalently c_H) and $\bar{w}$ are, the more likely it is that the separating equilibrium will be Pareto inefficient compared with the pooling equilibrium.

We can calculate the critical pooling wage $\hat{w}$ and the critical proportion of low productivity types $\hat{\pi}$ at which the two equilibria are equivalent. The high productivity type is indifferent between the separating equilibrium with $s^* = s_0$ and the pooling equilibrium when

$$\pi w_L + (1 - \pi)w_H = w_H - c_H s_0$$

substituting s_0 for the initial expression, we have

$$\hat{\pi} = c_H / c_L$$

So, whether the pooling equilibrium is Pareto superior to all separating equilibria depends only on the relative signalling cost and is unaffected by the relative productivity of the two types. If $\pi < \hat{\pi}$, pooling equilibrium is Pareto superior to all separating equilibrium.

1. **Example.** There are two types of worker, Alphas with productivity of 15 units and Gammas with productivity of 5 units. There is the same number of each type. Production takes place in perfectly competitive firms.

- (a) Assume that employers cannot assess the productivity of individual workers, but they do know the average productivity of the workforce. What will be the wage structure and output of the economy?

Firm's expected value is

$$EV = \frac{1}{2}(15 + 5) = 10.$$

The output of the economy will be $20N$ assuming that there are N α and γ workers and that all the workers are hired. The wages of the worker will be set equal to marginal productivity $w = 10$

- (b) Now suppose that the workers, who know their own type, can take a course. If they do, Alphas will obtain a degree, but Gammas will never pass, and this is generally known. If the monetary equivalent of the cost of effort put into taking the course is 2 units, show that there is a unique pure-strategy equilibrium, in which Alphas take the course and Gammas do not. What will be the wage structure and output of the economy, net of the cost of effort?

We assume that insofar as one person has the certification, the firm will assume that they can distinguish between α and γ workers and pay them according to their ‘supposed’ marginal productivity. **Furthermore, Employees have a belief that since Gammas cannot pass the course, those with the certificate are α type workers.** Suppose the worker is a γ . Then, suppose that at least one α takes the course. Then, his best response is to not take the course since $5 > 3$. Suppose that no α takes the course but at least one γ takes the course. His best response is to not take the course since $10 > 8$. Suppose that no one takes the course. His best response is to not take the course as well since $10 > 8$. So, γ worker’s dominant strategy is to not take the course.

Suppose the worker is a α . Then, suppose that at least one α takes the course. Then, his best response is to take the course since $13 > 5$. Suppose that no one else takes the course. His best response is to take the course since $13 > 5$. Suppose that no other α takes the course while at least one γ takes the course, his best response is to also take the course since $13 > 5$. So, α ’s dominant strategy is to take the course.

Given that the Nash Equilibrium is for α to take the course and for γ to not take the course, and that it is a dominant strategy Nash Equilibrium, it is a unique NE.

The wage structure is 15 for α and 5 for γ . The output, net of effort is, $18N$

- (c) If the cost of the course were 6 units, how would things change? (There are two pure-strategy equilibria – one in which everyone expects no-one to take the course, and one in which everyone expects just Alphas to take the course.)

Employers have a belief that that no one will take the course and that someone without the degree is an average type but someone with the degree is α . If a worker expects that no one takes the course, then if the worker is a γ they’ll choose to not take since $10 > 4$. If a worker is α , they’ll choose to not take since $10 > 9$. So, each type worker’s BR to others not doing anything is not taking the course. So, not taking the course, not taking the course is a NE. The wage structure here is 10 for γ , and 15 for graduates but no one is a graduate and the output is still $20N$.

Employers have a belief that those that take the course are α and those that do not are γ . If a worker expects that only α s take the course, then if they are a γ , they don’t take the course since $5 > -1$ while if they are α , they will take since $9 > 5$. So, all of α ’s best response to the expectation that only α s are taking the course is to take the course, and not taking the course for γ s. The wage structure here is 15 for α and 5 for γ and the output net of effort is $14N$.

- (d) Reverting to the situation where the cost of the course is 2 units, what would happen if individual workers did not know their own type? (Again, check two cases – one in which everyone expects no-one to take the course, and one in which everyone expects everyone to take the course. Assume that firms cannot observe whether or not a worker has taken the course, and that they can only distinguish between workers who have “Pass” certificates and those that do not – any worker that has taken the course and failed is indistinguishable from workers that haven’t taken the course.) Firms believe that no one takes the course. Thus, someone without the degree is an average degree type of wage 10, but someone with the degree is an α of wage 15. So, the expected value from taking the course is

$$EV = \frac{1}{2}(13 + 8) = 10.5 > 10$$

So there is a profitable deviation. Hence there is no pure strategy equilibrium where everyone does not take the course.

If everyone is expected to take the course, firms believe those with the certificate to be α and those without to be γ . Not taking the course yields a wage of 5 and taking the course yields a

wage of 15. Taking the course's expected value is 8 and so they will take the course. The wage structure becomes 15 for α and 5 for γ and the output net of effort is $20N - 4N = 16N$

2. **Example.** There are two types of workers: L-types with low productivity θ_L , and H-types with high productivity θ_H ; each workers knows their own type. Many risk-neutral firms compete for the services of the workers, but they cannot observe a worker's type.

- (a) Explain why some of the workers would like to be able to signal their type to the firms. What would make the signal credible?

They would like to signal their type so as to get a higher wage. Their signals would be credible if their payoff is $w(\theta_H) - c(\theta_H) > \frac{1}{2}(w(\theta_H) + w(\theta_L)) > w(\theta_H) - c(\theta_L)$ i.e. that the signal must be differentially costly between low skill and high skill workers.

Suppose that $\theta_L = 80$, $\theta_H = 100$, and that there are three times as many L-types as H-types; the workers have no outside opportunities.

- (b) What is the market equilibrium outcome if no signal is available?

The firm's expected value is $\frac{3}{4}(80) + \frac{1}{4}(100) = 60 + 25 = 85$. So, the firm will hire workers without knowledge of their type, and likely will end up with 3/4 type L workers and 1/4 type H workers given the population distribution. So, the equilibrium wage is $w = 85$ and the market output is also $85N$.

- (c) Suppose the workers can acquire an education, observable by the firms. This would cost 22 for an L-type, but only 12 for an H-type. Characterise the unique the market equilibrium.

There are four scenarios to go through:

- Everyone acquires education: the firm believes that those with an education signals an average type but those without an education signal a low type. So, they will offer the average wage of 85 for those with an education and 80 for those without. The payoff for L type is $85 - 22 = 63 < 80$ if they did not educate. Hence, there is a profitable deviation for L types
- Everyone does not acquire an education: the firm believes that those without an education signals an average type but those with an education signal a H type. So they will offer the average wage of 85 for those without an education and 100 for those with an education. In this case, the H type has a profitable deviation since $100 - 12 = 88 > 85$.
- L types acquire an education while H types do not: the firm believes that education signals an L type while no education signals a H type, paying 80 to those with an education and 100 to those without. Profitable deviation for an L type since to not educate yields $100 > 80 - 22 = 58$
- H types educate but not L types: the firm believes that education signals a H type while no education signals an L type. No profitable deviation since H types are offered 100 and their payoff is 88. If they did not educate, they would receive 80. L types are offered 80 and if they educated, they would receive $78 < 80$. So, no profitable deviation

The market equilibrium is therefore $w(\theta_H) = 100, w(\theta_L) = 80$ with output of $85N - 12N = 73N$

3. How would the analysis in part (c) change in education costs were 18 for an L-type, and 16 for an H-type?

Again, we verify four scenarios:

- Everyone acquires an education: the firm believes that those with an education signal an average type but those without signal a low type. So they will offer the average wage of 85 for those with an education and 80 for those without. The payoff for an L type to educate is $85 - 18 = 67 < 80$ if they did not educate. So, there is a profitable deviation for them
- Everyone does not acquire an education: the firm believes that those without an education signals an average type, but those with an education signals a H type. So, they will offer the average wage of 85 for those without an education and 100 for those with an education. Both types have no profitable deviation. If H -types educated, they will receive a payoff of $100 - 16 = 84 < 85$ while if L -types educated, they will receive a payoff of $100 - 18 = 82 < 85$. So, no profitable deviation.

- *L*-types do not educate but *H*-types do: the firm believes that those with an education signal a *H* type and those without signals a *L* type. So, they offer a wage of 100 to educated workers and 80 otherwise. *L* types have a profitable deviation to educate as their payoff otherwise is $100 - 18 = 82 > 80$.
- *H*-types do not educate but *L*-types do: the firm believes that those with education signal a *L*-type and *H*-type otherwise. So, they offer a wage of 80 to educated workers and 100 otherwise. *L*-types have a profitable deviation to not educate since their payoff will be $100 > 80 - 18$.

So, there is a unique equilibrium where no one educates, and the wage is 100 for educated workers and 85 otherwise (which is for everyone). The net output is $85N$

4. Is signalling socially wasteful?

Signalling may be socially wasteful since it reduces overall output. It may improve matching between firms and may thus be more efficient. If education is not a pure signal, it may increase output

Procedure to answer signalling equilibrium questions

1. Assess what the beliefs of the firm are given the scenario
2. Show that there is no profitable deviation or that there is such a profitable deviation
3. Fully characterise the equilibrium

7.1.3 Stiglitz and Rothschild's Model for Insurance: Screening

Model Set Up. We first make assumptions on the insurance firms: namely that (1) the market is perfectly competitive and hence they make 0 profits, (2) that if it is possible to offer an insurance contract which at least breaks even and would be preferred by buyers to contracts currently being offered, firms will do so, and that (3) insurance firms are risk neutral i.e. they only care about their expected profit. Taken together, what the assumption says is that if the firm offers a contract (α_1, α_2) where α_1 is the price of the contract and α_2 is the net payout that the insurance firm must make if a consumer faces an accident, then the firm faces

$$\max \pi = \alpha_1(1 - p) + \alpha_2 p = 0$$

where p denotes the probability of an accident for a consumer. Furthermore, we are going to assume that a consumer's type θ_i is private information to him and is not observable by the insurer directly. However, the insurer would know what a consumer of a certain type would buy i.e. if a contract for a lower risk buyer is available to a higher risk buyer, the higher risk buyer would purchase the low risk contract since it would be cheaper. This reveals to the insurer that the buyer is at least as risky as that type offered.

Model. The Rothschild-Stiglitz Model makes two claims

Proposition 7.2. A pooling equilibria will not exist in an insurance market

Proposition 7.3. It is not necessary that a separating equilibria exists in an insurance market.

Summary for the conditions of non-existence of equilibrium:

1. Too few high-risk individuals because it will make the cost of pooling low, driving out the separating contracts
2. Difference in accident probabilities not high because it will make the cost of pooling low, driving out the separating contracts
3. Risk attitudes of individuals, since it relates to the cost of separating. High costs of separating drives separating contracts out of the market
 - If individuals are risk-neutral, it never pays to pool since the expected loss is always higher by pooling with high risk individuals

- If they are infinitely risk-averse with utility functions $u(p, W_1, W_2) = \min\{W_1, W_2\}$, it always pays to pool

Corollary 7.1. There is a critical range $(\lambda^*, 1)$ of the proportion of low-risk individuals necessary for an equilibrium to exist

The following is from Gravelle and Rees.

‘Proof’ that there is not separating equilibrium.

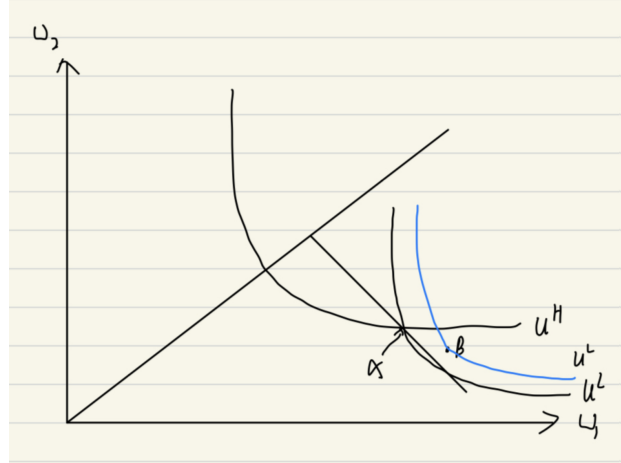
We are going to claim that there is no pooling equilibrium, or that if an insurance firm only offers a single contract for individuals, that there will not be an equilibrium. In this set up, the full information separating equilibrium cannot hold since low and high risks individuals receive the same amount of cover, but high risk individuals must pay a higher premium $P(\theta_H) = p(\theta_H)q > P(\theta_L) = p(\theta_L)q$. So, high risk types will choose the contract for the low risk types, making the contract not break even given the above inequality. Now, consider that the single contract is just $\alpha = (\alpha_1, \alpha_2)$ where α_1 is the premium that the insured must pay and that α_2 is the payout that the insured receives in the state of an accident. Because a single contract is offered, the firm considers the pooled probabilities i.e.

$$E[p] = \int_{-\infty}^{\infty} p(\theta_i) \phi(\theta_i) d\theta_i$$

So the firm faces

$$E[\pi] = \alpha_1(1 - E[p]) - (\alpha_2 - \alpha_1)E[p] = 0$$

Now, let us arbitrarily group the consumers together such that we have high-risk’ types where their $p(\theta_i) \geq \bar{p}$ and ‘low-risk’ types where their $p(\theta_i) < \bar{p}$ for some $\bar{p}$. Let us denote their average probabilities in both groups as p_H, p_L respectively, and that their share of the population is $(1 - \lambda, \lambda)$ respectively. Now we can construct a new contract β that is sufficiently near α such that the low-risk individuals prefer $\beta \succ \alpha$ but the high risks individuals prefer $\alpha \succ \beta$ as shown:



where w_1 represents the state with no accident and w_2 represents the state with accident. The reason why U^H and U^L must intersect in this way rather than overlapping is because their slopes must be different. Consider that the two MRS for $U^H = (1 - p_H)w_1 + p_H w_2$ and $U^L = (1 - p_L)w_1 + p_L w_2$ respectively:

$$MRS_H = \frac{\partial U^H(p_H, w_1, w_2) / \partial w_1}{\partial U^H(p_H, w_1, w_2) / \partial w_2} = \frac{1 - p_H}{p_H}$$

$$MRS_L = \frac{\partial U^L(p_L, w_1, w_2) / \partial w_1}{\partial U^L(p_L, w_1, w_2) / \partial w_2} = \frac{1 - p_L}{p_L}$$

and we know that by construction if $p_H > p_L$, then $\frac{1-p_H}{p_H} \neq \frac{1-p_L}{p_L}$. By construction, we also know that all low risk individuals go to β . This has two consequences. First, this leaves only high risk individuals choosing

α , which makes it unprofitable since $p_H > E[p]$ hence

$$E_\alpha[\pi] = \alpha_1(1 - p_H) - (\alpha_2 - \alpha_1)p_H < 0$$

Second, only low risk individuals choose β and since β is near α , we can claim that

$$E_\beta[\pi] = \beta_1(1 - p_L) - (\beta_2 - \beta_1)p_L > 0$$

This means that there is a **profitable deviation** for a firm to provide β instead of α , contradicting the fact that α is plausibly a pooling equilibrium. ■

‘Proof’ that there is no separating equilibrium if there are too few high cost individuals

We also want to claim that there is no separating equilibrium, and therefore no competitive equilibrium at all if $\lambda > \lambda^*$. Let us assume that there is a separating equilibrium between high and low types, where they are offered an amount of cover q_i and an insurance premium P_i where $i \in \{H, L\}$. So, there are two available contracts, $(P_H, q_H), (P_L, q_L)$. For the separating equilibrium to be stable, the insurance firms faces a self-selection constraint:

$$U^i(P_i, q_i) \geq U^i(P_j, q_j) \text{ where } i \neq j, i, j \in \{H, L\}$$

The constraints require that an individual of type i must achieve at least as high expected utility from the contract (P_i, q_i) designed for her type as from the contract designed for type j (P_j, q_j) . It is intuitive to see that the self-selection constraint will not be binding for the low risk individual since they will be offered a lower premium. Let us say that an agent begins with wealth w and faces a loss L if an accident occurs. So, we can write the constraint as

$$(1 - p_H)u(w - p_H q_H) + p_H u(w - L + q_H - p_H q_H) \geq (1 - p_H)u(w - p_L q_L) + p_H u(w - L + q_L - p_L q_L)$$

For $q_H < L, q_L < L$, the left hand side is increasing in q_H and the right hand side is increasing in q_L . Hence insurers will offer a contract to high risks which maximizes the left-hand side of the constraint. But the best breakeven contract for high risks has full cover: $q_H^* = L$. Thus the separating contract designed for high risks under adverse selection is the one they had under full information: full insurance at the fair premium. Given that $q_H = L$ satisfies the constraint, we consider the separating contract for low risks which maximises expected utility as

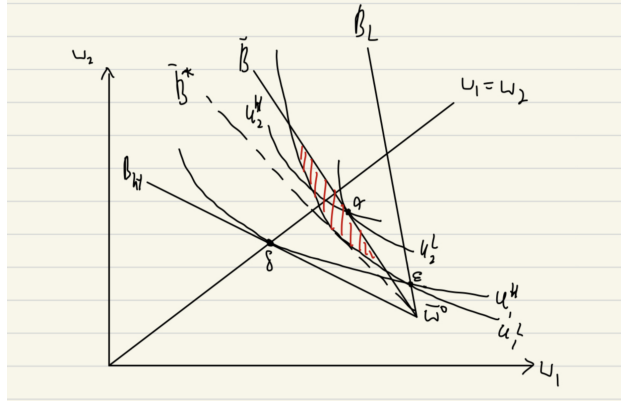
$$u(w - p_L L) = (1 - p_H)u(w - p_L q_L) + p_H u(w - L + q_L - p_L q_L)$$

To induce high-risk types to select the contract designed for them, the contract offered to low risks must be made sufficiently unattractive to high-risk types. Cover must be less than the full extent of the loss, since if $q_L = L$, high-risk types would prefer the contract designed for low risks. For a separating equilibrium not to exist, a pooling contract (P, q) must be able to drive the best possible separating contracts (δ, ε) off the market. So, it must at least breakeven

$$P \geq E[p]q$$

and attract the low-risks types away from their contract ε i.e.

$$(1 - p_L)u(w - P) + p_L u(w - L - P + q) \geq (1 - p_L)u(w - p_L q_L) + p_L u(w - L + q_L - p_L q_L)$$



First, note that slope of the breakeven line $\bar{B}w^0 = -(1 - E[p])/E[p]$, where $E[p] = \lambda p_L + (1 - \lambda)p_H$. The breakeven requirement on (P, q) is that it lies on or below the breakeven line $\bar{B}w^0$. The driving away requirement is that the pooling contract must lie on or above the low risk indifference curve u_1^L through ε . In the diagram, all pooling contracts in the red shaded area on or above u_1^L and on or below $\bar{B}w^0$ satisfy both requirements. Any contract in the shaded area drives the separating contracts off the markets. For pooling contracts to not drive off separating contracts, the breakeven line for pooling contracts must be below the low-risk indifference curve through ε . Since the slope of the pooling contract breakeven line is $-(1 - E[p])/E[p]$ and $E[p] = \lambda p_L + (1 - \lambda)p_H$, it will get flatter and eventually lie below the indifference curve u_1^L as λ falls. Thus, there will exist a critical proportion of λ^* of low risks in the population (i.e. θ_i is no longer normally distributed but skewed left), generating the pooling contracts breakeven line $\bar{B}^*w^0$ such that if $\lambda < \lambda^*$, the separating contracts are not driven from the market. So, if $\lambda \in (\lambda^*, 1)$, there exists a pooling contract which breaks even and attracts low risks from their separating contract at ε . But we have also established that no pooling contract can be an equilibrium, so no competitive equilibrium exists if there are too many low risk individuals. ■

7.1.4 Screening for Insurance Monopoly

Model Set Up

- There are high risk type and low risk type consumers (θ_H, θ_L) in $(\lambda, 1 - \lambda)$ proportions
- A monopolist can sell the contract at different qualities/dimensions and different prices. We denote this (d, p)
- High-risk buyers value the quality of the contract more. We denote this value coefficient $v_H > v_L$. Therefore, their utilities are

$$\begin{aligned} u_H(d, p) &= v_H d - p \\ u_L(d, p) &= v_L d - p \end{aligned}$$

- Profits for monopolists for each contract are denoted

$$\pi = p - c(d)$$

Full Information Case

If the monopoly was fully informed, the monopoly would offer two contracts

1. High-risk market: Find (d_H, p_H) to maximise $p_H - c(d_H)$, constrained by willingness to buy $v_H d_H \geq p_H$ which is the incentive rationality constraint. Then, the monopoly will extract all rents from the buyer by setting $v_H d_H = p_H$. To find the optimal d_H , the firm faces

$$\max v_H d_H - c(d_H)$$

and the first order condition is to set $v_H = c'(d_H)$

2. Low-risk market: Find (d_L, p_L) to maximise $p_L - c(d_L)$, constrained by willing to buy $v_L d_L \geq p_L$ which is the IR constraint. The monopoly extracts all rents $v_L d_L = p_L$. Then, the firm faces

$$\max v_L d_L - c(d_L)$$

which yields $v_L = c'(d_L)$.

Uninformed Case

The separating equilibrium cannot be sustained since notice that $v_H > v_L \implies v_H q_L > v_L q_L \implies v_H q_L - p_L > v_L q_L - p_L = 0$. So, the high-risk individual would be better off pretending to be a low-risk individual, since they would have 0 utility buying the high-risk contract.

The monopolist would engage in **screening** and allow the consumers to do this for free and design some contract to offer $(d_H, p_H), (d_L, p_L)$ and ensure that buyers would buy the product instead of nothing (IR) and that each type prefers their product (IC) and to ensure largest profits possible. So, the firm faces

$$\max \lambda(p_H - c(d_H)) + (1 - \lambda)(p_L - c(d_L))$$

subject to

$$v_H d_H - p_H \geq 0 \quad (\text{IRH})$$

This is the IR constraint for high-risk types.

$$v_L d_L - p_L \geq 0 \quad (\text{IRL})$$

This is the IR constraint for low-risk types.

$$v_H d_H - p_H \geq v_L d_L - p_L \quad (\text{ICH})$$

This is the IC for high-risk types

$$v_L d_L - p_L \geq v_L d_H - p_H \quad (\text{ICL})$$

This is the IC for low-risk types.

Conditions for Separating Types:

- IRH does not bind, IRL must bind with equality. Hence

$$p_L = v_L d_L$$

This is because the monopoly cannot extract all rents from everyone (only the low-risk types) or the high risk types will pretend to be low-risk types.

- ICH binds with equality while ICL does not bind and is slack. Hence

$$\begin{aligned} v_H d_H - p_H &= v_H d_L - p_L \\ p_H &= p_L + v_H(d_H - d_L) \\ p_H &= v_H d_H - (v_H - v_L)d_L \end{aligned}$$

ICL is always trivially satisfied since the low-risk type will never want to buy more insurance than is needed and the ICH will have to bind with equality since otherwise the firm is charging too little!

Hence, the firm faces

$$\max \lambda(p_H - c(d_H)) + (1 - \lambda)(p_L - c(d_L))$$

Our FOCs are

$$\begin{aligned} \frac{\partial \pi}{\partial d_H} &= \lambda[v_H - c'(d_H)] = 0 \\ \frac{\partial \pi}{\partial d_L} &= -\lambda[v_H - v_L] + (1 - \lambda)[v_L - c'(d_L)] = 0 \end{aligned}$$

Interpretation:

- The first condition is like that of full information so the quality of good of the product will be the same (but cheaper)
- The second generates lower quality than before, at its maximum possible price

Example. The queen of the Amazons is a monopolist selling statues to the people of Tracia. Lisipe knows that half of the Tracians are very devoted to Afrodita:

- $\lambda = \frac{1}{2}$
- High valuation: $3d - p$ where p is price paid
- Low valuation: $2d - p$
- $C(d) = d^2$

How should Lisipe price the statues?

Solution. For the full information case, Lisipe faces

$$\begin{aligned} \max(3d_H - d_H^2) \\ \max(2d_L - d_L^2) \end{aligned}$$

So, $d_H = 1.5$ and extract all rents, so $p_H = 4.5$ and $d_L = 1, p_L = 2$. Total profits is $0.5 \times 2.25 + 0.5 \times 1 = 1.625$. If the monopolist does not have full information and engages in screening,

- IRL: $p_L = 2d_L$
- ICH: $p_H = p_L + v_H[d_H - d_L] = 3d_H - d_L$
- Maximise $\frac{1}{2}(3d_H - d_L - d_H^2) + \frac{1}{2}(2d_L - d_L^2)$
- FOC(H): $3 - 2d_H = 0$ leads to $d_H = 1.5$
- FOC(L): $-\frac{1}{2} + 1 - d_L = 0$ leads to $d_L = \frac{1}{2}$
- $p_L = 1, p_H = 4$
- Total profits are $0.5 \times 1.75 + 0.5 \times 0.75 = 1.25$

Example. All households have the same wealth w , are risk averse, and face a potential loss $K < w$. Half of the households are low-risk and have a probability $\frac{1}{4}$ of incurring the loss. Half of the households are high-risk, with a probability $\frac{1}{2}$ of incurring the loss.

Insurance companies operate in a competitive market, are risk-neutral, and have no administrative costs.

1. Suppose that types are observable by insurers. Regulators 1 and 2 propose that insurance companies set difference prices for different households p_L and p_H per pound of cover. While Regulator 1 proposes that all households must fully insure, Regulator 2 proposes that they can partially insure (for no more than their total loss K). Explain why, in equilibrium, households may find both regulations equivalent. What are the equilibrium prices p_L, p_H ?

Let q denote cover. The firm's profit function is $\pi_L = p_L q - \frac{1}{4}q$. Due to perfect competition and risk-neutrality, $\pi_L = 0$, hence $p_L = \frac{1}{4}$. The same can be said for $p_H = \frac{1}{2}$. Since the price p equals the insurance premium, the insurance is actuarially fair and so consumers will always fully insure even if allowed to partially insure.

2. Regulators 3 and 4 propose that insurance companies must set the same price for all households, p per pound of cover.

- (a) Regulator 3 proposes that all households must fully insure. What is the equilibrium price p ?
The expected profit is

$$\pi = pq - \left(\frac{1}{2} \cdot \frac{1}{4}q + \frac{1}{2} \cdot \frac{1}{2}q\right) = pq - \frac{3}{8}q$$

Setting profits to 0, we yield $p = \frac{3}{8}$

- (b) Regulator 4 proposes allowing households to partially insure (for no more than the total loss K). Denote by $q(p)$ the low-risk household's demand for insurance and assume that $q'(p) < 0$ holds for prices $\frac{1}{2} \geq p \geq \frac{1}{4}$. Verify whether there is a unique equilibrium price, which may be written as a weighted average of $\frac{1}{2}$ and $\frac{1}{4}$, at which insurance companies break even in expectation. If $p = \frac{1}{4}$, equilibrium cannot exist since a loss will be made on high risk households as they will all fully insure. Consider,

$$\pi_H = \frac{1}{4}q - \frac{1}{2}q < 0$$

If $p = \frac{1}{2}$, then low risk individuals will fully insure while high-risk individuals partially insure. Then,

$$\pi = \frac{1}{2}(q + K) - (\frac{1}{2}q + \frac{1}{4}K) > 0$$

This does not satisfy equilibrium in perfect competition. So, if an equilibrium exists, it must be that

$$\frac{1}{2} \geq p \geq \frac{1}{4}$$

Hence, we write expected profits and set it to zero

$$E[\pi] = \frac{1}{2}(p - \frac{1}{4})q(p) + \frac{1}{2}(p - \frac{1}{2})K = 0$$

Rearranging gives us

$$p = \frac{\frac{1}{4}q(p) + \frac{1}{2}K}{q(p) + K}$$

Example. Cornelia's wealth consists of 2 million ducats at home and 8 million ducats abroad. She is an expected utility maximiser with utility $u(w) = \ln(w)$ on final wealth w measured in millions of ducats. Cornelia knows that her wealth abroad will be confiscated with probability $p = \frac{1}{3}$ and wants to take out insurance against this adverse event.

Insurance companies are aware of Cornelia's initial wealth position and her utility of final wealth is $\ln(w)$, but they are unsure about the probability that her foreign wealth will be confiscated. They think the probability of loss could be either $\frac{1}{3}$ or $\frac{1}{2}$. Currently, they offer insurance against foreign wealth loss at premium $\pi = \frac{1}{2}$. That is, Cornelia can buy unit coverage for an up-front payment of $\pi = \frac{1}{2}$ ducats. She can buy any coverage at this unit price.

1. Verify that Cornelia's final wealth when she buys x amount of coverage is $w_N = 10 - 0.5x$ in case her foreign wealth is not confiscated and it is $w_A = 2 + 0.5x$ in case her foreign wealth is confiscated. Write down her expected utility and find the level of coverage that she chooses.

Cornelia's wealth is generally

$$w = 2 + 1\{\text{no confiscation}\} \cdot 8 - 0.5x + 1\{\text{confiscated}\} \cdot x$$

Thus, $w_N = 10 - 0.5x$ and $w_A = 2 + 0.5x$. Her expected utility is

$$u(w) = \frac{1}{3}\ln(w_A) + \frac{2}{3}\ln(w_N) = \frac{1}{3}\ln(2 + 0.5x) + \frac{2}{3}\ln(10 - 0.5x)$$

Cornelia faces

$$\max_x u(x) = \frac{1}{3}\ln(2 + 0.5x) + \frac{2}{3}\ln(10 - 0.5x)$$

The FOC is

$$\frac{du}{dx} = \frac{1}{3} \cdot \frac{1/2}{2 + 0.5x} - \frac{2}{3} \cdot \frac{1/2}{10 - 0.5x} = 0 \implies x = 4$$

2. What would be the premium charged by competitive insurers if they knew that Cornelia's probability of loss was $\frac{1}{3}$ rather than $\frac{1}{2}$ and how much coverage would she buy in that case?

The insurer faces

$$\max(\pi x - \frac{1}{3}x)$$

Since in a perfectly competitive market, profits are 0, we get $\pi = \frac{1}{3}$ so Cornelia would fully insure at $x = 8$ since the insurance is actuarially fair.

3. The insurers, not knowing whether Cornelia's probability of loss is $\frac{1}{2}$ or $\frac{1}{3}$, offer her two options: either full coverage at premium $\pi = \frac{1}{2}$, or limited coverage $\bar{x}$ at premium $\bar{\pi} = \frac{1}{3}$. Under what condition on the value of $\bar{x}$ does this work as screening insurance? Verify whether $\bar{x} = 2, 3$ or 4 satisfy the condition. Cornelia has two options.

Case 1: Full coverage at $\pi = \frac{1}{2}$. Then $u(x) = p \cdot \ln(10 - 0.5x) + (1 - p)\ln(2 + 0.5x)$. When $x = 8$, $u(8) = \ln 6$.

Case 2: Partial coverage at $\bar{\pi} = \frac{1}{3}$. Since full insurance at $\pi = \frac{1}{2}$ yields a sure utility of $\ln 6$, $\bar{x}$ must satisfy

$$\frac{1}{2}\ln(10 - \frac{1}{3}\bar{x}) + \frac{1}{2}\ln(10 + \frac{2}{3}\bar{x}) = \ln 6$$

Remark: This condition is obtained for the following reasons

- We only consider the higher risk consumer because the Incentive Compatibility Constraint for the low risk consumer is not going to bind
- Due to the competitive nature of the market, the IC for the higher risk consumer must bind. In a competitive insurance market, firms cannot impose contracts that make any individual strictly worse off than their utility without insurance or their utility at the market alternative. If a high-risk individual were offered a strictly worse utility from the limited coverage plan, they would refuse to purchase it and look for alternatives, or other firms could step in with a more attractive offer.

Hence, only $\bar{x} = 3$ satisfies this condition.

4. Is Cornelia better off in any of the limited coverage screening contracts with premium $\bar{\pi} = \frac{1}{3}$ found in (3) or by buying any coverage she likes at premium $\pi = \frac{1}{2}$? Which arrangement is more profitable for the insurers? Explain.

If $\bar{x} = 3$, then $u(w) = \ln 6$. If she chooses $\pi = \frac{1}{2}$, then

$$u = \frac{2}{3}\ln(8) + \frac{1}{3}\ln(4) < \ln 6$$

The inequality follows from the property of convexity. It would be better to contract in (3) arrangement. Insurance companies are worse off because they no longer make a profit on her insurance.

7.2 Moral Hazard

In the Asymmetric Information case, we consider problems of hidden types. Moral Hazard considers problems of hidden actions i.e. the amount of effort in work.

Definition 7.6 (Moral Hazard). Moral Hazard is a market situation in which (i) the outcome of a contract depends upon specific actions of one of the parties contracting but (ii) the other party cannot verify or monitor such actions, remaining uninformed

7.2.1 The Principal-Agent Problem

Model Set Up:

1. A risk-neutral firm (the principal)
2. A risk-averse agent who can choose the amount of effort $e \in \{0, 1\}$ to exert
3. A risk-averse agent has the option of doing nothing and has a reservation utility $\bar{U}$ with a quasilinear utility function of the form

$$U(m, e) = u(m) - c(e)$$

where $c(1) > c(0)$

4. Effort e affects the probability of the higher wage outcome. In other words,

$$\begin{aligned} p_L \pi_H + (1 - p_L) \pi_L & \text{ if } e = 0 \\ p_H \pi_H + (1 - p_H) \pi_L & \text{ if } e = 1 \end{aligned}$$

where $p_H > p_L$

However, the firm can condition the payment of a wage depending on the outcome. So, the firm will write a contract of the form

$$\{w_L | \pi_L, w_H | \pi_H\}$$

This basically means that the agent's expected wage is calculated as

$$w = p_i w_i + (1 - p_{-i}) w_{-i} \text{ where } i \in \{L, H\}$$

which is also the expected cost to the firm which the firm wants to minimise.

We now define two additional constraints for the firm.

First, we want the contract to be better than doing nothing. This is called the Individual Rationality Constraint.

Definition 7.7 (Individual Rationality Constraint). The IR constraint is defined as

$$(1 - p_H)u(w_L) + p_H u(w_H) - c_H \geq \bar{U}$$

Second, we want the contract to induce high effort. This is called the Incentive Compatibility Constraint.

Definition 7.8 (Incentive Compatibility Constraint). The IC constraint is defined as

$$(1 - p_H)u(w_L) + p_H u(w_H) - c_H \geq (1 - p_L)u(w_L) + p_L u(w_H) - c_L$$

The firm therefore faces the following problem when trying to induce e_H

$$\begin{aligned} & \min (1 - p_H)w_L + p_H w_H \\ & \text{subject to} \\ & (1 - p_H)u(w_L) + p_H u(w_H) - c_H \geq \bar{U} \\ & (1 - p_H)u(w_L) + p_H u(w_H) - c_H \geq (1 - p_L)u(w_L) + p_L u(w_H) - c_L \end{aligned}$$

Why the IC and IR constraints bind with equality

1. If the (IC) fails to be an equality, we are transferring too much risk. We can make the two payments more similar and yet satisfy (IC). But then, we can pay a bit less in expectation because the agent is risk-averse and we are assigning less risk.
2. If the (IR) fails to be an equality, we are paying too much. We can pay a bit less in both scenarios, taking a bit more from the winning case, likely, to respect the (IC).

If the firm wants to induce e_L , we just set $w = \bar{U} + c_L$.

The problem is in deciding whether inducing e_L or e_H will give higher expected profits.

Therefore, the firm is always worse off and pays agency cost.

Definition 7.9 (Agency Cost). Agency cost, c , is the difference between the profits of the firm in the full-information case and in the moral-hazard case

Example. A risk-neutral principal can hire a risk-averse agent to undertake a project. There are two possible outcomes for the gross profit of the principal, $\pi_L < \pi_H$; assume throughout that $\pi_L = 60$. There are also two possible effort levels that the agent can exert, $e = 0$ or 1 ; if $e = 0$, the probability of π_H is only $2/5$, but if $e = 1$, the probability of π_H increases to $3/5$.

The agent's utility from receiving a wage w and exerting effort e is $\sqrt{w} - e$, and the agent has a reservation utility of 8 .

1. *Observable effort*

In this case, w can depend on e directly, and the contract will specify the effort level (0 or 1) and the wage to be paid by the principal to the agent. Let w_e denote the optimal wage associated with effort e .

- What are the w_0 and w_1 ?
- When $\pi_H = 195$, is the optimal contract $\{0, w_0\}$ or $\{1, w_1\}$?
- What about when $\pi_H = 165$?
- What if $\pi_H = 135$?

The principal's utility is given by $\pi_i - w_e$. The principal faces

$$\max \pi_i - w_e \quad \text{s.t.} \quad \sqrt{w} - e \geq 8$$

since it is decreasing in w_e , we know that the constraint binds with equality. So we consider two cases: when $e = 0$, $\sqrt{w} = 8 \implies w_0 = 64$. When $e = 1$, $\sqrt{w_e} = 9$, $w_e = 81$.

When $\pi_H = 195$, we have when $e = 0$, $\frac{2}{5}(195) + \frac{3}{5}(60) - 64 = 50$, and when $e = 1$, $\frac{3}{5}(195) + \frac{2}{5}(60) - 81 = 60$. So the optimal contract is $\{1, w_1\}$.

When $\pi_H = 165$, we have when $e = 0$, $\frac{2}{5}(165) + \frac{3}{5}(60) - 64 = 38$, and when $e = 1$, $\frac{3}{5}(165) + \frac{2}{5}(60) - 81 = 42$. The optimal contract is $\{1, w_1\}$.

When $\pi_H = 135$, we have when $e = 0$, $\frac{2}{5}(135) + \frac{3}{5}(60) - 64 = 26$ and when $e = 1$, $\frac{3}{5}(135) + \frac{2}{5}(60) - 81 = 24$. The optimal contract is $\{0, w_0\}$.

2. *Unobservable effort*

Now, the contract between the two parties cannot have w depend on e directly, but it can depend on the outcome π_j , $j = L, H$. Thus a contract will be of the form $\{w(\pi_L), w(\pi_H)\}$, where $w(\pi_j)$ is the wage paid by the principal to the agent if the gross profit is π_j .

- Explain why the optimal way to induce $e = 0$ is for the principal to offer the agent w_0 (from part (a)) regardless of the outcome π .
- By considering the Individual Rationality and Incentive Compatibility constraints, determine the optimal way to induce $e = 1$. What is the expected wage in this case?
- What is the optimal contract when $\pi_H = 195$?
- ...and when $\pi_H = 165$?
- ...and finally if $\pi_H = 135$?

This is because at $w_0 = 64$, the principal is offering just enough such that the worker will exert $e = 0$ because it is above his reservation utility. Giving him a higher wage at any of the states will just be giving more than what is needed, as the worker will just exert $e = 0$ regardless.

The IR constraint is

$$\frac{3}{5}\sqrt{w(\pi_H)} + \frac{2}{5}\sqrt{w(\pi_L)} - 1 \geq 8 \implies 3\sqrt{w(\pi_H)} + 2\sqrt{w(\pi_L)} \geq 45$$

The IC constraint is

$$\frac{3}{5}\sqrt{w(\pi_H)} + \frac{2}{5}\sqrt{w(\pi_L)} - 1 \geq \frac{2}{5}\sqrt{w(\pi_H)} + \frac{3}{5}\sqrt{w(\pi_L)} \implies \sqrt{w(\pi_H)} - \sqrt{w(\pi_L)} \geq 5$$

The principal's maximisation problem is

$$\max \frac{3}{5}(\pi_H - w(\pi_H)) + \frac{2}{5}(\pi_L - w(\pi_L)) \quad \text{s.t. IR and IC}$$

The solution must just be when IR and IC bind with equality since the firm's objective function is decreasing in wage. So, we have

$$\begin{aligned}\sqrt{w(\pi_H)} - \sqrt{w(\pi_L)} &= 5 \\ 3\sqrt{w(\pi_H)} + 2\sqrt{w(\pi_L)} &= 45\end{aligned}$$

Solving we have

$$\begin{aligned}3(5 + \sqrt{w(\pi_H)}) + 2\sqrt{w(\pi_L)} &= 45 \\ 15 + 5\sqrt{w(\pi_H)} &= 45 \\ w(\pi_L) &= 36 \\ w(\pi_H) &= 121\end{aligned}$$

The expected wage is

$$E(w) = \frac{2}{5}(36) + \frac{3}{5}(121) = 87$$

We either have contracts $\{64, 64\}_{e=0}$ or $\{36, 121\}_{e=1}$.

When $\pi_H = 195$,

$$\frac{3}{5}(165 - 121) + \frac{2}{5}(60 - 36) = 54 > \frac{2}{5}195 + \frac{3}{5}60 - 64 = 50$$

So, $\{36, 121\}_{e=1}$ is optimal

When $\pi_H = 165$,

$$\frac{3}{5}(165 - 121) + \frac{2}{5}(60 - 36) = 36 < \frac{2}{5}(165) + \frac{3}{5}(6) - 64 = 38$$

Optimal contract is $\{64, 64\}_{e=0}$.

When $\pi_H = 135$,

$$\frac{2}{5}(135) + \frac{3}{5}(60) - 64 = 26 > \frac{3}{5}(135 - 121) + \frac{2}{5}(60 - 36) = 18$$

So, the optimal contract is $\{64, 64\}_{e=0}$.

3. For each value of π_H that you have considered, compare the optimal contract in part (b) with that in part (a).

When $\pi_H = 195$, the optimal contract from (b) inducing $e = 1$ is more expensive than (a) because effort is not observable, so (b)'s contract is second best. When $\pi_H = 165$, the optimal contract from (b) is to not induce $e = 1$ but $e = 0$, while (a) is $e = 1$, likely because the cost of inducing $e = 1$ in (b) is much higher. When $\pi_H = 135$, (b) and (a) contracts are equivalent because they both induce 0 effort.

Example. A risk-neutral principal can hire a risk-averse agent to undertake a project. There are two possible outcomes for the gross profit of the principal, $\pi_L = 20$ and $\pi_H = 50$. There are also two possible effort levels that the agent can exert, $e = 0$ or 1 ; if $e = 0$, the probability of π_H is only $1/3$, but if $e = 1$, the probability of π_H increases $2/3$.

The agent's utility from receiving a wage w and is exerting effort e is $\sqrt{w} - e$, and the agent has a reservation utility of $\bar{u} = 2$

1. Assume that effort is observable.

What wage will the principal offer if she wants to induce low effort?

What wage will she offer if she wants to induce high effort?

What contract is optimal for the principal?

If $e = 0$, the IR is $\sqrt{w} \geq 2$. Since the objective function of the principal is decreasing in wage, she will offer $w = 4$. If $e = 1$, $\sqrt{w} - 1 \geq 2 \implies \sqrt{w} \geq 3$. For the same reason, $w = 9$. If $e = 1$, the payoff for the principal is $\frac{2}{3}(50) + \frac{1}{3}(20) - 9 = 31$. If $e = 0$, $\frac{1}{3}(50) + \frac{2}{3}(20) - 4 = 26$. So the optimal contract is $\{1, 31\}$.

2. (a) What is the optimal way for the principal to induce low effort?

- (b) What is the optimal way for the principal to induce high effort?

What is the expected wage in this case?

What is the agency cost?

- (c) What is the optimal contract?

a. To induce low effort, she will just offer $w = 4$ regardless of the state of profit. This is because it is compatible with the individual rationality constraint and the incentive compatibility constraint.

b. IR is $\frac{2}{3}\sqrt{w(\pi_H)} + \frac{1}{3}\sqrt{w(\pi_L)} - 1 \geq 2 \implies 2\sqrt{w(\pi_H)} + \sqrt{w(\pi_L)} \geq 9$. The IC is $\frac{2}{3}\sqrt{w(\pi_H)} + \frac{1}{3}\sqrt{w(\pi_L)} - 1 \geq \frac{1}{3}\sqrt{w(\pi_H)} + \frac{2}{3}\sqrt{w(\pi_L)}$. This eventuates into $\sqrt{w(\pi_H)} - \sqrt{w(\pi_L)} \geq 3$. Since IC and IR binds,

$$2(3 + \sqrt{w(\pi_H)}) + \sqrt{w(\pi_L)} = 9$$

this gives $w(\pi_L) = 1, w(\pi_H) = 16$. Expected wage is $\frac{2}{3} \times 16 + \frac{1}{3} = 11$. The agency cost is $11 - 9 = 2$.

- c. Consider $\{4, 4\}_{e=0}$,

$$\frac{1}{3}(50) + \frac{2}{3}(20) - 4 = 26$$

Consider $\{16, 1\}_{e=1}$,

$$\frac{2}{3}(50) + \frac{1}{3}(20) - 11 = 29$$

So, $\{16, 1\}_{e=1}$ is optimal.

3. Continue to assume that effort is unobservable, but suppose the two possible outcomes for the gross profit of the principal become $\pi_L = 24$ and $\pi_H = 42$. How does the optimal contract differ from that in part (2)?

Again consider $\{4, 4\}_{e=0}$,

$$\frac{1}{3}(42) + \frac{2}{3}(24) - 4 = 26$$

If $\{16, 1\}_{e=1}$,

$$\frac{2}{3}(42) + \frac{1}{3}(24) - 11 = 25$$

Optimal is $\{4, 4\}_{e=0}$. The optimal contract is now inducing no effort instead because the profit difference between the high and low states (adjusted for probability) does not justify the agency cost of inducing effort, in contrast to (2).

8 A Précis of Mathematics for Microeconomics

8.1 Basic Calculus

Definition A.1 (Derivative): For a function $f : \mathbb{R} \rightarrow \mathbb{R}$, the derivative of f at $x_0 \in \mathbb{R}$ is:

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

Definition A.2 (Differentiability): A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable if $f'(x)$ exists at any point $x \in \mathbb{R}$.

In other words, the graph of a differentiable function has a non-vertical tangent point at each interior point in its domain.

Theorem A.1 (First Order Taylor Approximation): If $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable, then for any $x_0 \in \mathbb{R}$,

$$f(x) = f(x_0) + (x - x_0)f'(x_0) + R_1(x, x_0)$$

where $\lim_{x \rightarrow x_0} \frac{R_1(x, x_0)}{x - x_0} = 0$.

Hence, the Taylor Polynomial of degree 1 of f around x_0 is

$$y = f(x_0) + (x - x_0)f'(x_0)$$

This is nothing else than the tangent line of f at x_0 .

Theorem A.2 (Second Order Taylor Approximation): If $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable, then for any $x \in \mathbb{R}$,

$$f(x) = f(x_0) + (x - x_0)f'(x_0) + \frac{1}{2!}(x - x_0)^2 f''(x_0) + R_2(x, x_0)$$

where $\lim_{x \rightarrow x_0} \frac{R_2(x, x_0)}{(x - x_0)^2} = 0$.

So, the Taylor Polynomial of degree 2 of f around x_0 is

$$y = f(x_0) + (x - x_0)f'(x_0) + \frac{1}{2!}(x - x_0)^2 f''(x_0)$$

Generally, the Taylor Expansion for any function is

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)$$

8.2 Sets and functions

Definition B.1 (Convex Sets): $S \subset \mathbb{R}^n$ is a convex set if for all $\mathbf{x}_1, \mathbf{x}_2 \in S$, we have

$$t\mathbf{x}_1 + (1 - t)\mathbf{x}_2 \in S$$

for all t in the interval $[0, 1]$.

Definition B.2 (Convexity and Concavity): Suppose that X is a convex subset of $\mathbb{R}^n$, $f : X \rightarrow \mathbb{R}$ is:

- convex if for all $x, y \in X$ and $t \in [0, 1]$,

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y)$$

- concave if for all $x, y \in X$ and $t \in [0, 1]$,

$$f(tx + (1 - t)y) \geq tf(x) + (1 - t)f(y)$$

Or equivalently, f is concave iff $-f$ is convex. Proof is obvious.

If the inequalities are strict, then we say that the functions are strictly concave or convex. Note that the functions need not be smooth/differentiable.

Definition B.3 (Local Convexity and Concavity for scalar-valued functions of one variable): Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function. Then,

- f is convex iff

$$f(y) \geq f(x) + f'(x)(y - x)$$

or equivalently,

$$f''(x) \geq 0, \forall x \in \mathbb{R}$$

- f is concave iff

$$f(y) \leq f(x) + f'(x)(y - x)$$

or equivalently,

$$f''(x) \leq 0, \forall x \in \mathbb{R}$$

Local concavity at a point, which means there is some neighbourhood containing x whereby the function is convex/concave, utilises the second definition for x_0 .

Definition B.4 (Local Convexity and Concavity for scalar-valued functions of two variables): Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a differentiable function. Then,

- f is (weakly) convex iff for all (x, y) :

$$f_{xx} \geq 0, f_{yy} \geq 0 \quad \text{and} \quad f_{xx}f_{yy} - f_{xy}^2 \geq 0$$

- f is (weakly) concave iff for all (x, y) :

$$f_{xx} \leq 0, f_{yy} \leq 0 \quad \text{and} \quad f_{xx}f_{yy} - f_{xy}^2 \geq 0$$

For strict convexity/concavity, we use strict inequalities.

Concavity at a point: If at (x_0, y_0) , $f_{xx} < 0$, $f_{yy} < 0$ and $f_{xx}f_{yy} - f_{xy}^2 > 0$, then f is concave at (x_0, y_0) .

Proposition B.1 (Properties of concave functions): Consider $f : \mathbb{R}^n \rightarrow \mathbb{R}, g : \mathbb{R}^n \rightarrow \mathbb{R}$ and $\alpha > 0$:

1. f is concave iff $-f$ is convex
2. If f is concave, αf is also concave
3. If f and g are concave, then $f + g$ is also concave
4. f is linear iff it is both concave and convex

Note: if a function is not concave, it does not mean that it is convex.

Note: it is easy to see that the above hold for convex functions as well by applying Proposition B.1.1 to each of them.

Definition B.5 (Cardinal Properties): A property $(*)$ is cardinal if whenever f has property $(*)$, and $\alpha > 0, \beta \in \mathbb{R}$ then $\alpha f + \beta$ also has property $(*)$. In other words, a property is cardinal if it is preserved by increasing linear transformations.

Proposition B.2: Concavity and convexity are cardinal properties.

Proof. Notice that β is linear, and by Proposition B.1.2 that αf retains convexity or concavity. Hence, by Proposition B.1.3, the property (*) is retained by increasing linear transformations. ■

Definition B.6 (Ordinal Properties): A property (*) is ordinal if whenever f has property (*), and $g : \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing, then $g \circ f : \mathbb{R}^n \rightarrow \mathbb{R}$ also has property (*). In other words, a property is ordinal if it is preserved by increasing transformations.

Non-example (Concavity): Concavity is not an ordinal property. Consider $f(x) = \sqrt{x}$ which is concave, and $g(y) = y^4$ which is an increasing function. Then $h(x) = g \circ f(x) = (\sqrt{x})^4 = x^2$ which is a convex function. So concavity is not preserved by increasing transformations.

Definition B.7 (Quasi-concave/Quasi-convex sets): Suppose that X is a convex subset of $\mathbb{R}^n$. A function $f : X \rightarrow \mathbb{R}$ is:

- quasi-concave if for all $x, y \in X$ for all $t \in [0, 1]$,

$$f(tx + (1 - t)y) \geq \min\{f(x), f(y)\}$$

- quasi-convex if for all $x, y \in X$ and all $t \in [0, 1]$,

$$f(tx + (1 - t)y) \leq \max\{f(x), f(y)\}$$

Strict quasi-convexity and concavity is obtained by replacing weak inequalities with strict inequalities.

Note: all concave functions are quasi-concave and all convex functions are quasi-convex but the converse does not hold.

Proposition B.3 (Properties of Quasi-concave functions): Consider $f : \mathbb{R}^n \rightarrow \mathbb{R}, g : \mathbb{R} \rightarrow \mathbb{R}$:

1. f is quasi-concave iff $-f$ is quasi-convex
2. If f is quasi-concave and h is increasing, $h \circ f$ is quasi-concave
3. All monotonic functions are quasi-concave and quasi-convex

Note: a function could be convex and quasi-concave e.g. $f(x) = e^x$. The sum of quasi-concave functions does not need to be quasi-concave.

Proposition B.4 (Characterisation of quasi-convexity/concavity): A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is:

- quasi-concave iff the upper contour set $\{x \in \mathbb{R}^n : f(x) \geq a\}$ is convex for all $a \in \mathbb{R}$
- quasi-convex iff the lower contour set $\{x \in \mathbb{R}^n : f(x) \leq a\}$ is convex for all $a \in \mathbb{R}$

This characterisation is useful because quasi-convexity/concavity is a global property that cannot be transformed into a local property.

Application: The utility function u is quasi-concave iff the indifference curves are convex.

8.3 Multivariate calculus and Optimisation

8.3.1 Multivariate Calculus

Definition C.1 (Partial Derivatives): Consider a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Then the derivative at the point $\mathbf{a}$ with respect to the i -th variable x_i is

$$\frac{\partial}{\partial x_i} f(\mathbf{a}) = \lim_{h \rightarrow 0} \frac{f(a_1, a_2, \dots, a_i + h, \dots, a_n) - f(a_1, \dots, a_i, \dots, a_n)}{h}$$

Theorem C.1 (Young's Theorem/Clairaut-Schwartz's Theorem:) If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ has second order partial derivatives,

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}) = \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{x})$$

Definition C.2 (Total Differential): Consider $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ with continuous partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$, then the total differential is given by:

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

8.3.2 Unconstrained Optimisation

Theorem C.2 (Unconstrained Optimisation for single-variable functions): Consider a function $f : \mathbb{R} \rightarrow \mathbb{R}$

- **First Order Condition:** If f has an optimal point at x_0 , then $f'(x_0) = 0$. Such a point x_0 is a critical point.
- **Second Order Conditions:** Suppose $f'(x_0) = 0$. Then
 1. If f is concave at x_0 i.e. $f''(x_0) < 0$, x_0 is a local maximum
 2. If f is convex at x_0 i.e. $f''(x_0) > 0$, x_0 is a local minimum
 3. If $f''(x_0) = 0$, then need to continue investigating (i.e. might be a saddle point)
- **Sufficient Conditions for Global Optima:**
 1. If f is concave (or strictly quasi-concave), and x_0 is a local maximum, then x_0 is a global maximum.

Proof (for the concave case). AFC that x_0 is a local maximum but not a global maximum, then there exists an x_1 s.t. $f(x_0) < f(x_1)$. But by the definition of concavity, for all $\alpha \in (0, 1)$:

$$f(\alpha x_0 + (1 - \alpha)x_1) \geq \alpha f(x_0) + (1 - \alpha)f(x_1) > f(x_0)$$

implying that x_0 is not a local maximum because there are values of $x_i \in U$, a neighbourhood around x_0 , s.t. $f(x_i) > f(x_0)$, contradicting our assumption. ■

2. If f is convex (or strictly quasi-convex), and x_0 is a local minimum, then x_0 is a global minimum.

Proof (for the convex case). AFC that x_0 is a local minimum but not a global minimum. Then there exists an x_1 s.t. $f(x_0) > f(x_1)$. But by the definition of convexity, for all $\alpha \in (0, 1)$:

$$f(\alpha x_0 + (1 - \alpha)x_1) \leq \alpha f(x_0) + (1 - \alpha)f(x_1) < f(x_0)$$

since $f(x_1) < f(x_0)$. This implies that there are values $x_i \in U$, a neighbourhood around x_0 , s.t. $f(x_i) < f(x_0)$, contradicting our assumption. ■

Theorem C.3 (Unconstrained Optimisation for two-variable functions): Consider a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$.

- **First Order Conditions:** If f has an optimal point at (x_0, y_0) , then

$$\frac{\partial f}{\partial x}(x_0, y_0) = 0, \quad \frac{\partial f}{\partial y}(x_0, y_0) = 0$$

where (x_0, y_0) is the critical point of f

- **Second Order Conditions:** A critical point (x_0, y_0) of f is

1. A local maximum if f is concave at (x_0, y_0) :

$$\frac{\partial^2 f}{\partial x^2}(x_0, y_0) < 0,$$

$$\frac{\partial^2 f}{\partial y^2}(x_0, y_0) < 0, \text{ and}$$

$$\frac{\partial^2 f}{\partial x^2}(x_0, y_0) \frac{\partial^2 f}{\partial y^2}(x_0, y_0) > \left(\frac{\partial^2 f}{\partial x \partial y}(x_0, y_0) \right)^2$$

2. A local minimum if f is convex at (x_0, y_0) :

$$\frac{\partial^2 f}{\partial x^2}(x_0, y_0) > 0,$$

$$\frac{\partial^2 f}{\partial y^2}(x_0, y_0) > 0, \text{ and}$$

$$\frac{\partial^2 f}{\partial x^2}(x_0, y_0) \frac{\partial^2 f}{\partial y^2}(x_0, y_0) > \left(\frac{\partial^2 f}{\partial x \partial y}(x_0, y_0) \right)^2$$

3. A saddle point if

$$\frac{\partial^2 f}{\partial x^2}(x_0, y_0) \frac{\partial^2 f}{\partial y^2}(x_0, y_0) < \left(\frac{\partial^2 f}{\partial x \partial y}(x_0, y_0) \right)^2$$

- **Sufficient Conditions for Global Optima:**

1. If f is strictly quasi-concave (or concave), and (x_0, y_0) is a local maximum, then (x_0, y_0) is a global maximum
2. If f is strictly quasi-convex (or convex), and (x_0, y_0) is a local minimum, then (x_0, y_0) is a global minimum

Theorem C.4 (Unconstrained Optimisation for n-variable functions): Consider a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$.

- **First Order Conditions:** If f has an optimal point at $\mathbf{a}$, then

$$\nabla f(\mathbf{a}) = \mathbf{0}$$

where $\nabla f(\mathbf{a})^T = (\frac{\partial f}{\partial x_1}(\mathbf{a}), \dots, \frac{\partial f}{\partial x_n}(\mathbf{a}))$

- **Second Order Conditions:**

1. If f is concave at a critical point $\mathbf{a}$, then $\mathbf{a}$ is a local maximum
2. If f is convex at a critical point $\mathbf{a}$, then $\mathbf{a}$ is a local minimum

Remark: The condition that f is concave at $\mathbf{a}$ is equivalent to the Hessian Matrix being negative semidefinite at $\mathbf{a}$; the condition that f is convex at $\mathbf{a}$ is equivalent to the Hessian Matrix being positive semidefinite at $\mathbf{a}$.

- **Global Conditions:**

1. If f is strictly quasi-concave (or concave) and $\mathbf{a}$ is a local maximum, then $\mathbf{a}$ is a global maximum
2. If f is strictly quasi-convex (or convex), and $\mathbf{a}$ is a local minimum, then $\mathbf{a}$ is a global minimum.

8.3.3 Constrained Optimisation

Theorem C.5 (Lagrange's Method): Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $g : \mathbb{R}^n \rightarrow \mathbb{R}$ are functions and consider the problem of finding a local maximum or minimum of f on the region where $g(\mathbf{x}) = c$. If a local extremum of f on the constraint region occurs at the point $\mathbf{a}$, then either

- $\nabla g(\mathbf{a}) = \mathbf{0}$, or
- $\nabla f(\mathbf{a}) - \lambda \nabla g(\mathbf{a})$, where λ is a constant that depends on $\mathbf{a}$

Remark. This is equivalent to maximising the following equation for a function f representing well-behaved preferences

$$\mathcal{L}(x, y, \lambda) = f(\mathbf{x}) + \lambda(c - g(\mathbf{x}))$$

where $f(\mathbf{x})$ is the objective function and $g(\mathbf{x}) = c$ is the constraint.

The Lagrangian can be generalised to when there are n variables and m constraints whereby we have the problem

$$\begin{aligned} \max f(\mathbf{x}) \\ \text{s.t. } \mathbf{g}(\mathbf{x}) = \mathbf{c} \end{aligned}$$

whereby $\mathbf{g}$ is a vector-valued constraint function i.e. $\mathbf{g}(\mathbf{x}) = (g_1(\mathbf{x}), g_2(\mathbf{x}), \dots, g_m(\mathbf{x}))$. Then our Lagrangian is

$$\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) + \lambda \cdot (\mathbf{c} - \mathbf{g}(\mathbf{x}))$$

where λ is a vector of Lagrangian multipliers i.e. $\lambda = (\lambda_1, \dots, \lambda_m)$.

Remark: The key insight of Lagrange's method is that for every constrained optimisation problem, there is a corresponding unconstrained optimisation problem that we can solve.

Theorem C.6 (Second Order Conditions): Suppose (x, y, λ) solve the FOCs. If f is quasi-concave, and g is quasi-convex, then (x, y) solves the constrained maximisation problem.

Theorem C.7 (Complementary Slackness Conditions): Suppose we face the maximisation problem (P)

$$\max \pi(q) \quad \text{s.t.} \quad q \geq 0$$

If q^* is a solution to (P), then

$$q^* \geq 0; \quad \pi'(q^*) \leq 0; \quad q^* \pi'(q^*) = 0$$

SOC are equivalent. If π is concave, then a solution to the Complementary Slackness Conditions is a global maximum.

We have an analogous set of conditions for the minimisation problem (P*). Suppose we face the minimisation problem (P*)

$$\min \pi(q) \quad \text{s.t.} \quad q \geq 0$$

If q^* is a solution to (P*), then

$$q^* \geq 0; \quad \pi'(q^*) \geq 0; \quad q^* \pi'(q^*) = 0$$

Example. Suppose Anna and Bob share a flat and they consume two types of goods: a private good x_i and a public good q , where $i \in \{a, b\}$. They have an income of 100 and face the same prices. Assume that the price of the goods are 1. Anna's preferences are given by $u_a(x_a, q) = 2\ln(x_a) + \ln(q)$, and Bob's preferences are given by $u_b(x_b, q) = \ln(x_b) + 2\ln(q)$. How much of the public good will each purchase? Anna's problem is

$$\begin{aligned} \max_{x_a, q_a} \quad & 2\ln(x_a) + \ln(q_a + q_b) \\ \text{s.t.} \quad & x_a + q_a = 100 \\ & q_a \geq 0 \end{aligned}$$

Analogously, Bob's problem is

$$\begin{aligned} \max_{x_b, q_b} \quad & \ln(x_b) + 2\ln(q_a + q_b) \\ \text{s.t.} \quad & x_b + q_b = 100 \\ & q_b \geq 0 \end{aligned}$$

We can transform the problem into a problem with a single variable by substituting the budget constraint. Anna's problem becomes

$$\begin{aligned} \max_{q_a} \quad & 2\ln(100 - q_a) + \ln(q_a + q_b) \\ \text{s.t.} \quad & q_a \geq 0 \end{aligned}$$

Bob's problem becomes

$$\begin{aligned} \max_{q_b} \quad & \ln(100 - q_b) + 2\ln(q_a + q_b) \\ \text{s.t.} \quad & q_b \geq 0 \end{aligned}$$

The complementary slackness conditions are:

1. For Anna:

$$q_a \geq 0; \quad -\frac{2}{100 - q_a} + \frac{1}{q_a + q_b} \leq 0; \quad q_a \left(-\frac{2}{100 - q_a} + \frac{1}{q_a + q_b}\right) = 0$$

2. For Bob:

$$q_b \geq 0; \quad -\frac{2}{100 - q_b} + \frac{1}{q_a + q_b} \leq 0; \quad q_b \left(-\frac{2}{100 - q_b} + \frac{1}{q_a + q_b}\right) = 0$$

We can check that both $q_a > 0$ and $q_b > 0$ and $q_a = q_b = 0$ are not solutions since they contradict the CS conditions. Further, $q_b = 0, q_a > 0$ is not a solution. The only solution is $q_a = 0, q_b > 0$ where $q_b = \frac{200}{3}$. ■

Theorem C.8 (Kuhn Tucker Conditions): Consider that we face the following problem (P**):

$$\begin{aligned} \max \quad & f(x, y) \\ \text{s.t.} \quad & g(x, y) = b \\ & x \geq 0 \\ & y \geq 0 \end{aligned}$$

This is just maximising our Langrangian

$$\mathcal{L}(x, y, \lambda) = f(x, y) + \lambda(b - g(x, y))$$

subject to $x, y \geq 0$.

Then, we yield the Kuhn Tucker Conditions: If (x, y) solves the problem (P**), then there exists λ such that

$$\begin{aligned} x \geq 0; \quad \mathcal{L}_x(x, y, \lambda) &\leq 0; \quad x\mathcal{L}_x(x, y, \lambda) = 0 \\ y \geq 0; \quad \mathcal{L}_y(x, y, \lambda) &\leq 0; \quad y\mathcal{L}_y(x, y, \lambda) = 0 \\ g(x, y) &= b \end{aligned}$$

Second Order Conditions are similar to ones with equality constraints; simply check that the Langrangian is concave/quasi-concave.

For minimisation problems

$$\begin{aligned} \min \quad & f(x, y) \\ \text{s.t.} \quad & g(x, y) = b \\ & x \geq 0 \\ & y \geq 0 \end{aligned}$$

If (x, y) solves the problem, then there exists λ such that

$$\begin{aligned} x \geq 0; \quad \mathcal{L}_x(x, y, \lambda) &\geq 0; \quad x\mathcal{L}_x(x, y, \lambda) = 0 \\ y \geq 0; \quad \mathcal{L}_y(x, y, \lambda) &\geq 0; \quad y\mathcal{L}_y(x, y, \lambda) = 0 \\ g(x, y) &= b \end{aligned}$$

SOC is that the Langrangian is convex or quasi-convex.

8.4 Applications and Duality

8.4.1 The Envelope Theorem

Theorem D.1 (The Envelope Theorem): Suppose we have an optimisation problem where the agent chooses a decision variable x to maximise an objective function $f(x, \theta)$ which depends on both x and θ . Formally, we have

$$V(\theta) = \max_x f(x, \theta)$$

If $x^*(\theta)$ is the optimal choice of x that maximises $f(x, \theta)$ for a given θ , then the derivative of the value function $V(\theta) = f(x^*(\theta), \theta)$ with respect to θ is given by

$$\frac{dV(\theta)}{d\theta} = \frac{df(x^*(\theta))}{d\theta} = \frac{\partial f}{\partial \theta}(x^*(\theta), \theta)$$

Proof. By using the total differential of $f(x^*(\theta), \theta)$, we have

$$df(x^*(\theta), \theta) = \frac{\partial f}{\partial x}(x^*(\theta), \theta)dx^*(\theta) + \frac{\partial f}{\partial \theta}(x^*(\theta), \theta)d\theta$$

By using a similar argument to Riemann in defining integrals, we treat $d\theta$ as an infinitesimally small quantity. So, dividing by $d\theta$, we yield:

$$\frac{df(x^*(\theta), \theta)}{d\theta} = \frac{\partial f}{\partial x}(x^*(\theta), \theta) \frac{dx^*(\theta)}{d\theta} + \frac{\partial f}{\partial \theta}(x^*(\theta), \theta)$$

The first term is known as the indirect effect through $x^*(\theta)$ and the second term is the direct effect. Since $x^*(\theta)$ satisfies the first order conditions by assumption, $\frac{\partial f}{\partial x}(x^*(\theta), \theta) = 0$. So, at the optimum, the indirect effect is zero and only the direct matters. Hence

$$\frac{dV(\theta)}{d\theta} = \frac{df(x^*(\theta))}{d\theta} = \frac{\partial f}{\partial \theta}(x^*(\theta), \theta)$$

■

The envelope theorem gives us two useful results in consumer and producer theory: Sheppard's Lemma and Roy's identity.

Sheppard's Lemma: The demand for a particular good i at a given utility level u and given prices $\mathbf{p}$ equals the derivative of the expenditure function with respect to the price of a given good:

$$h_i(\mathbf{p}, u) = \frac{\partial e(\mathbf{p}, u)}{\partial p_i}$$

Proof. Here

■

Roy's identity

8.4.2 The Duality Theorem

Definition D.1 (Marshallian Demand): Our Marshallian demand is derived from solving

$$v(\mathbf{p}, y) = \max_{\mathbf{x} \in \mathbb{R}_+^n} u(\mathbf{x}) \text{ s.t. } \mathbf{p} \cdot \mathbf{x} \leq y$$

The function $v(\mathbf{p}, y)$ is called our indirect utility function.

Definition D.2 (Hicksian Demand): Our Hicksian demand is derived from solving

$$e(\mathbf{p}, u) \equiv \min_{\mathbf{x} \in \mathbb{R}_+^n} \mathbf{p} \cdot \mathbf{x} \text{ s.t. } u(\mathbf{x}) \geq u$$

for all $\mathbf{p} \gg \mathbf{0}$, where $e(\mathbf{p}, u)$ is the expenditure function (as the minimum value function).

Assumption D.1 (Consumer Preferences): The consumer's preference relation $\succsim$ is complete, transitive, continuous, strictly monotonic, and strictly convex on $\mathbb{R}_+^n$. Therefore, it can be represented by a real-valued utility function u that is continuous, strictly increasing and strictly quasi-concave on $\mathbb{R}_+^n$.

Theorem D.2 (Duality between Marshallian and Hicksian Demand): Under Assumption D.1, we have the following relations between the Hicksian and Marshallian demand functions for $\mathbf{p} \gg \mathbf{0}, y \geq 0, u \in \mathcal{U}$ and $i = 1, \dots, n$:

1. $x_i(\mathbf{p}, y) = x_i^h(\mathbf{p}, v(\mathbf{p}, y))$
2. $x_i^h(\mathbf{h}, u) = x_i(\mathbf{p}, e(\mathbf{p}, u))$

Remark: (From Jehle and Reny): Relation 1 says that the Marshallian demand at prices $\mathbf{p}$ and income y equals the Hicksian demand at prices $\mathbf{p}$ and the utility level that is the maximum that can be achieved at prices $\mathbf{p}$ and income y . Relation 2 says that the Hicksian demand at any price $\mathbf{p}$ and utility level u is the same as the Marshallian demand at those prices and an income level equal to the minimum expenditure

necessary at those prices to achieve that utility level.

Roughly, Theorem D.2 says that the solutions to the problem in Definition D.1 are also solutions to the problem in Definition D.2. More precisely, if $\mathbf{x}^*$ solves (D.1) at $(\mathbf{p}, y)$, the theorem says that $\mathbf{x}^*$ solves (D.2) at $(\mathbf{p}, u)$, where $u = u(\mathbf{x}^*)$. Conversely, if $\mathbf{x}^*$ solves (D.2) at $(\mathbf{p}, u)$, then $\mathbf{x}^*$ solves (D.1) at $(\mathbf{p}, y)$ where $y = \mathbf{p} \cdot \mathbf{x}^*$.

8.5 Utility functions

Definition D.3 (Well-behaved Utilities): A utility function is well-behaved if it is continuous, increasing, and strictly quasi-concave.

This means that for the utility function u : (1) it has a solution (because it is continuous); (2) its FOCs are sufficient (strictly quasi-concave); (3) the budget constraint has to be exhausted i.e. no interior solutions (increasing in x_1, x_2); (4) only one solution (strictly quasi-concave).

Note: If a utility function u is well-behaved, this means that even when we face inequality constraints such as

$$v(\mathbf{p}, y) = \max_{\mathbf{x} \in \mathbb{R}_+^n} u(\mathbf{x}) \quad \text{s.t.} \quad \mathbf{p} \cdot \mathbf{x} \leq y$$

because the utility function is increasing in $x_1, \dots, x_n$, the maximum must bind when equality holds.

Types of Utility Functions

1. Cobb-Douglas Utilities:

$$u(x_1, x_2) = x_1^a x_2^b, \quad a > 0, b > 0$$

Properties:

- Continuous
- Increasing
- Strictly quasi-concave

2. Perfect Substitutes:

$$u(x_1, x_2) = ax_1 + bx_2, \quad a > 0, b > 0$$

Properties:

- Continuous
- Increasing
- Quasi-concave, since linear, but not strictly quasi-concave

3. Perfect Complements:

$$u(x_1, x_2) = \min\{ax_1, bx_2\}, \quad a > 0, b > 0$$

Properties:

- Continuous
- Increasing
- Quasi-concave
- Not differentiable

4. Constant Elasticity of Substitution:

$$u(x_1, x_2) = (ax_1^\rho + bx_2^\rho)^{\frac{1}{\rho}}, \quad a > 0, b > 0, \rho \leq 1, \rho \neq 0$$

Properties:

- Continuous

- Increasing
- Strictly quasi-concave for $\rho < 1$

This is actually the general form of the previous three:

- If $\rho = 1$, perfect substitutes
- As $\rho \rightarrow 0$, Cobb Douglas
- As $\rho \rightarrow -\infty$, perfect complements

5. Quasi-Linear Utilities:

$$u(x_1, x_2) = v(x_1) + x_2 \quad \text{with } v'(\cdot) > 0, v''(\cdot) < 0$$

Properties:

- Continuous
- Increasing
- Strictly quasi-concave

8.6 Elementary Probability

Definition E.1 (Probability Space and axioms): A probability space is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ where

1. Ω is a set, called the sample space
2. $\mathcal{F}$ is the power set (i.e. collection of subsets) of Ω , called events, satisfying the axioms:
 - (a) $\Omega \in \mathcal{F}$
 - (b) If $A \in \mathcal{F}$, then also $A^c \in \mathcal{F}$
 - (c) If $\{A_i, i \in I\}$ is a finite or countably infinite collection of members of $\mathcal{F}$, then $\bigcup_{i \in I} A_i \in \mathcal{F}$.
3. $\mathbb{P}$ is a probability measure, which is a function $\mathbb{P} : \mathcal{F} \rightarrow \mathbb{R}$ satisfying the axioms:
 - (a) For all $A \in \mathcal{F}$, $\mathbb{P}(A) \geq 0$
 - (b) $\mathbb{P}(\Omega) = 1$
 - (c) If $\{A_i, i \in I\}$ is a finite or countably infinite collection of members of $\mathcal{F}$, and $A_i \cap A_j = \emptyset$, for $i \neq j$, then $\mathbb{P}(\bigcup_{i \in I} A_i) = \sum_{i \in I} \mathbb{P}(A_i)$

Definition E.2 (Probability Mass Function): The probability mass function of a discrete r.v. X is the function $p_X = P(X = x)$. Note that this is positive if x is in the support of X and 0 otherwise.

Theorem E.1 (Validity of PMF): Let X be a discrete r.v. with support $x_1, x_2, \dots$. The PMF p_X of X must satisfy the following criteria:

- Non-negative: $p_X(x) > 0$ if $x = x_j$ for some j , and $p_X(x) = 0$ otherwise;
- Sums to 1: $\sum_{j=1}^{\infty} p_X(x_j) = 1$

Definition E.3 (Probability Density Function): The probability density function of a continuous r.v. X is the derivative f of the Cumulative Density Function of X , given by $f(x) = F'(x)$. The support of X , and of its distribution, is the set of all x where $f(x) > 0$.

Theorem E.2 (PDF to CDF): Let X be a continuous r.v. with PDF f . Then, the CDF of X is given by

$$F(x) = \int_{-\infty}^x f(t) dt$$

Hence, the PDF of X must integrate to 1.

Definition E.4 (Conditional Probability):

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

or the continuous version

$$f(A|B) = \frac{f(A \cap B)}{f(B)}$$

Theorem E.3 (Bayes Rule): Bayes Rule is given by

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B)}$$

or the continuous version

$$f(A|B) = \frac{f(B|A)f(A)}{f(B)}$$

Definition E.5 (Expectation): The expectation operator $E(X)$ is given by

- $E(X) = \sum_{-\infty}^{\infty} x\mathbb{P}(X = x)$ in the discrete case
- $E(X) = \int_{-\infty}^{\infty} xf(x)$ in the continuous case

Proposition E.1 (Properties of Expectation): For a r.v. X and Y , we have

1. Linearity: $E(X + Y) = E(X) + E(Y)$
2. Affinity: $E(aX + c) = aE(X) + c$

Theorem E.4 (Jensen's Inequality): Let X be a random variable. If g is a convex function, then $E(g(X)) \geq g(E(X))$. If g is a concave function, then $E(g(X)) \leq g(E(X))$. In both cases, the only way the equality can hold is if there are constants a, b s.t. $g(X) = a + bX$.

Proof. (From Blitzstein and Hwang) We proceed by geometric argument. If g is convex, then all lines that are tangent to g lie below g . In particular, let $\mu = E(X)$ and consider the tangent line at the point $(\mu, g(\mu))$. Denoting this line by $a + bx$ we have $g(x) \geq a + bx$ for all x by convexity, so $g(X) \geq a + bX$. Taking expectations of both sides

$$E(g(X)) \geq E(a + bX) = a + bE(X) = a + b\mu = g(\mu) = g(E(\mu))$$

as desired. If g is concave, then $h = -g$ is convex, so we can apply what we just proved to h to see that the inequality will be reversed (multiply the inequality by -1) from the convex case.

Lastly, assume that the inequality holds in the convex case. Let $Y = g(X) - a - bX$. Then Y is a non-negative r.v. with $E(Y) = 0$, so $\mathbb{P}(Y = 0) = 1$. So the equality holds iff $\mathbb{P}(g(X) = a + bX) = 1$. For the concave case, use the same argument with $Y = a + bX - g(X)$. ■

Theorem E.5 (Law of the Unconscious Statistician):

- **Discrete Case:** If X is a discrete r.v. and g is a function $\mathbb{R}$ to $\mathbb{R}$, then

$$E(g(X)) = \sum_x g(x)\mathbb{P}(X = x)$$

- **Continuous case:** If X is a continuous r.v. and g is a function $\mathbb{R}$ to $\mathbb{R}$, then

$$E(g(X)) = \int_{-\infty}^{\infty} g(x)f(x)dx$$

Definition E.6 (Variance): The variance of a r.v. X is given by

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = E[(X - E(X))^2]$$

Theorem E.5 (Law of Iterated Expectations/Adam's Law): For any r.v.s X, Y ,

$$E[E(X|Y)] = E(X)$$

Theorem E.6 (Eve's Law): For any r.v.s X, Y

$$\text{Var}(Y) = E(\text{Var}(Y|X)) + \text{Var}(E(Y|X))$$

Definition E.7 (Covariance): The Covariance between r.v.s X and Y is

$$\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))] = E(XY) - E(X)E(Y)$$

Theorem E.7 (Independence): If X and Y are independent, then they are uncorrelated.

Converse is false: let $X \sim \mathcal{N}(0, 1)$ and let $Y = X^2$. Then $E(XY) = E(X^3) = 0$ because the odd moments of the standard Normal distribution are equal to 0 by symmetry. Thus, X and Y are uncorrelated but are certainly not independent. ■

Proposition E.2 (Key Properties of Covariance):

1. $\text{Cov}(X, X) = \text{Var}(X)$
2. $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
3. $\text{Cov}(X, c) = 0$ for all c
4. $\text{Cov}(aX, Y) = a\text{Cov}(X, Y)$
5. $\text{Cov}(X + Y, Z + W) = \text{Cov}(X, Z) + \text{Cov}(X, W) + \text{Cov}(Y, Z) + \text{Cov}(Y, W)$
6. $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$.
7. For n r.v.s $X_1, \dots, X_n$,

$$\text{Var}(X_1 + \dots + X_n) = \text{Var}(X_1) + \dots + \text{Var}(X_n) + 2 \sum_{i < j} \text{Cov}(X_i, X_j)$$

9 Technical Appendix: Proofs of Theorems

Many of the proofs below rely on topology (Katukani fixed point theorem etc) and real analysis (convergence of subsequences etc). These proofs are not really meant to be understood in detail but are given for completeness and can be useful to be seen in full.

9.1 Proofs in General Equilibrium

Existence of Walrasian Equilibrium. Consider an exchange economy. Suppose that the consumers' preferences are continuous, strongly monotonic and convex and so satisfies Proposition 1.1 and Proposition 1.2, and that $\mathbf{w} \gg \mathbf{0}$. Then, there exists a price vector $\mathbf{p}^* \gg \mathbf{0}$ such that $\mathbf{z}(\mathbf{p}^*) \leq \mathbf{0}$.

Before proceeding with the proof, MWG specifies two further properties of aggregate excess demand to (i) - (iii) in **Proposition 1.2**: (iv) there is an $s > 0$ s.t. $z_k(\mathbf{p}) > -s$ for every commodity k and all p (there cannot be unlimited excess supply i.e. always some non-trivial level of demand); and (v) If $\mathbf{p}^n \rightarrow \mathbf{p}$, where $\mathbf{p} \neq \mathbf{0}$ and $p_k = 0$ for some k , then

$$\max\{z_1(\mathbf{p}^n), \dots, z_n(\mathbf{p}^n)\} \rightarrow \infty$$

MWG's version of the theorem is as follows (it merely adds these two conditions which are provable):

Theorem 9.1 (Existence of Walrasian Equilibrium (MWG)). Suppose that $\mathbf{z}(\mathbf{p})$ is a function defined for all strictly positive price vectors $\mathbf{p} \in \mathbb{R}_{++}^n$ and satisfying all five conditions of the aggregate excess demand function. Then, the system of equations $\mathbf{z}(\mathbf{p}) = \mathbf{0}$ has a solution. Hence, a Walrasian equilibrium exists in any pure exchange economy in which $\sum_{i \in \mathcal{I}} \mathbf{w}^i \gg \mathbf{0}$ and every consumer has continuous, strictly convex, and strongly monotone preferences

The proof is mainly an application of Kakutani's fixed point theorem in five steps. In the first two, construct a certain correspondence $\mathbf{f}(\cdot)$ from Δ to Δ . In the third, we argue that any fixed point of $\mathbf{f}(\cdot)$, that is, any $\mathbf{p}^*$ with $\mathbf{p}^* \in \mathbf{f}(\mathbf{p}^*)$, has $\mathbf{z}(\mathbf{p}^*) = \mathbf{0}$. The fourth step proves that $\mathbf{f}(\cdot)$ is convex valued and upper hemicontinuous (or equivalently, has a closed graph). Finally, the fifth step applies Kakutani's fixed point theorem to show that a $\mathbf{p}^*$ with $\mathbf{p}^* \in \mathbf{f}(\mathbf{p}^*)$ necessarily exists. For notational clarity, in defining the set $\mathbf{f}(\mathbf{p}) \subset \Delta$, MWG denote the vectors that are elements of $\mathbf{f}(\mathbf{p})$ by $\mathbf{q}$.

Proof. (From Mas-Colell, Whinston and Green)

Begin by normalising prices in a convenient way. We denote by

$$\Delta = \left\{ \mathbf{p} \in \mathbb{R}_{++}^n : \sum_k p_k = 1 \right\}$$

the unit simplex in $\mathbb{R}^n$. Because the function $\mathbf{z}(\cdot)$ is homogenous of degree 0, we can restrict ourselves, in our search for an equilibrium, to price vectors in Δ . Note however, that the function $\mathbf{z}(\cdot)$ is well defined only for price vectors in the set

$$\text{Int}\Delta = \{\mathbf{p} \in \Delta : p_k > 0 \text{ for all } k\}$$

Step 1: Construction of the fixed point correspondence for $\mathbf{p} \in \text{Int}\Delta$.

Whenever $\mathbf{p} \gg \mathbf{0}$, we let

$$\mathbf{f}(\mathbf{p}) = \{\mathbf{q} \in \Delta : \mathbf{z}(\mathbf{p}) \cdot \mathbf{q} \geq \mathbf{z}(\mathbf{p}) \cdot \mathbf{q}' \text{ for all } \mathbf{q}' \in \Delta\}$$

Basically, given the current proposal price $\mathbf{p}$ the counter-proposal assigned by the correspondence function $\mathbf{f}(\cdot)$ is any price vector $\mathbf{q}$ that, among the permissible price vectors, maximises the value of the excess demand vector. Intuitively, $\mathbf{f}(\cdot)$ is a rule that adjusts current prices in a direction that eliminates any excess demand, the correspondence $\mathbf{f}(\cdot)$ as defined above assigns the highest prices to the commodities with the highest excess demand. In particular, we have

$$\mathbf{f}(\mathbf{p}) = \{\mathbf{q} \in \Delta : \mathbf{q}_k = 0 \text{ if } z_k(\mathbf{p}) < \max\{z_1(\mathbf{p}), \dots, z_n(\mathbf{p})\}\}$$

Observe that if $\mathbf{z}(\mathbf{p}) \neq \mathbf{0}$ for $\mathbf{p} \gg \mathbf{0}$, then because of Walras' law, we have $z_k(\mathbf{p}) < 0$ for some k and $z_{k'}(\mathbf{p}) > 0$ for some $k' \neq k$. Thus, for such a $\mathbf{p}$, any $\mathbf{q} \in \mathbf{f}(\mathbf{p})$ has $q_k = 0$ for some k . Therefore, if $\mathbf{z}(\mathbf{p}) \neq \mathbf{0}$, then $\mathbf{f}(\mathbf{p}) \subset \text{Boundary}\Delta = \Delta \setminus \text{Int}\Delta$. In contrast, if $\mathbf{z}(\mathbf{p}) = \mathbf{0}$ then $\mathbf{f}(\mathbf{p}) = \Delta$.

Step 2: Construction of the fixed-point correspondence for $\mathbf{p} \in \text{Boundary}\Delta$.

If $\mathbf{p} \in \text{Boundary}\Delta$, we let

$$\mathbf{f}(\mathbf{p}) = \{\mathbf{q} \in \Delta : \mathbf{q} \cdot \mathbf{p} = 0\} = \{\mathbf{q} \in \Delta : q_k = 0 \text{ if } p_k > 0\}$$

Because $p_k = 0$ for some k , we have $\mathbf{f}(\mathbf{p}) \neq \emptyset$. Note also that with this construction, no price from $\text{Boundary}\Delta$ can be a fixed point; that is, $\mathbf{p} \in \text{Boundary}\Delta$ and $\mathbf{p} \in \mathbf{f}(\mathbf{p})$ cannot occur because $\mathbf{p} \cdot \mathbf{p} > 0$ while $\mathbf{p} \cdot \mathbf{q} = 0$ for all $\mathbf{q} \in \mathbf{f}(\mathbf{p})$.

Step 3: A fixed point $\mathbf{f}(\cdot)$ is an equilibrium.

Suppose that $\mathbf{p}^* \in \mathbf{f}(\mathbf{p}^*)$. As we pointed out in step 2, we must have $\mathbf{p}^* \notin \text{Boundary}\Delta$. Therefore, $\mathbf{p}^* \gg \mathbf{0}$. If $\mathbf{z}(\mathbf{p}^*) \neq \mathbf{0}$, then as in step 1 that $\mathbf{f}(\mathbf{p}^*) \subset \text{Boundary}\Delta$, which is incompatible with $\mathbf{p}^* \in \mathbf{f}(\mathbf{p}^*)$ and $\mathbf{p}^* \gg \mathbf{0}$. Hence, if $\mathbf{p}^* \in \mathbf{f}(\mathbf{p}^*)$ we must have $\mathbf{z}(\mathbf{p}^*) = \mathbf{0}$.

Step 4: The fixed-point correspondence is convex-valued, and upper hemicontinuous.

This is most of the work of the proof. To establish convex-valuedness, note that, both when $\mathbf{p}^* \in \text{Int}\Delta$ and when $\mathbf{p} \in \text{Boundary}\Delta$, $\mathbf{f}(\mathbf{p})$ equals a level set of a linear function defined on the convex set Δ [that is, a set of the form $\{\mathbf{q} \in \Delta : \lambda \cdot \mathbf{q} = k\}$ for some scalar k and vector $\lambda \in \mathbb{R}^n$], and so it is convex.

To establish upper hemicontinuity (i.e. has a closed graph), consider sequences $\mathbf{p}^m \rightarrow \mathbf{p}, \mathbf{q}^m \rightarrow \mathbf{q}$ with $\mathbf{q}^m \in \mathbf{f}(\mathbf{p}^m)$ for all m . We have to show that $\mathbf{q} \in \mathbf{f}(\mathbf{p})$. There are two cases: $\mathbf{p} \in \text{Int}\Delta$ and $\mathbf{p} \in \text{Boundary}\Delta$. If $\mathbf{p} \in \text{Int}\Delta$, then $\mathbf{p}^m \gg \mathbf{0}$ for m sufficiently large. From $\mathbf{q}^m \cdot \mathbf{z}(\mathbf{p}^m) \geq \mathbf{q}' \cdot \mathbf{z}(\mathbf{p}^m)$ for all $\mathbf{q}' \in \Delta$ and the continuity of $\mathbf{z}(\cdot)$, we get $\mathbf{q} \cdot \mathbf{z}(\mathbf{p}) \geq \mathbf{q}' \cdot \mathbf{z}(\mathbf{p})$ for all $\mathbf{q}'$. That is, $\mathbf{q} \in \mathbf{f}(\mathbf{p})$.

Now suppose that $\mathbf{p} \in \text{Boundary}\Delta$. Take any k with $p_k > 0$. We shall argue that for m sufficiently large, we have $q_k^m = 0$ and therefore it must be that $q_k = 0$; from this, $\mathbf{q} \in \mathbf{f}(\mathbf{p})$ follows. Because $p_k > 0$, there is an $\varepsilon > 0$ such that $p_k^m > \varepsilon$ for m sufficiently large. If in addition $\mathbf{p}^m \in \text{Boundary}\Delta$ then $q_k^m = 0$ by the definition of $\mathbf{f}(\mathbf{p}^m)$. If instead $\mathbf{p}^m \gg \mathbf{0}$, then the boundary conditions (iv) and (v) of aggregate excess demand come into play. They imply that for m sufficiently large, we must have

$$z_k(\mathbf{p}^m) < \max\{z_1(\mathbf{p}^m), \dots, z_n(\mathbf{p}^m)\}$$

and therefore that again $q_k^m = 0$. To prove the above inequality, note that by condition (v) the right hand side of the above expression goes to infinity with m because $\mathbf{p} \in \text{Boundary}\Delta$, some prices go to zero as $m \rightarrow \infty$. But the left hand side is bounded above because if it is positive, then

$$z_k(\mathbf{p}^m) \leq \frac{1}{\varepsilon} p_k^m z_k(\mathbf{p}^m) = -\frac{1}{\varepsilon} \sum_{k' \neq k} p_{k'}^m z_{k'}(\mathbf{p}^m) < \frac{s}{\varepsilon} \sum_{k' \neq k} p_{k'}^m < \frac{s}{\varepsilon}$$

where s is the bound in excess supply given by condition (iv). In summary, for $\mathbf{p}^m$ close enough to $\text{Boundary}\Delta$, the maximal demand corresponds to some of the commodities whose price is close to zero. Therefore, we conclude that, for large m , any $\mathbf{q}^m \in \mathbf{f}(\mathbf{p}^m)$ will put nonzero weight only on commodities whose prices approach zero. But this guarantees $\mathbf{p} \cdot \mathbf{q} = 0$, and so $\mathbf{q} \in \mathbf{f}(\mathbf{p})$.

Step 5: A fixed point exists.

Kakutani's fixed point theorem says that a convex-valued, upper hemicontinuous correspondence from a nonempty, compact, convex set into itself has a fixed point. Since Δ is a nonempty, compact, and convex set, and since $\mathbf{f}(\cdot)$ is a convex-valued upperhemicontinuous correspondence from Δ to Δ , we conclude that there is a $\mathbf{p}^* \in \Delta$ with $\mathbf{p}^* \in \mathbf{f}(\mathbf{p}^*)$ ■

9.2 Proofs in Welfare Economics and Social Choice Theory

Theorem 9.2 (Arrow's Impossibility Theorem). If there are at least three social states in X , then there is no social welfare function f that simultaneously satisfies conditions U , WP , IIA and D .

The strategy of the proof is to show that first three conditions imply not-D. Note that axiom U, the unrestricted domain axiom, is used in each step whenever we choose or alter preferences profile under consideration, since it ensures that every such profile of preferences is admissible.

Proof. (Geanakoplos, 1996) **Step 1:** Consider any social state, c . Suppose each individual places state c at the bottom of his ranking. By WP, the social ranking must place c at the bottom as well.

Step 2: Imagine now moving c to the top of individual 1's ranking, leaving the ranking of all other states unchanged. Next, do the same with individual 2: move c to the top of 2's ranking. Continue doing this one individual at a time, keeping in mind that each of these changes in individual preferences might have an effect on the social ranking. Eventually, c will be the top of every individual's ranking and so must be the top of the social ranking by WP. Consequently, there must be a first time during this process that the social ranking of c increase. Let individual n be the first such that raising c to the top of his ranking causes the social ranking of c to increase.

We claim that when c moves to the top of individual n 's ranking, the social ranking of c not only increases but c also moves to the top of the social ranking. To see this, assume for contradiction that the social ranking of c increases, but not to the top; i.e. αRc and $cR\beta$ for some states $\alpha, \beta \neq c$.

Now, because c is either at the bottom or at the top of every individual's ranking, we can change each individual i 's preferences so that $\beta \succ^i \alpha$, while leaving the position c unchanged for that individual. But this produces our desired contradiction, because, on the one hand, $\beta \succ^i \alpha$ for every individual implies by WP that β must be strictly preferred to α according to the social ranking; i.e. $\beta R\alpha$. But on the other hand, because the rankings of c relative to α and β have not changed in any individual's ranking (since c is either at the top or bottom), IIA implies that the social rankings of c relative to α and β must be unchanged i.e. αRc and βRc . By transitivity, this implies $\alpha R\beta$ which contradicts $\beta R\alpha$ established earlier. So c must move to the top of the social rankings.

R^1	R^2	$\dots$	R^n	$\dots$	R^N	R
c	c	$\dots$	c	$\dots$	x''	c
x	x'	$\dots$		$\dots$	y''	
y	y'				.	.
.	.				.	.
.	.				.	.
.	.				.	.
w	w'	$\dots$		$\dots$	c	w'''

Fig 9.1. Axioms WP, U, and IIA yield a pivotal individual

Step 3: Consider now any two distinct social states $a, b \neq c$. In Fig 9.1, change the profile of preferences as follows: change individual n 's ranking so that $a \succ^n c \succ^n b$ and for every other individual rank a and b in any way so long as the position of c is unchanged for that individual. Let us call this profile 1. Note that, in the new profile of preferences the ranking of a to c is the same for every individual as it was just before raising c to the top of individual n 's ranking in Step 2. Therefore, by IIA, the social ranking of a and c must be the same as it was at that moment, since individual n in profile 1 ranks c below a i.e. he did not change his relative ranking of c and a . So this means that $a Rc$ in profile 1 because at the moment, c was still at the bottom of the social ranking. We can apply IIA because in both profiles (profile 1 and the profile right before applying Step 2) the relative individual rankings of a and c remained unchanged for everyone.

Similarly, in the new profile of preferences (profile 1), the ranking of c to b is the same for every individual as it was just after raising c to the top of individual n 's ranking in Step 2 (including the n -th individual's own rankings of c and b). Therefore, by IIA, the social ranking of c and b must be the same as it was in the profile right after applying step 2. But this means that cRb because right after applying step 2, c had just risen to the top of the social ranking. Again, we can apply IIA to conclude that cRb in profile 1 because the relative individual rankings of b and c remained unchanged across profiles.

So, because $a Rc$ and cRb , we may conclude that aRb . Note that no matter how others rank a and b , the

social ranking agrees with individual n 's ranking. By IIA, and because a and b were arbitrary, we may therefore conclude that for all social states $a, b \neq c$ that

$$a \succ^n b \implies aRb$$

That is, individual n is a dictator on all pairs of social states not involving c .

Step 4: This step shows that individual n is in fact a dictator. Let a be distinct from c . We may repeat the above steps with a playing the role of c to conclude that some individual is a dictator on all pairs not involving a . However, recall that individual n 's ranking of c (bottom or top) in Fig 9.1 affects the social ranking of c (bottom or top). Hence, it must be individual n who is the dictator on all pairs not involving a . Because a was an arbitrary state distinct from c , and together with our previous conclusion about individual n , we conclude that there is an individual i such that for all x, y in X , $x \succ^i y$ implies xRy regardless of the preferences P^j of all other individuals $j \neq i$. So, individual n is a dictator. ■

9.3 Proofs in Applied Welfare

Proposition 9.1. The Clarke-Groves Mechanism is strategy-proof.

Proof. Here ■

9.4 Proofs in Game Theory

Theorem (Existence of Nash Equilibrium): A Nash equilibrium exists in the game $\mathcal{G}_N = \langle I, \{S_i\}, \{u_i(\cdot)\} \rangle$ if for all $i = 1, \dots, I$: (i) S_i is a nonempty, convex, and compact subset of some Euclidean space $\mathbb{R}^M$, and (ii) $u_i(s_1, \dots, s_I)$ is continuous in $s_1, \dots, s_I$ and quasiconcave in s_i .

Mas-Colell, Whinston and Green's proof relies on the following Lemma:

Lemma: If the sets $S_1, \dots, S_I$ are nonempty, S_i is compact and convex, and $u_i(\cdot)$ is continuous in $(s_1, \dots, s_I)$ and quasiconcave in s_i , then player i 's best-response correspondence $b_i(\cdot)$ is nonempty, convex-valued, and upper hemicontinuous.

Proof. Note first that $b_i(s_{-i})$ is the set of maximisers of the continuous function $u_i(\cdot, s_{-i})$ on the compact set S_i . Hence, it is nonempty. The convexity of $b_i(s_{-i})$ follows because the set of maximisers of a quasiconcave function (here, the function $u_i(\cdot, s_{-i})$) on a convex set (S_i) is convex. Finally, for upper hemicontinuity, we need to show that for any sequence $(s_i^n, s_{-i}^n) \rightarrow (s_i, s_{-i})$ such that $s_i^n \in b_i(s_{-i}^n)$ for all n , we have $s_i \in b_i(s_{-i})$. To see this, note that for all n , $u_i(s_i^n, s_{-i}^n) \geq u_i(s_i', s_{-i}^n)$ for all $s_i' \in S_i$. Therefore, by the continuity of $u_i(\cdot)$, we have $u_i(s_i, s_{-i}) \geq u_i(s_i', s_{-i})$. ■

Now we turn to the actual proof.

Proof. Define the correspondence $b : S \rightarrow S$ by

$$b(s_1, \dots, s_I) = b_1(s_{-1}) \times \dots \times b_I(s_{-I})$$

Note that $b(\cdot)$ is a correspondence from the nonempty, convex and compact set $S = S_1 \times \dots \times S_I$ to itself. By the above Lemma, $b(\cdot)$ is a nonempty convex-valued, and upper hemicontinuous correspondence. Thus, all the conditions of the Kakutani fixed point theorem are satisfied. Hence, there exists a fixed point for this correspondence, a strategy profile $s \in S$ such that $s \in b(s)$. The strategies at this fixed point constitute a Nash equilibrium because by construction, $s_i \in b_i(s_{-i})$ for all $i = 1, \dots, I$. ■

Theorem (Existence of mixed strategy Nash Equilibrium): Every game $\mathcal{G}_N = \langle I, \{\Delta(S_i)\}, \{u_i(\cdot)\} \rangle$ in which the sets $S_1, \dots, S_I$ have a finite number of elements has a mixed strategy Nash equilibrium.

Proof. The game $\mathcal{G}_N = \langle I, \{\Delta(S_i)\}, \{u_i(\cdot)\} \rangle$, viewed as a game with strategy sets $\{\Delta(S_i)\}$ and payoff functions $u_i(\sigma_1, \dots, \sigma_I) = \sum_{s \in S} (\prod_{k=1}^I \sigma_k(s_k)) u_i(s)$ for all $i = 1, \dots, I$ satisfies all the assumptions of the above theorem. Hence, this is a direct corollary of that result. ■

10 Mathematical Theorems for proofs

Brouwer's Fixed Point Theorem: For any continuous function f mapping a nonempty convex set to itself, there exists a point x_0 such that $f(x_0) = x_0$.

Applications: Game Theory, General Equilibrium.

Kakutani Fixed point theorem: Let S be a nonempty, compact and convex subset of the Euclidean space $\mathbb{R}^n$. Let $\phi : S \rightarrow 2^S$ be a set-valued function on S with the following properties:

- ϕ has a closed graph (is upper hemicontinuous)
- $\phi(x)$ is nonempty and convex for all $x \in S$

Then ϕ has a fixed point.

Applications: General Equilibrium.

Hyperplane Separation Theorem: Let A and B be two disjoint nonempty convex subsets of $\mathbb{R}^n$. Then there exists a nonzero vector v and a real number c such that

$$\langle x, v \rangle \geq c \text{ and } \langle y, v \rangle \leq c$$

for all x in A and y in B i.e. the hyperplane $\langle \cdot, v \rangle = c$ with the normal vector v separates A and B .

If both sets are closed, then the separation can be strict, such that

$$\langle x, v \rangle > c_1 \text{ and } \langle y, v \rangle < c_2$$

where $c_1 > c_2$.

Applications: Second Fundamental Welfare Theorem.